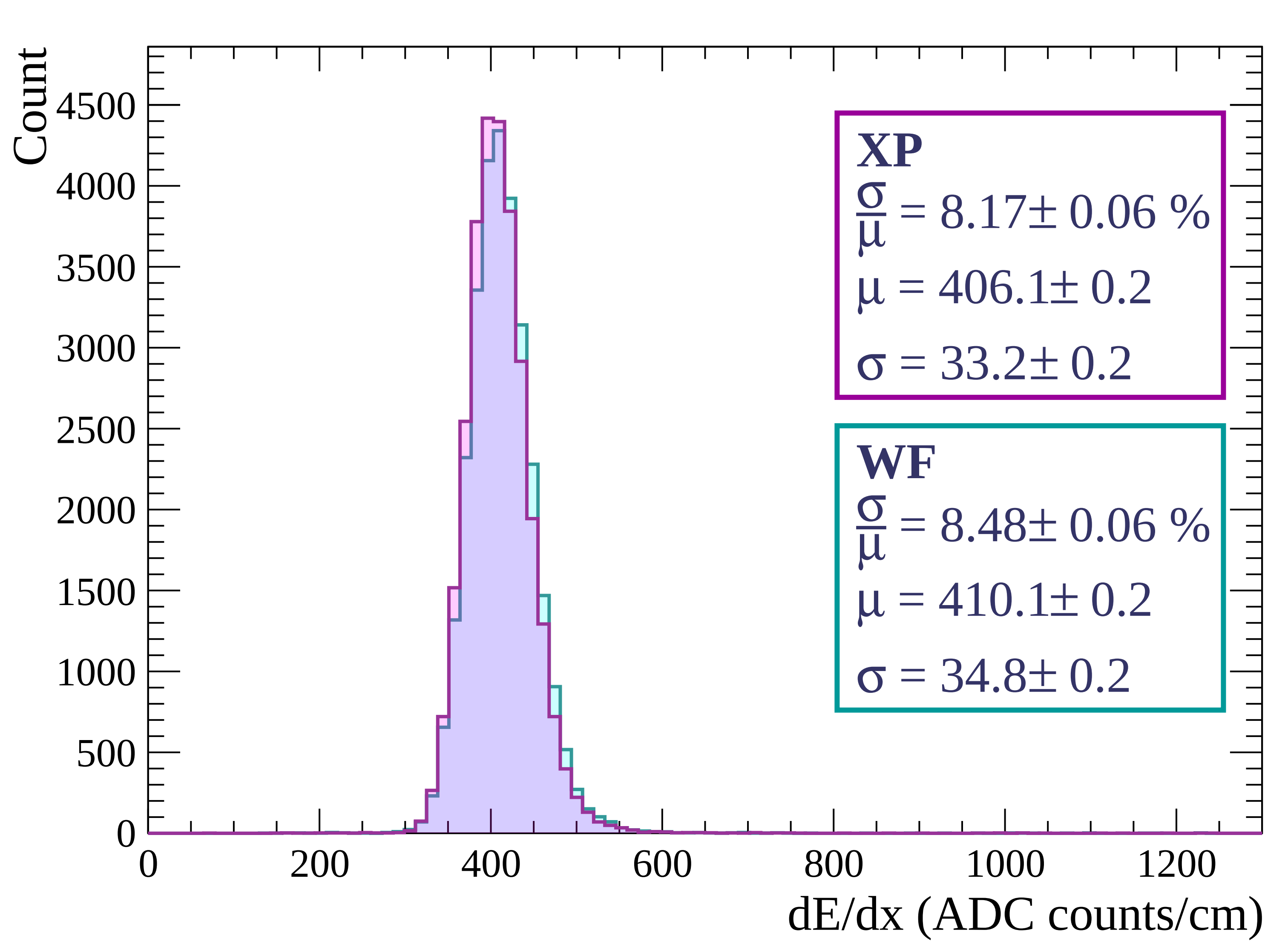
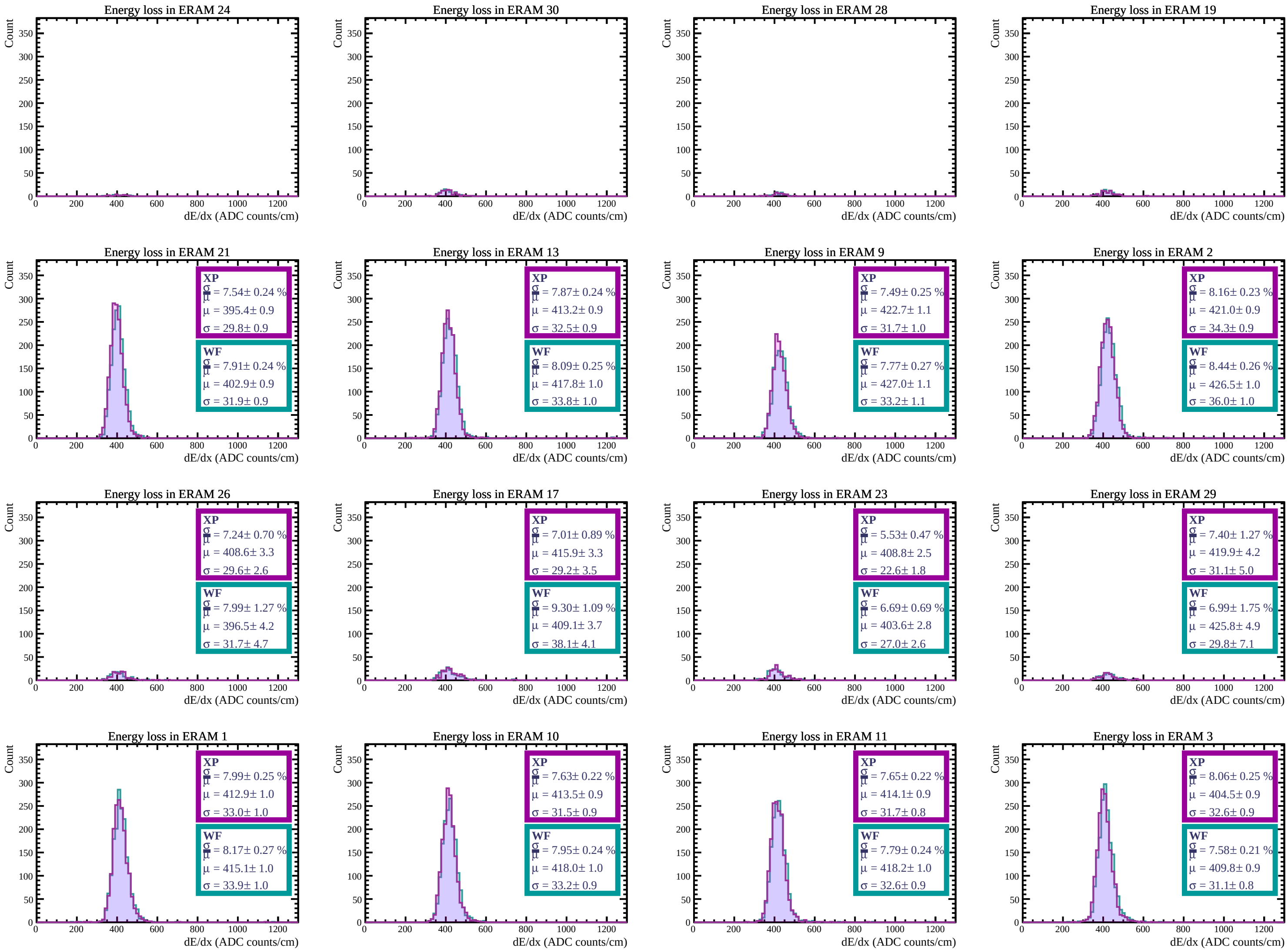
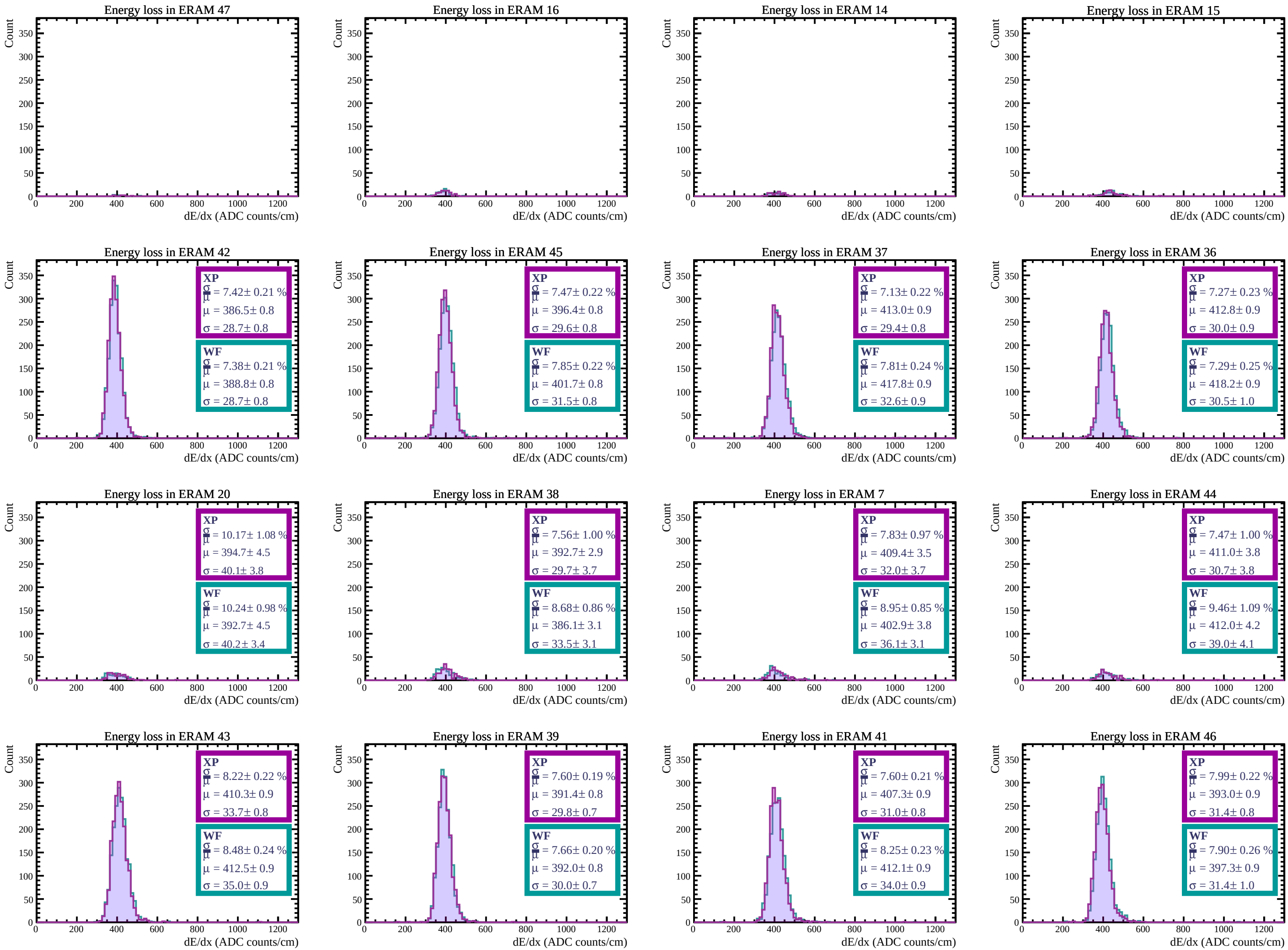
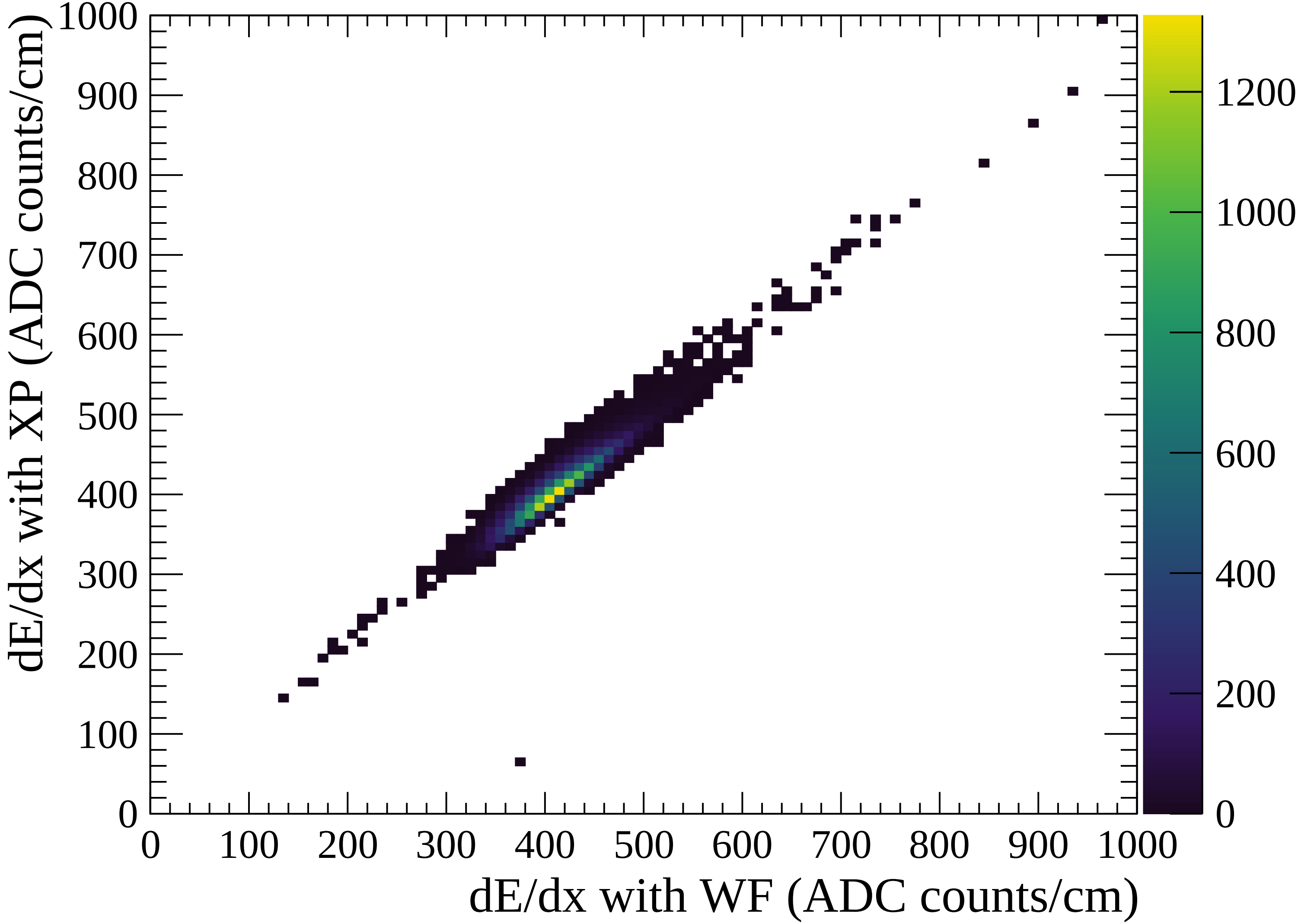
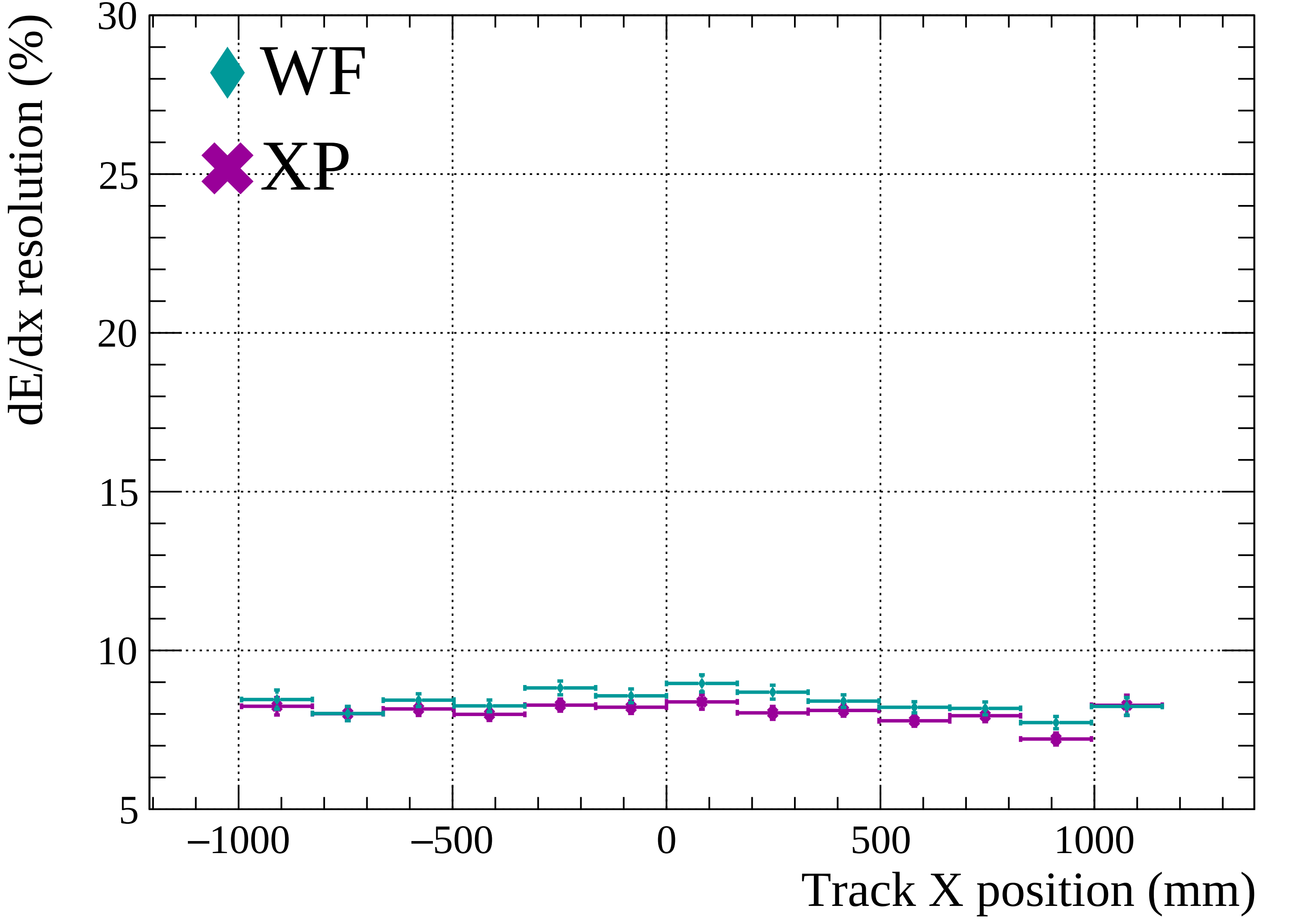

Count
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.17± 0.06 %
μ = 406.1± 0.2
σ = 33.2± 0.2
WF
$\frac{\sigma}{\mu}$ = 8.48± 0.06 %
μ = 410.1± 0.2
σ = 34.8± 0.2


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
XP
σ/μ = 7.54± 0.24 %
μ = 395.4± 0.9
σ = 29.8± 0.9
WF
σ/μ = 7.91± 0.24 %
μ = 402.9± 0.9
σ = 31.9± 0.9
Energy loss in ERAM 13
XP
σ/μ = 7.87± 0.24 %
μ = 413.2± 0.9
σ = 32.5± 0.9
WF
σ/μ = 8.09± 0.25 %
μ = 417.8± 1.0
σ = 33.8± 1.0
Energy loss in ERAM 9
XP
σ/μ = 7.49± 0.25 %
μ = 422.7± 1.1
σ = 31.7± 1.0
WF
σ/μ = 7.77± 0.27 %
μ = 427.0± 1.1
σ = 33.2± 1.1
Energy loss in ERAM 2
XP
σ/μ = 8.16± 0.23 %
μ = 421.0± 0.9
σ = 34.3± 0.9
WF
σ/μ = 8.44± 0.26 %
μ = 426.5± 1.0
σ = 36.0± 1.0
Energy loss in ERAM 26
XP
σ/μ = 7.24± 0.70 %
μ = 408.6± 3.3
σ = 29.6± 2.6
WF
σ/μ = 7.99± 1.27 %
μ = 396.5± 4.2
σ = 31.7± 4.7
Energy loss in ERAM 17
XP
σ/μ = 7.01± 0.89 %
μ = 415.9± 3.3
σ = 29.2± 3.5
WF
σ/μ = 9.30± 1.09 %
μ = 409.1± 3.7
σ = 38.1± 4.1
Energy loss in ERAM 23
XP
σ/μ = 5.53± 0.47 %
μ = 408.8± 2.5
σ = 22.6± 1.8
WF
σ/μ = 6.69± 0.69 %
μ = 403.6± 2.8
σ = 27.0± 2.6
Energy loss in ERAM 29
XP
σ/μ = 7.40± 1.27 %
μ = 419.9± 4.2
σ = 31.1± 5.0
WF
σ/μ = 6.99± 1.75 %
μ = 425.8± 4.9
σ = 29.8± 7.1
Energy loss in ERAM 1
XP
σ/μ = 7.99± 0.25 %
μ = 412.9± 1.0
σ = 33.0± 1.0
WF
σ/μ = 8.17± 0.27 %
μ = 415.1± 1.0
σ = 33.9± 1.0
Energy loss in ERAM 10
XP
σ/μ = 7.63± 0.22 %
μ = 413.5± 0.9
σ = 31.5± 0.9
WF
σ/μ = 7.95± 0.24 %
μ = 418.0± 1.0
σ = 33.2± 0.9
Energy loss in ERAM 11
XP
σ/μ = 7.65± 0.22 %
μ = 414.1± 0.9
σ = 31.7± 0.8
WF
σ/μ = 7.79± 0.24 %
μ = 418.2± 1.0
σ = 32.6± 0.9
Energy loss in ERAM 3
XP
σ/μ = 8.06± 0.25 %
μ = 404.5± 0.9
σ = 32.6± 0.9
WF
σ/μ = 7.58± 0.21 %
μ = 409.8± 0.9
σ = 31.1± 0.8
Count
dE/dx (ADC counts/cm)

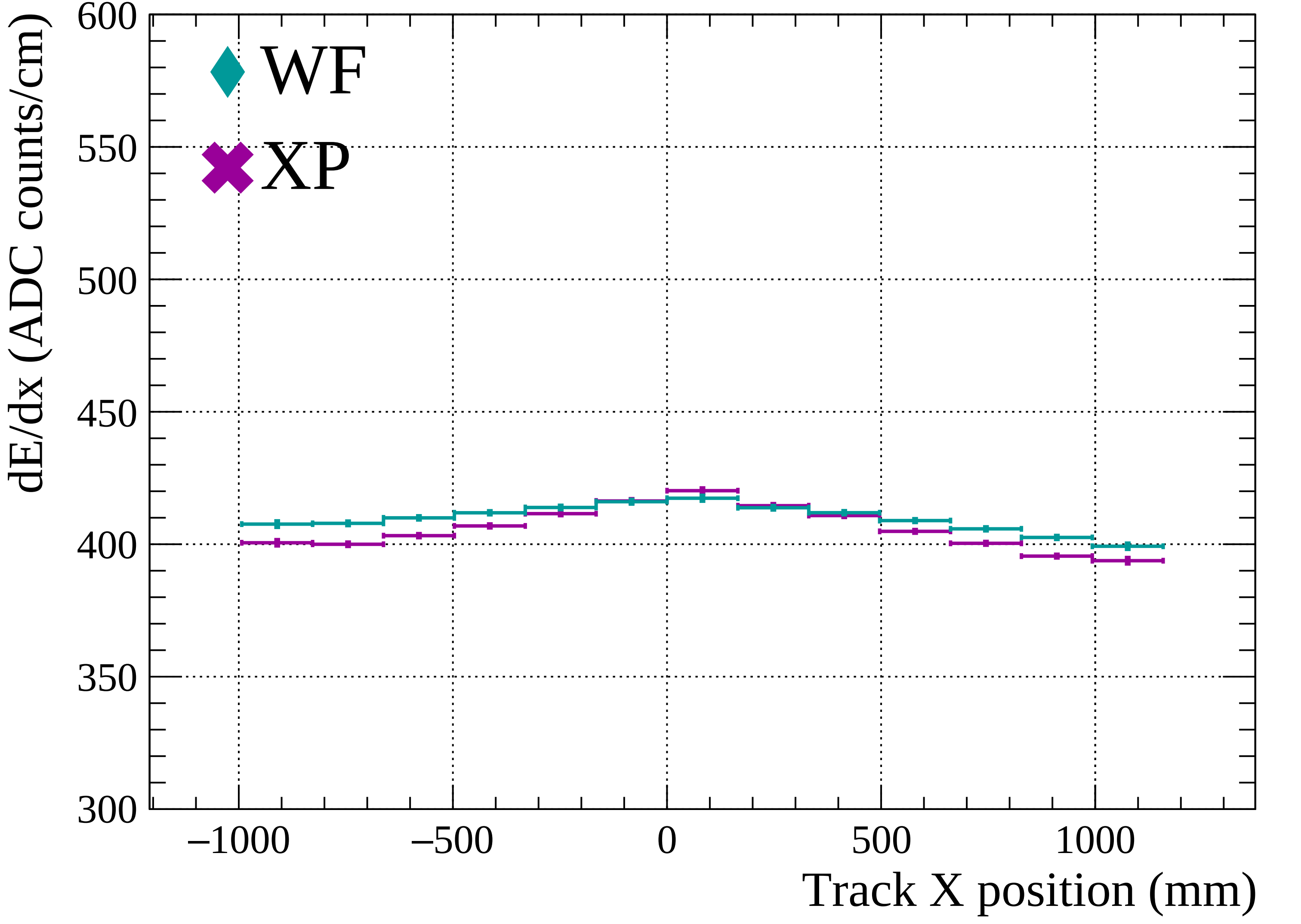

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
XP
σ/μ = 7.42± 0.21 %
μ = 386.5± 0.8
σ = 28.7± 0.8
WF
σ/μ = 7.38± 0.21 %
μ = 388.8± 0.8
σ = 28.7± 0.8
Energy loss in ERAM 45
XP
σ/μ = 7.47± 0.22 %
μ = 396.4± 0.8
σ = 29.6± 0.8
WF
σ/μ = 7.85± 0.22 %
μ = 401.7± 0.8
σ = 31.5± 0.8
Energy loss in ERAM 37
XP
σ/μ = 7.13± 0.22 %
μ = 413.0± 0.9
σ = 29.4± 0.8
WF
σ/μ = 7.81± 0.24 %
μ = 417.8± 0.9
σ = 32.6± 0.9
Energy loss in ERAM 36
XP
σ/μ = 7.27± 0.23 %
μ = 412.8± 0.9
σ = 30.0± 0.9
WF
σ/μ = 7.29± 0.25 %
μ = 418.2± 0.9
σ = 30.5± 1.0
Energy loss in ERAM 20
XP
σ/μ = 10.17± 1.08 %
μ = 394.7± 4.5
σ = 40.1± 3.8
WF
σ/μ = 10.24± 0.98 %
μ = 392.7± 4.5
σ = 40.2± 3.4
Energy loss in ERAM 38
XP
σ/μ = 7.56± 1.00 %
μ = 392.7± 2.9
σ = 29.7± 3.7
WF
σ/μ = 8.68± 0.86 %
μ = 386.1± 3.1
σ = 33.5± 3.1
Energy loss in ERAM 7
XP
σ/μ = 7.83± 0.97 %
μ = 409.4± 3.5
σ = 32.0± 3.7
WF
σ/μ = 8.95± 0.85 %
μ = 402.9± 3.8
σ = 36.1± 3.1
Energy loss in ERAM 44
XP
σ/μ = 7.47± 1.00 %
μ = 411.0± 3.8
σ = 30.7± 3.8
WF
σ/μ = 9.46± 1.09 %
μ = 412.0± 4.2
σ = 39.0± 4.1
Energy loss in ERAM 43
XP
σ/μ = 8.22± 0.22 %
μ = 410.3± 0.9
σ = 33.7± 0.8
WF
σ/μ = 8.48± 0.24 %
μ = 412.5± 0.9
σ = 35.0± 0.9
Energy loss in ERAM 39
XP
σ/μ = 7.60± 0.19 %
μ = 391.4± 0.8
σ = 29.8± 0.7
WF
σ/μ = 7.66± 0.20 %
μ = 392.0± 0.8
σ = 30.0± 0.7
Energy loss in ERAM 41
XP
σ/μ = 7.60± 0.21 %
μ = 407.3± 0.9
σ = 31.0± 0.8
WF
σ/μ = 8.25± 0.23 %
μ = 412.1± 0.9
σ = 34.0± 0.9
Energy loss in ERAM 46
XP
σ/μ = 7.99± 0.22 %
μ = 393.0± 0.9
σ = 31.4± 0.8
WF
σ/μ = 7.90± 0.26 %
μ = 397.3± 0.9
σ = 31.4± 1.0
Count
dE/dx (ADC counts/cm)

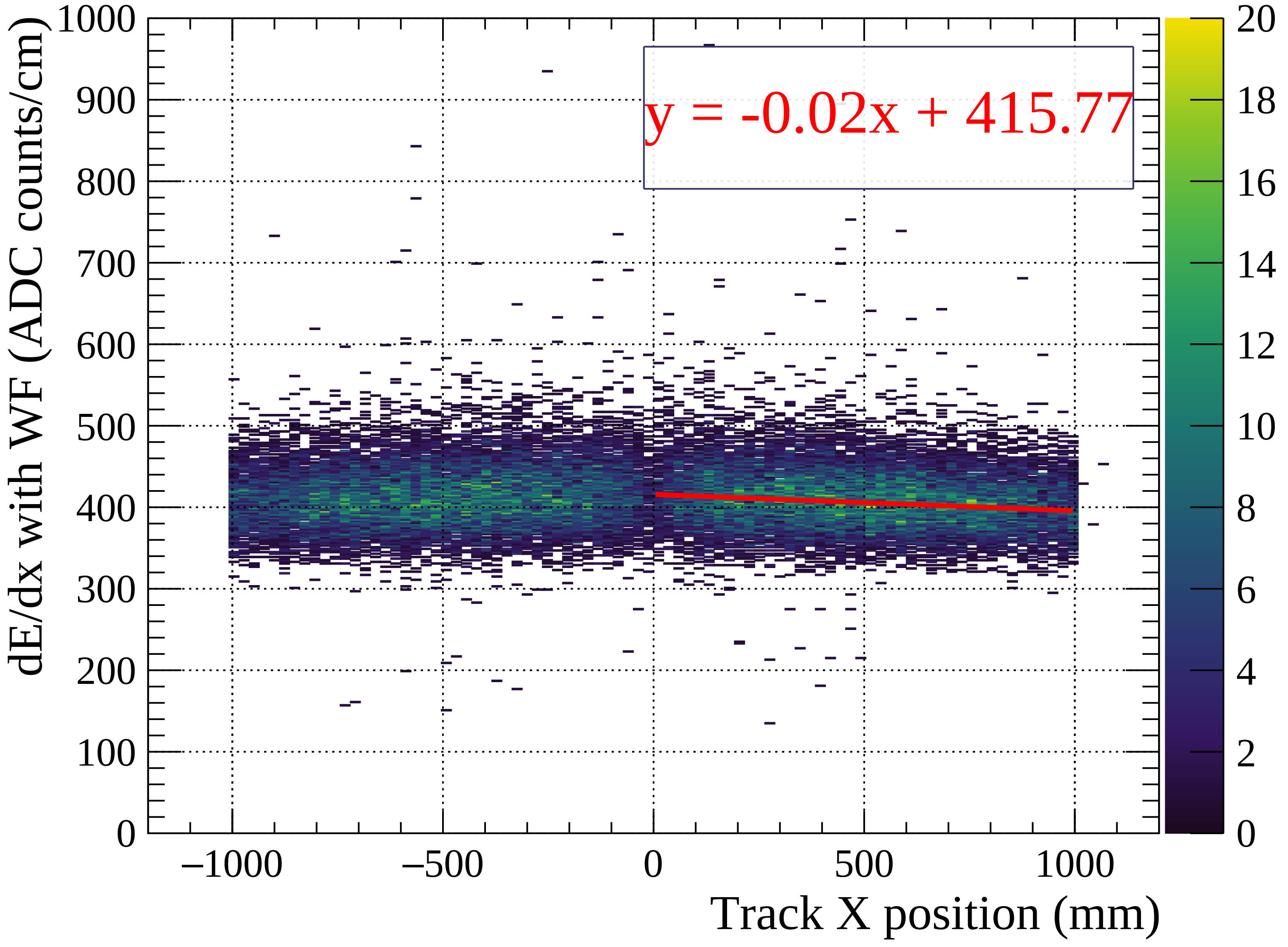


dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
1200

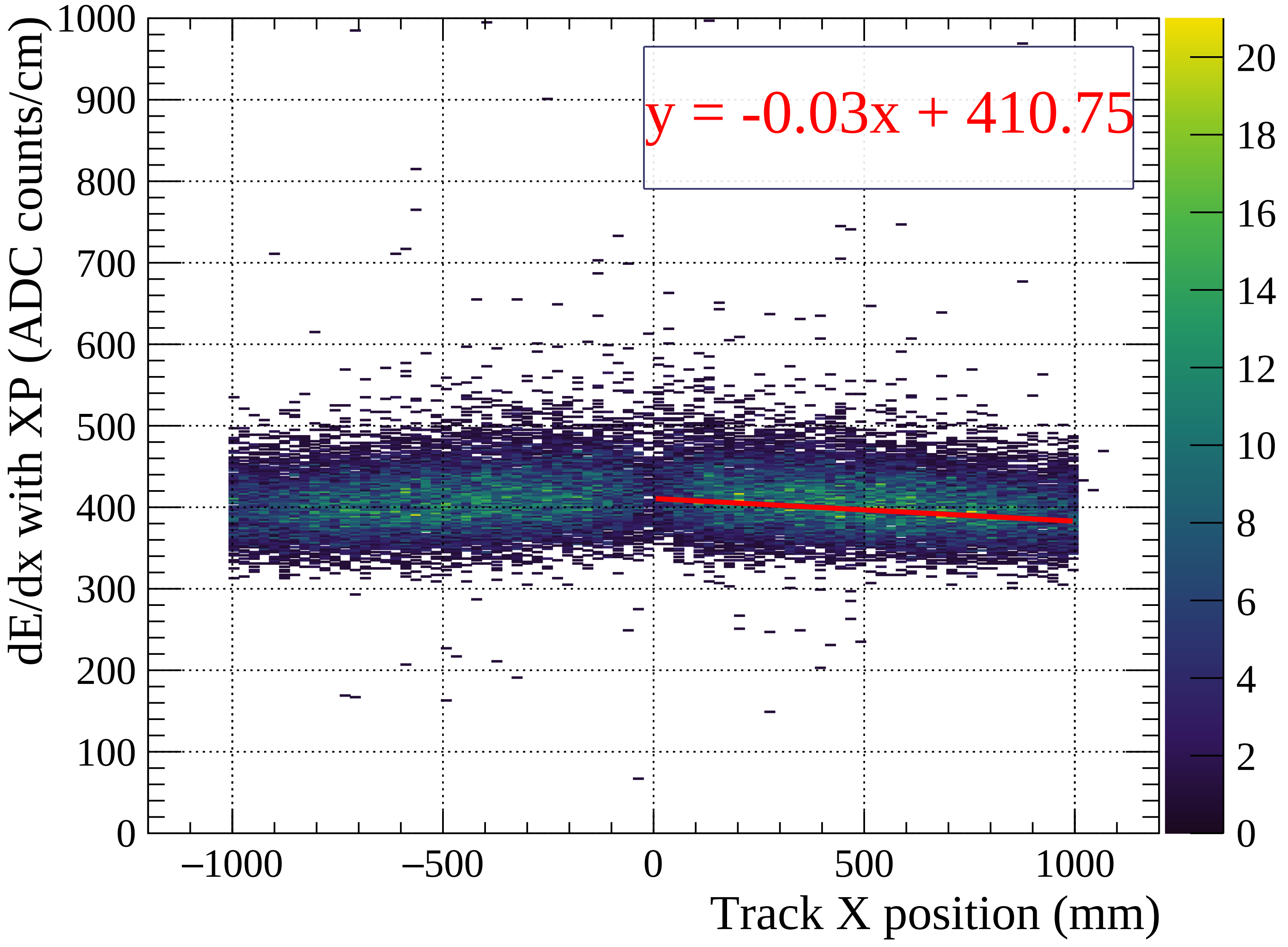


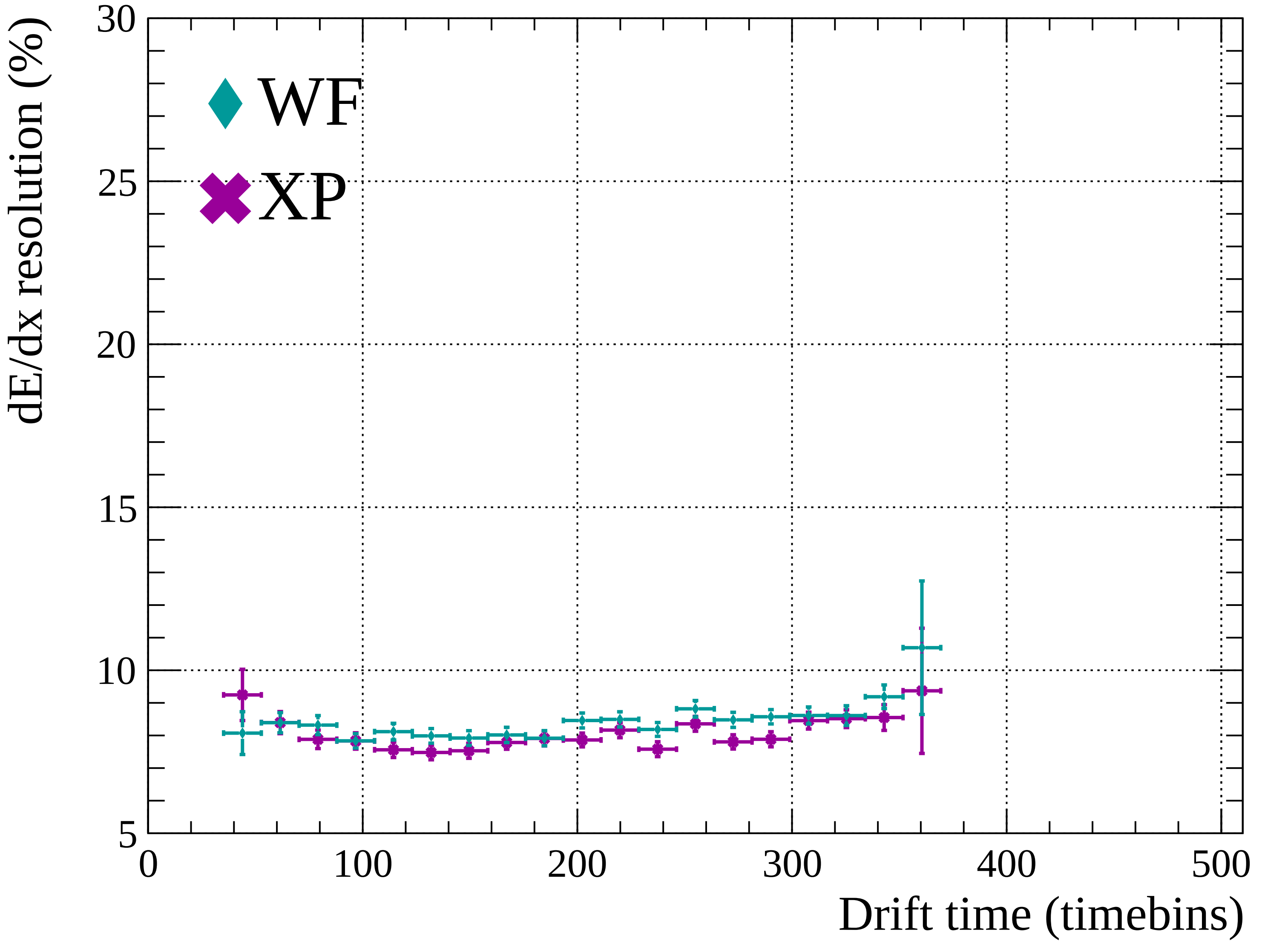
WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000

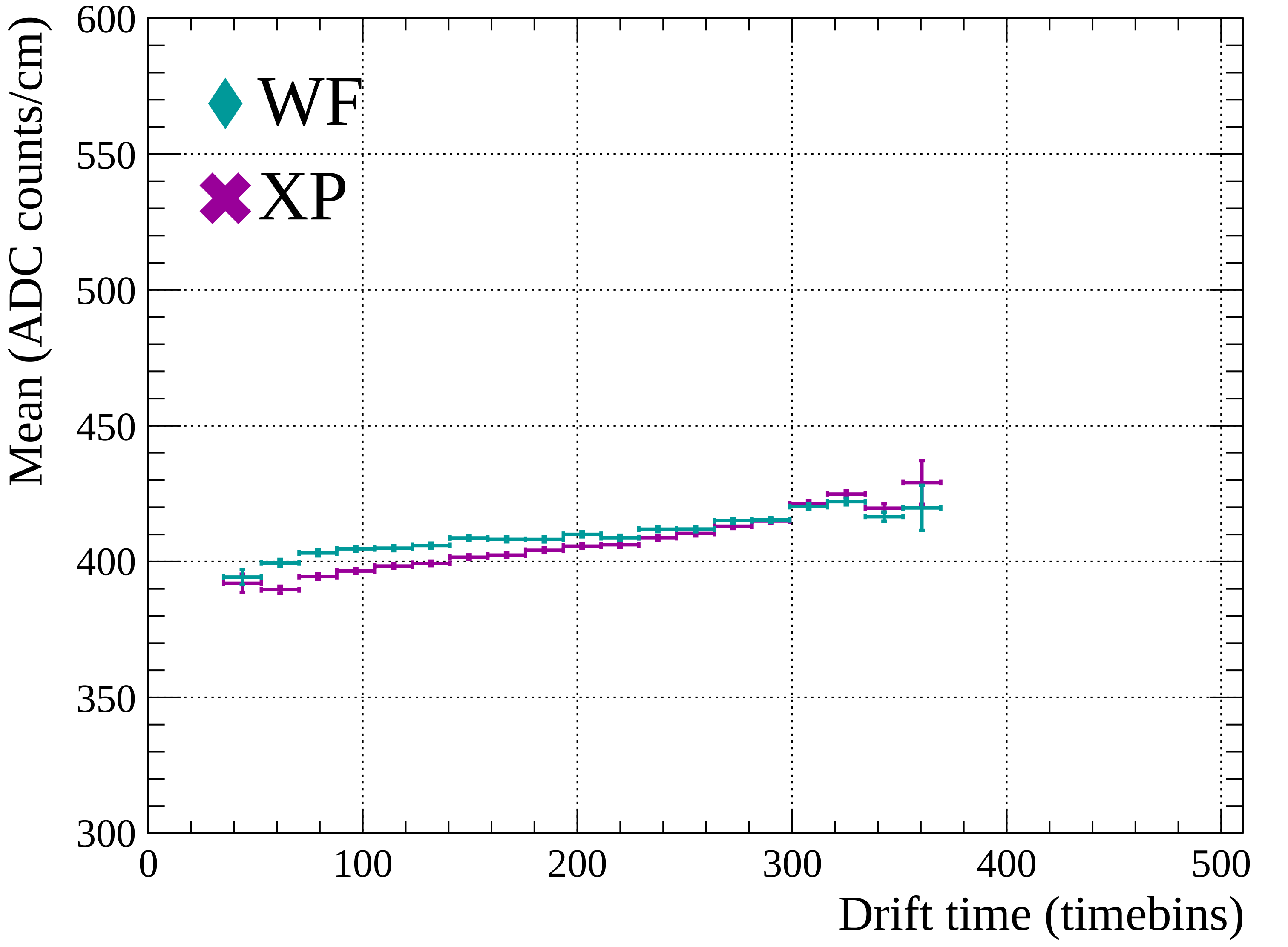

WF
XP
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)
dE/dx (ADC counts/cm)

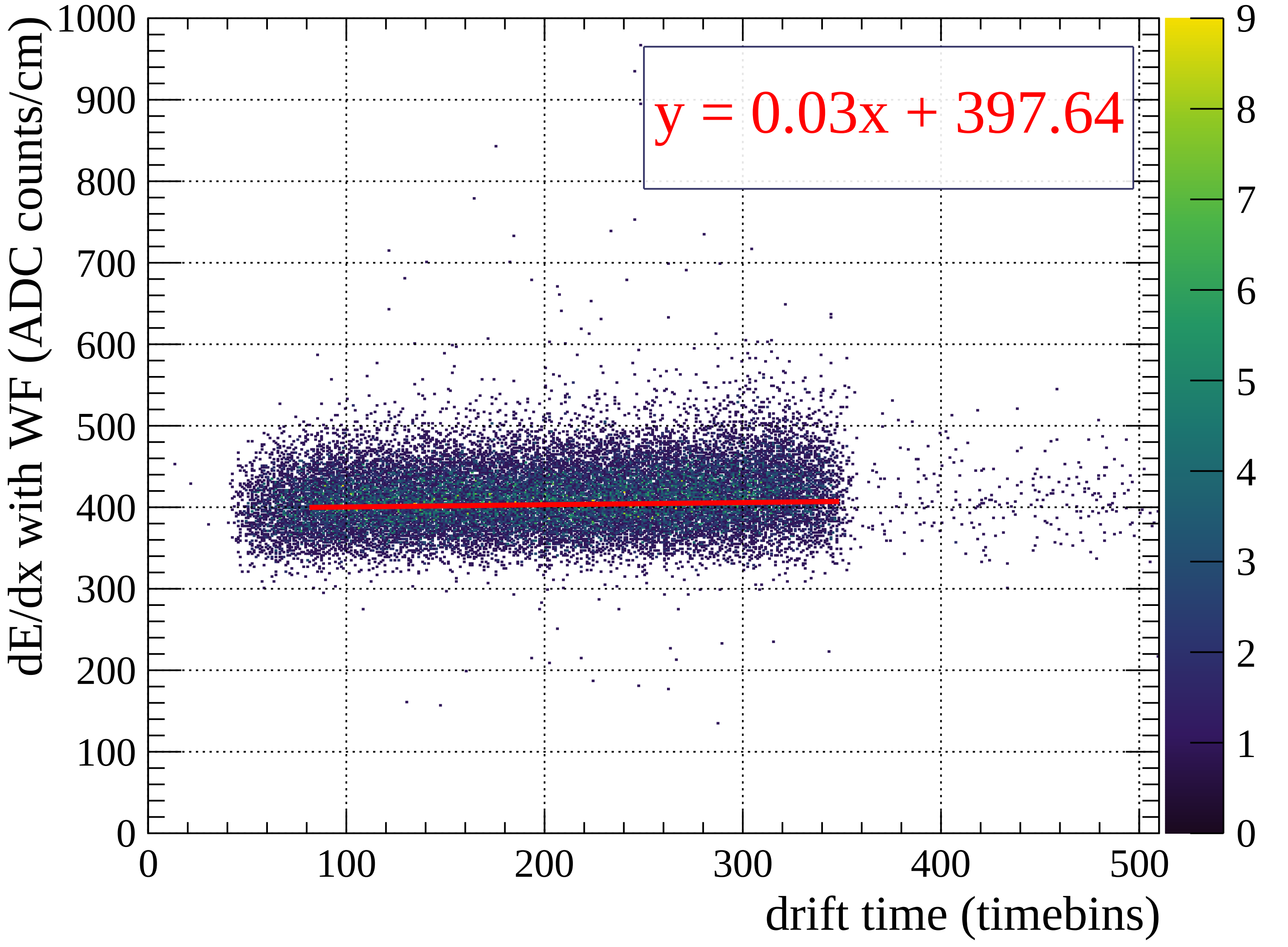


y = -0.02x + 415.77
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
2
4
6
8
10
12
14
16
18
20

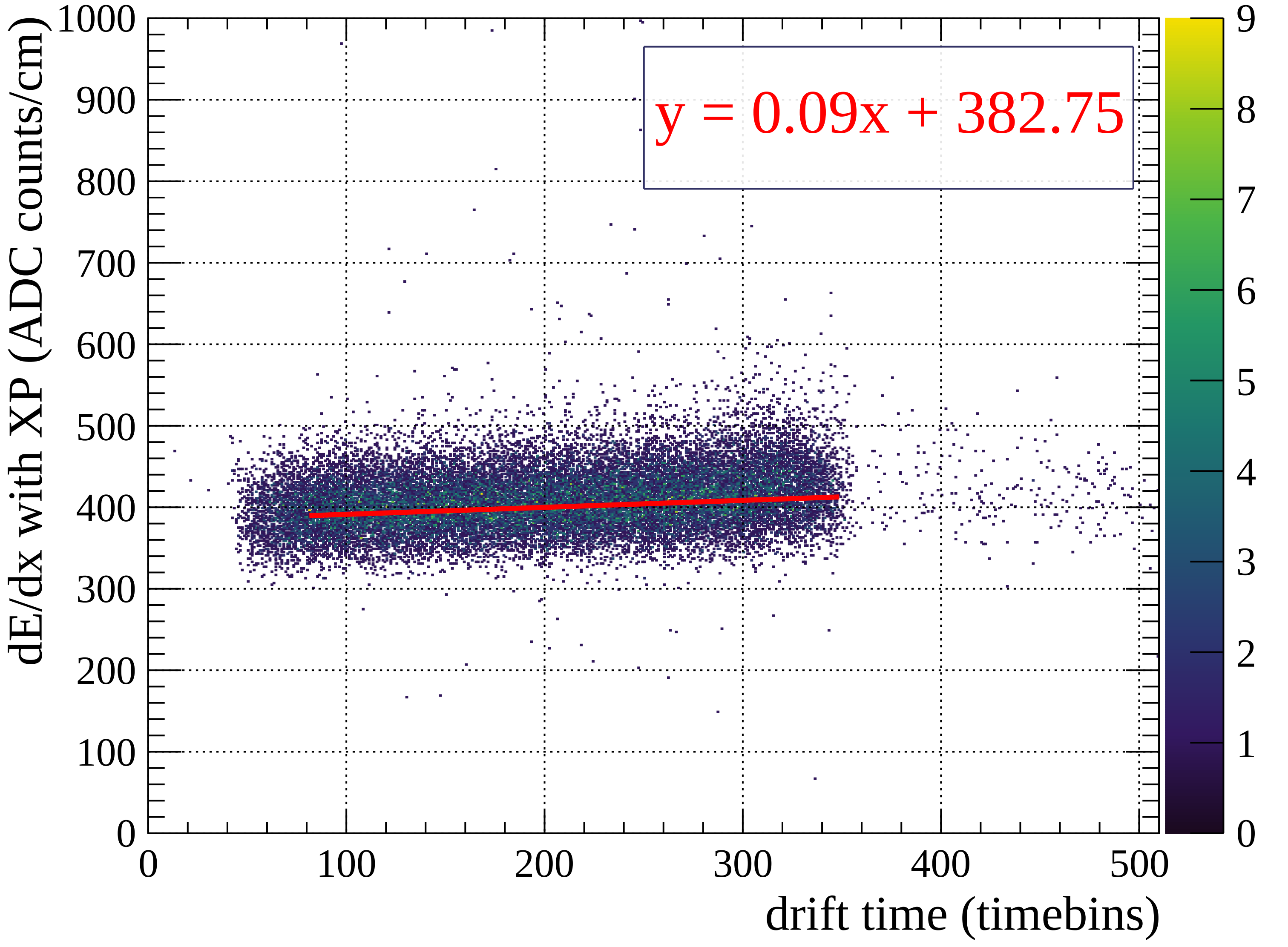

y = -0.03x + 410.75
dE/dx with XP (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
20
18
16
14
12
10
8
6
4
2
0

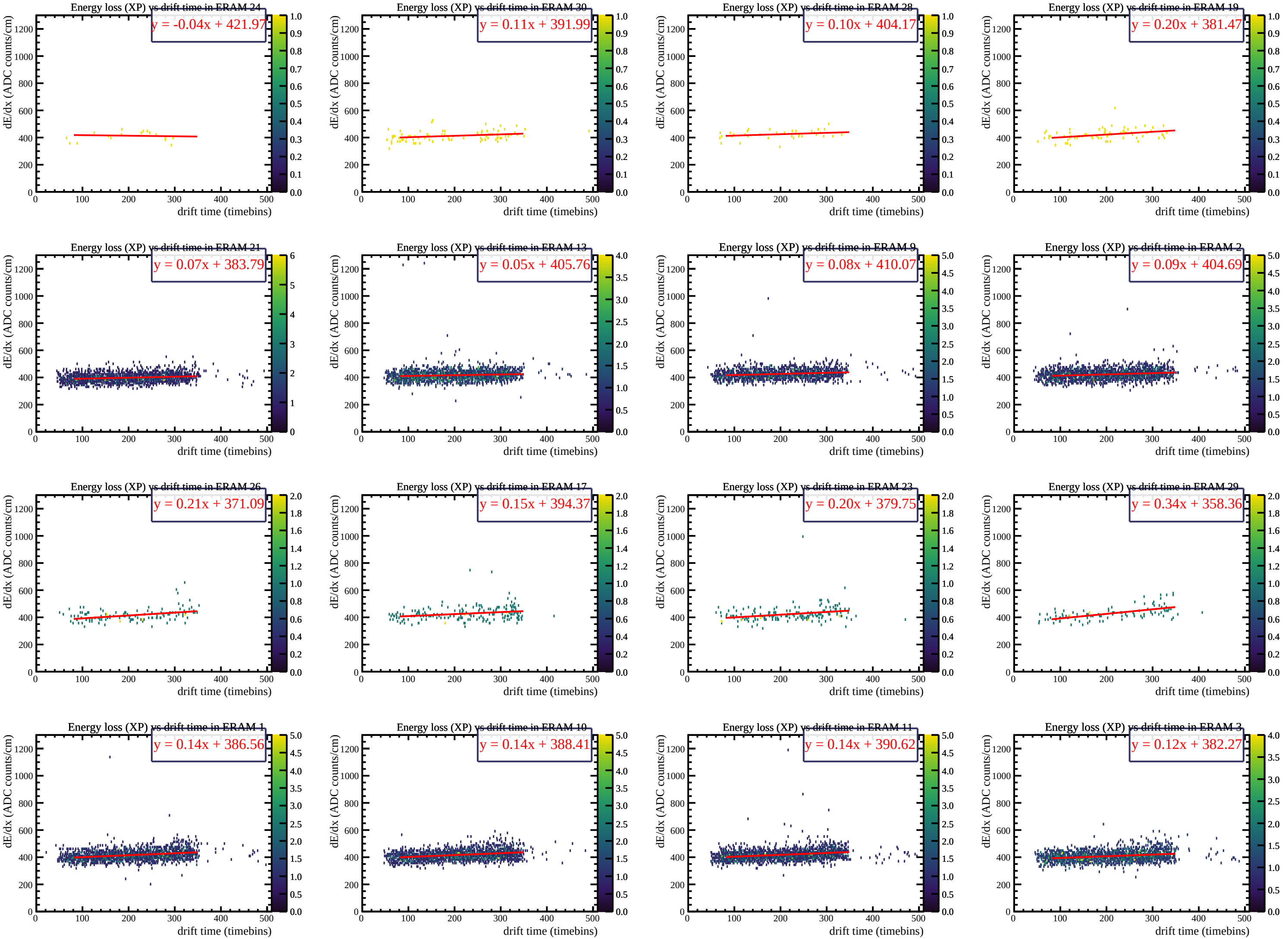

WF
XP
dE/dx resolution (%)
Drift time (timebins)
30
25
20
15
10
5
0
100
200
300
400
500

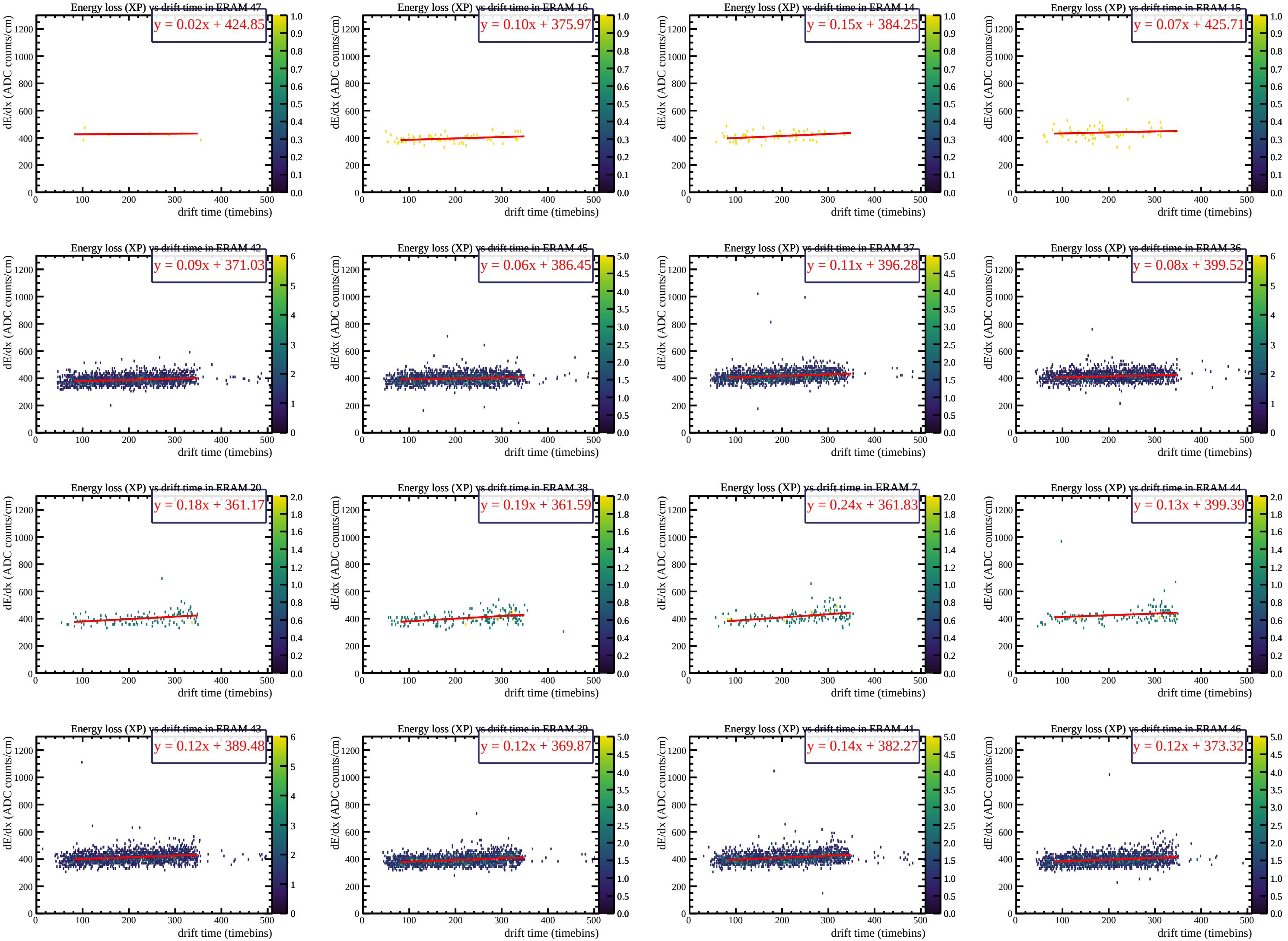

WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500



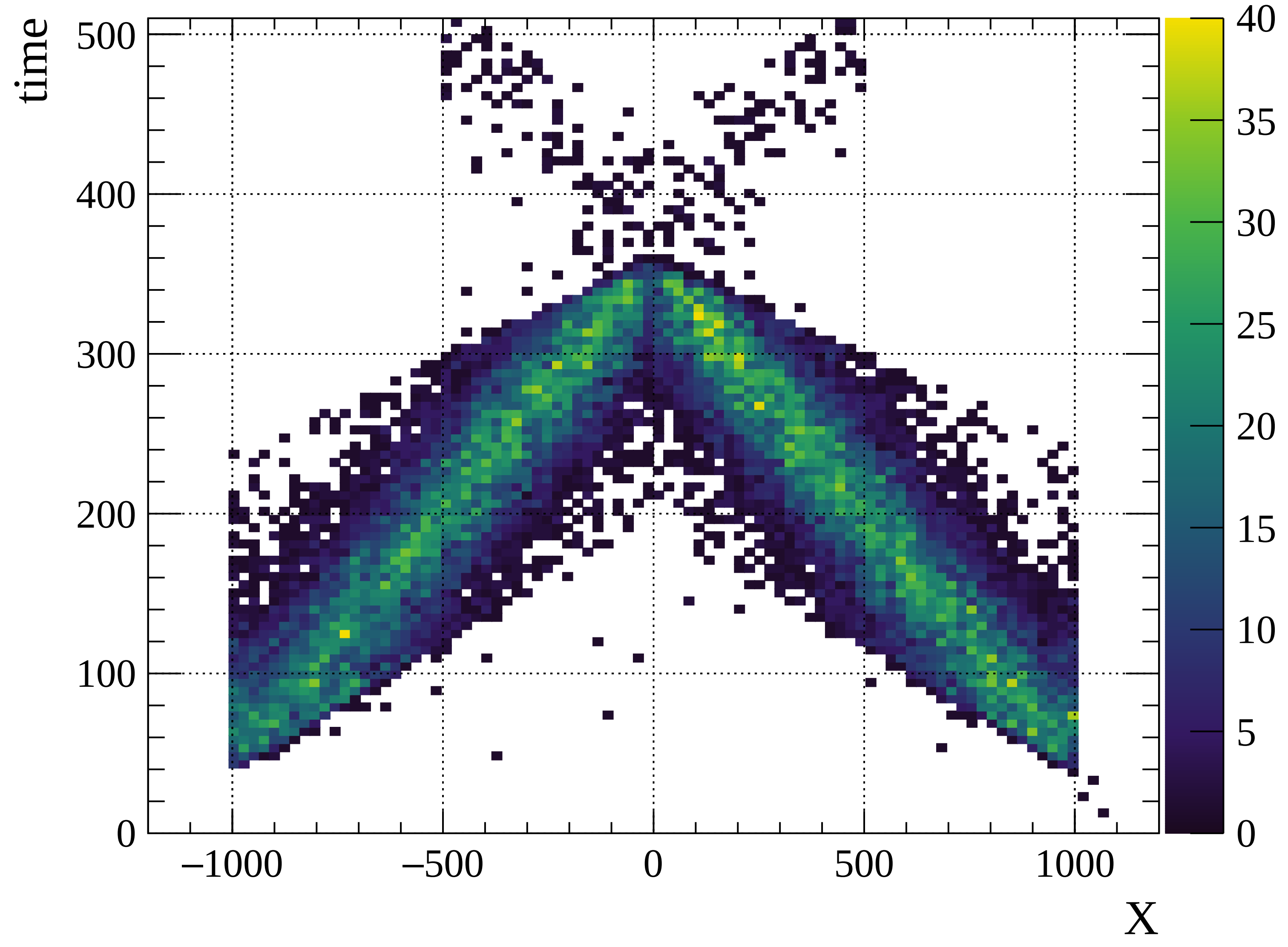
y = 0.03x + 397.64
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6
7
8
9

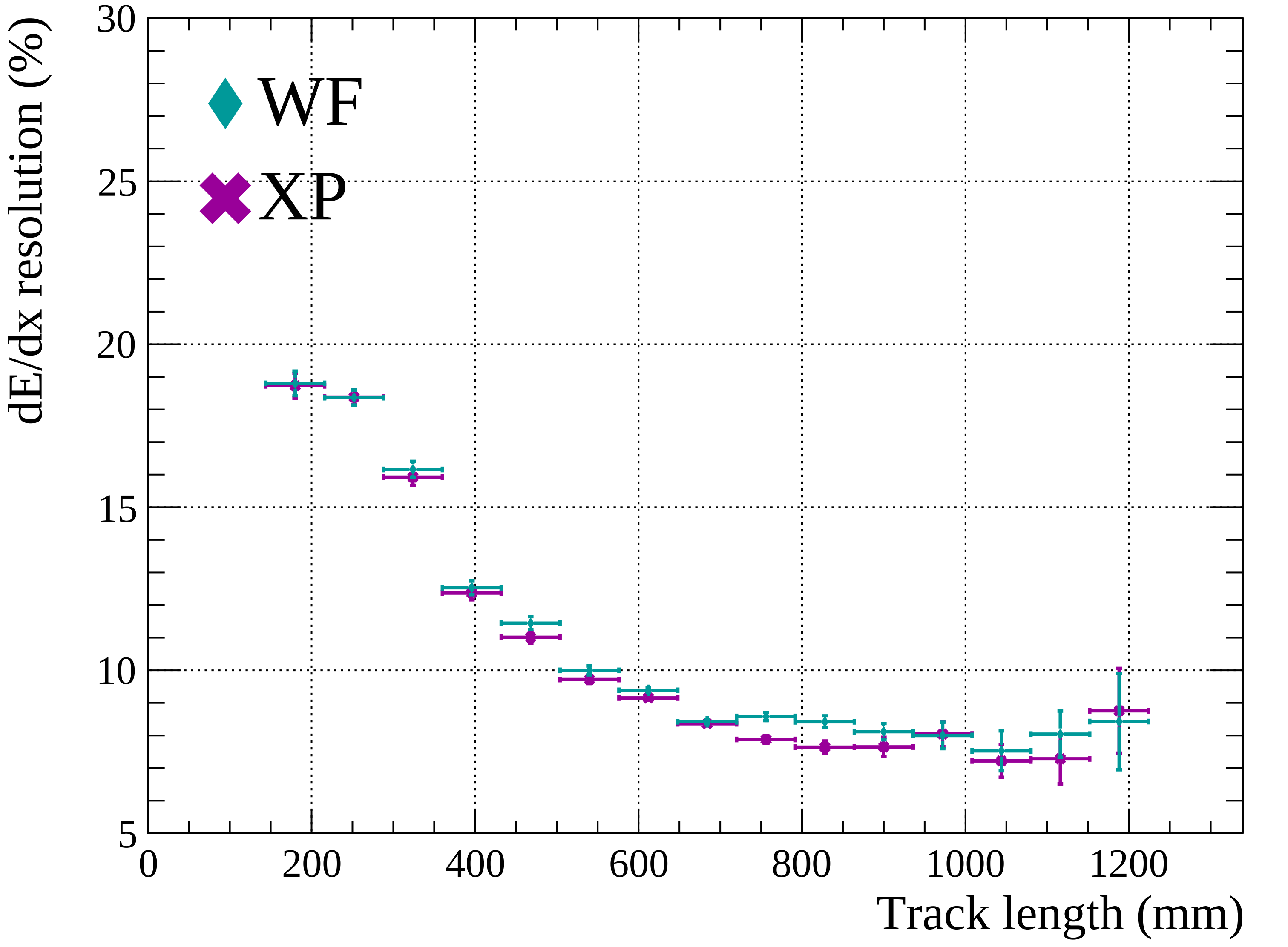

y = 0.09x + 382.75
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6
7
8
9

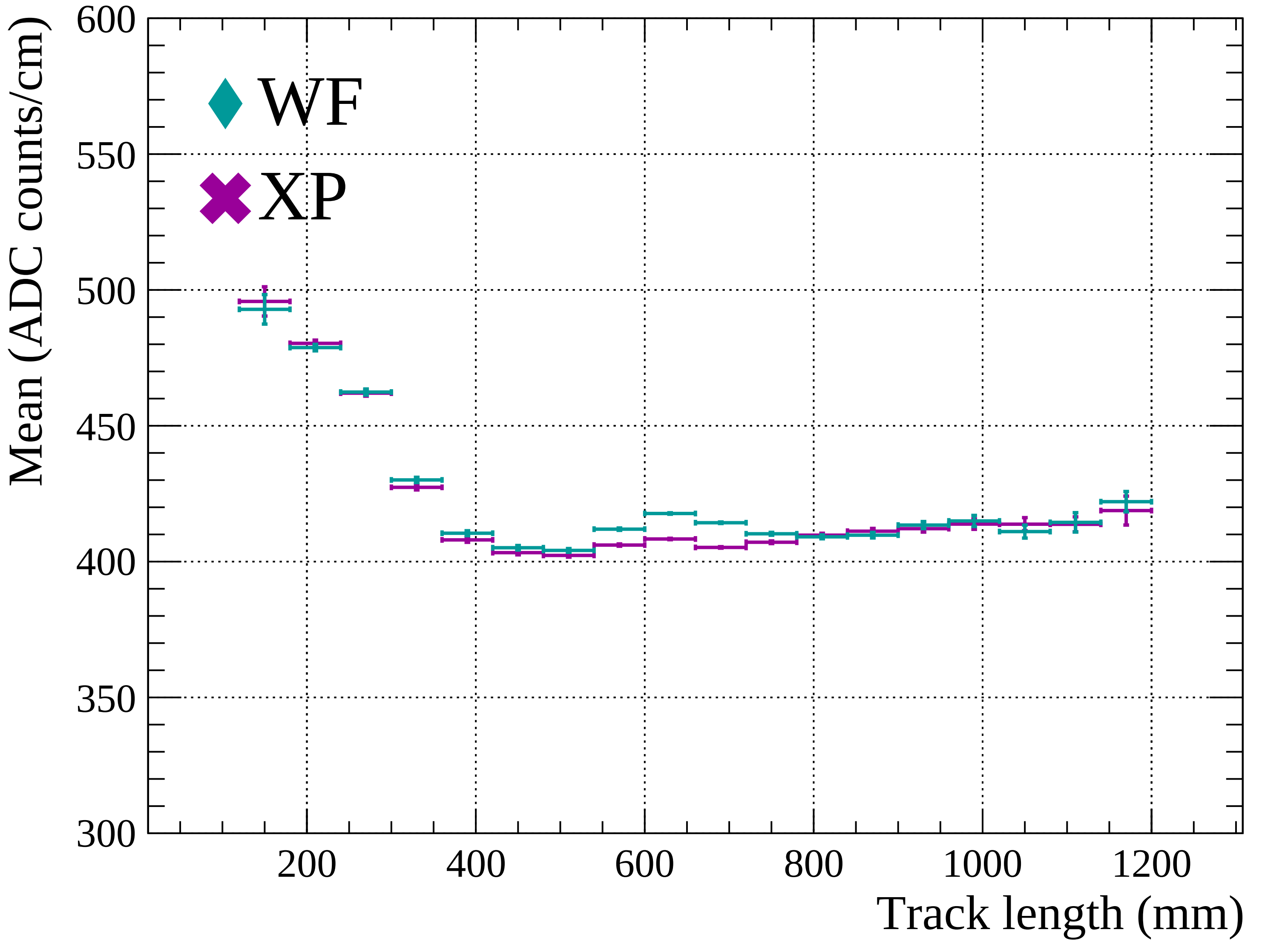

Energy loss (XP) vs drift time in ERAM 24
y = -0.04x + 421.97
Energy loss (XP) vs drift time in ERAM 30
y = 0.11x + 391.99
Energy loss (XP) vs drift time in ERAM 28
y = 0.10x + 404.17
Energy loss (XP) vs drift time in ERAM 19
y = 0.20x + 381.47
Energy loss (XP) vs drift time in ERAM 21
y = 0.07x + 383.79
Energy loss (XP) vs drift time in ERAM 13
y = 0.05x + 405.76
Energy loss (XP) vs drift time in ERAM 9
y = 0.08x + 410.07
Energy loss (XP) vs drift time in ERAM 2
y = 0.09x + 404.69
Energy loss (XP) vs drift time in ERAM 26
y = 0.21x + 371.09
Energy loss (XP) vs drift time in ERAM 17
y = 0.15x + 394.37
Energy loss (XP) vs drift time in ERAM 23
y = 0.20x + 379.75
Energy loss (XP) vs drift time in ERAM 29
y = 0.34x + 358.36
Energy loss (XP) vs drift time in ERAM 1
y = 0.14x + 386.56
Energy loss (XP) vs drift time in ERAM 10
y = 0.14x + 388.41
Energy loss (XP) vs drift time in ERAM 11
y = 0.14x + 390.62
Energy loss (XP) vs drift time in ERAM 3
y = 0.12x + 382.27
dE/dx (ADC counts/cm)
drift time (timebins)



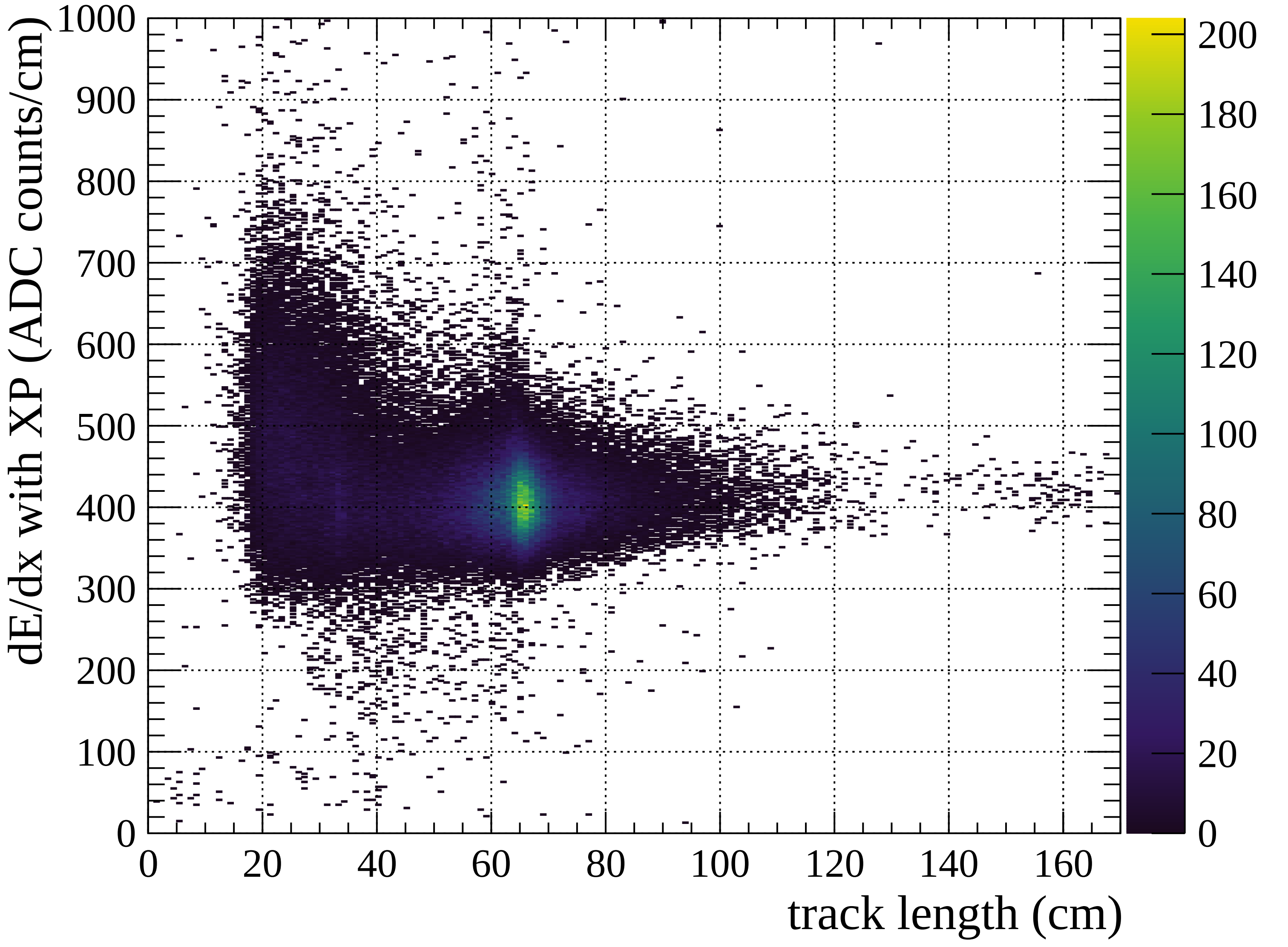
Energy loss (XP) vs drift time in ERAM 47
y = 0.02x + 424.85
Energy loss (XP) vs drift time in ERAM 16
y = 0.10x + 375.97
Energy loss (XP) vs drift time in ERAM 14
y = 0.15x + 384.25
Energy loss (XP) vs drift time in ERAM 15
y = 0.07x + 425.71
Energy loss (XP) vs drift time in ERAM 42
y = 0.09x + 371.03
Energy loss (XP) vs drift time in ERAM 45
y = 0.06x + 386.45
Energy loss (XP) vs drift time in ERAM 37
y = 0.11x + 396.28
Energy loss (XP) vs drift time in ERAM 36
y = 0.08x + 399.52
Energy loss (XP) vs drift time in ERAM 20
y = 0.18x + 361.17
Energy loss (XP) vs drift time in ERAM 38
y = 0.19x + 361.59
Energy loss (XP) vs drift time in ERAM 7
y = 0.24x + 361.83
Energy loss (XP) vs drift time in ERAM 44
y = 0.13x + 399.39
Energy loss (XP) vs drift time in ERAM 43
y = 0.12x + 389.48
Energy loss (XP) vs drift time in ERAM 39
y = 0.12x + 369.87
Energy loss (XP) vs drift time in ERAM 41
y = 0.14x + 382.27
Energy loss (XP) vs drift time in ERAM 46
y = 0.12x + 373.32
dE/dx (ADC counts/cm)
drift time (timebins)


time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
40
35
30
25
20
15
10
5
0

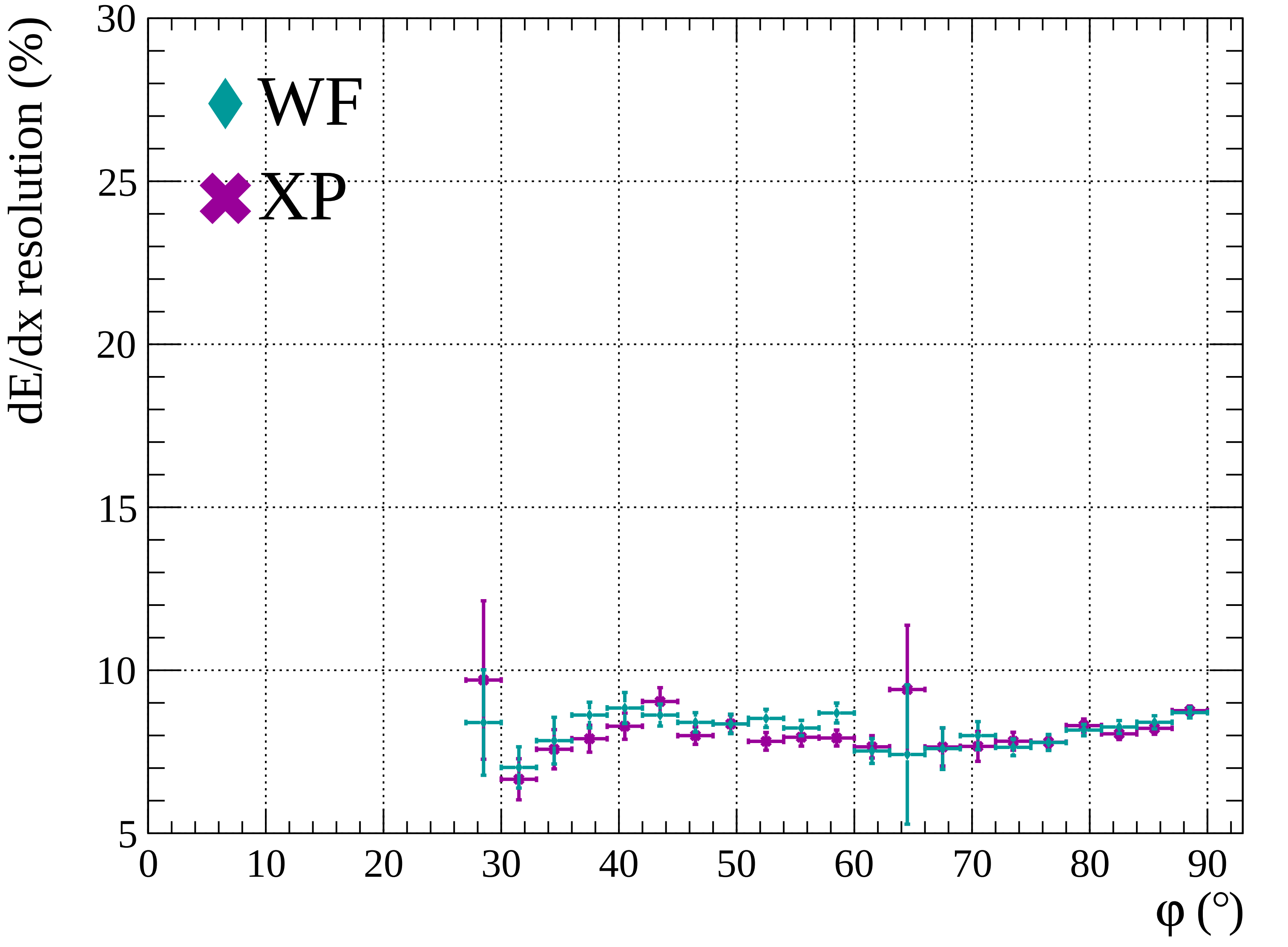

WF
XP
dE/dx resolution (%)
Track length (mm)
5
10
15
20
25
30
0
200
400
600
800
1000
1200

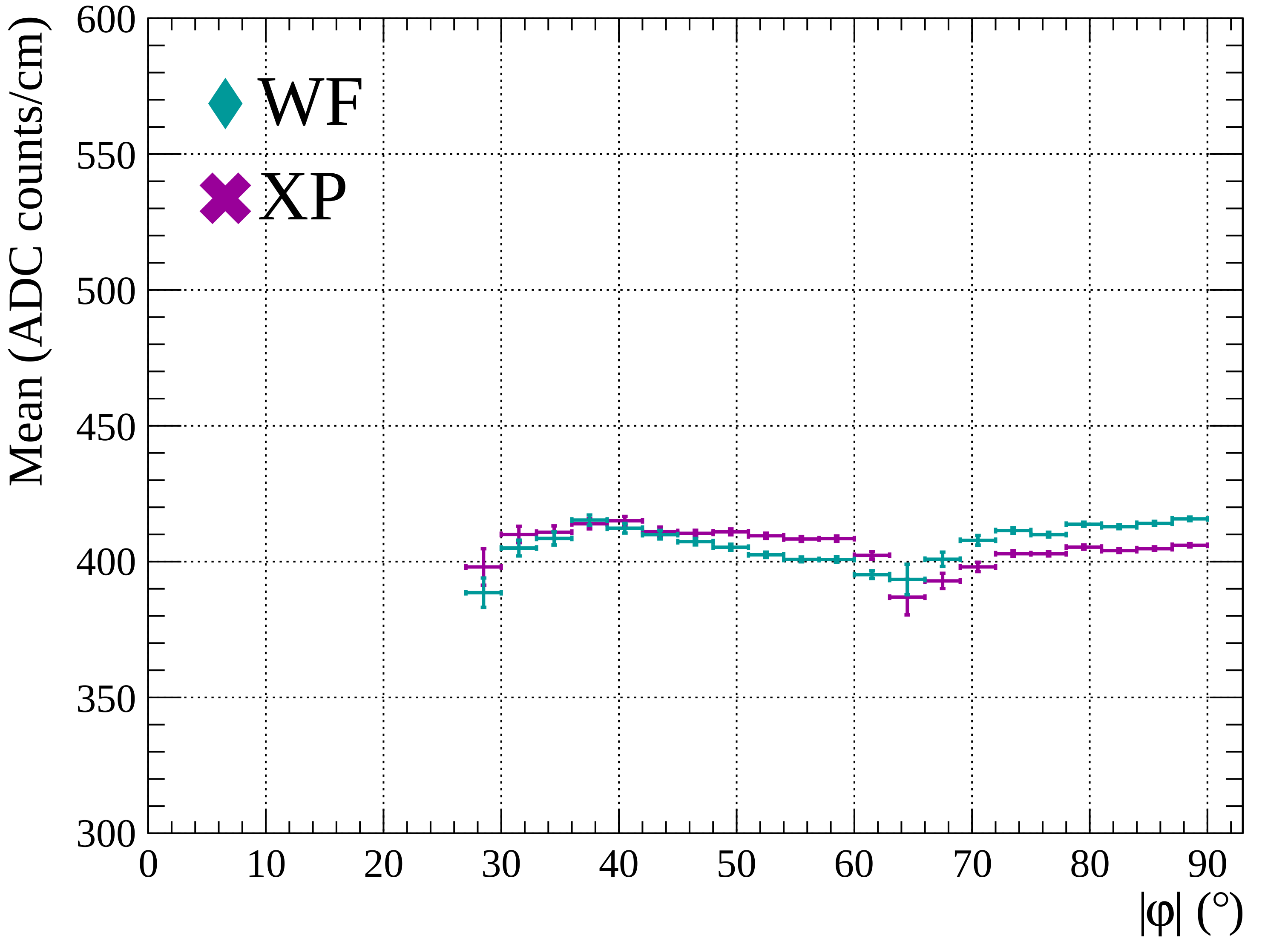


WF
XP
Mean (ADC counts/cm)
Track length (mm)
600
550
500
450
400
350
300
200
400
600
800
1000
1200

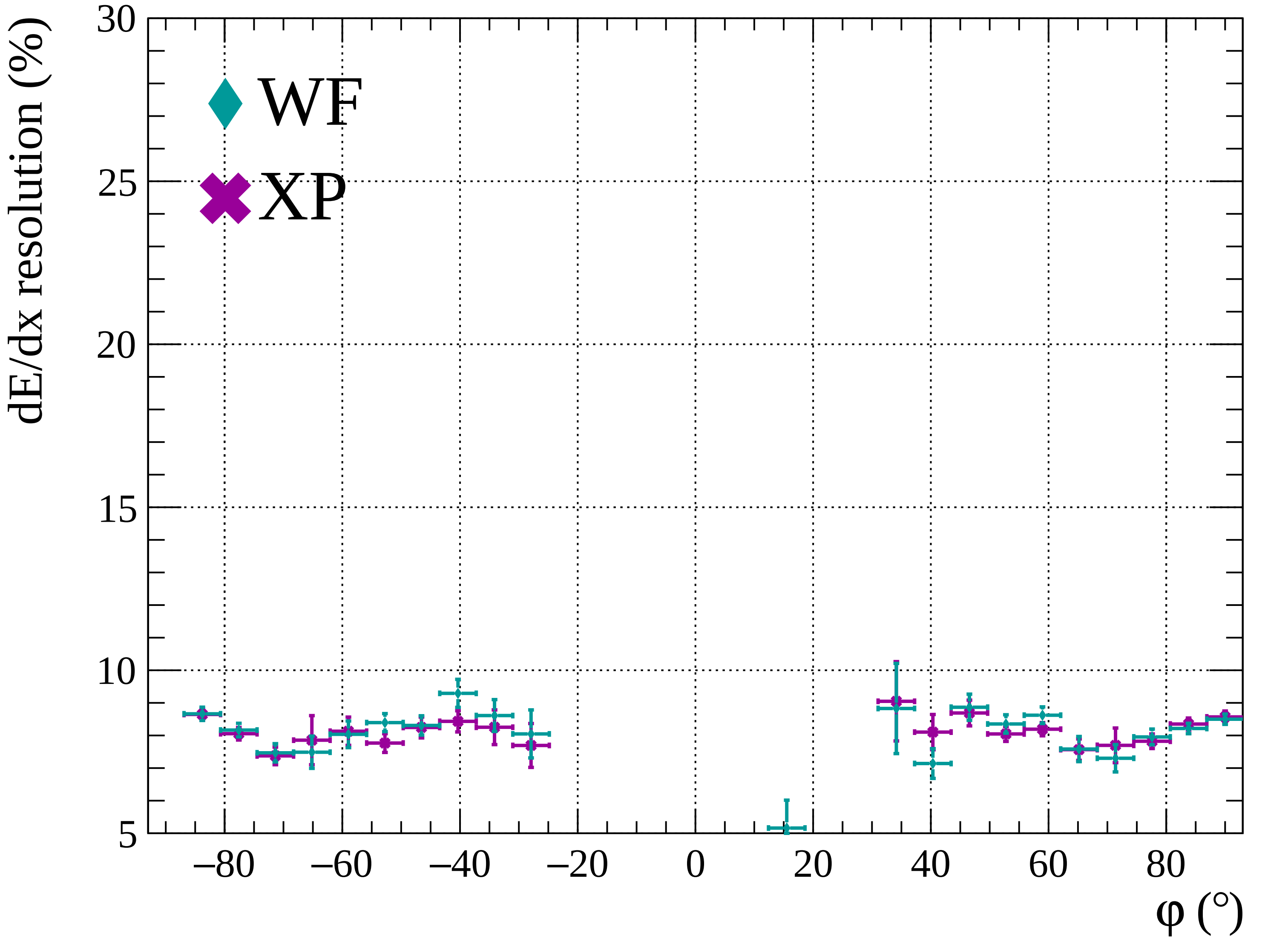


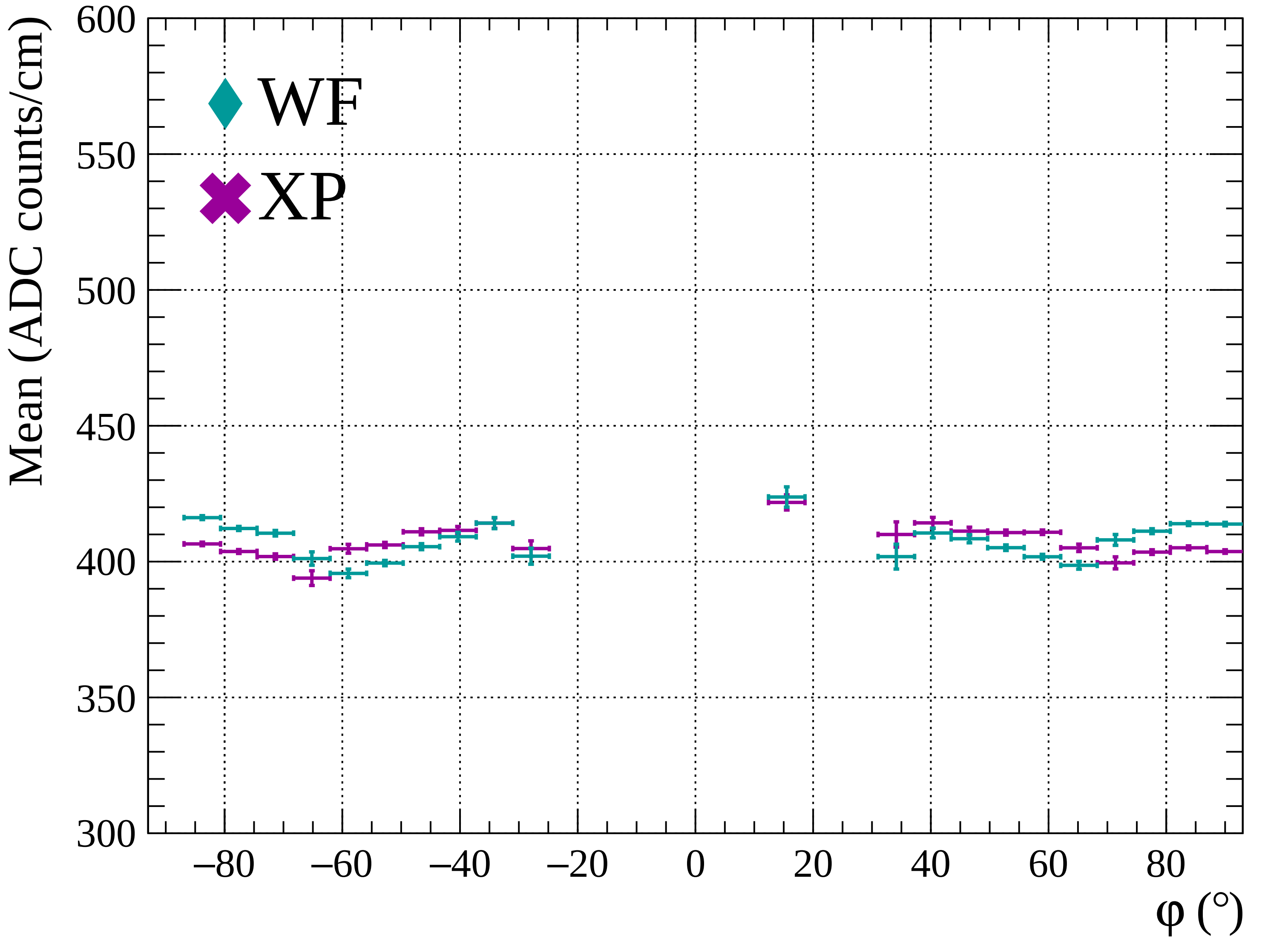
dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
20
40
60
80
100
120
140
160
180
200


dE/dx resolution (%)
WF
XP
φ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90

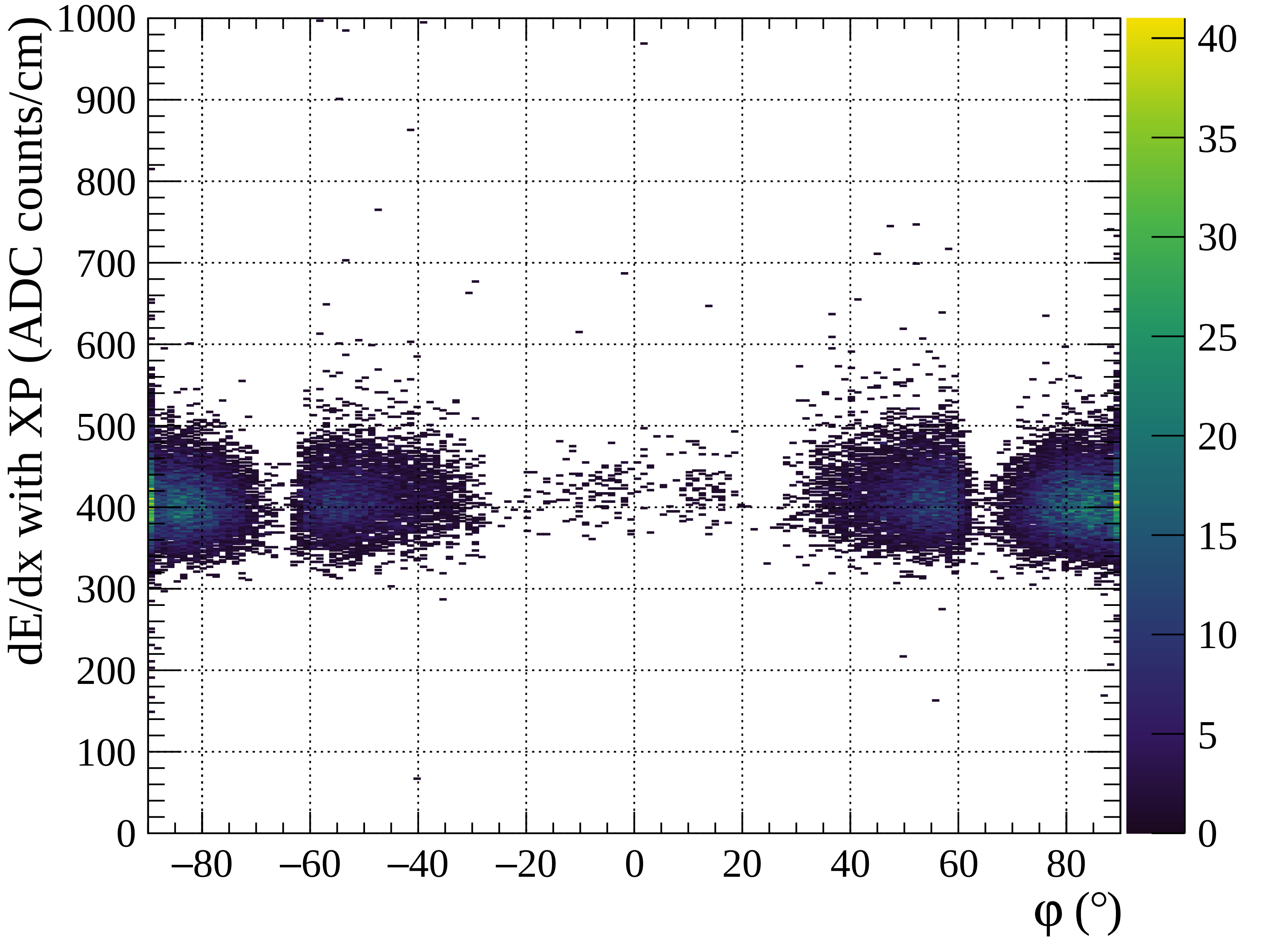


Mean (ADC counts/cm)
WF
XP
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
|φ| (°)

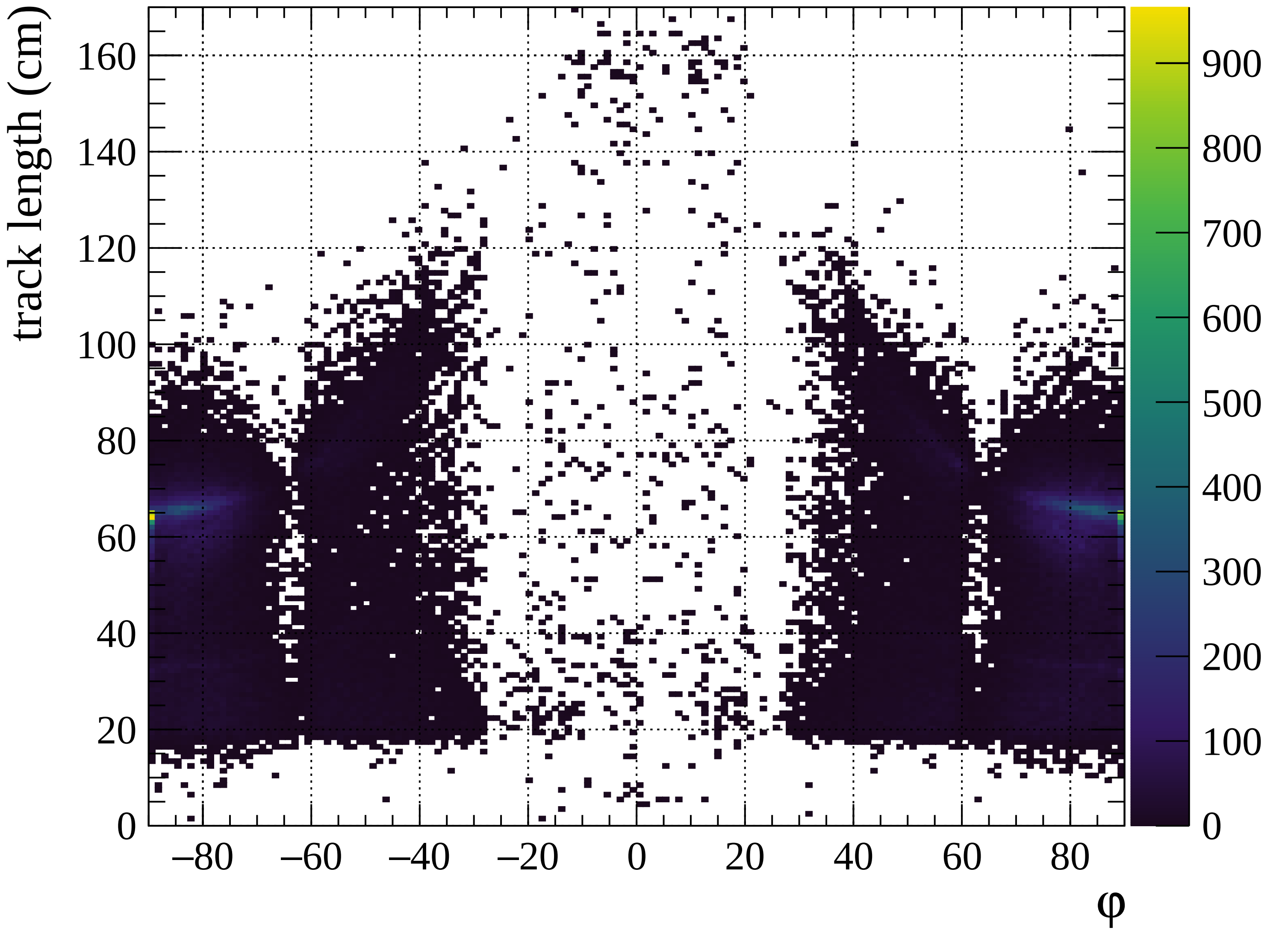

dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

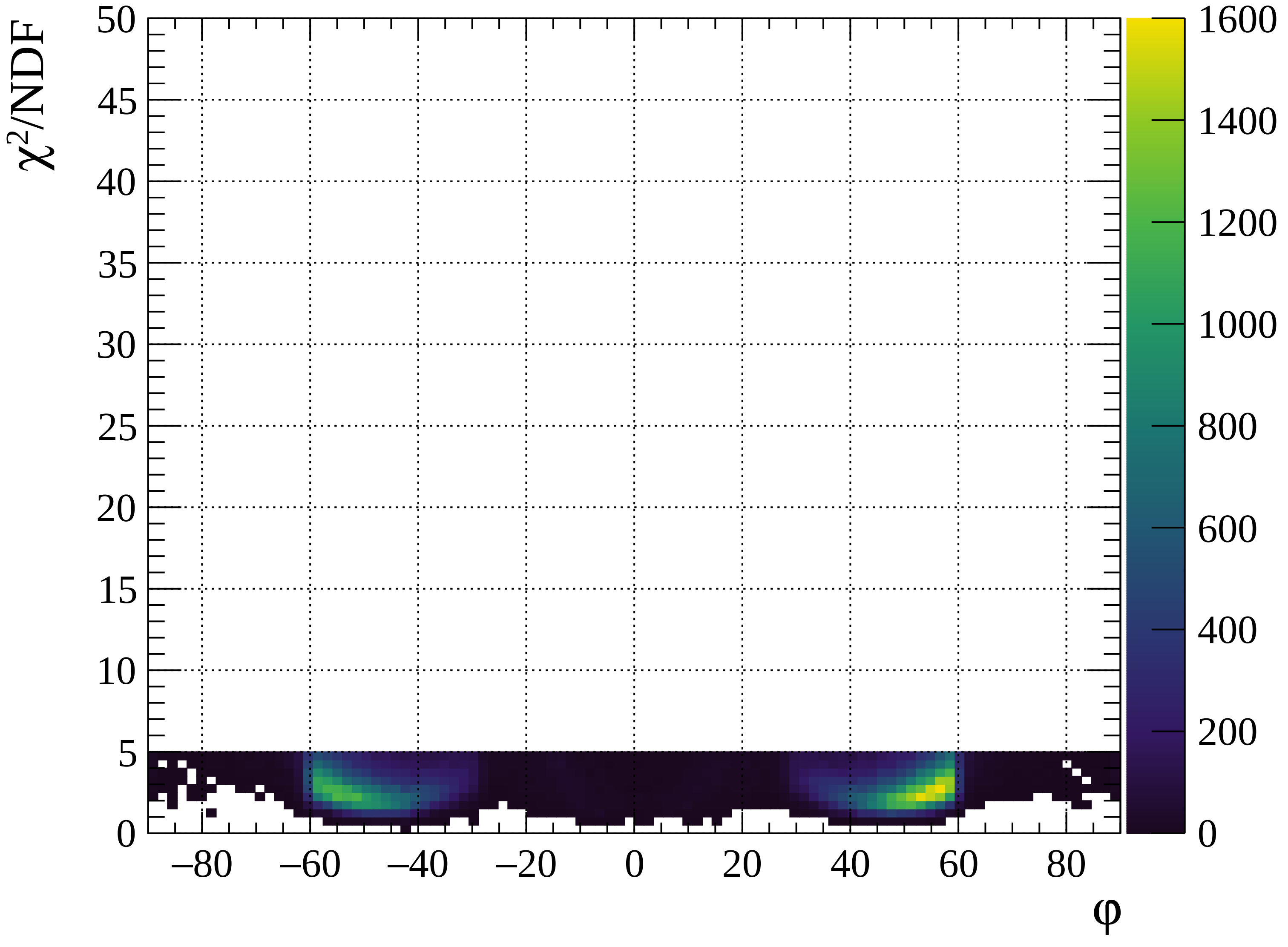


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

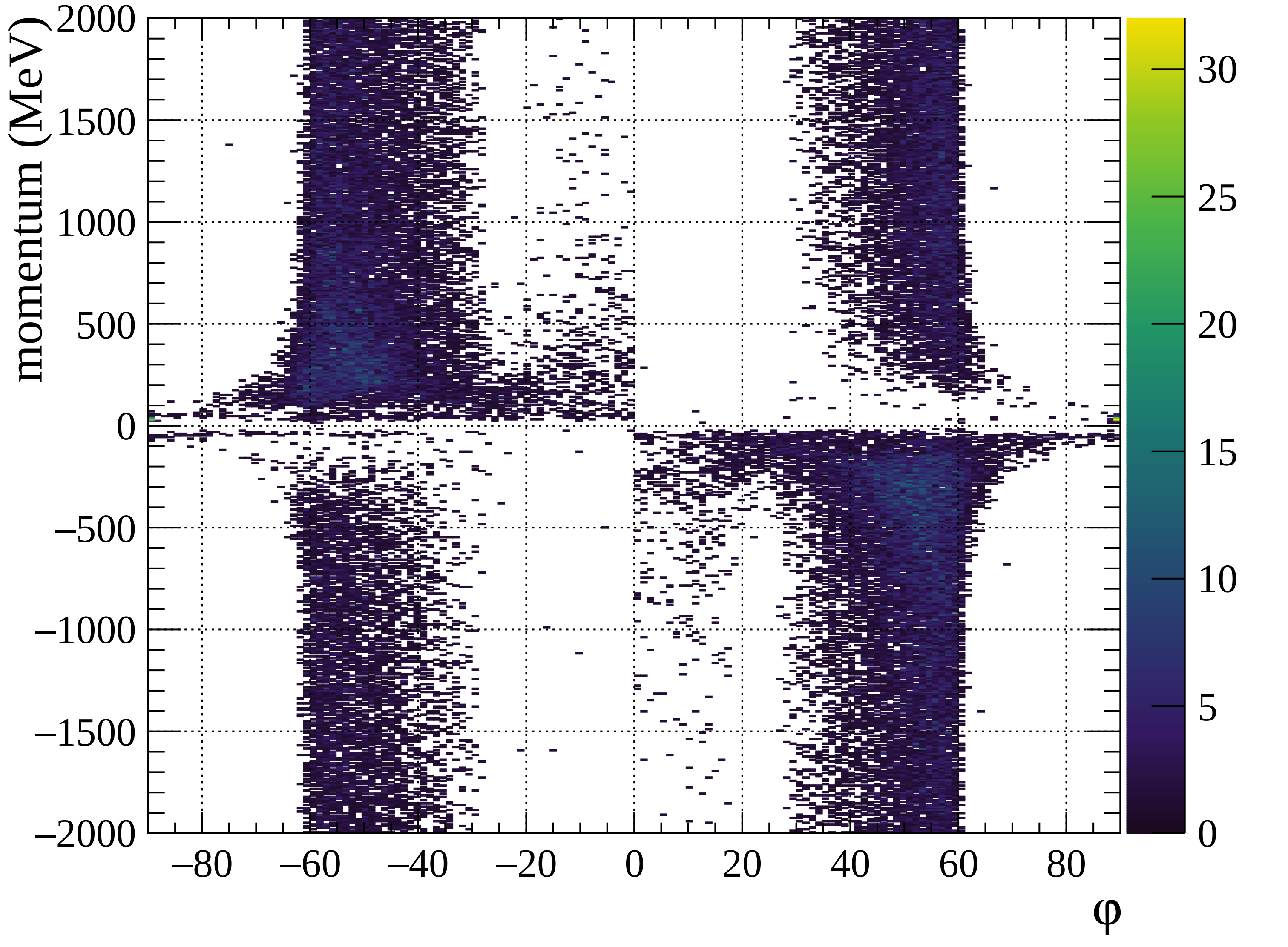


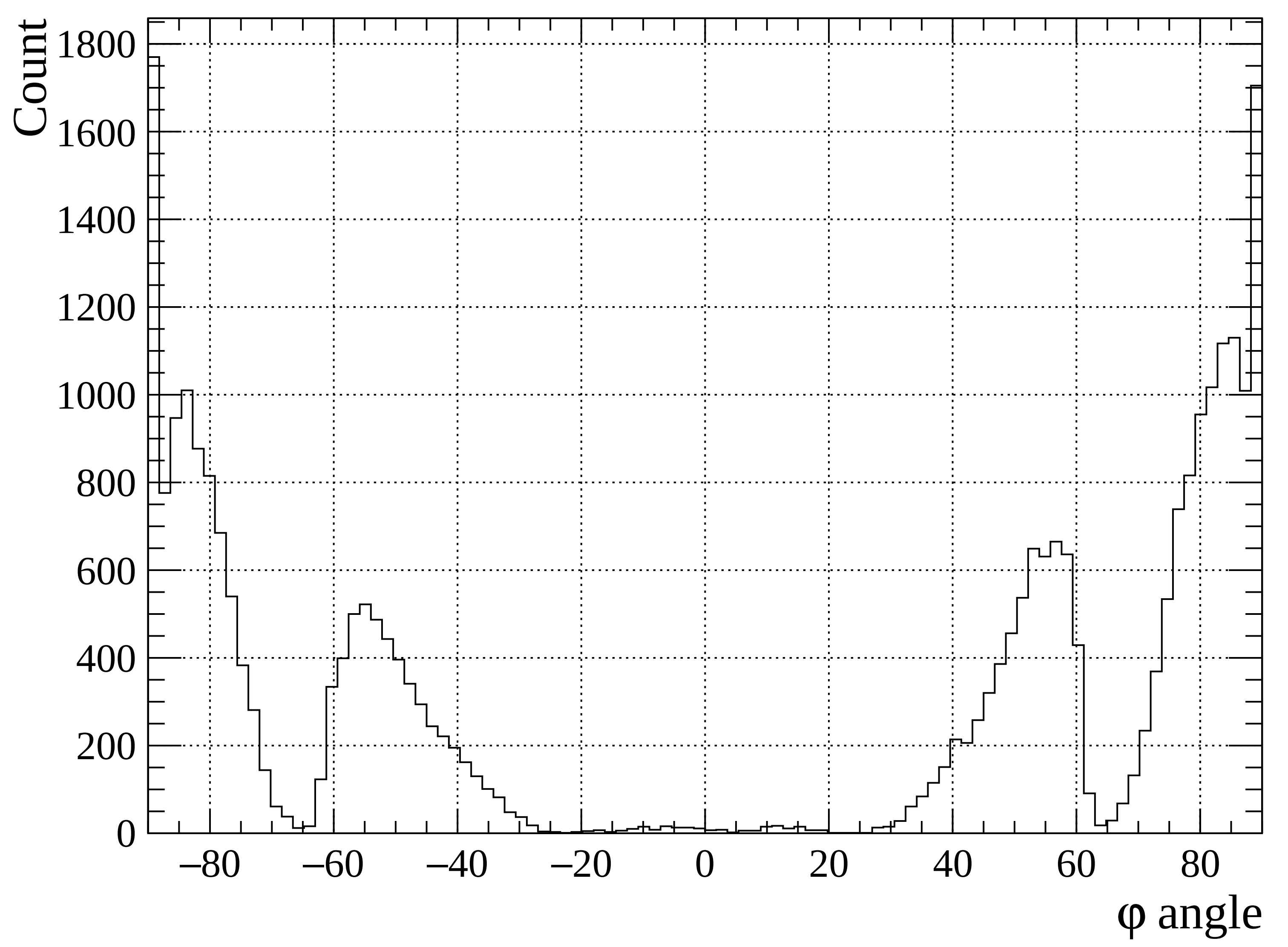
dE/dx with XP (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
φ (°)
0
5
10
15
20
25
30
35
40

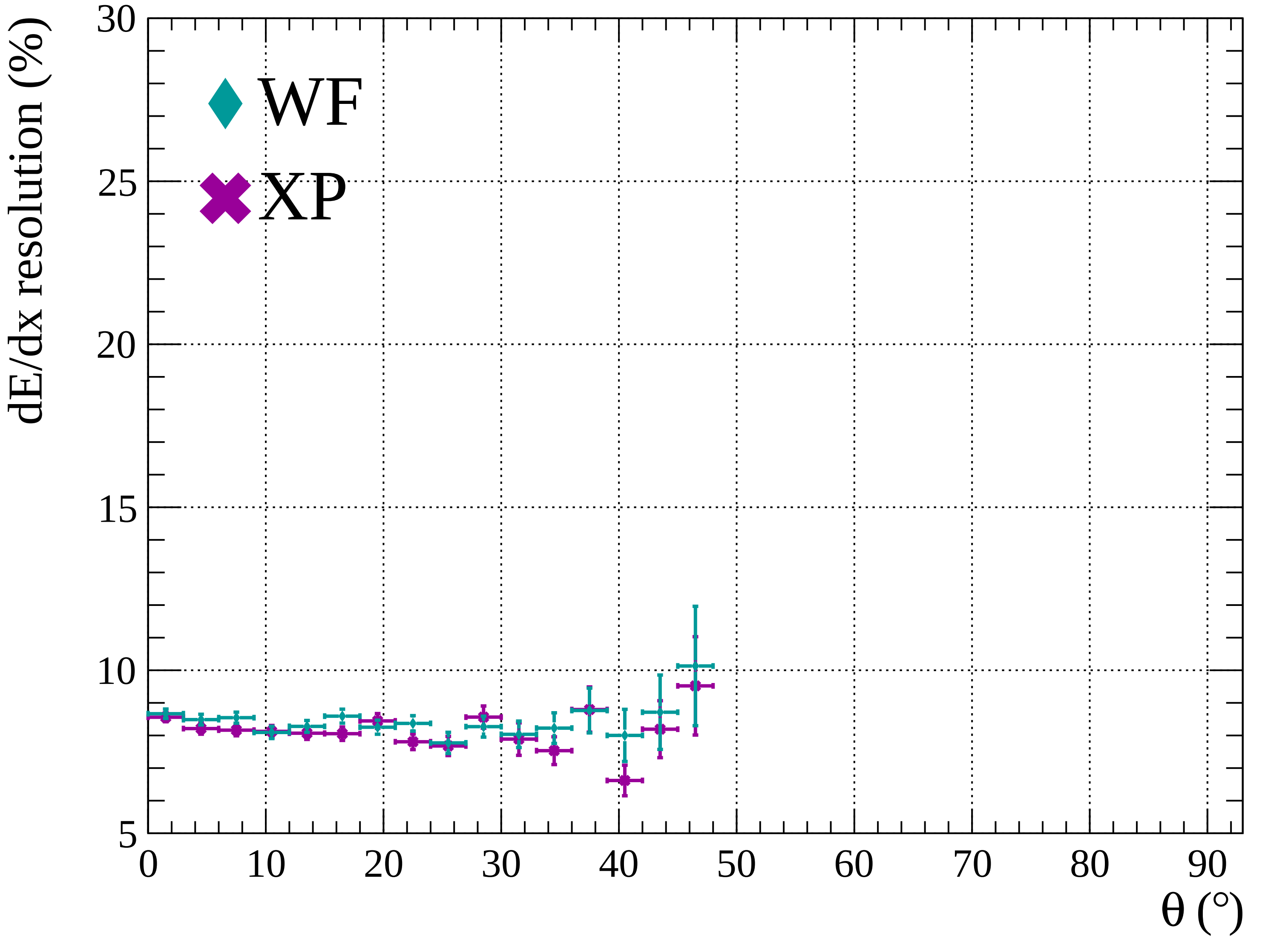

track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
100
200
300
400
500
600
700
800
900



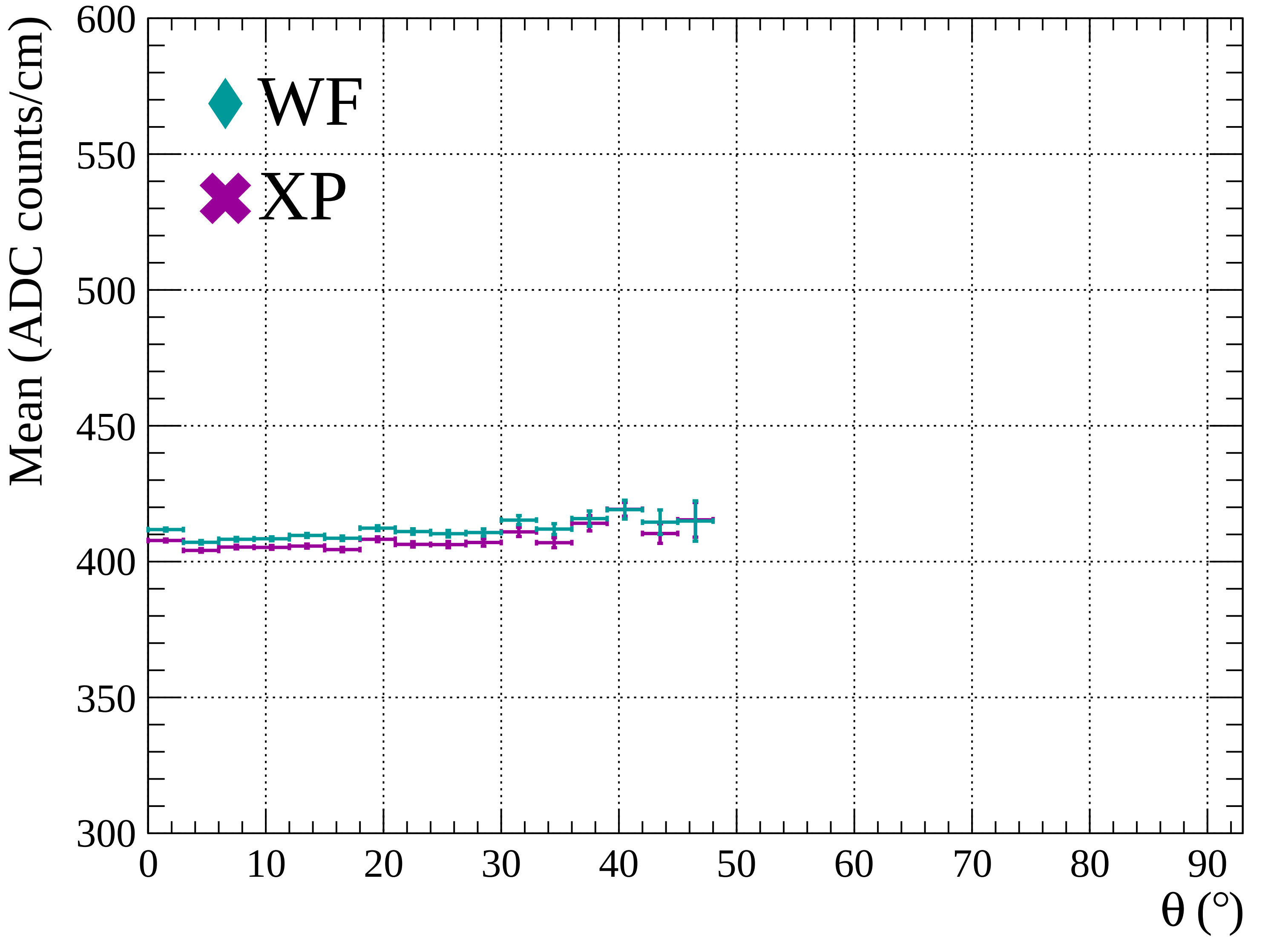
χ²/NDF
φ
0
5
10
15
20
25
30
35
40
45
50
−80
−60
−40
−20
0
20
40
60
80
0
200
400
600
800
1000
1200
1400
1600


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
30
25
20
15
10
5
0

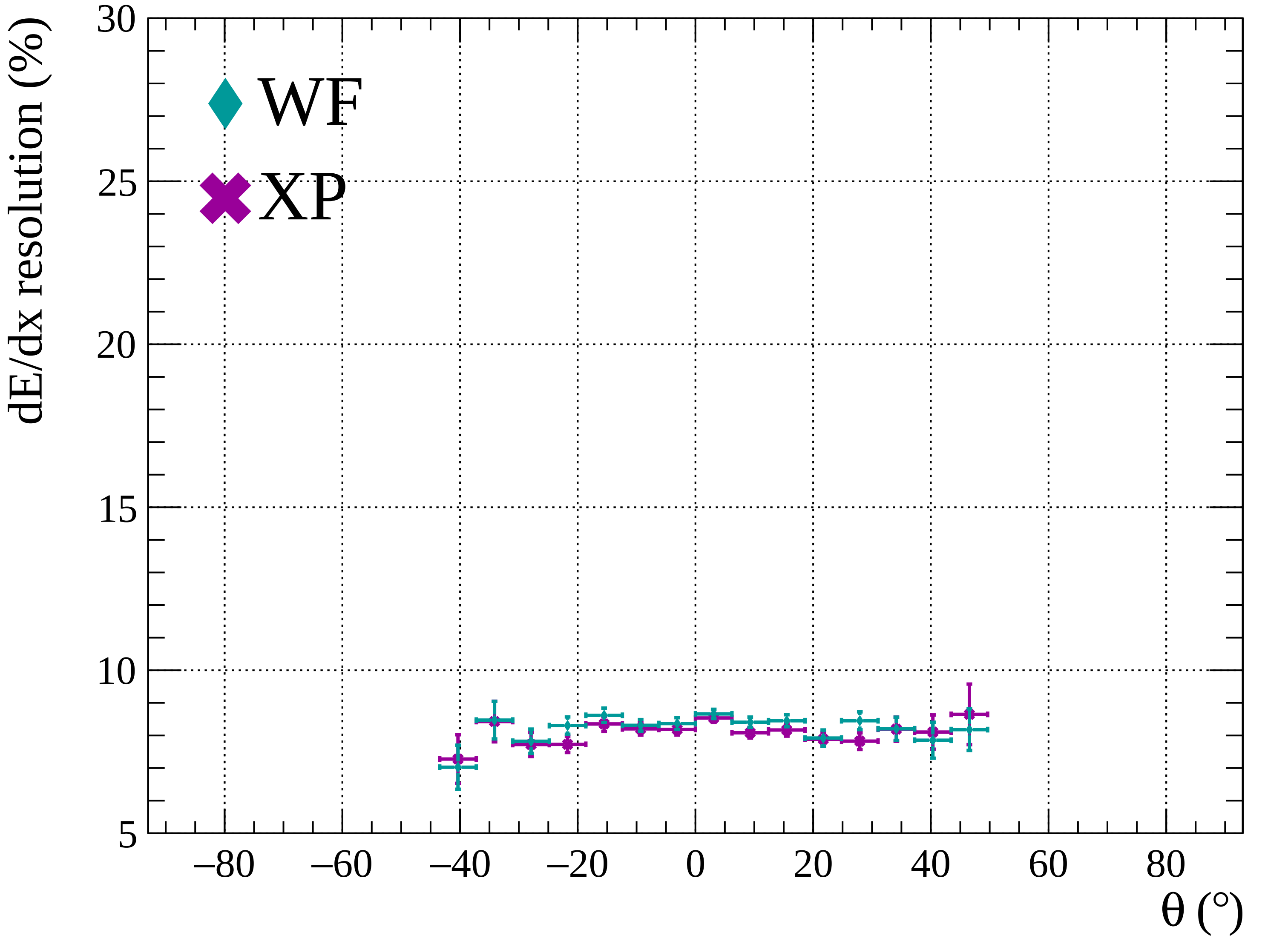

Count
1800
1600
1400
1200
1000
800
600
400
200
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



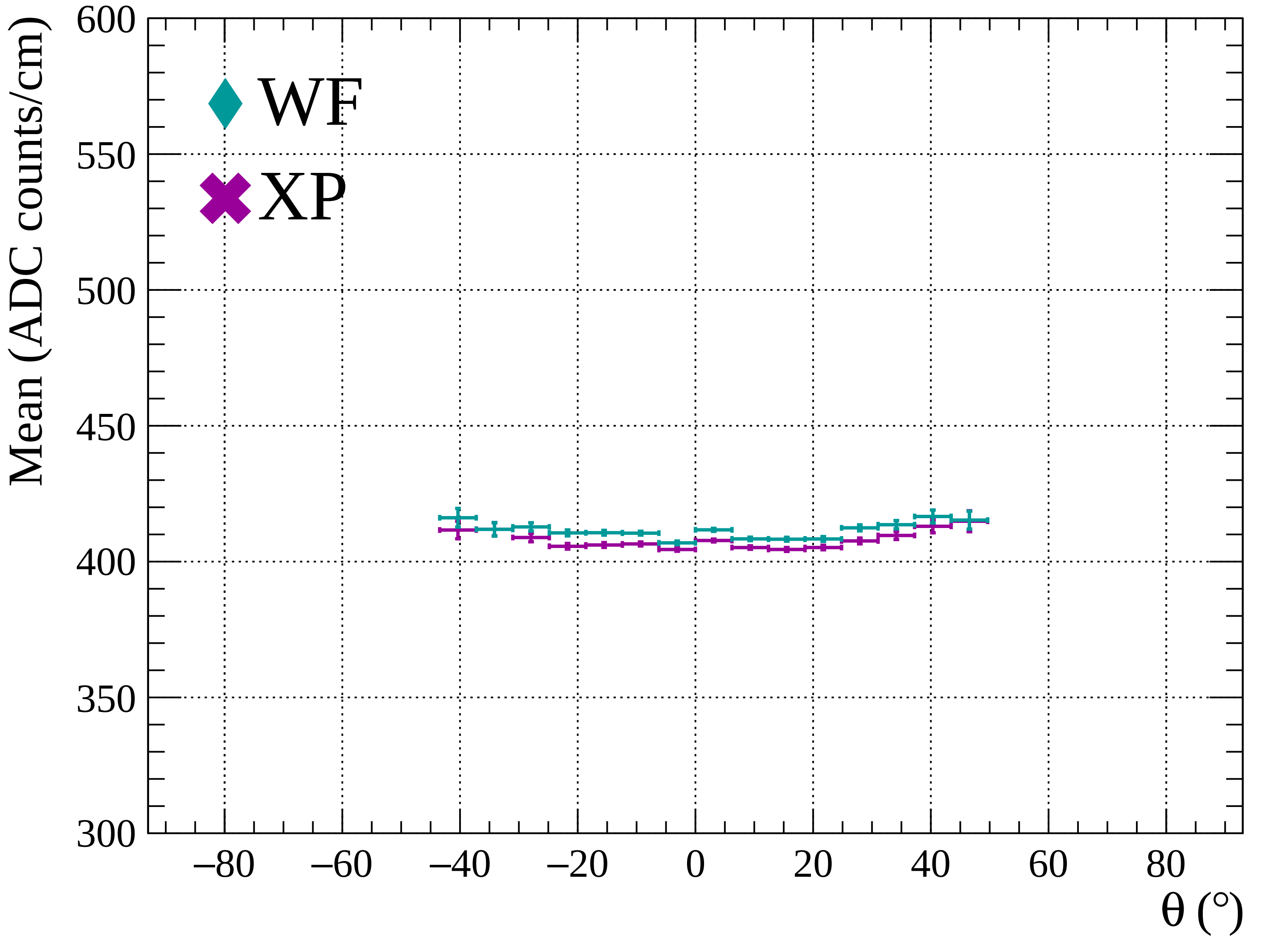
WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



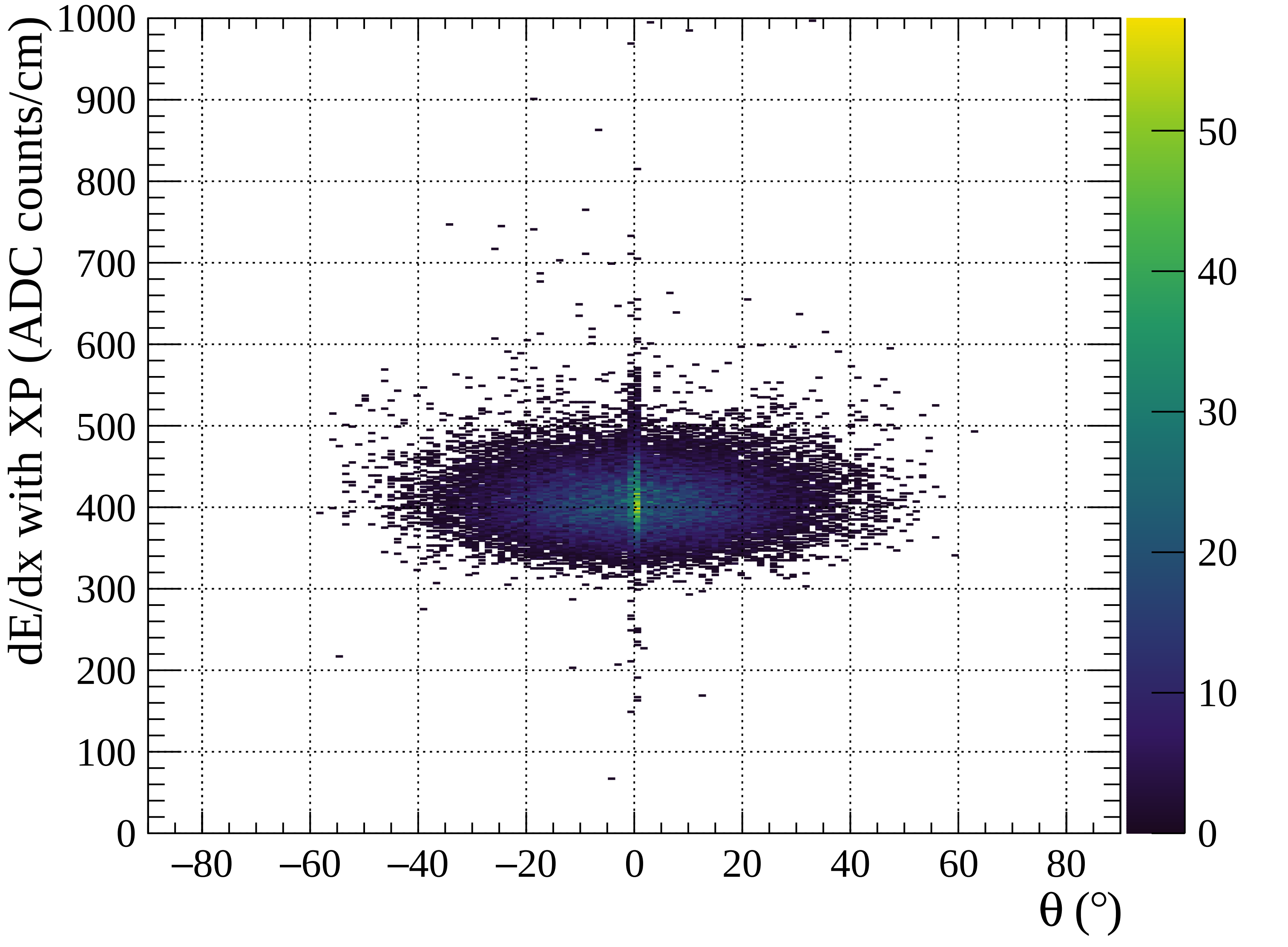
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
θ (°)



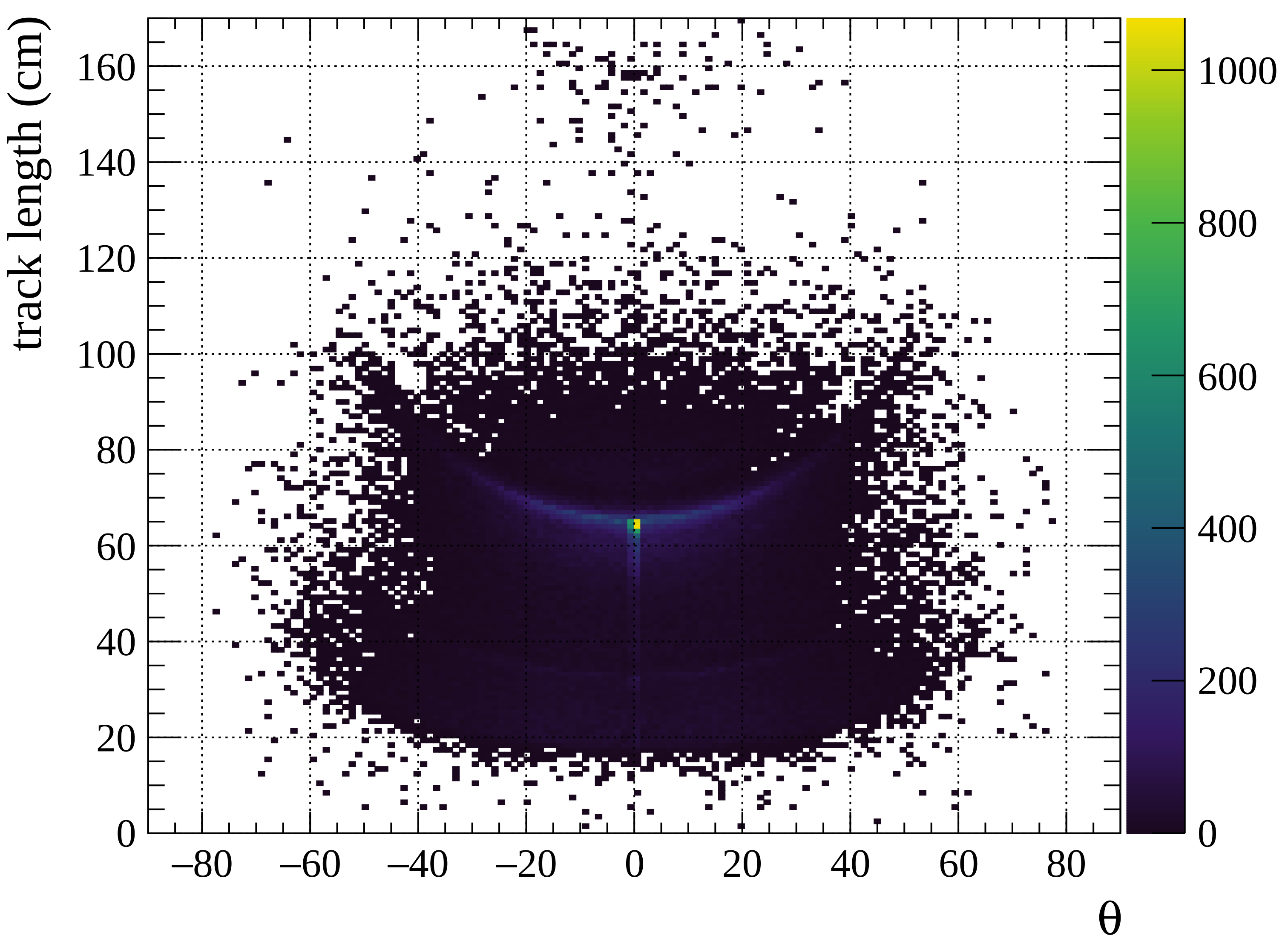
WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

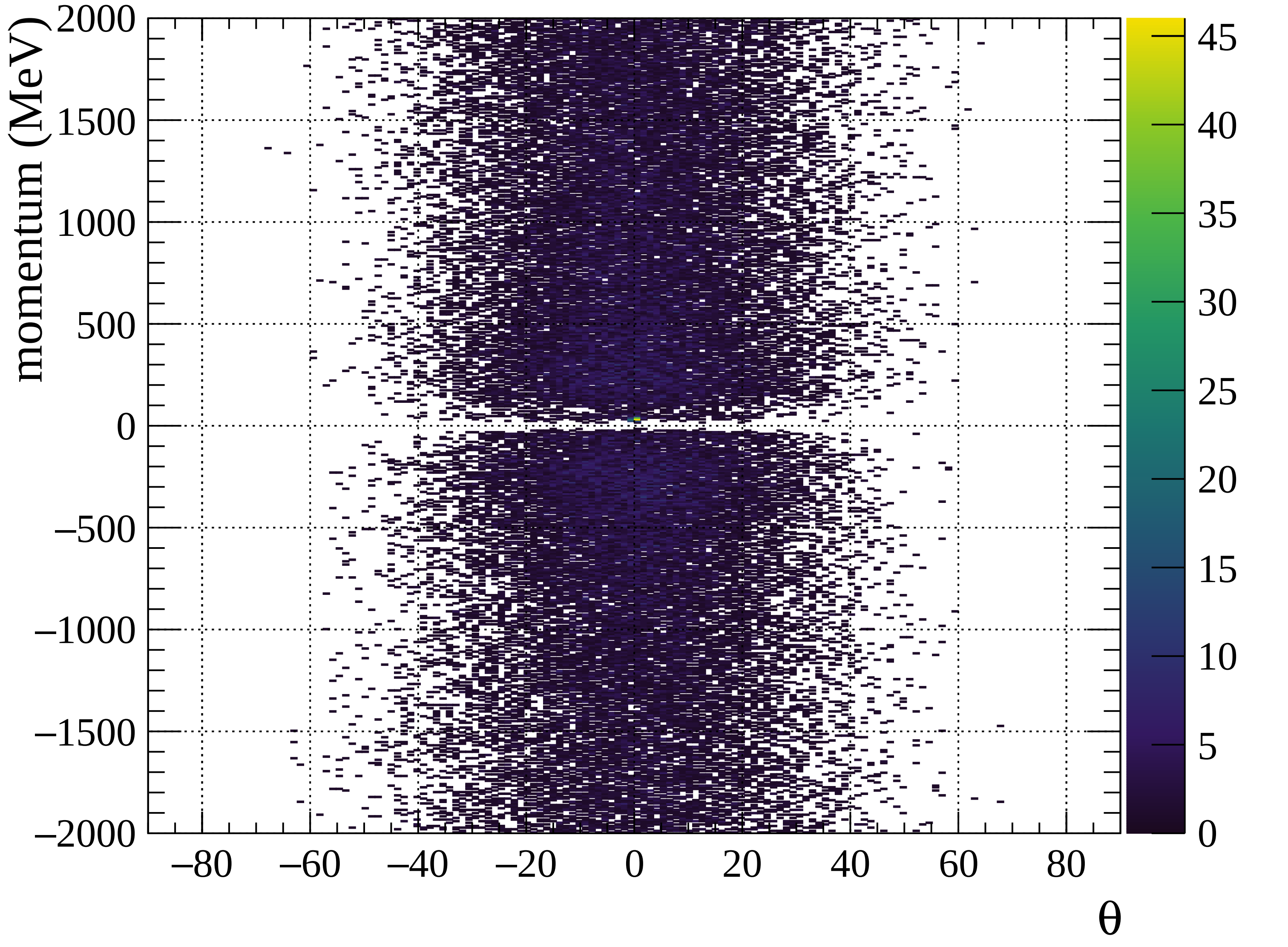


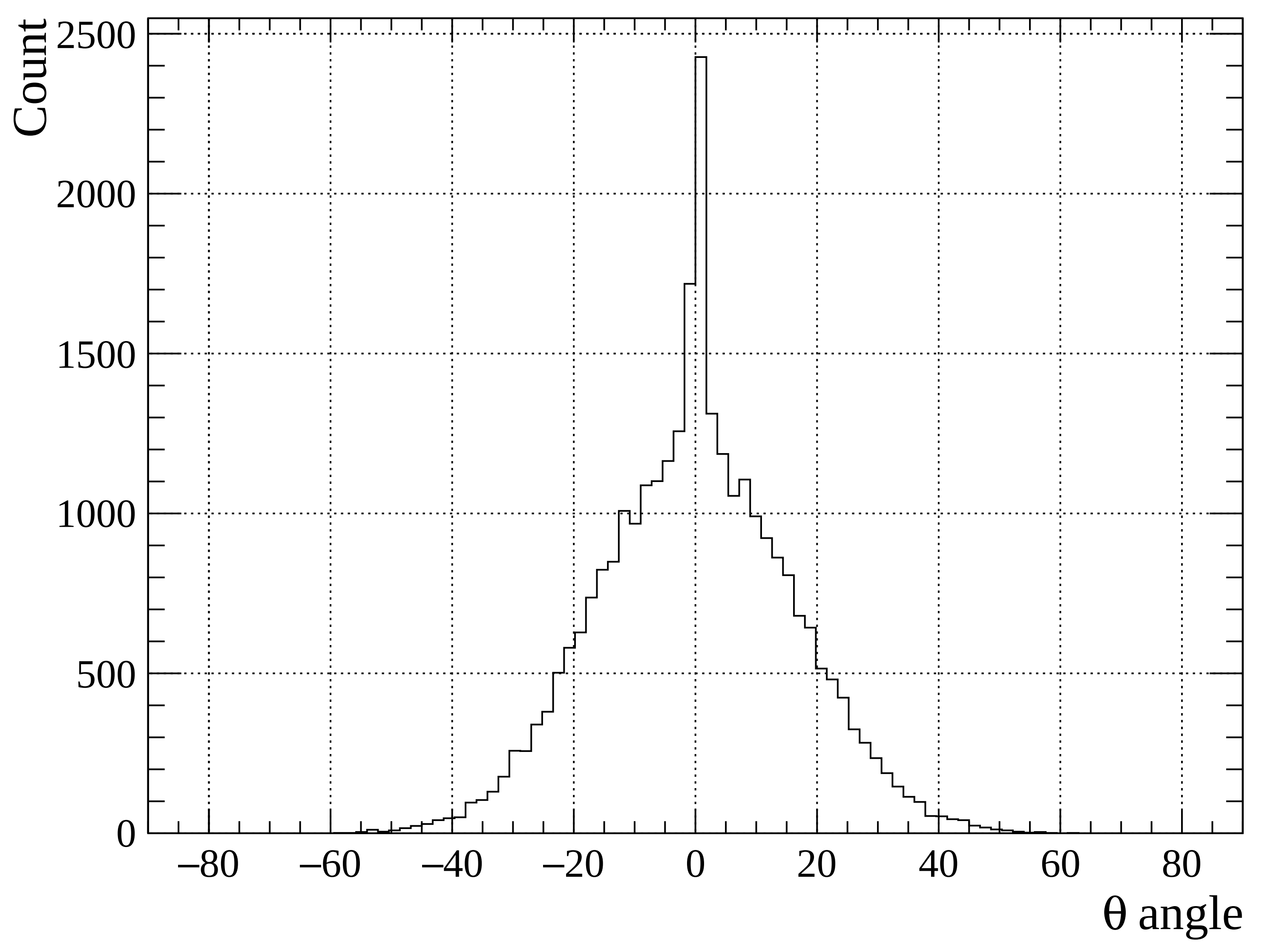
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)



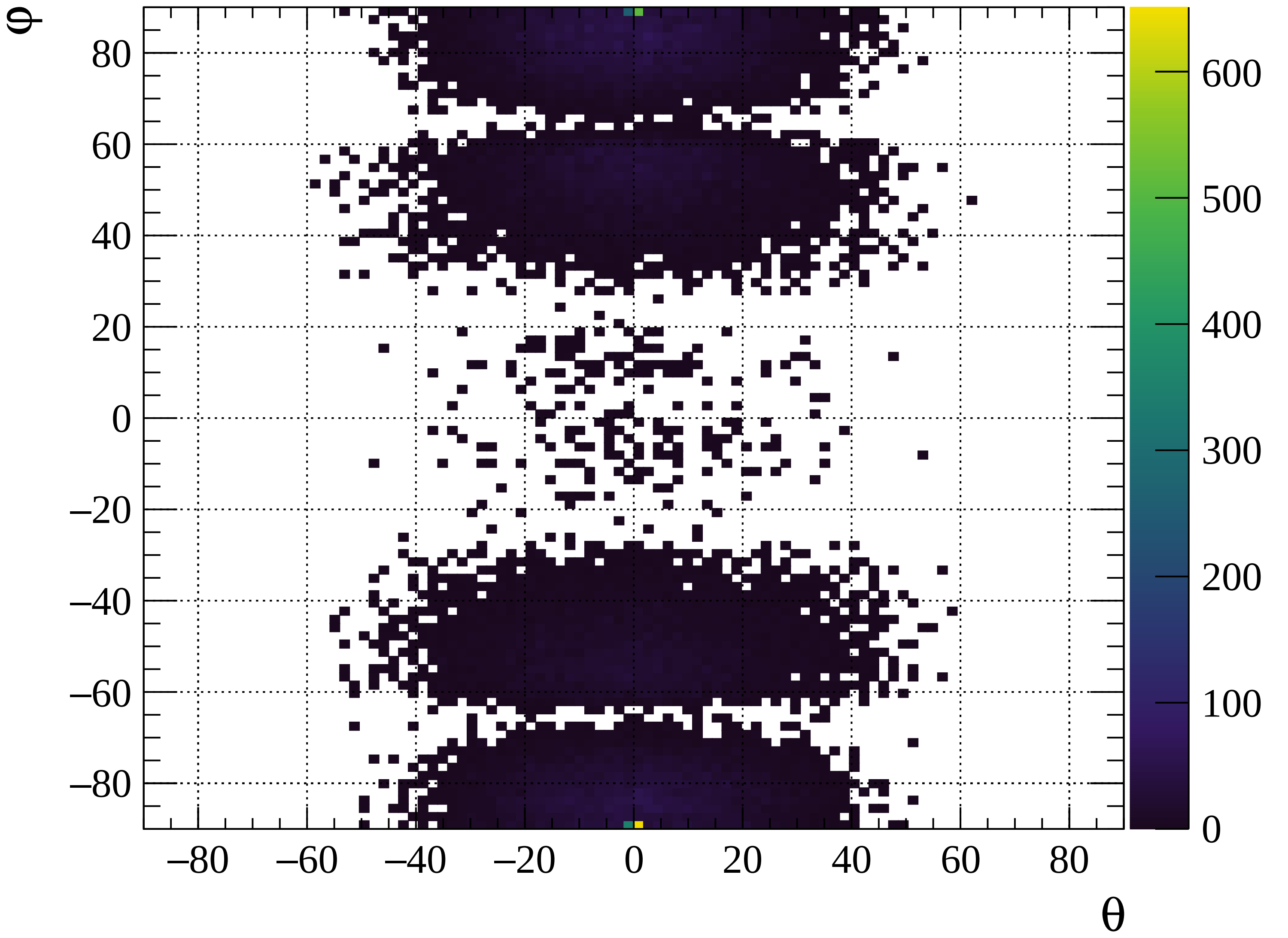
dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
50
40
30
20
10
0

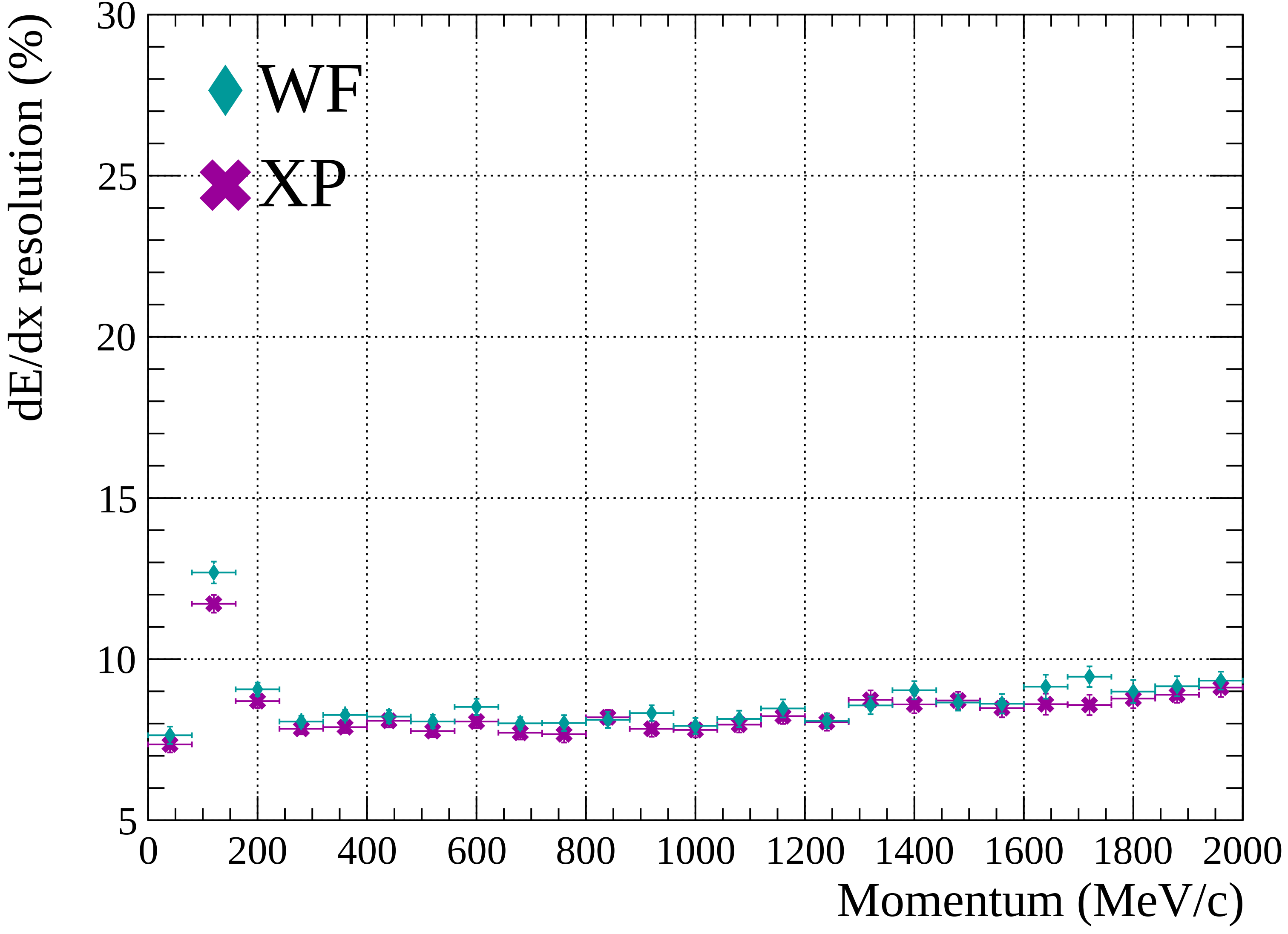

track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
θ
0
200
400
600
800
1000


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
45
40
35
30
25
20
15
10
5
0

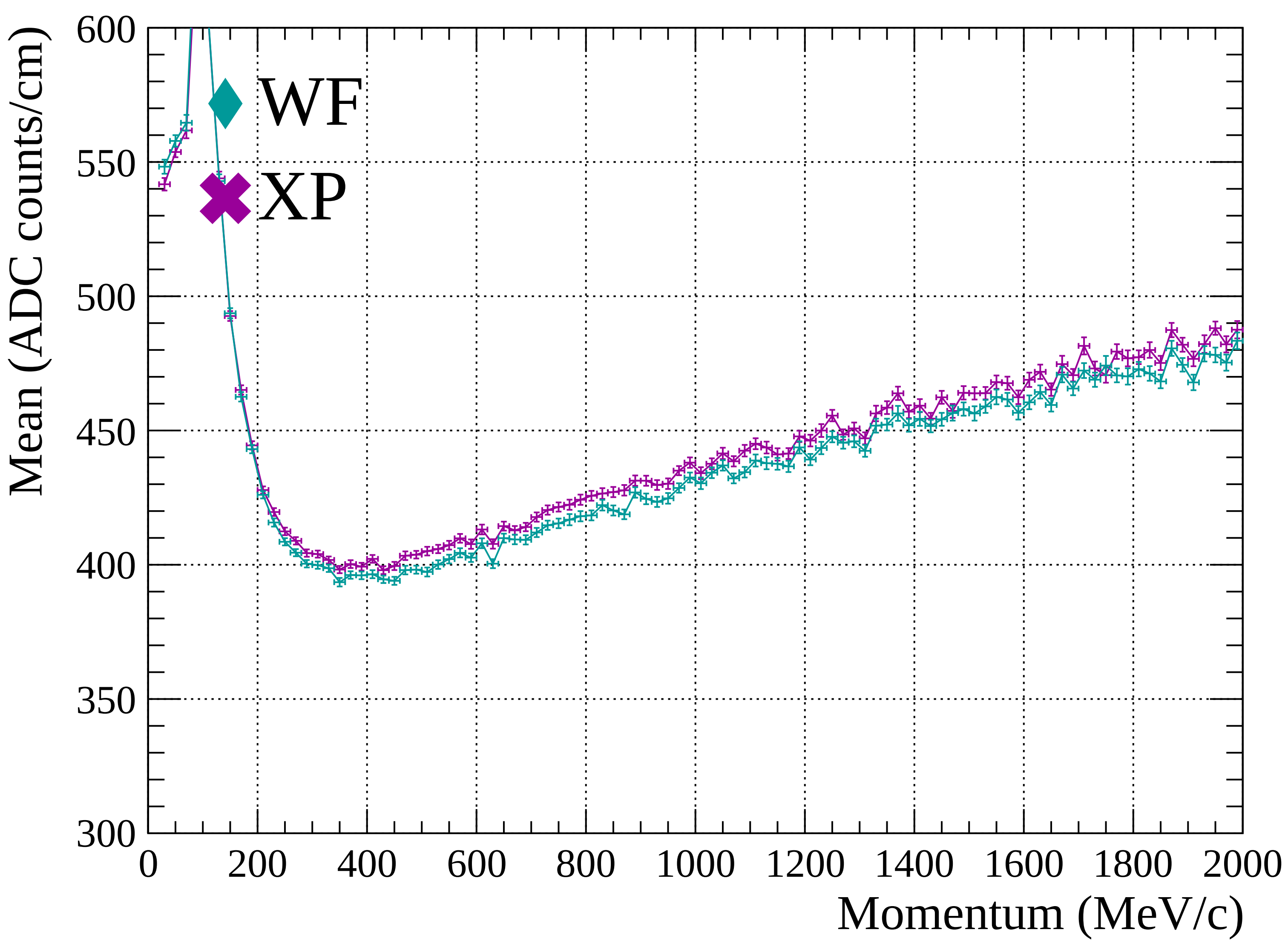


Count
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

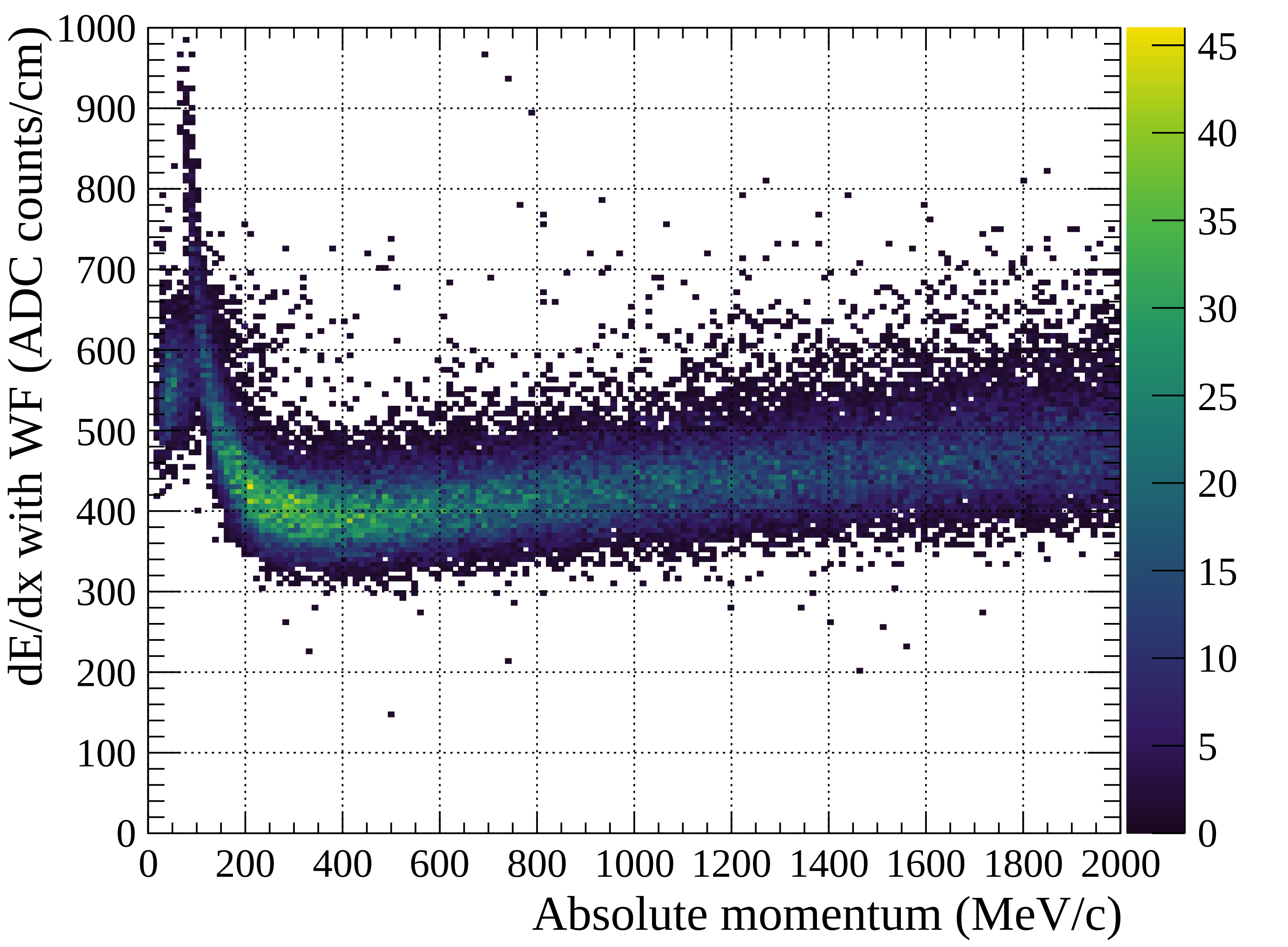

φ
θ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
0
100
200
300
400
500
600

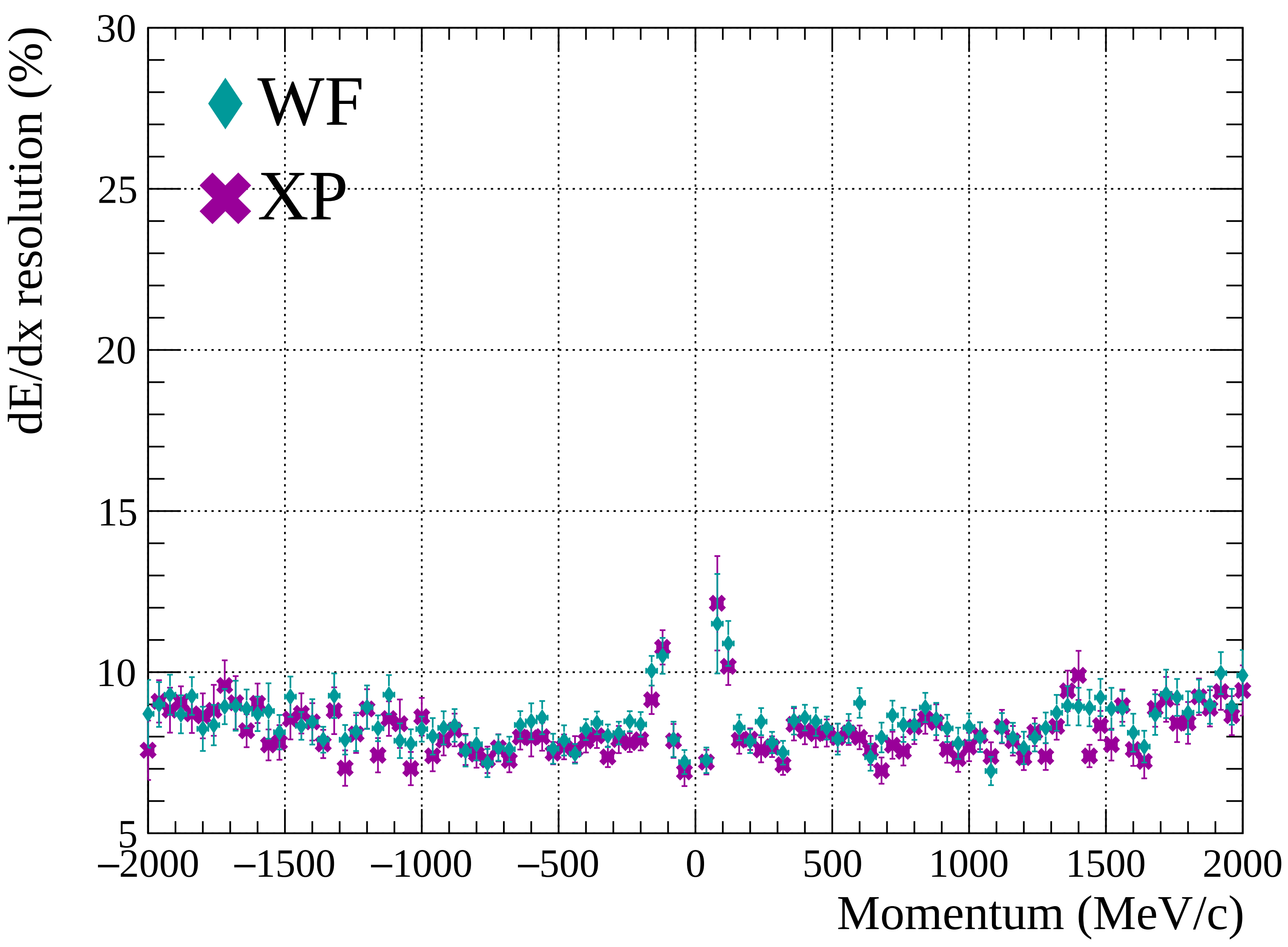


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400
1600
1800
2000

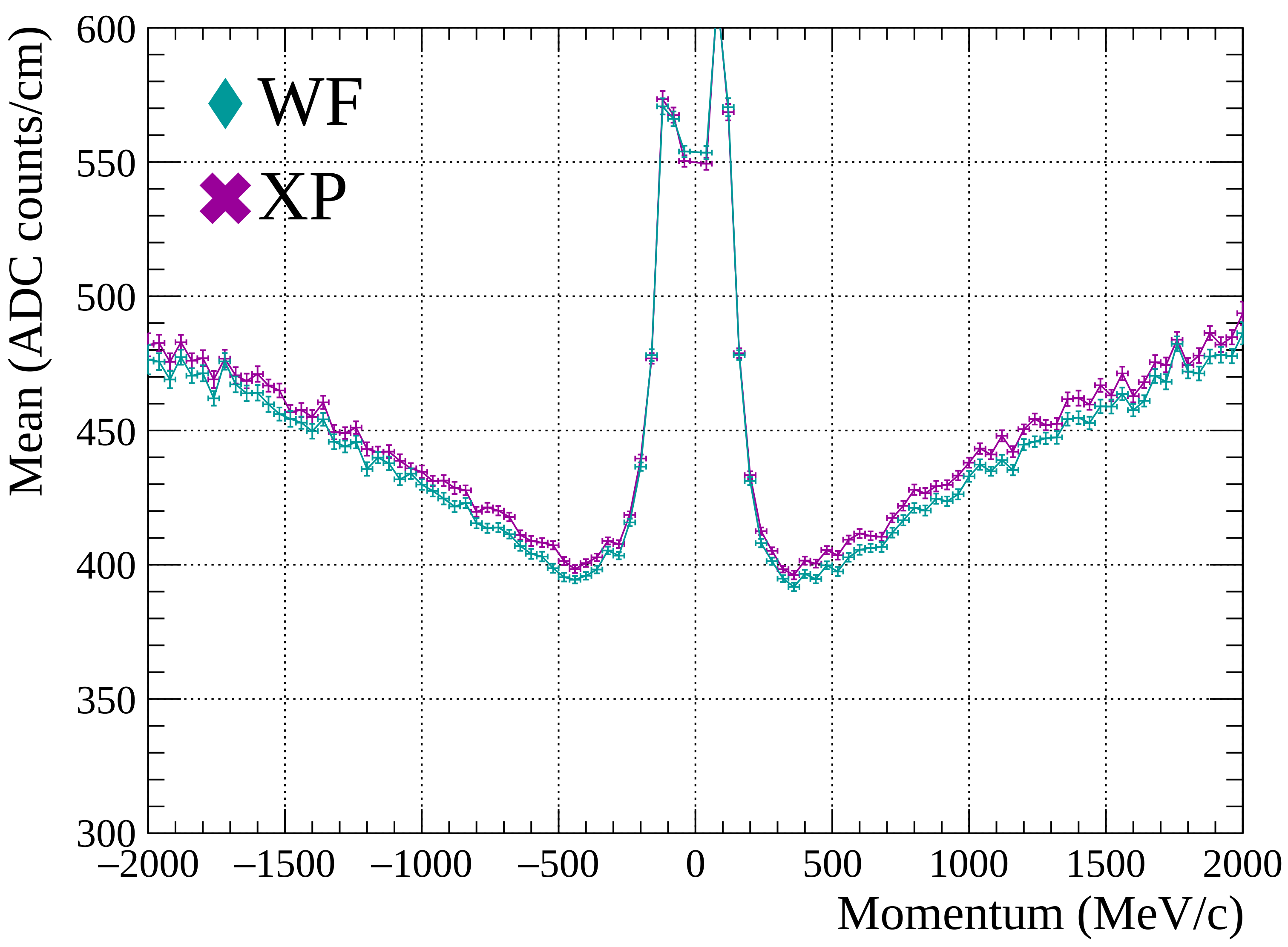


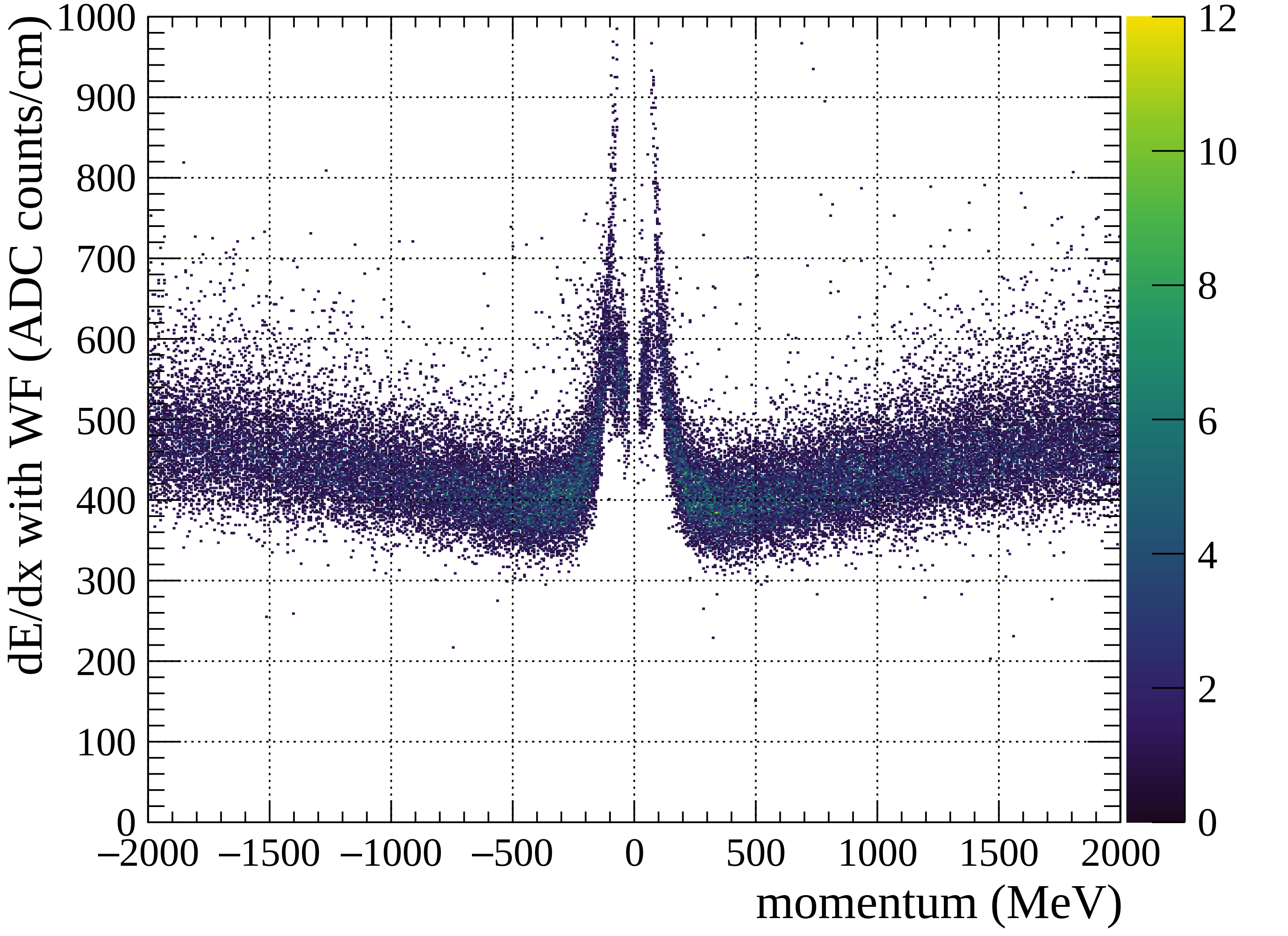
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000

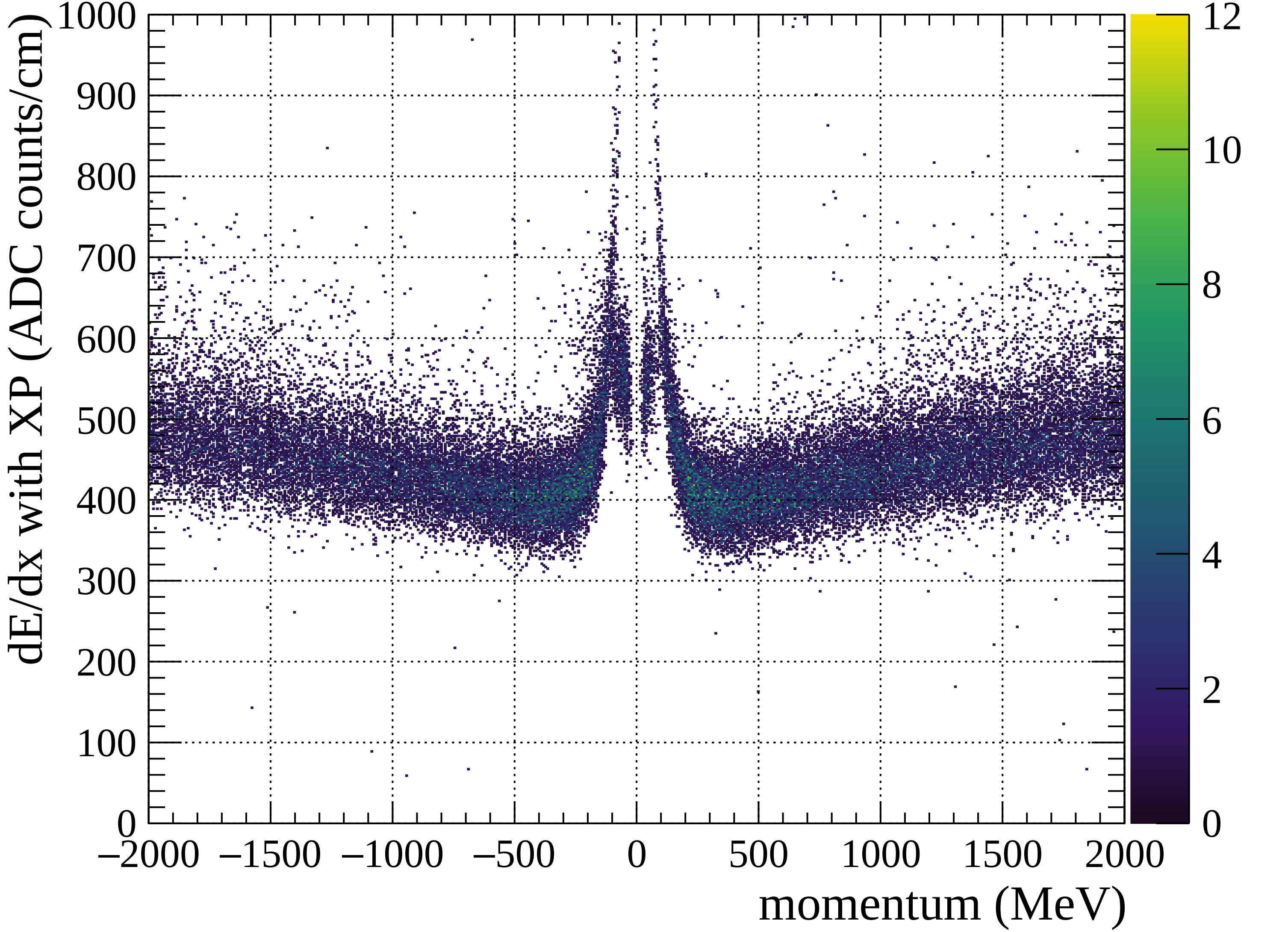

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25
30
35
40
45

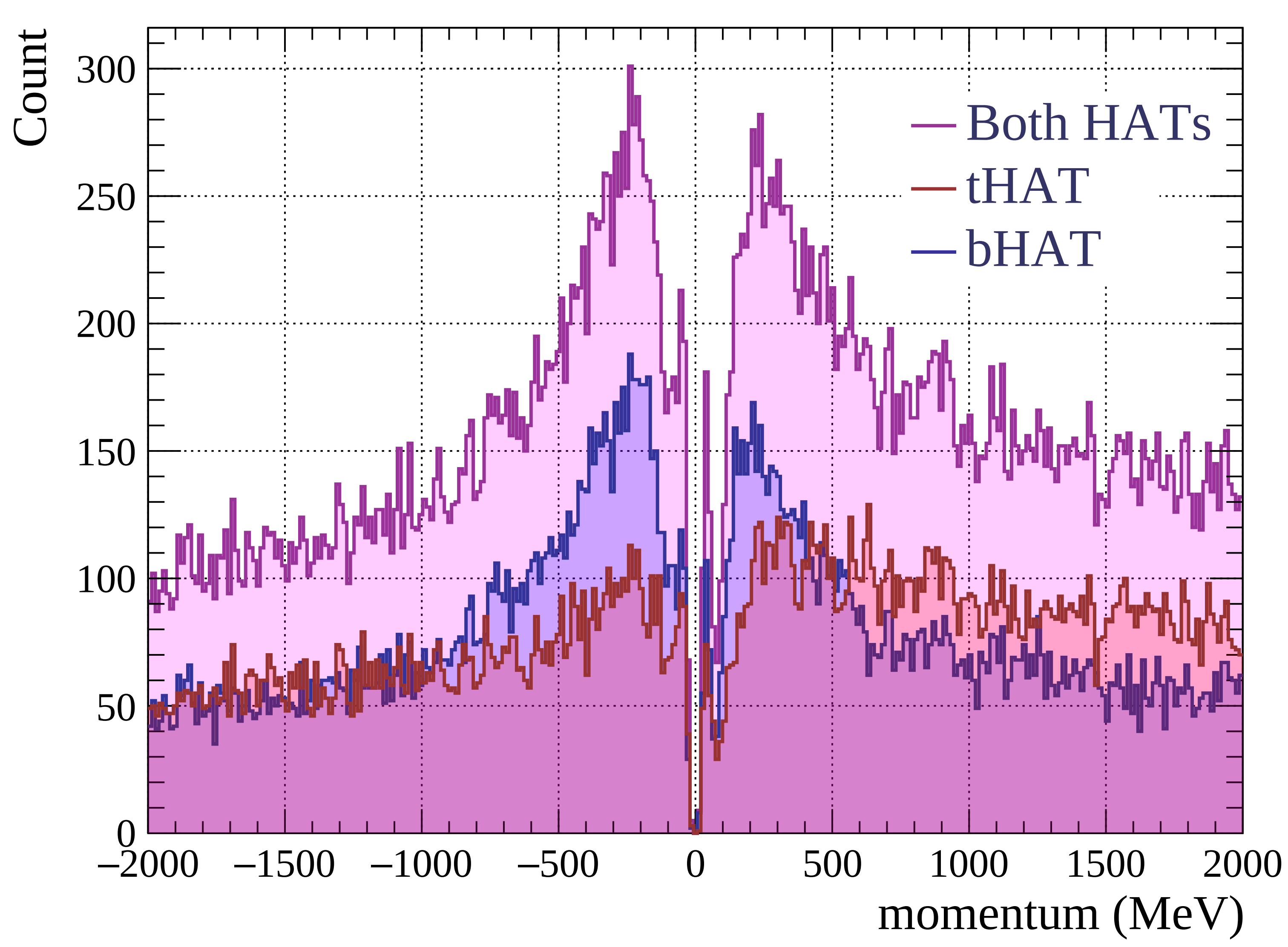


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

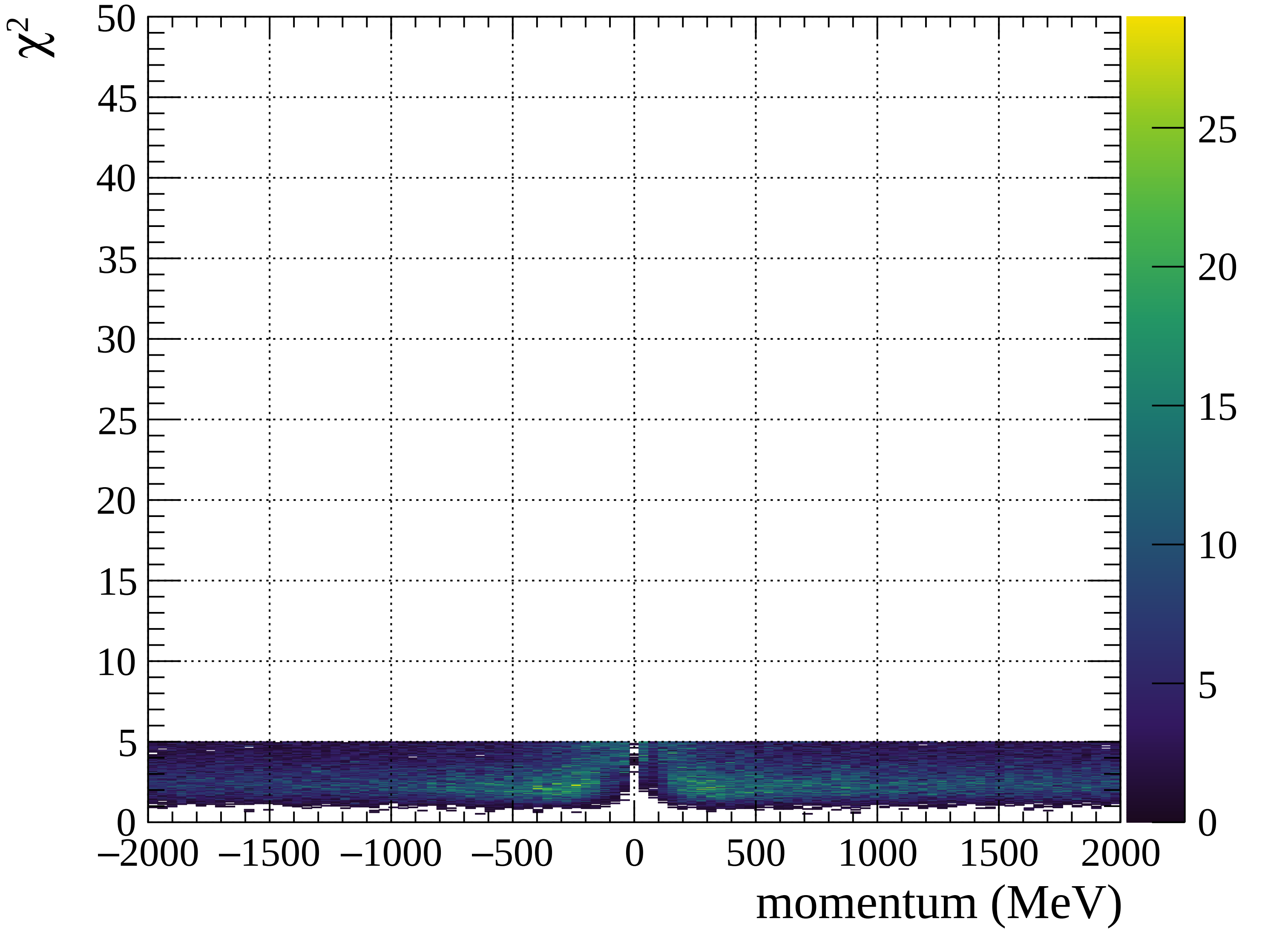

WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
12
10
8
6
4
2
0

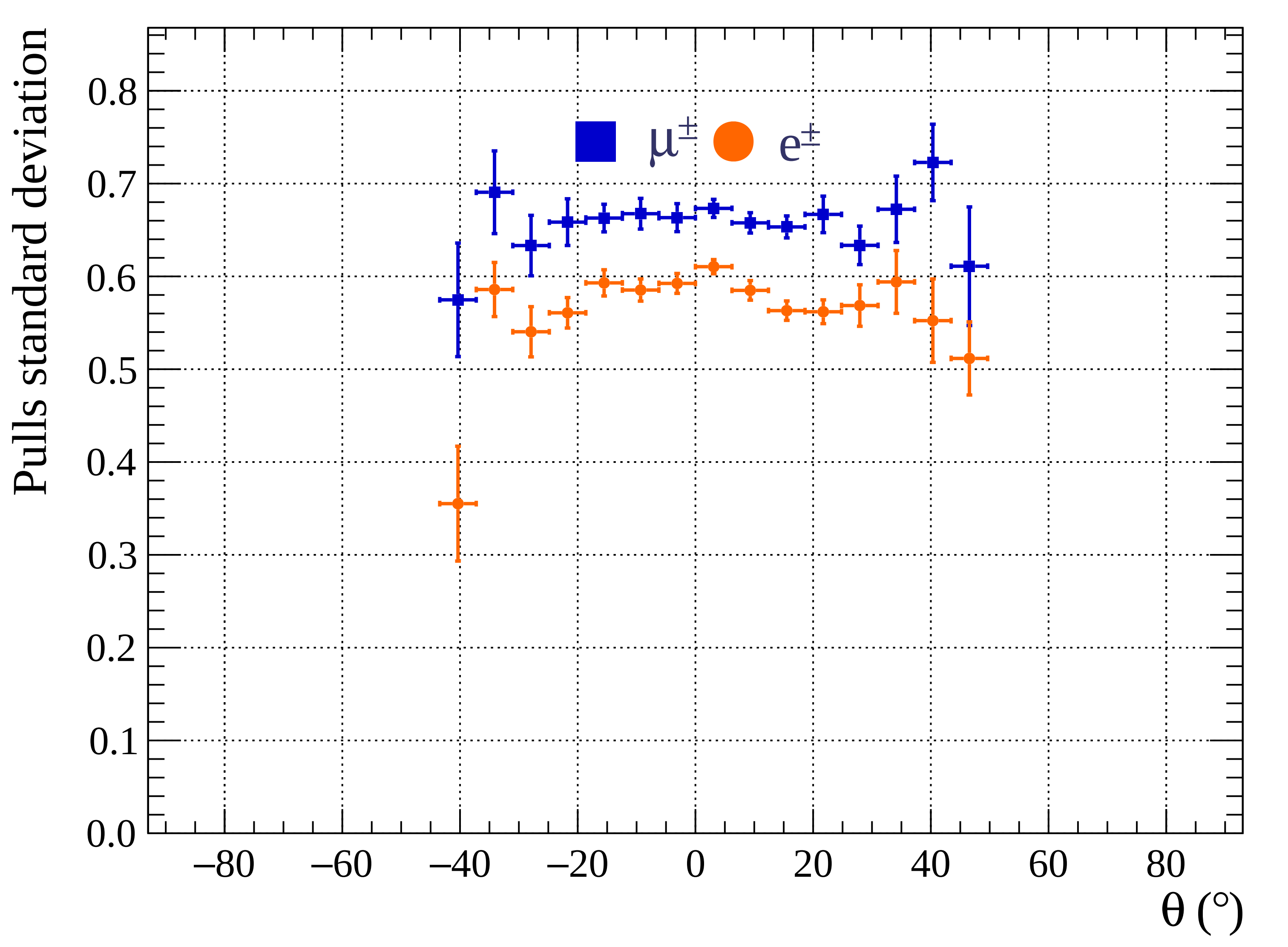

dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12



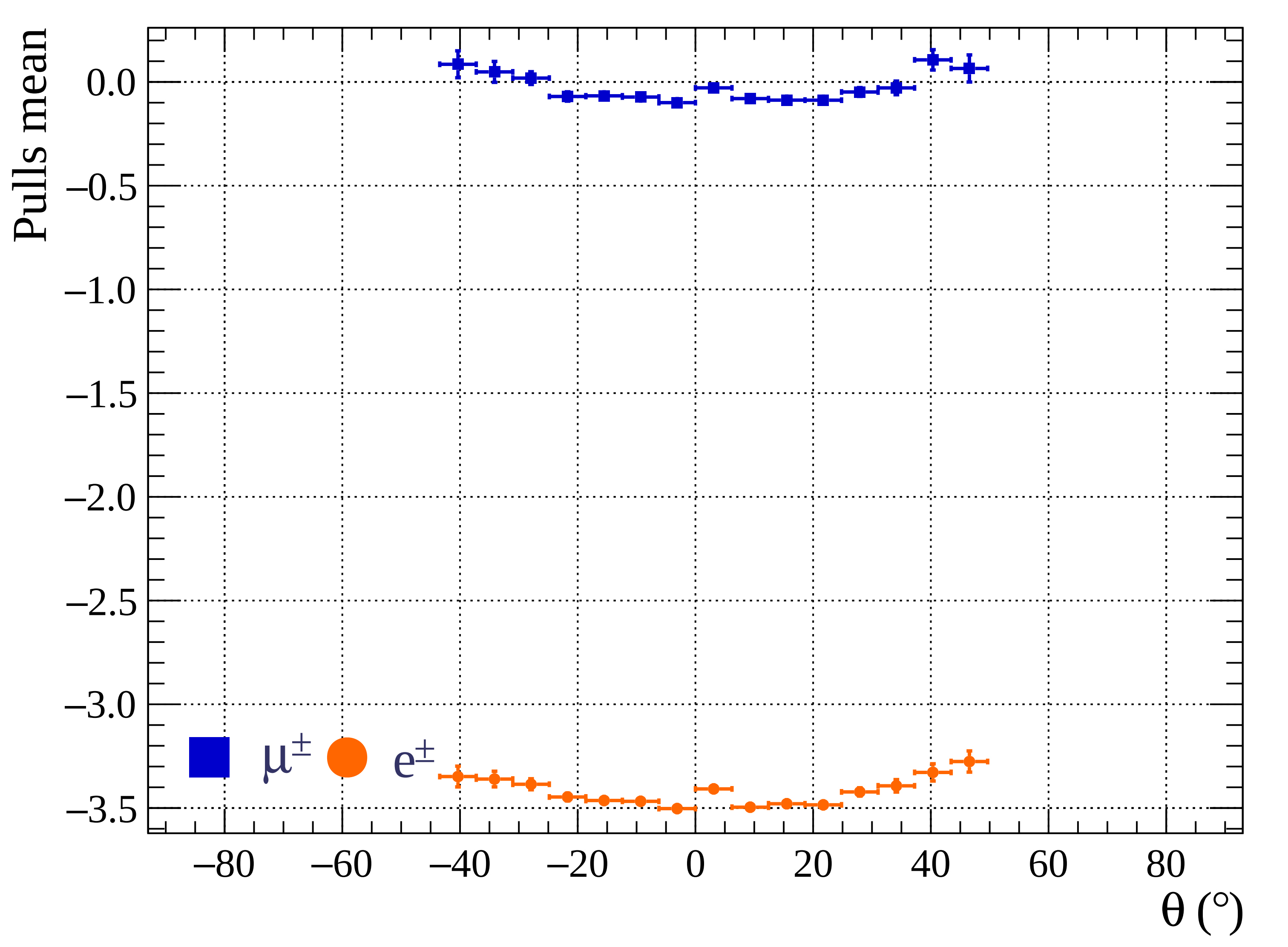
Count
300
250
200
150
100
50
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



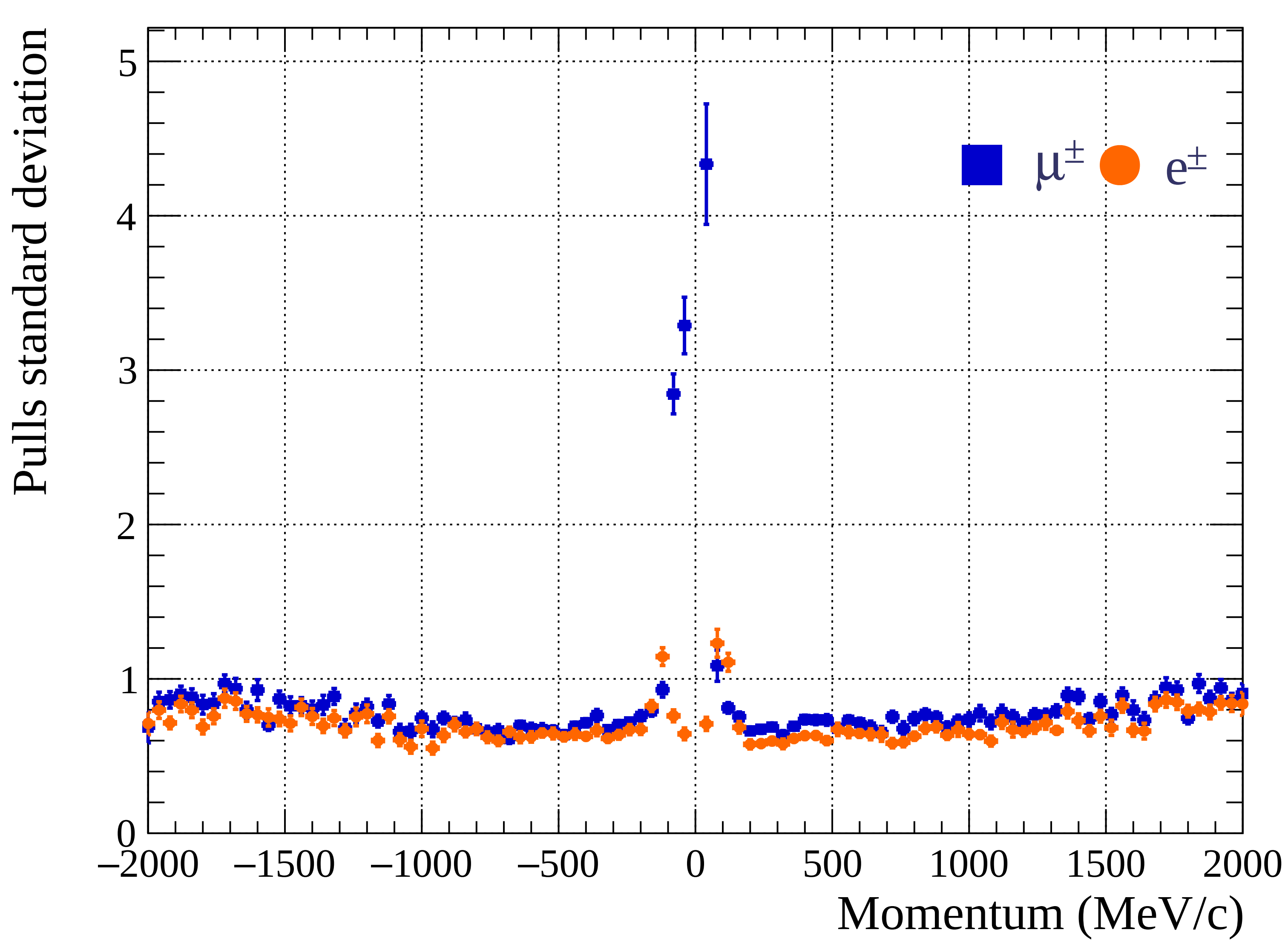
χ^2
0
5
10
15
20
25
30
35
40
45
50
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
0
5
10
15
20
25



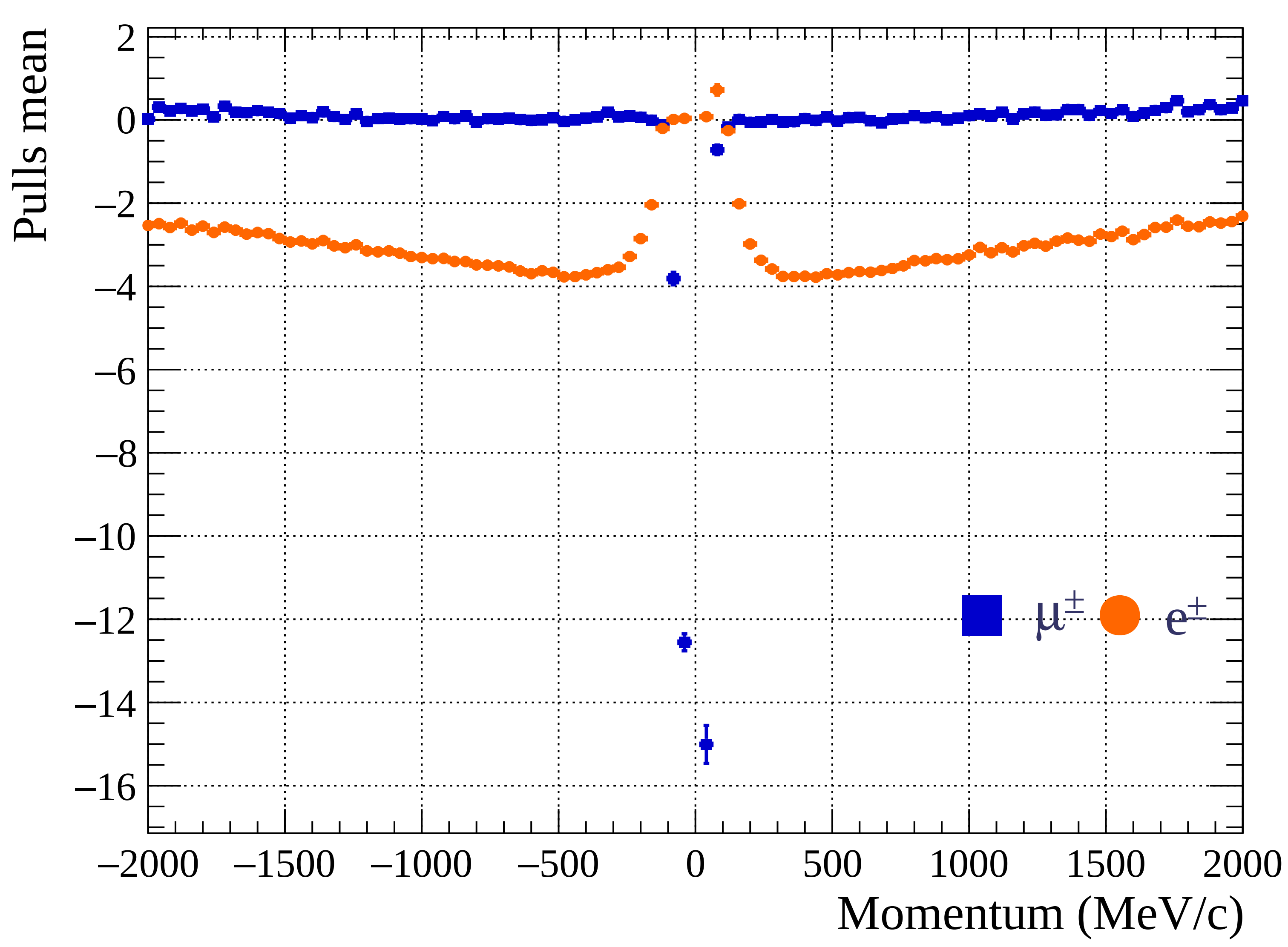
Pulls standard deviation
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



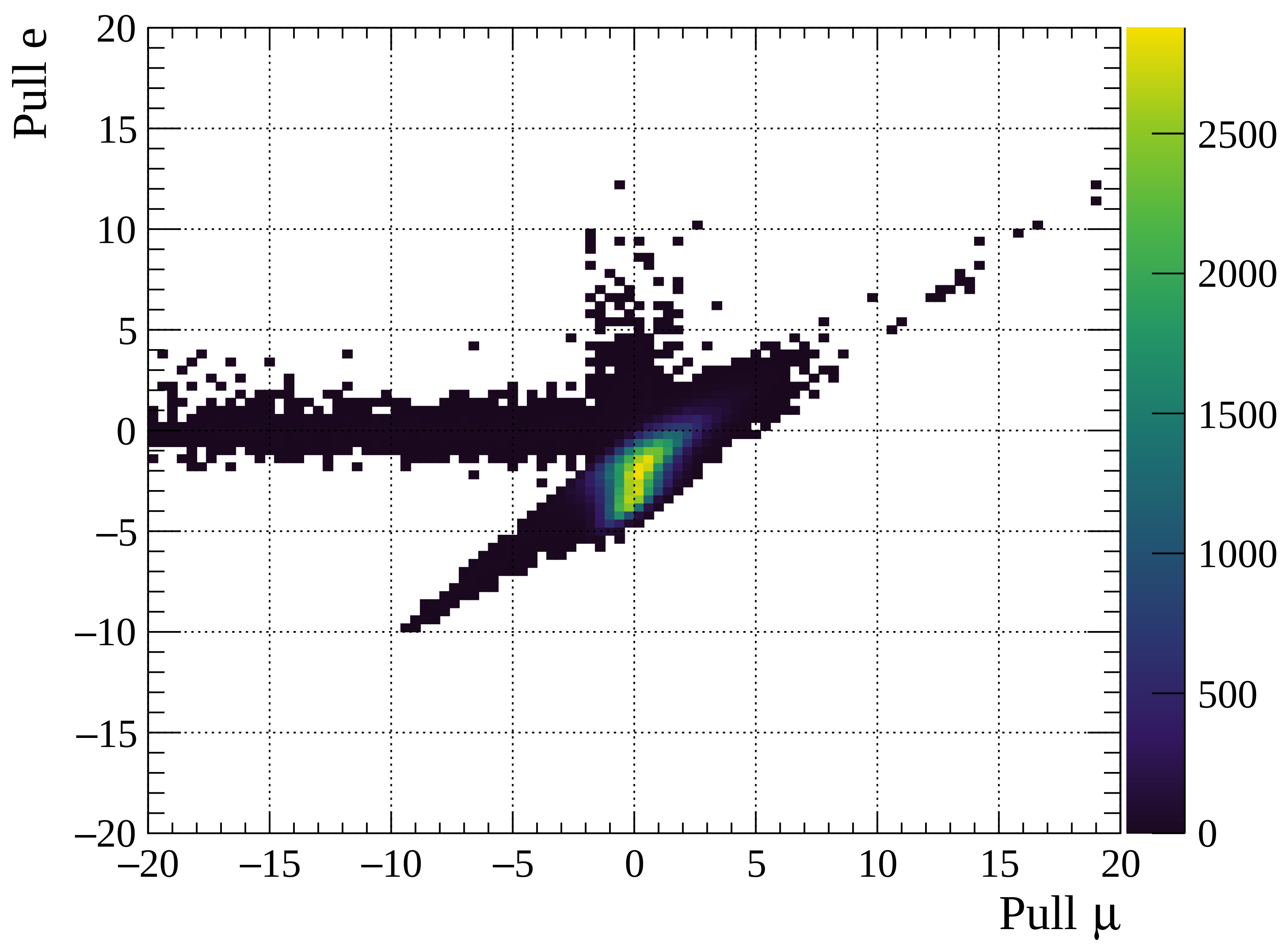
Pulls mean
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)



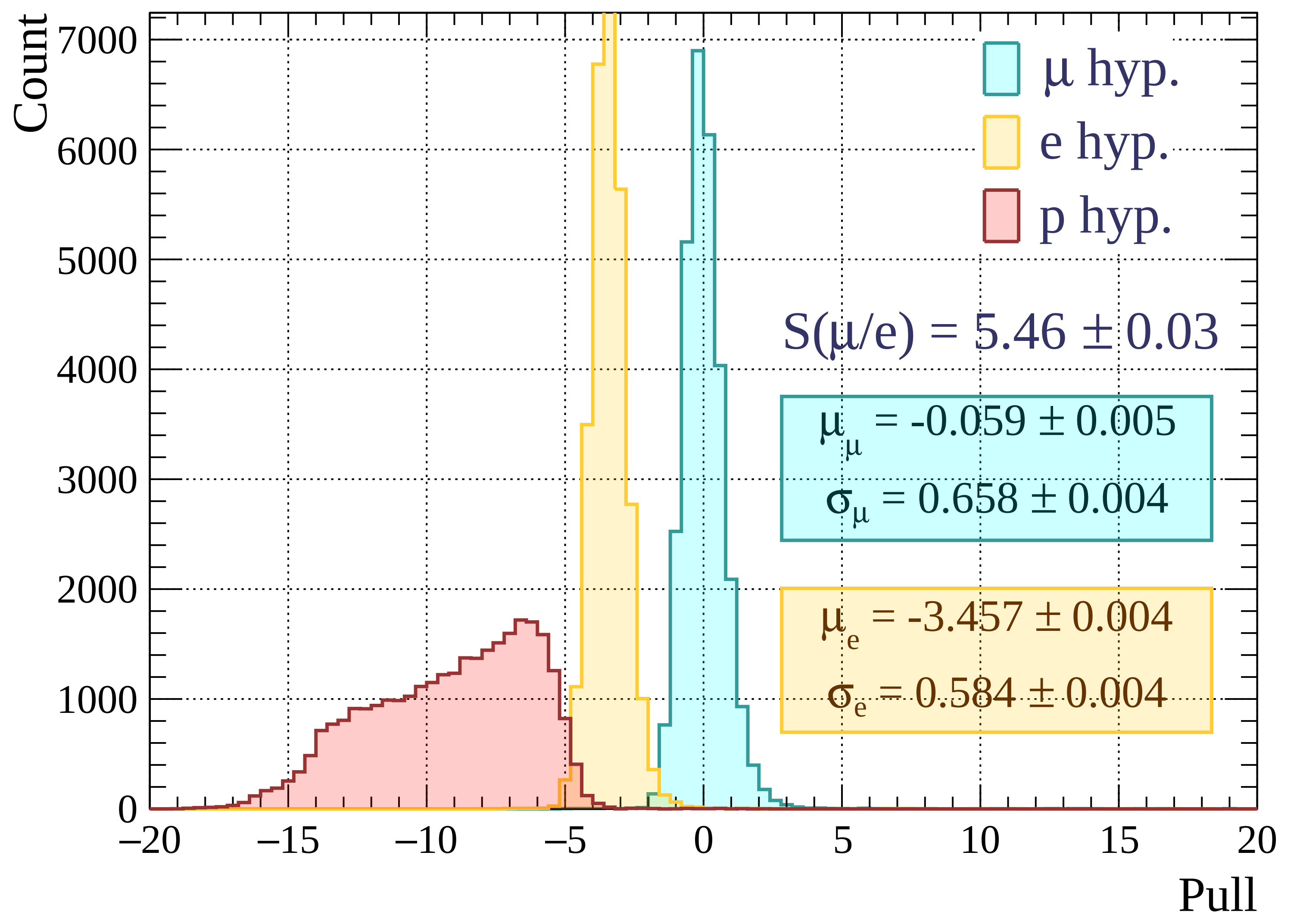
Pulls standard deviation
0
1
2
3
4
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



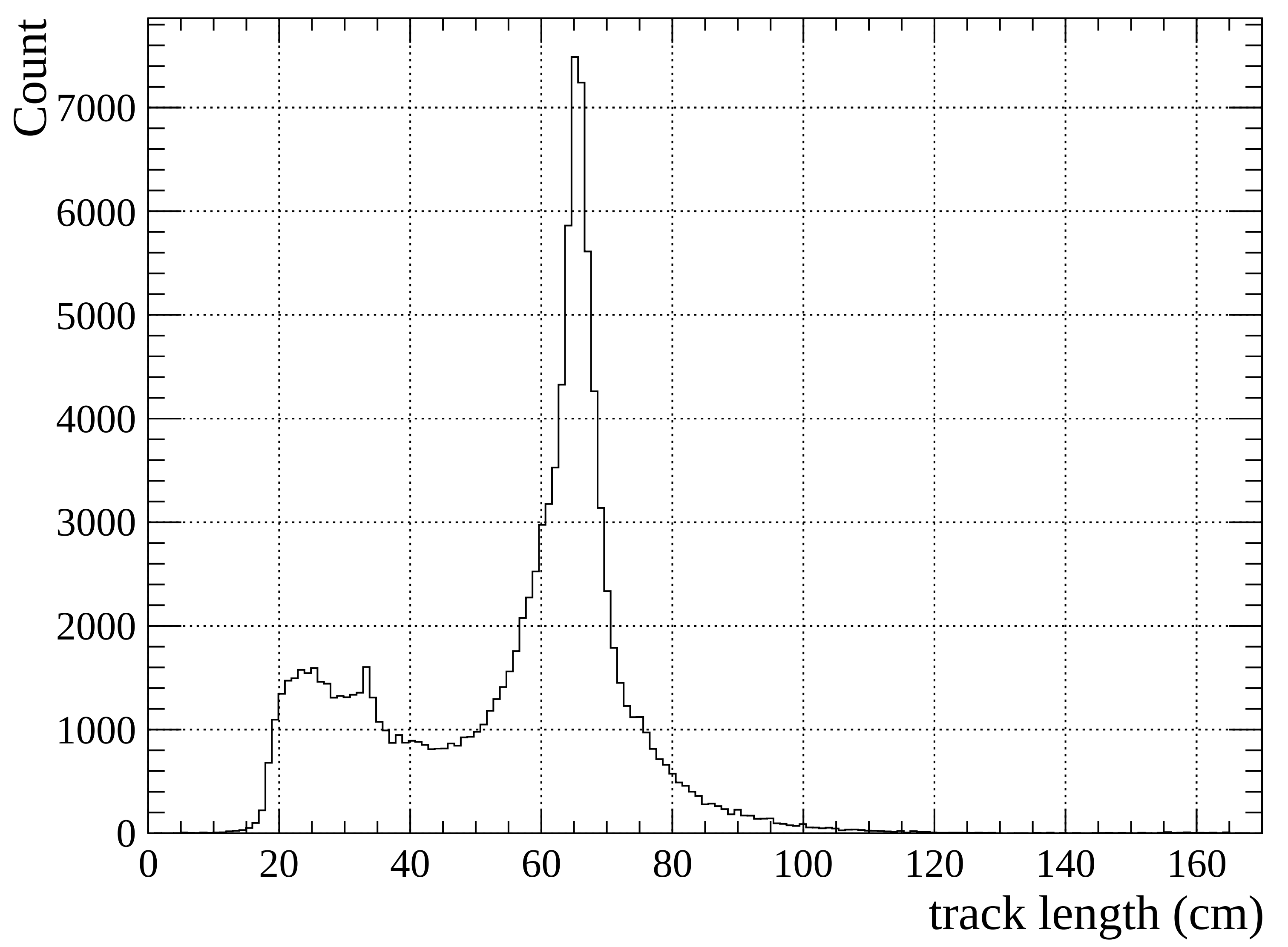
Pulls mean
2
0
−2
−4
−6
−8
−10
−12
−14
−16
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

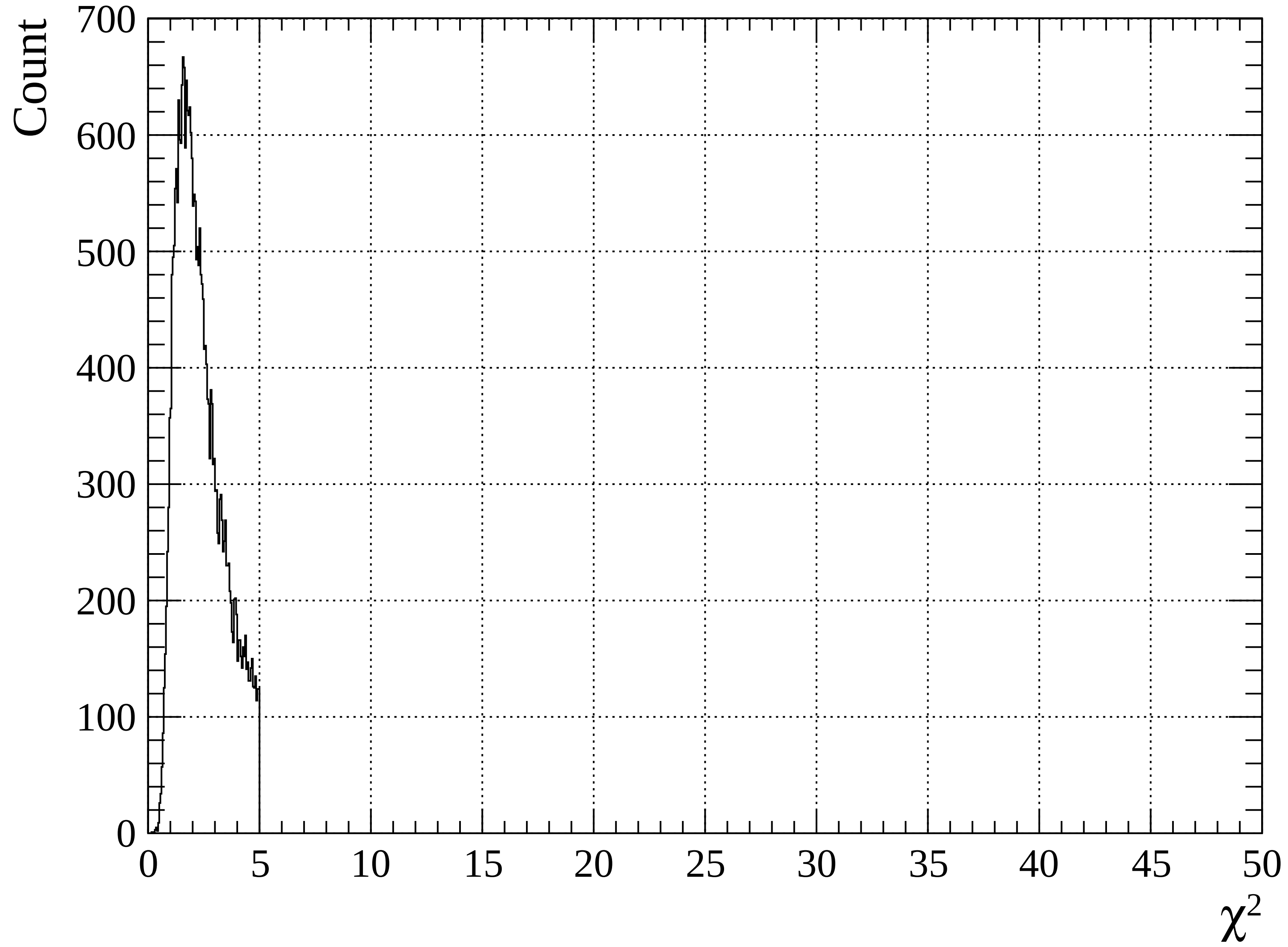


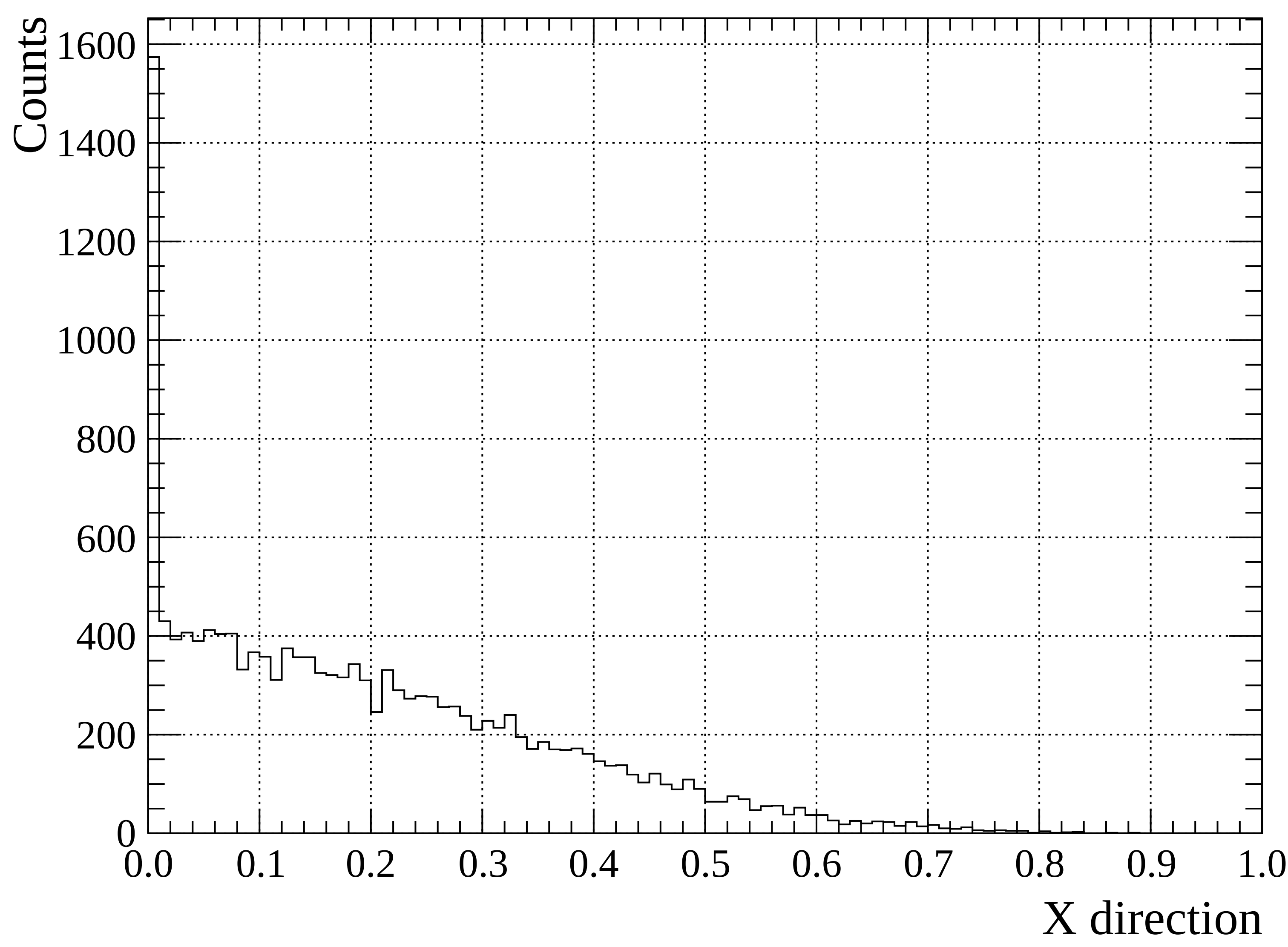
Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
2500
2000
1500
1000
500
0

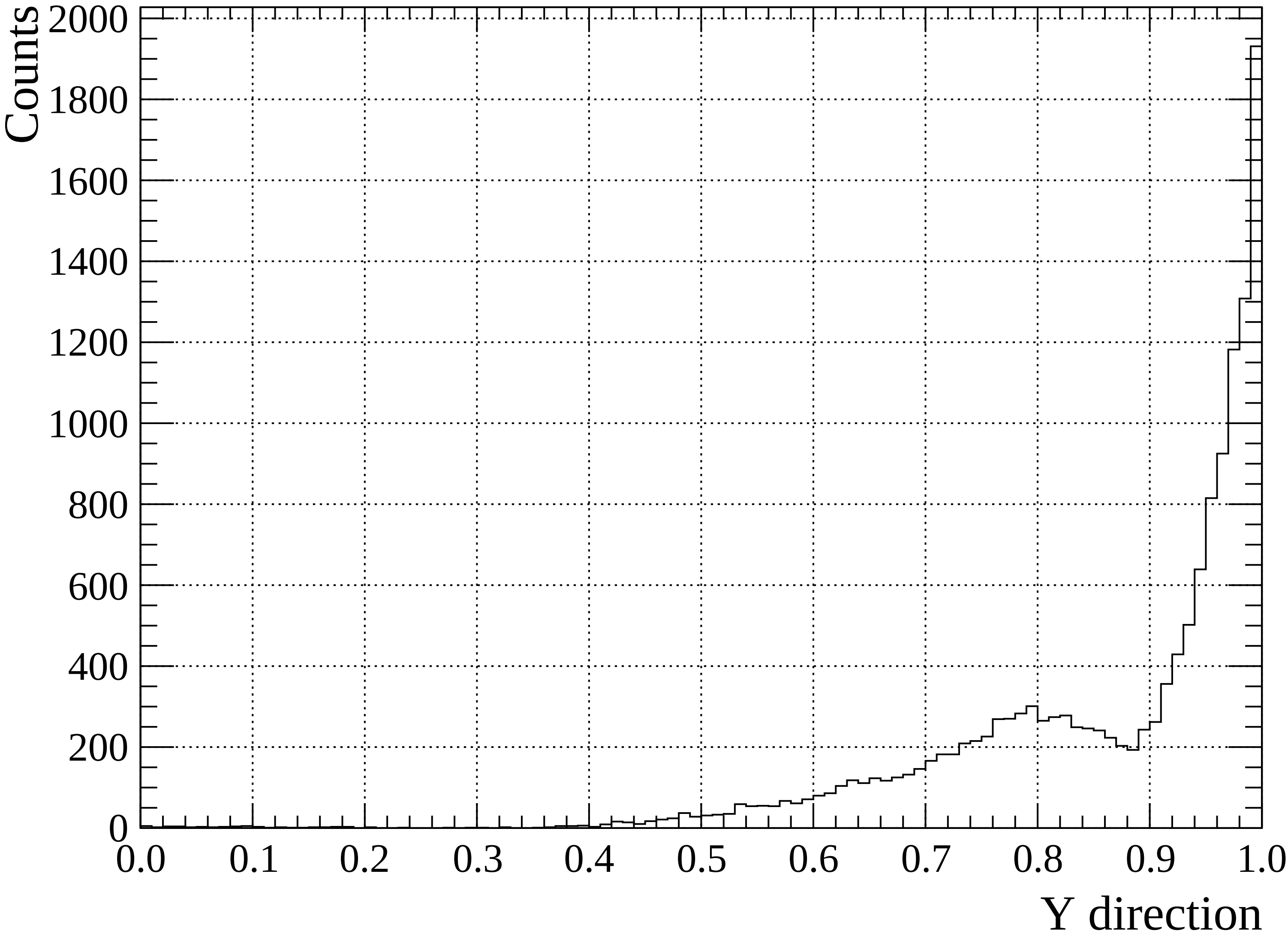


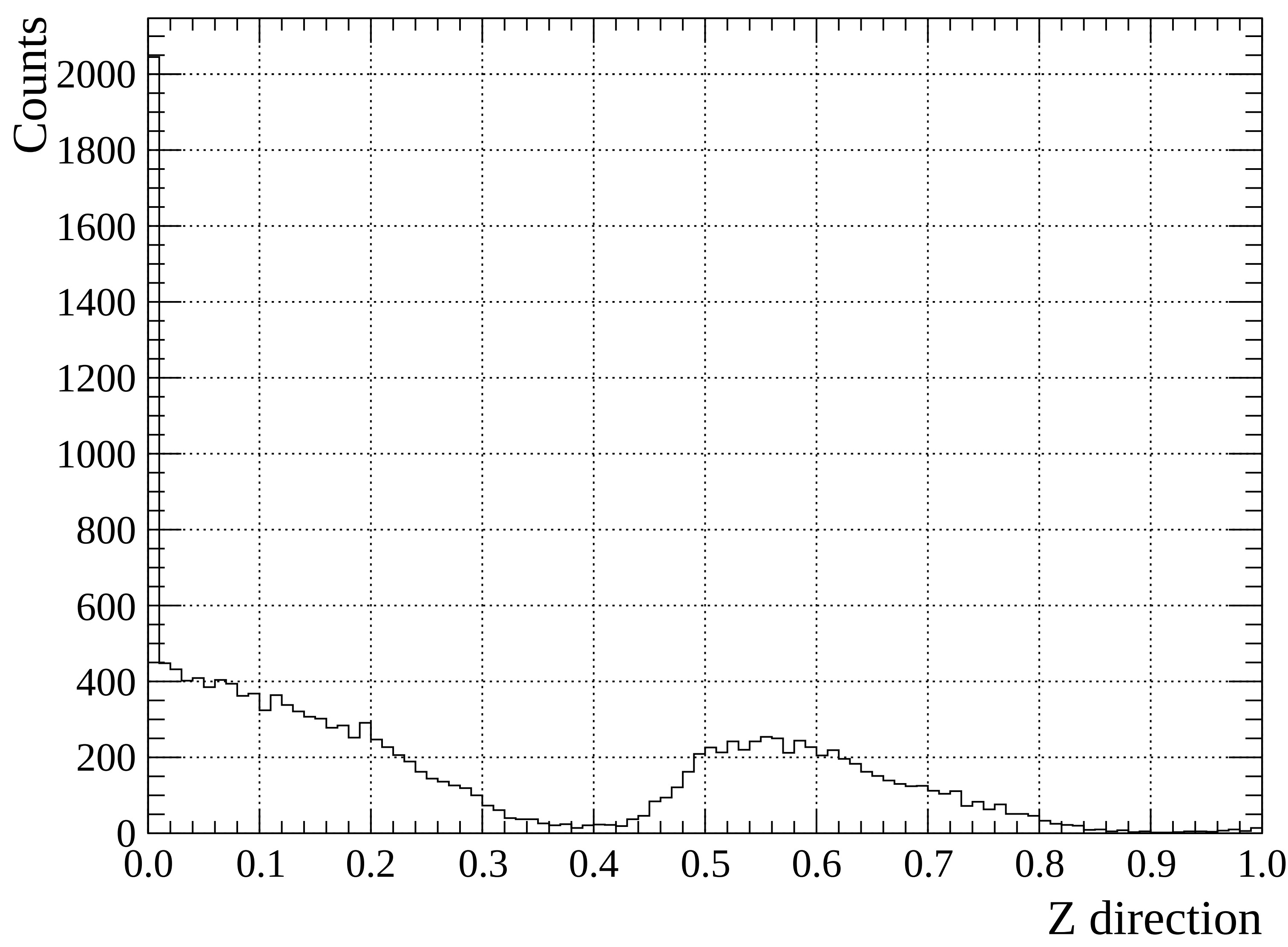
Count
7000
6000
5000
4000
3000
2000
1000
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 5.46 ± 0.03
μ_μ = -0.059 ± 0.005
σ_μ = 0.658 ± 0.004
μ_e = -3.457 ± 0.004
σ_e = 0.584 ± 0.004

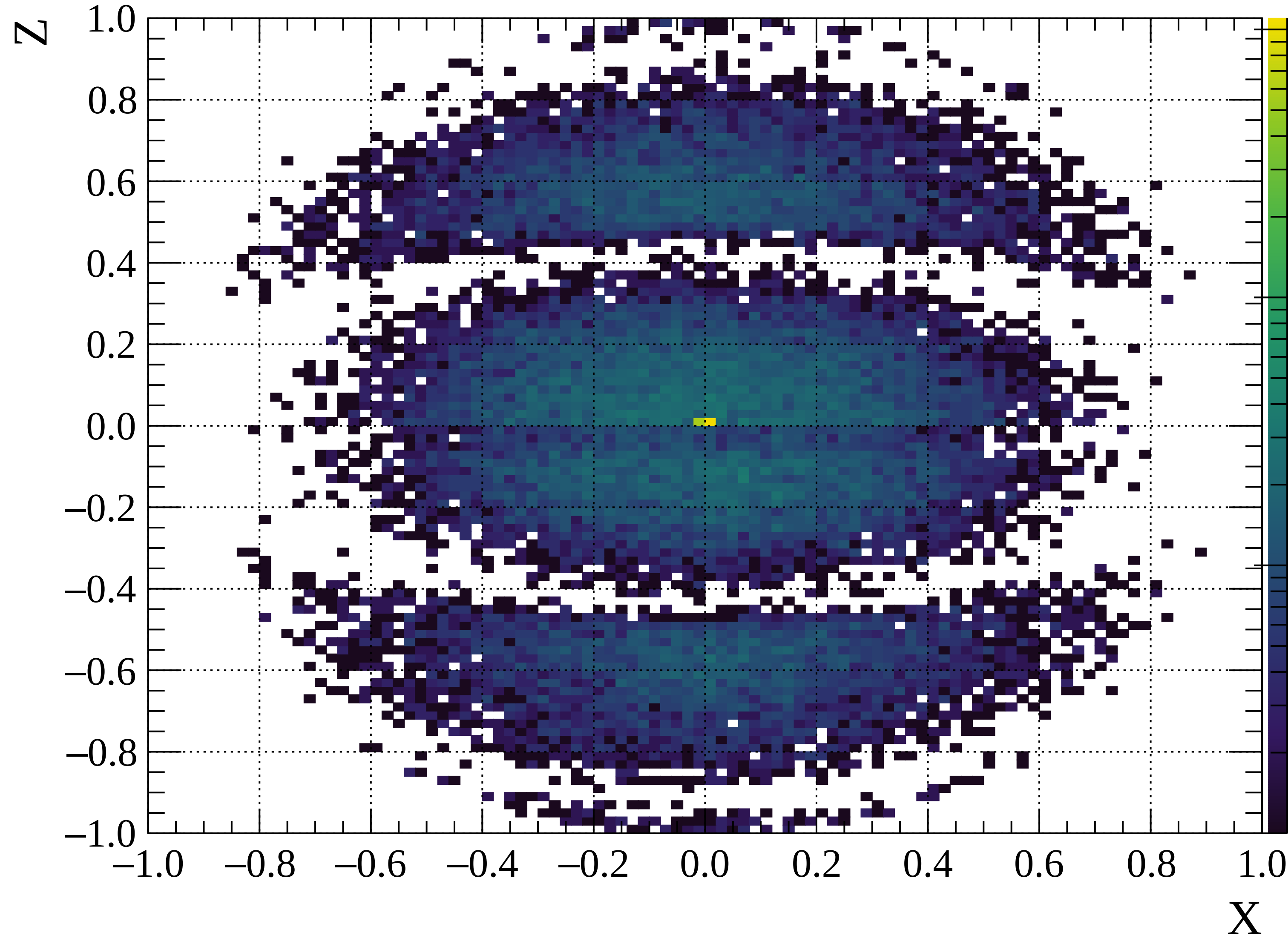

Count
7000
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

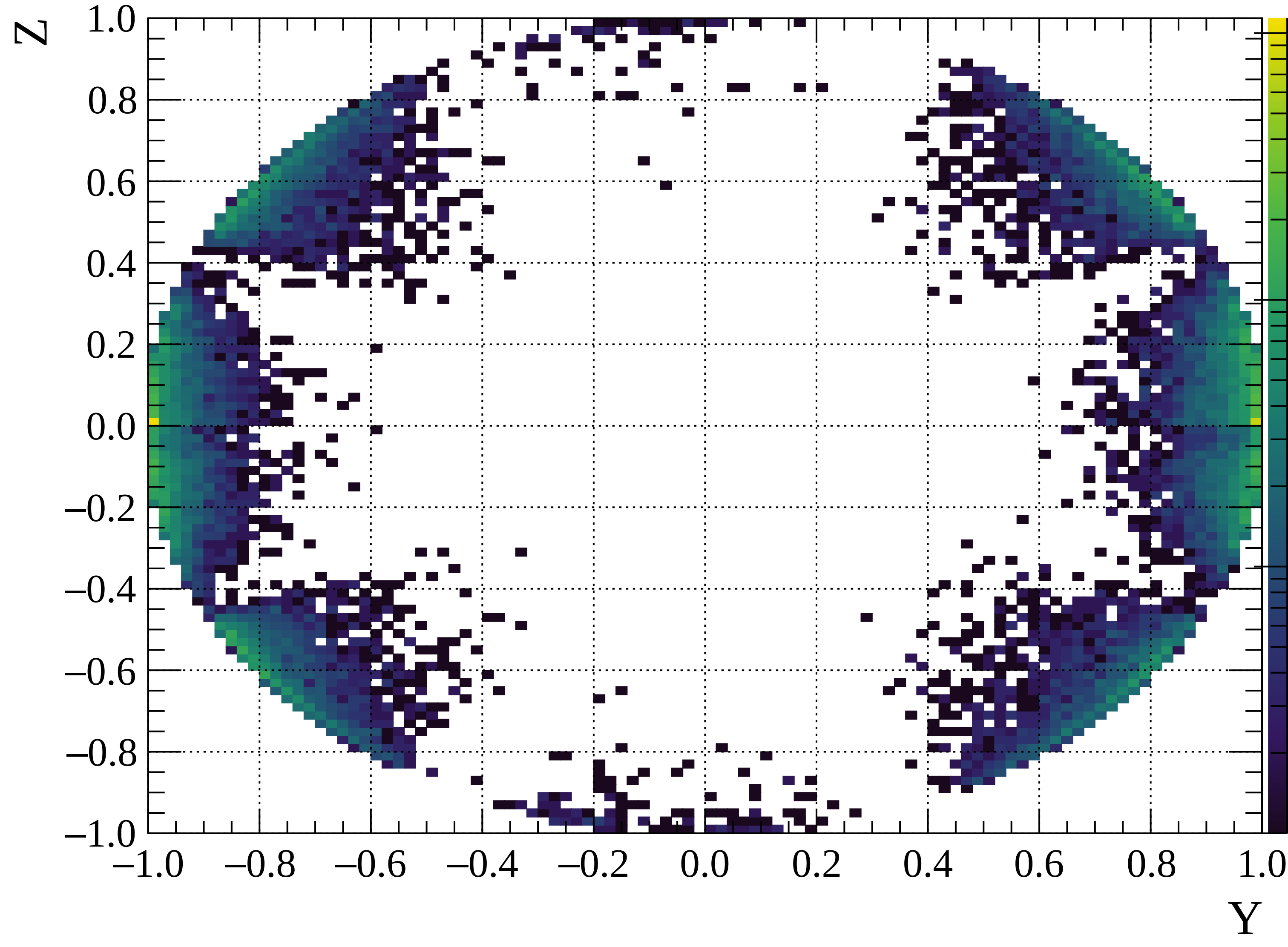

Count
700
600
500
400
300
200
100
0
0
5
10
15
20
25
30
35
40
45
50
χ^2


Counts
1600
1400
1200
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

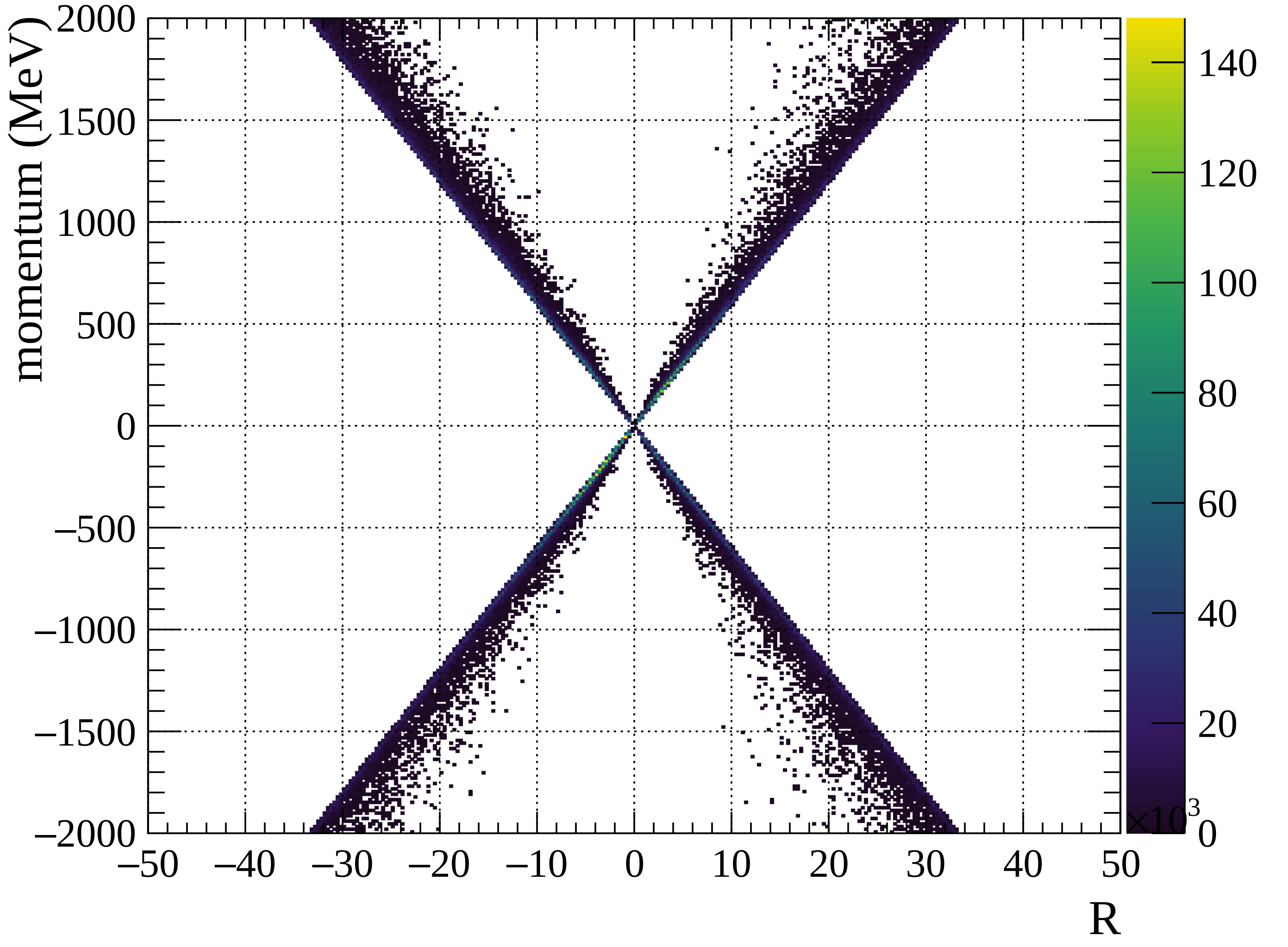

Counts
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction


Counts
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction

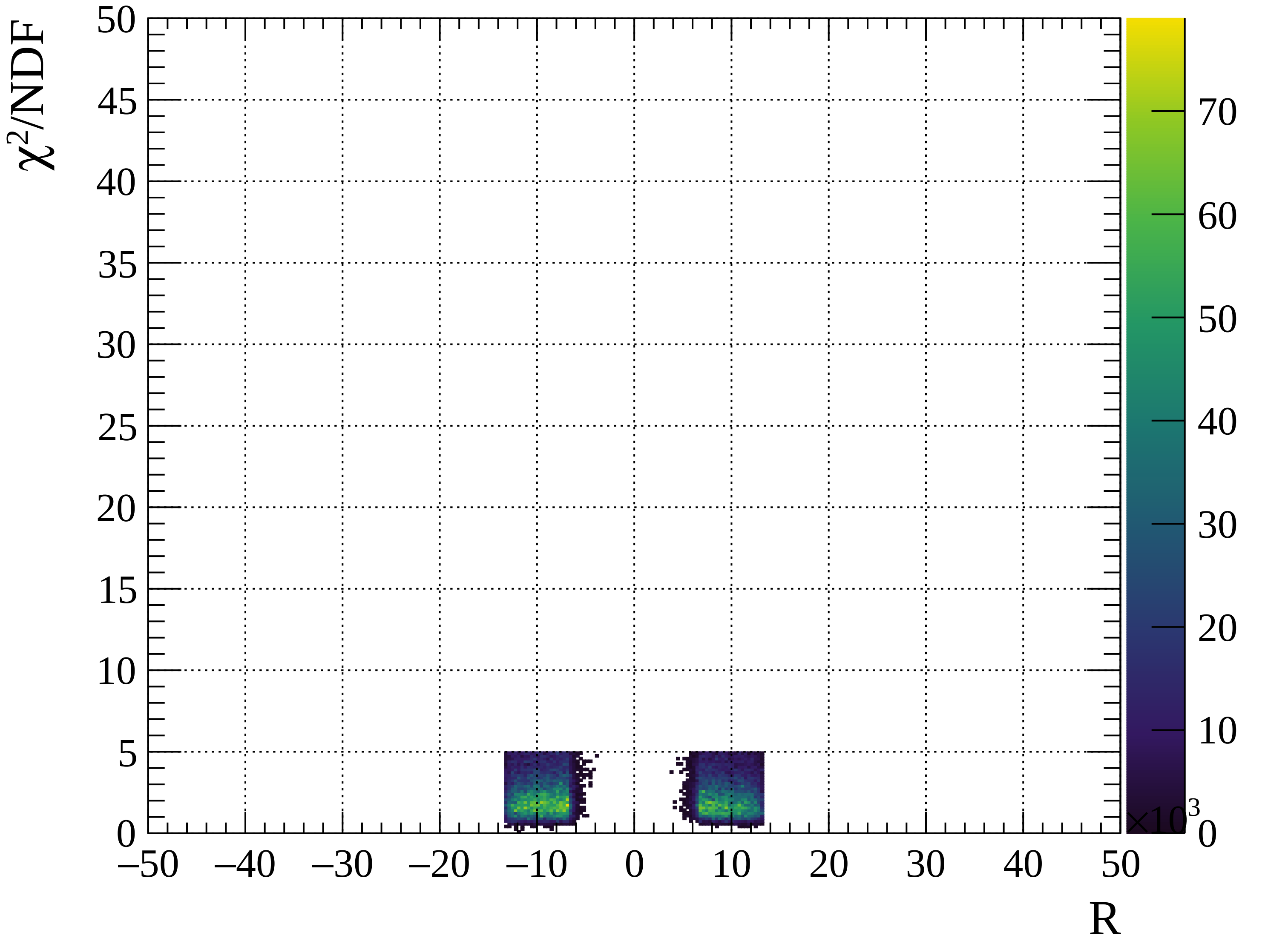

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



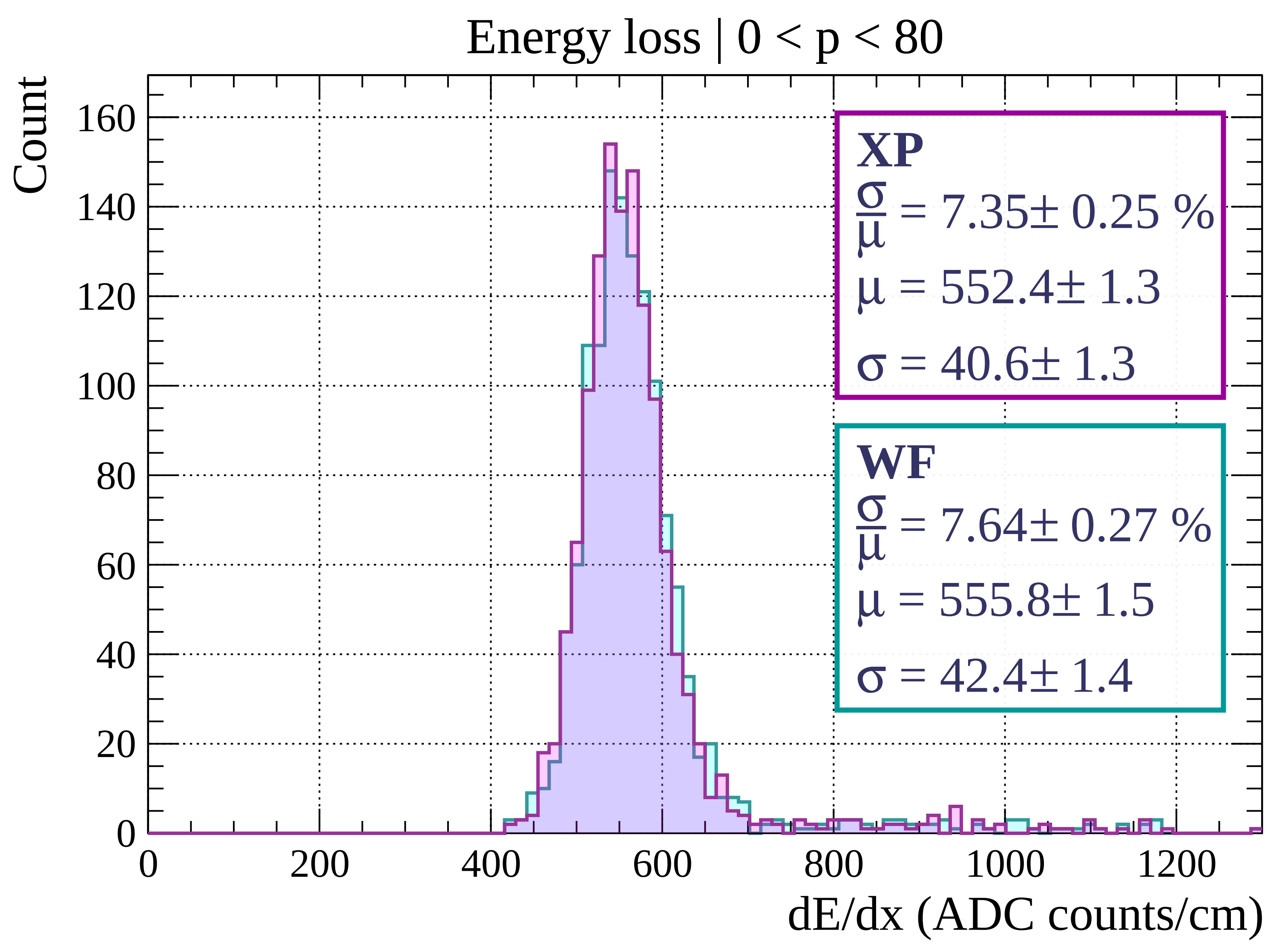
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y



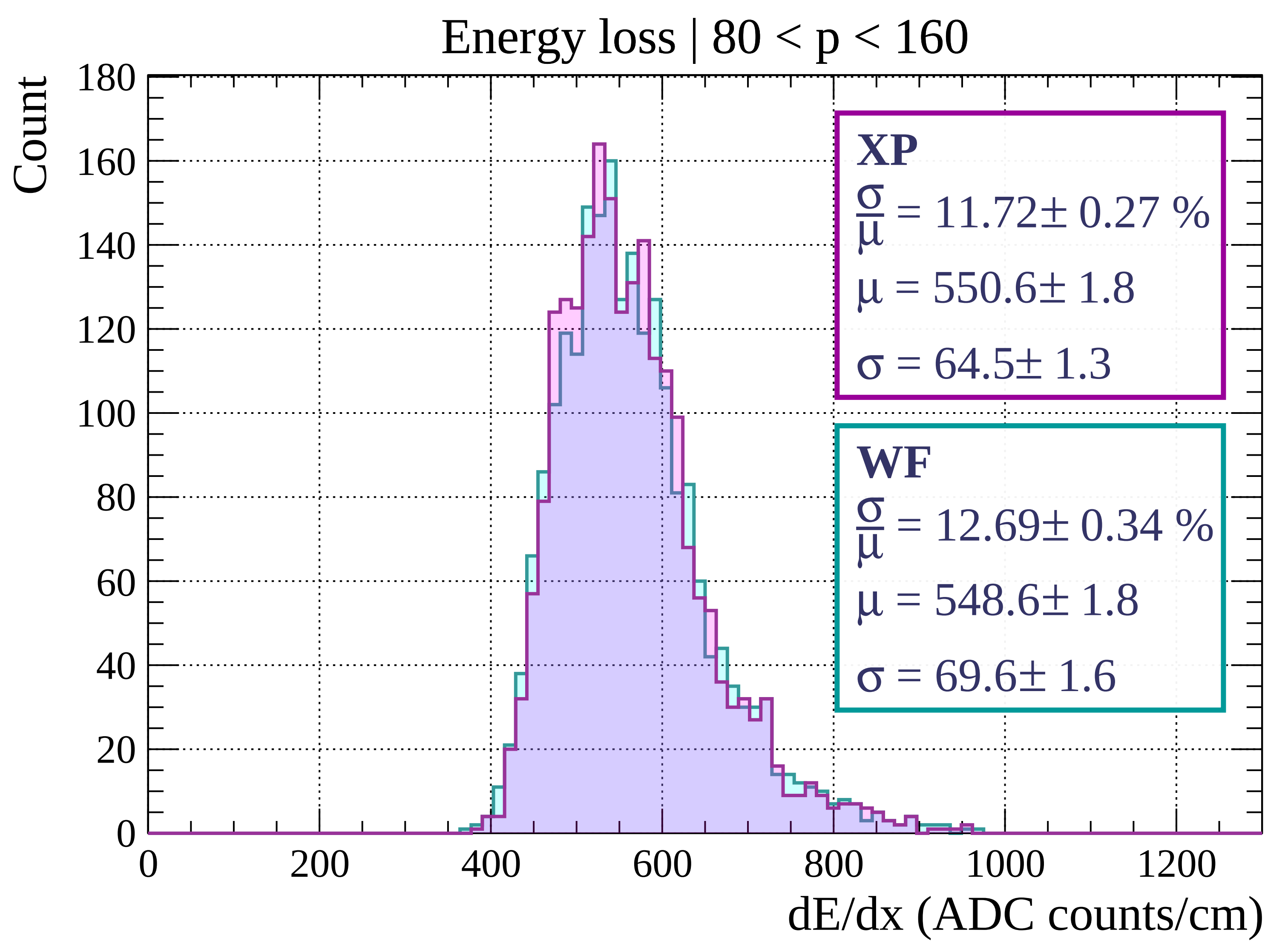
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
140
120
100
80
60
40
20
0
×10^3



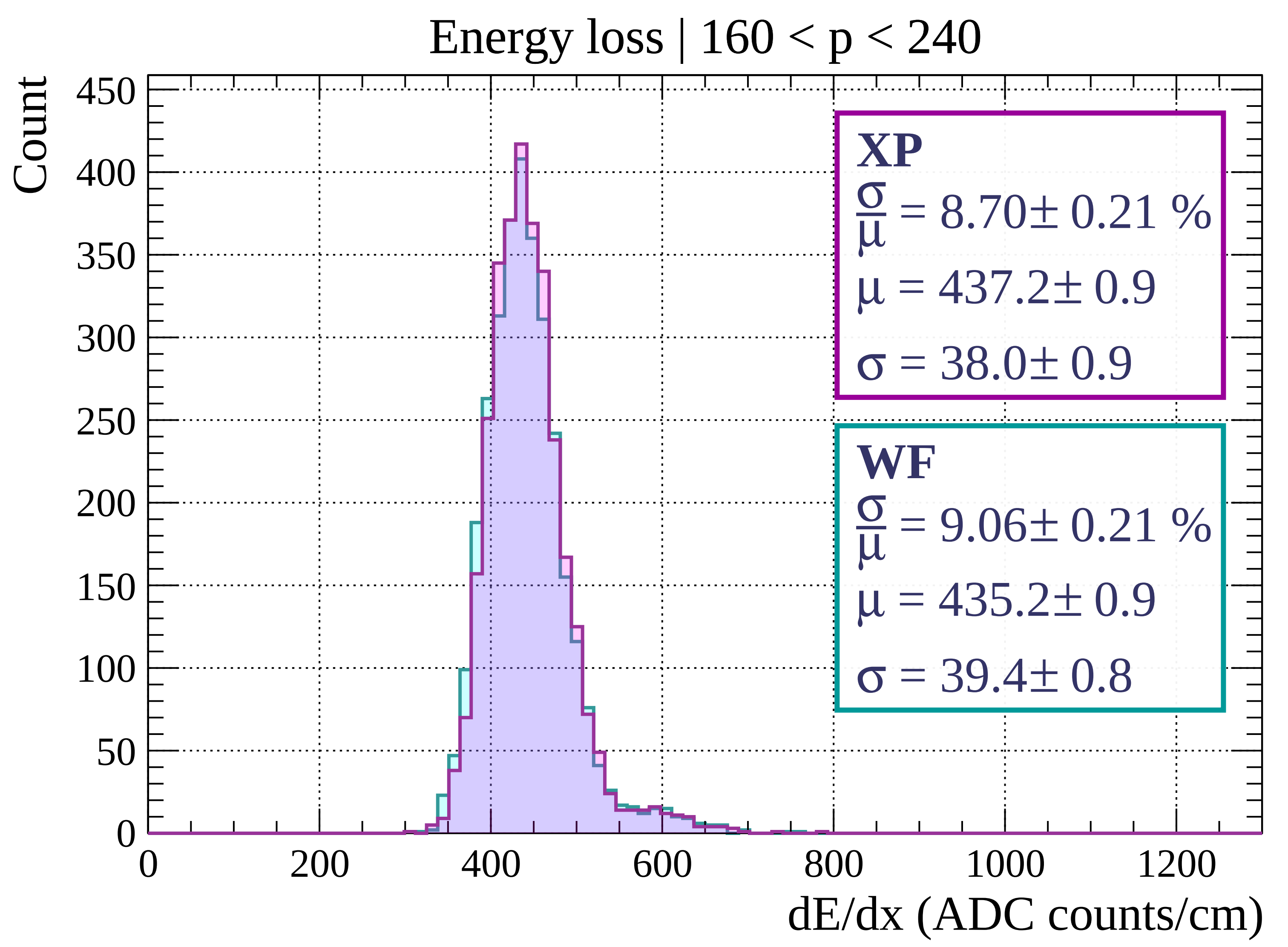
χ^2/NDF
R
$\times10^3$



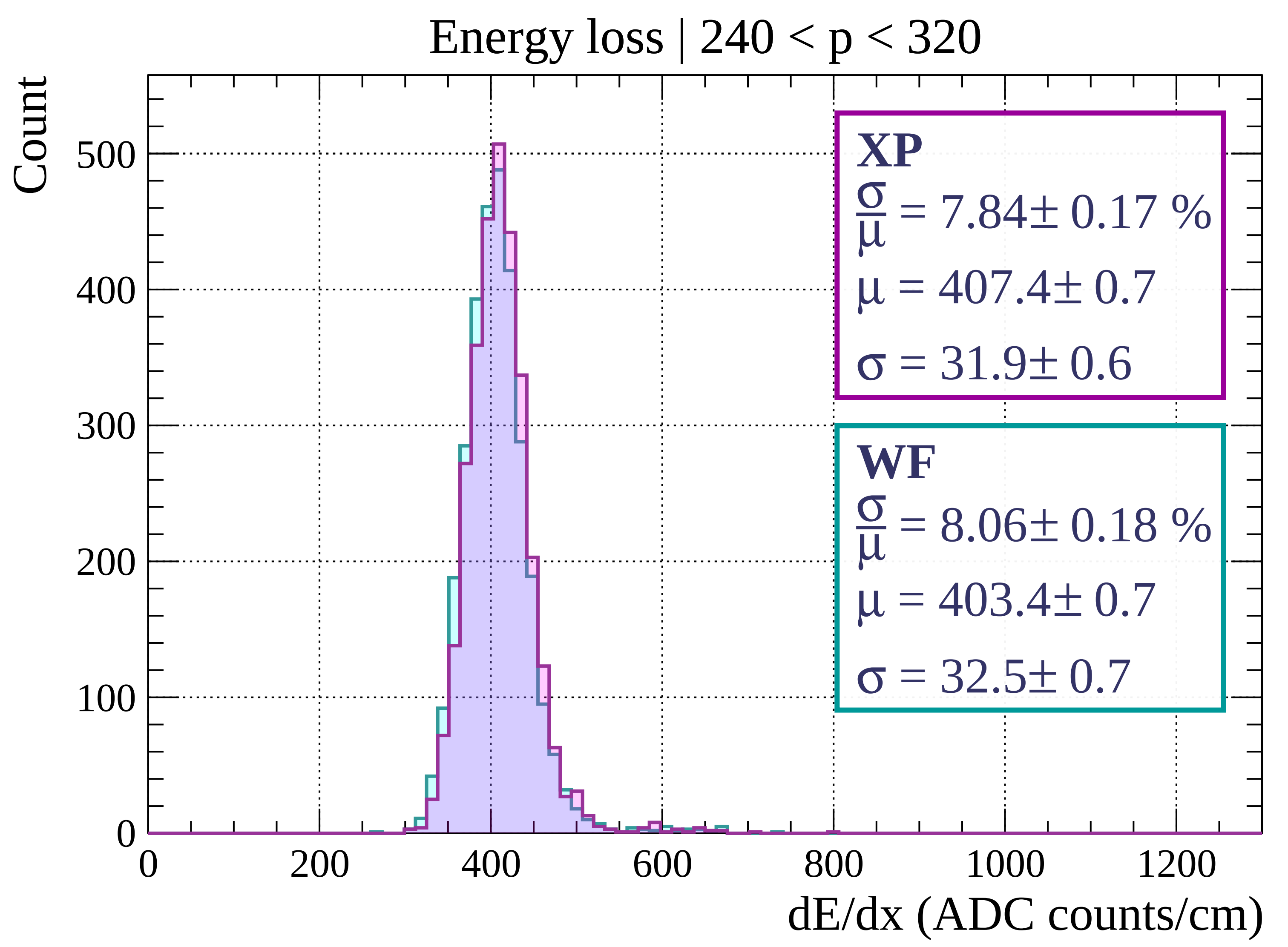
Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 7.35± 0.25 %
μ = 552.4± 1.3
σ = 40.6± 1.3
WF
$\frac{\sigma}{\mu}$ = 7.64± 0.27 %
μ = 555.8± 1.5
σ = 42.4± 1.4

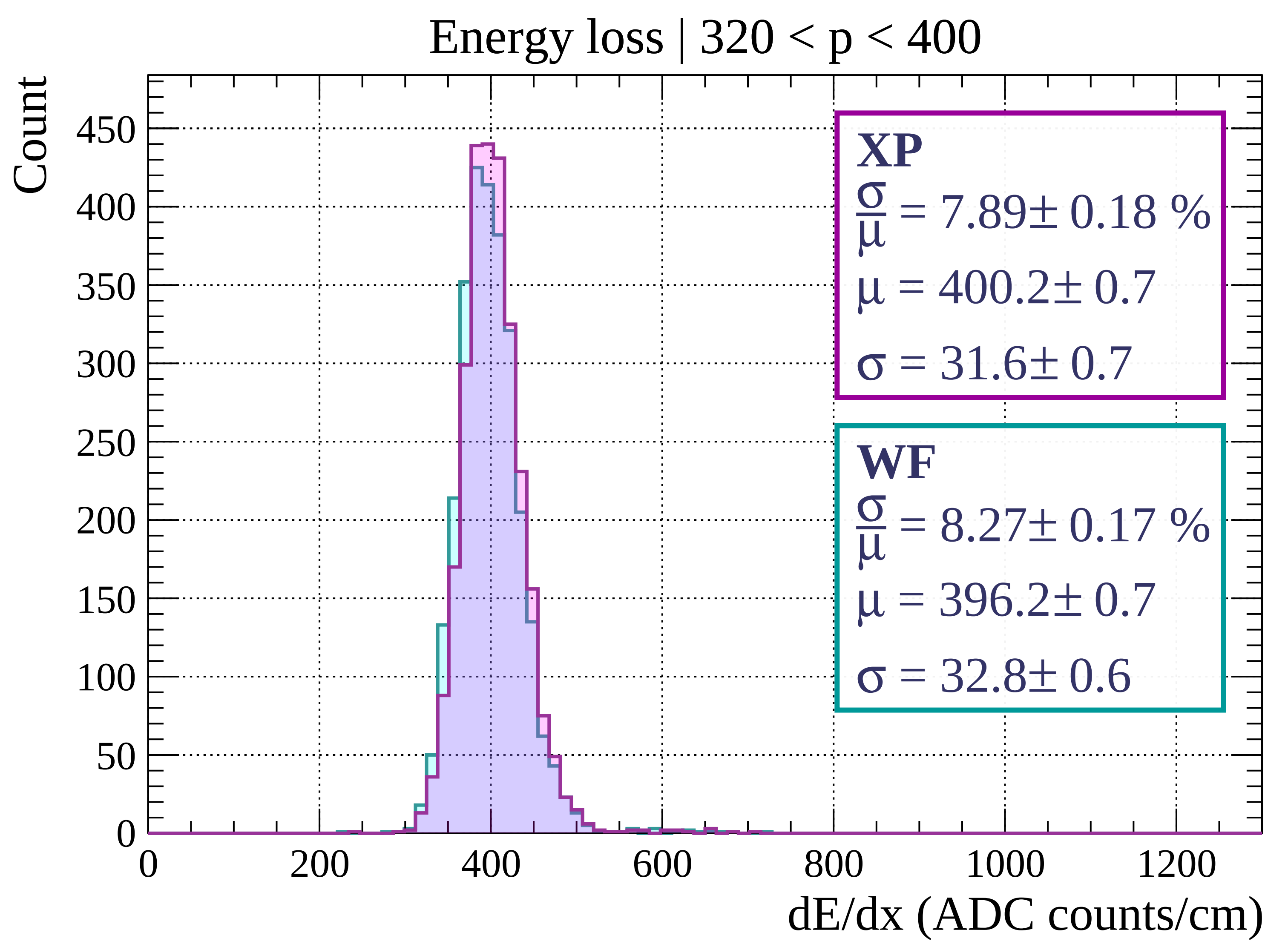


Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 11.72± 0.27 %
μ = 550.6± 1.8
σ = 64.5± 1.3
WF
σ/μ = 12.69± 0.34 %
μ = 548.6± 1.8
σ = 69.6± 1.6

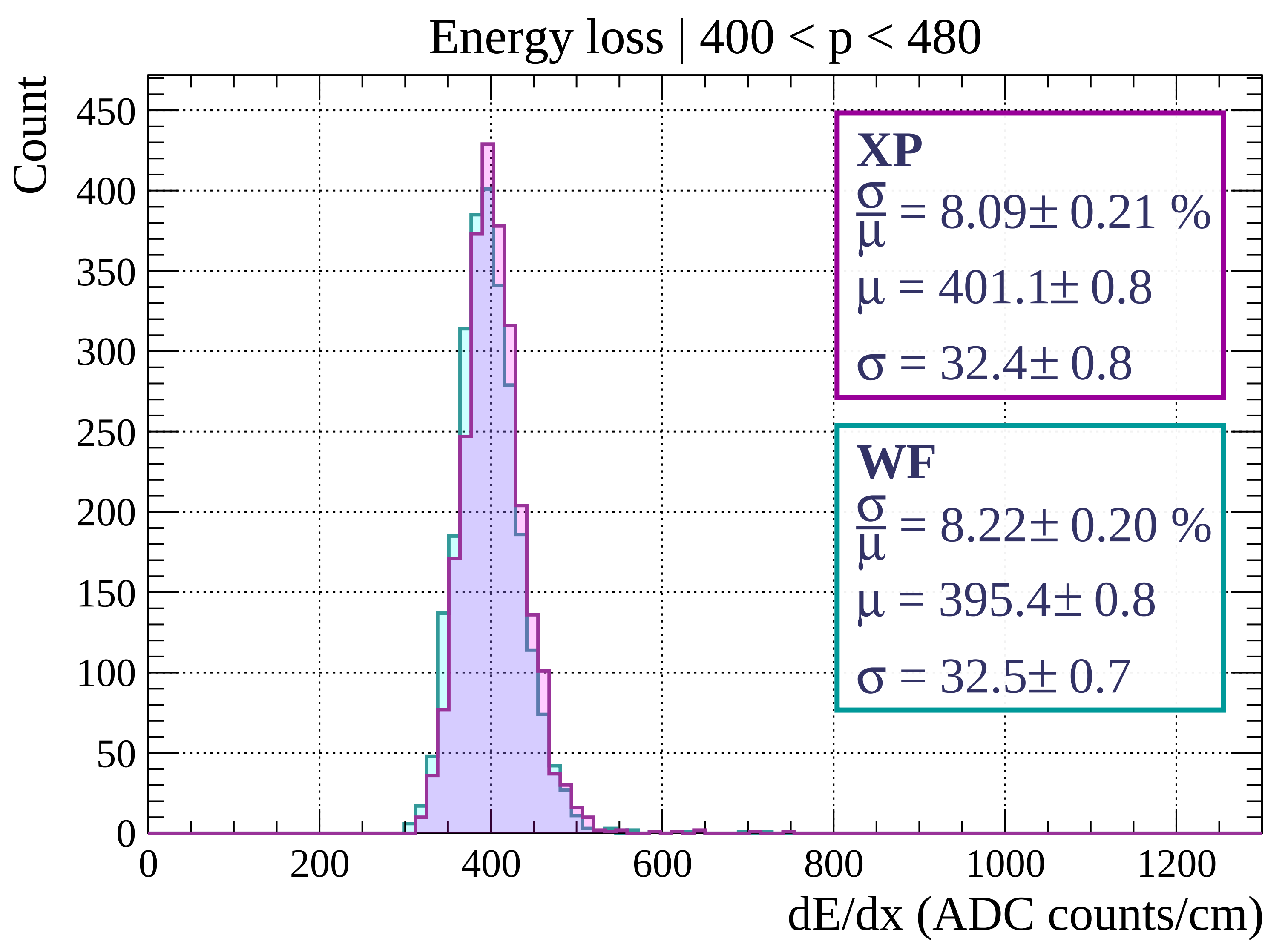


Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.70± 0.21 %
μ = 437.2± 0.9
σ = 38.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.06± 0.21 %
μ = 435.2± 0.9
σ = 39.4± 0.8

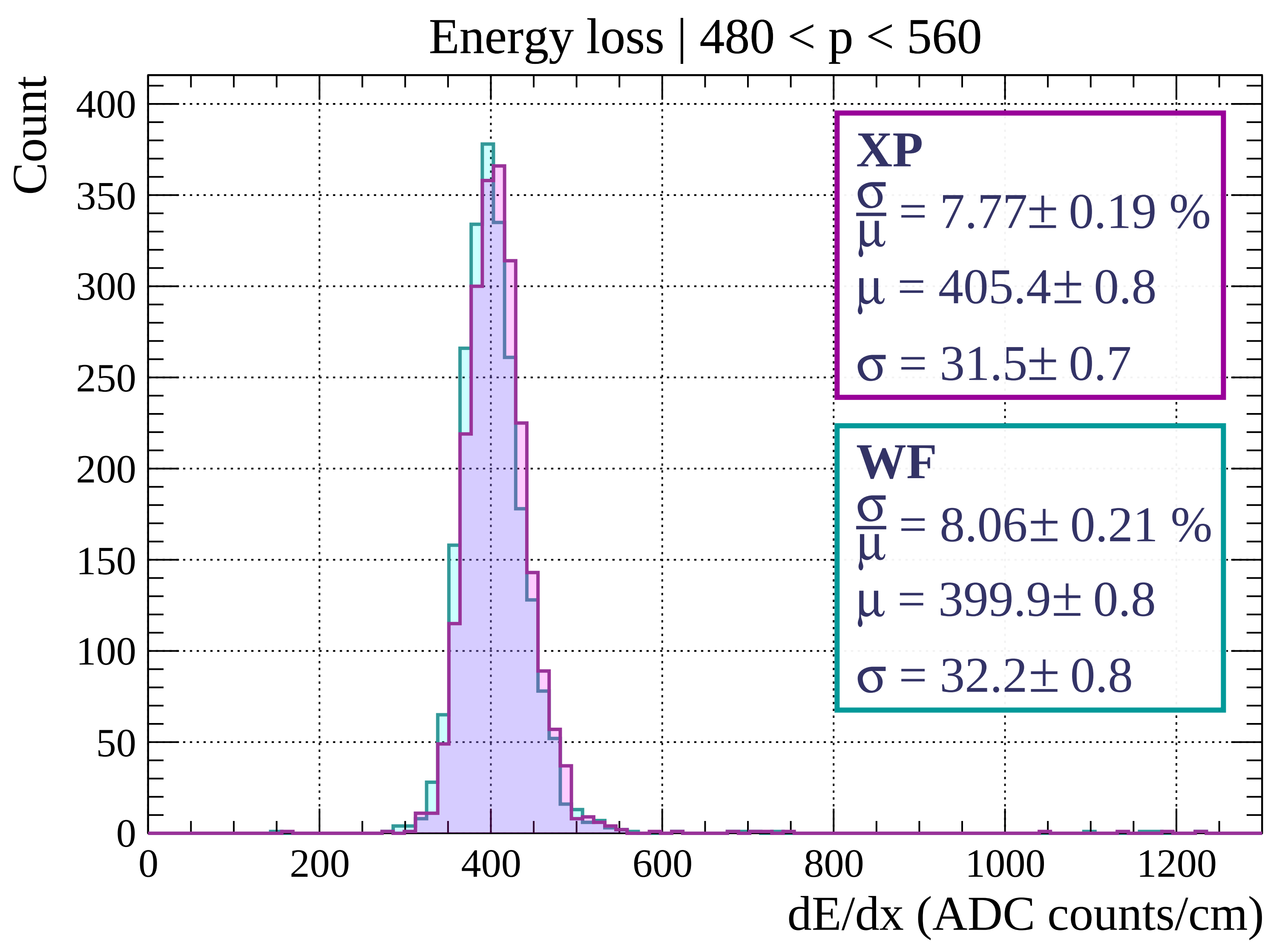

Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.84± 0.17 %
μ = 407.4± 0.7
σ = 31.9± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.06± 0.18 %
μ = 403.4± 0.7
σ = 32.5± 0.7
0
100
200
300
400
500
0
200
400
600
800
1000
1200



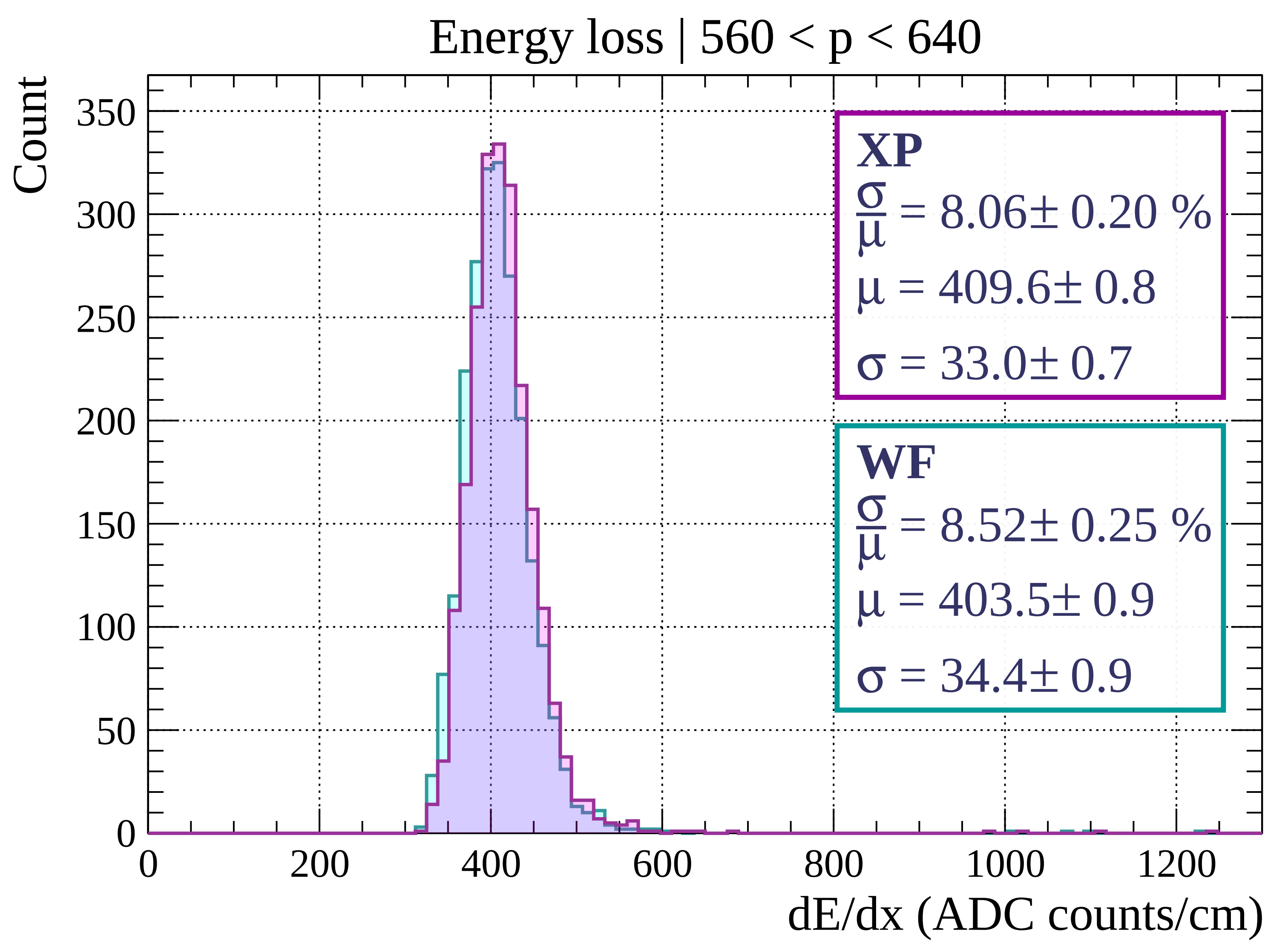
Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.89± 0.18 %
μ = 400.2± 0.7
σ = 31.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.27± 0.17 %
μ = 396.2± 0.7
σ = 32.8± 0.6



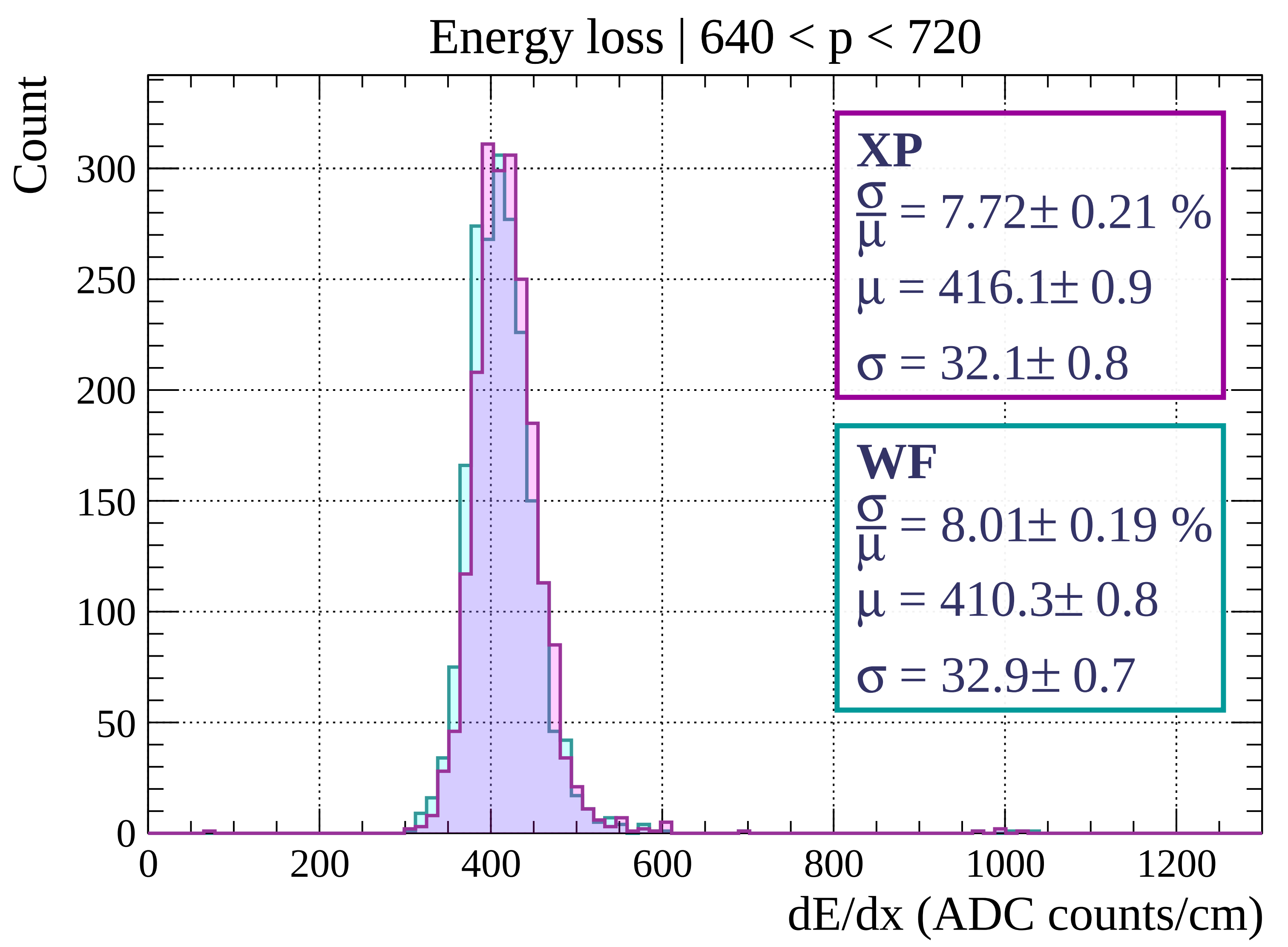
Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.21 %
μ = 401.1± 0.8
σ = 32.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.22± 0.20 %
μ = 395.4± 0.8
σ = 32.5± 0.7

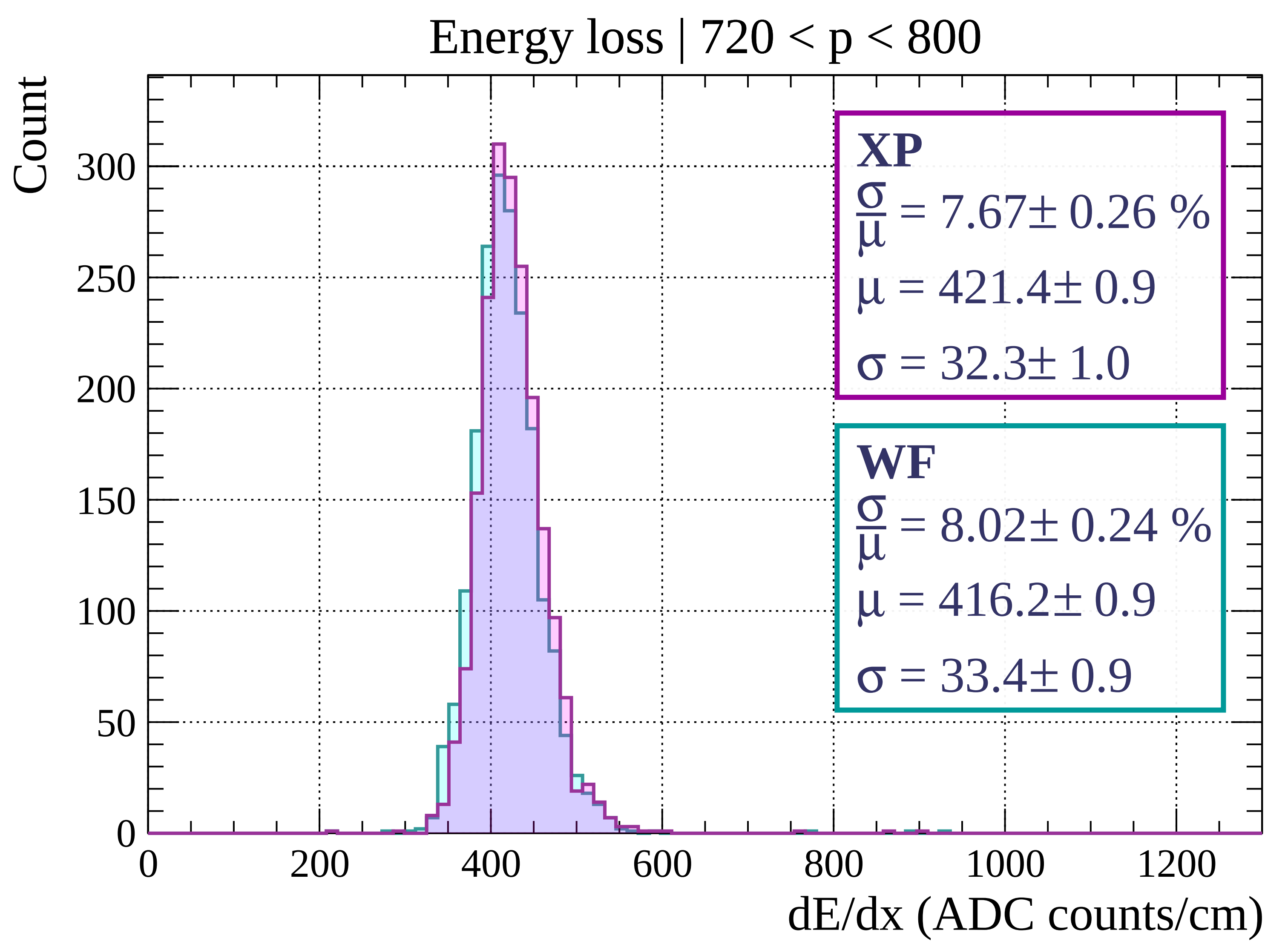


Energy loss | 480 < p < 560
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.77± 0.19 %
μ = 405.4± 0.8
σ = 31.5± 0.7
WF
σ/μ = 8.06± 0.21 %
μ = 399.9± 0.8
σ = 32.2± 0.8
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400

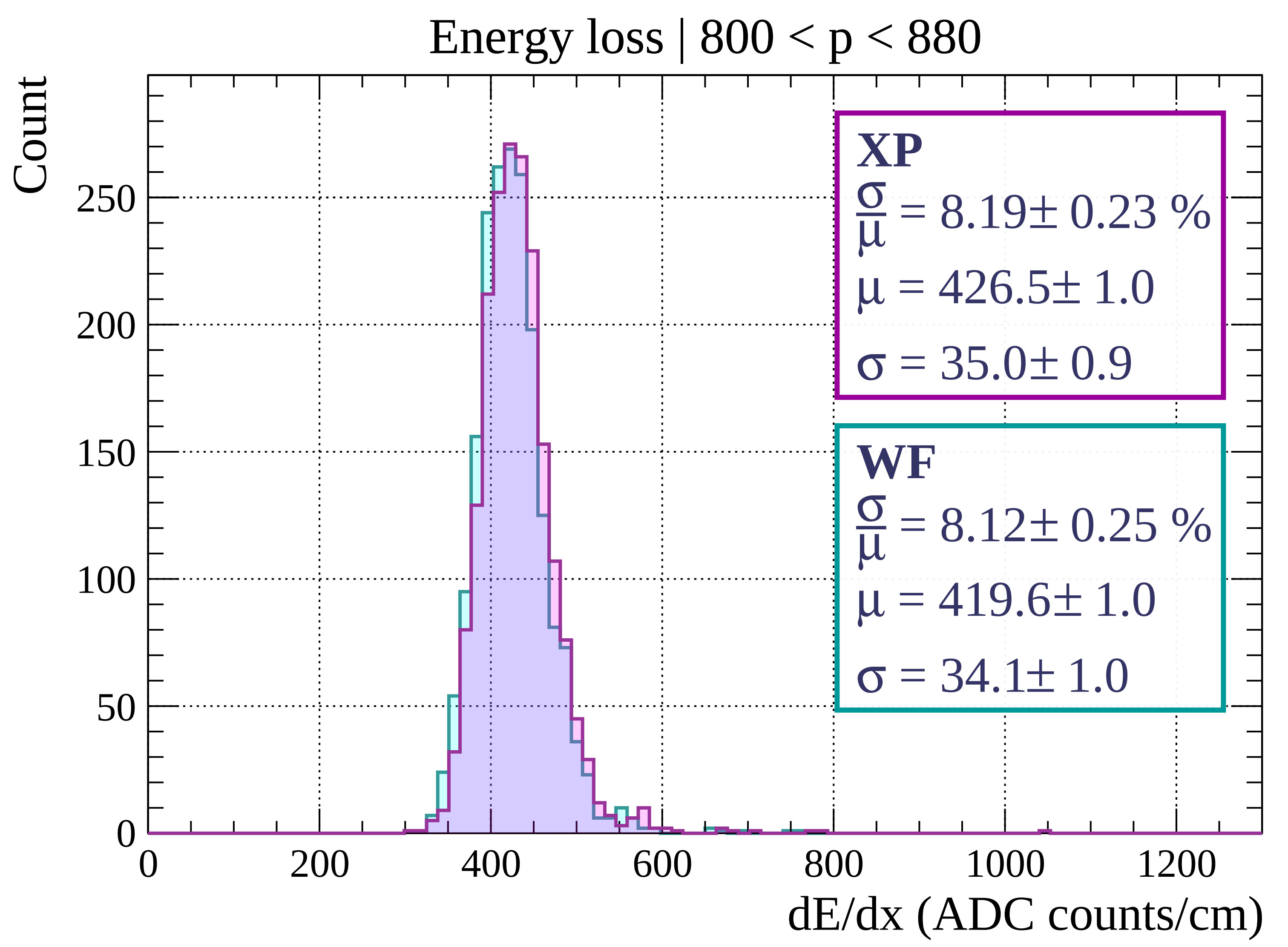


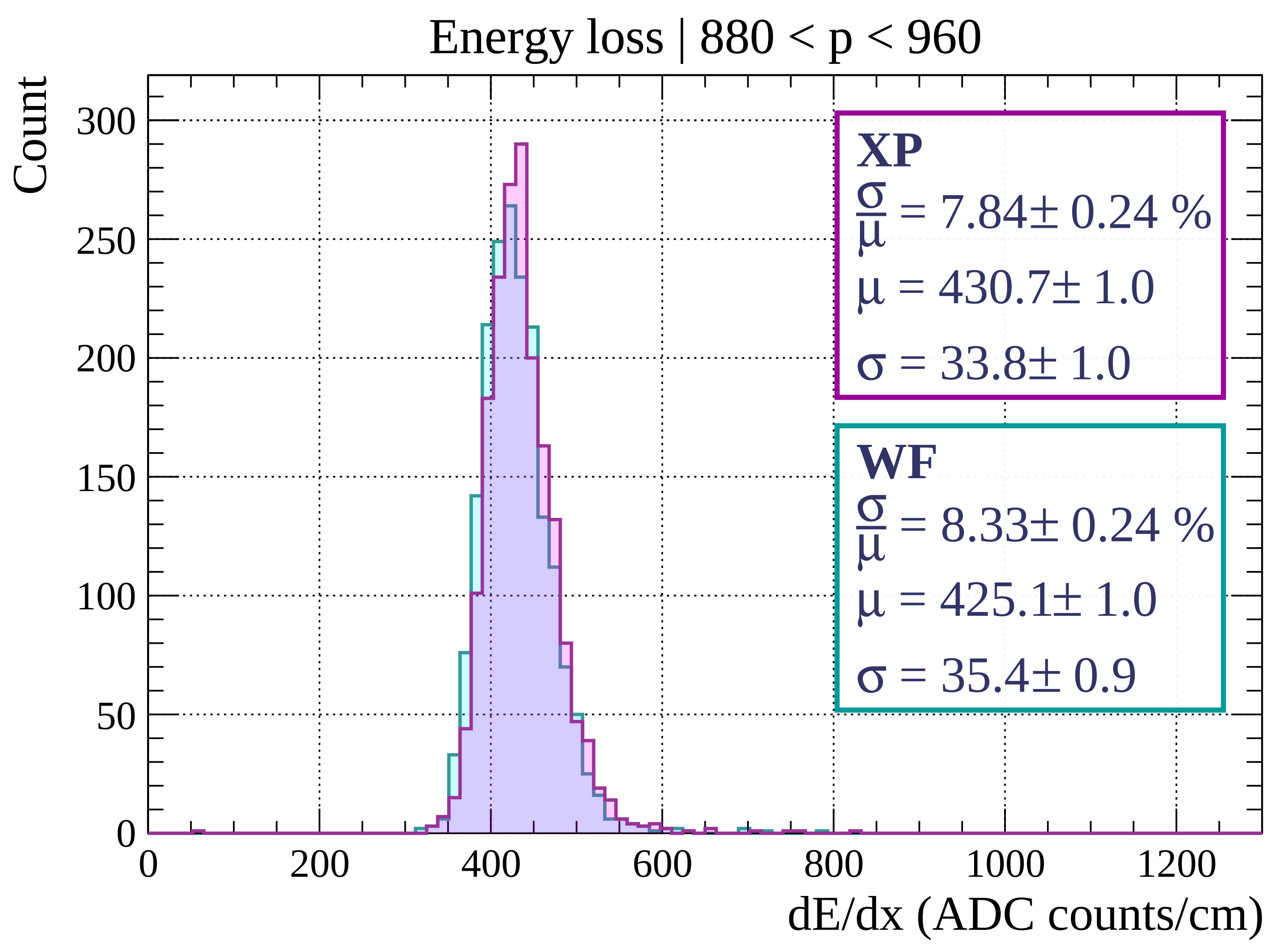
Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.06 \pm 0.20$ %
$\mu = 409.6 \pm 0.8$
$\sigma = 33.0 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 8.52 \pm 0.25$ %
$\mu = 403.5 \pm 0.9$
$\sigma = 34.4 \pm 0.9$

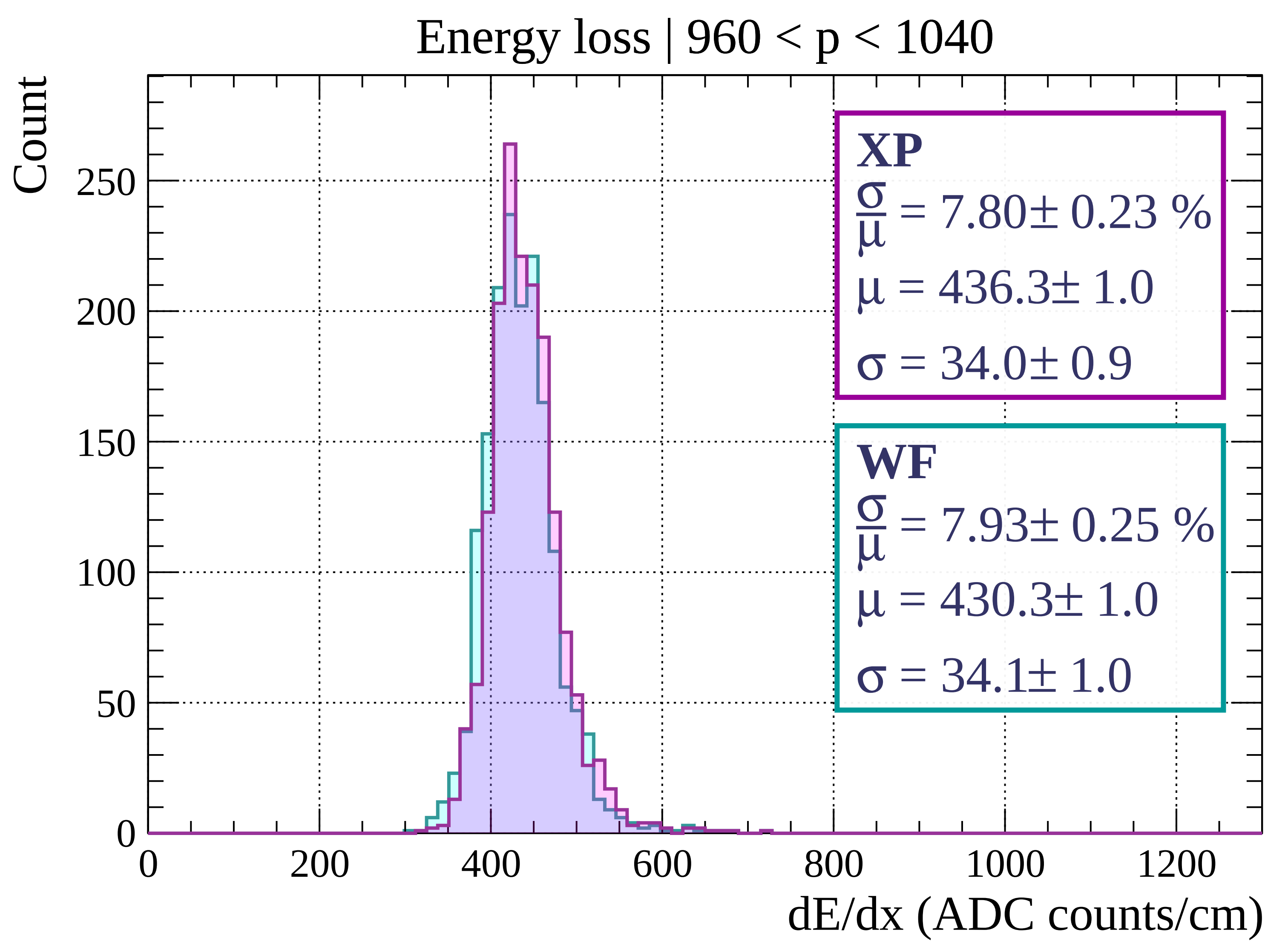

Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.72± 0.21 %
μ = 416.1± 0.9
σ = 32.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.01± 0.19 %
μ = 410.3± 0.8
σ = 32.9± 0.7
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

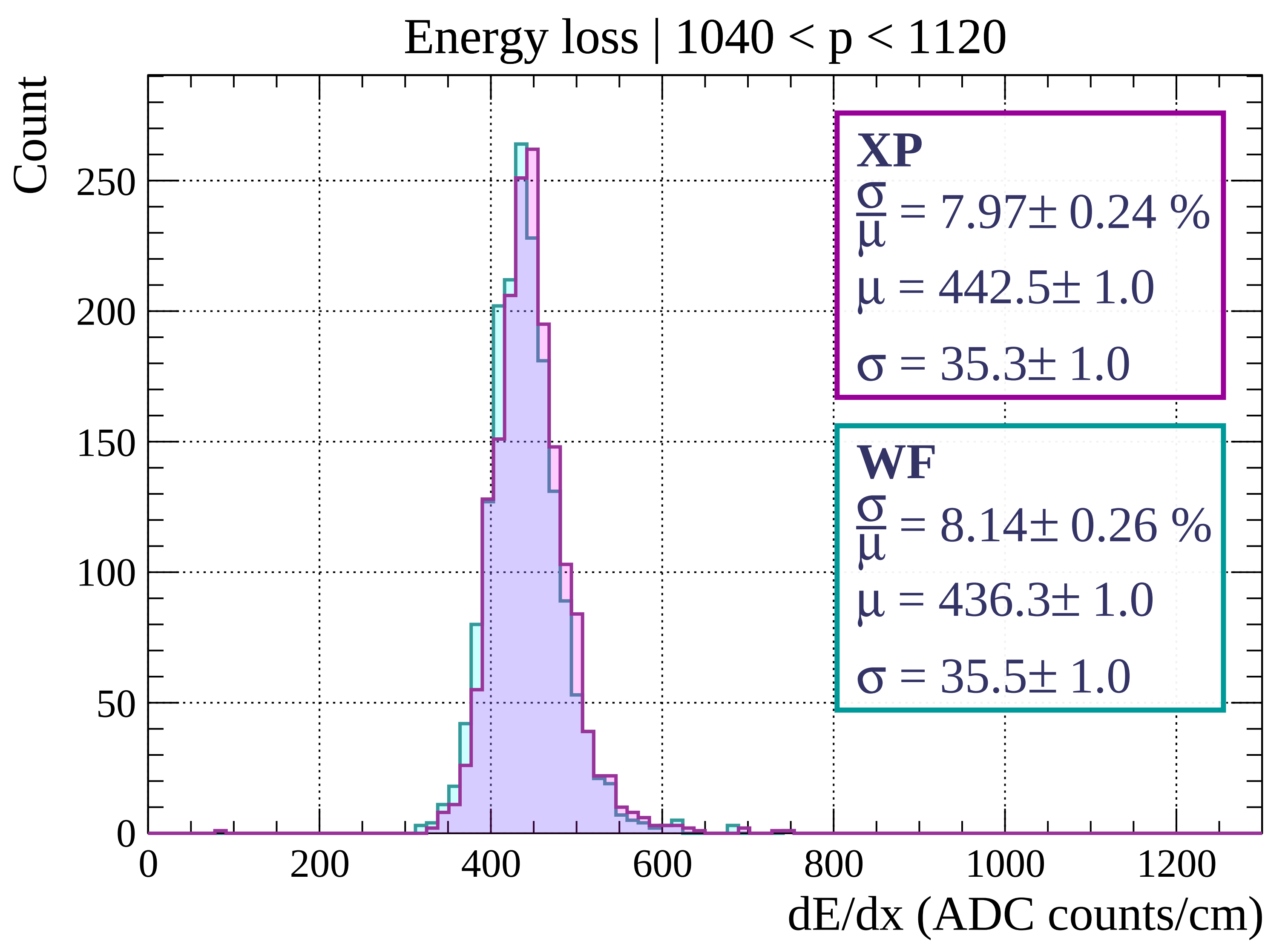


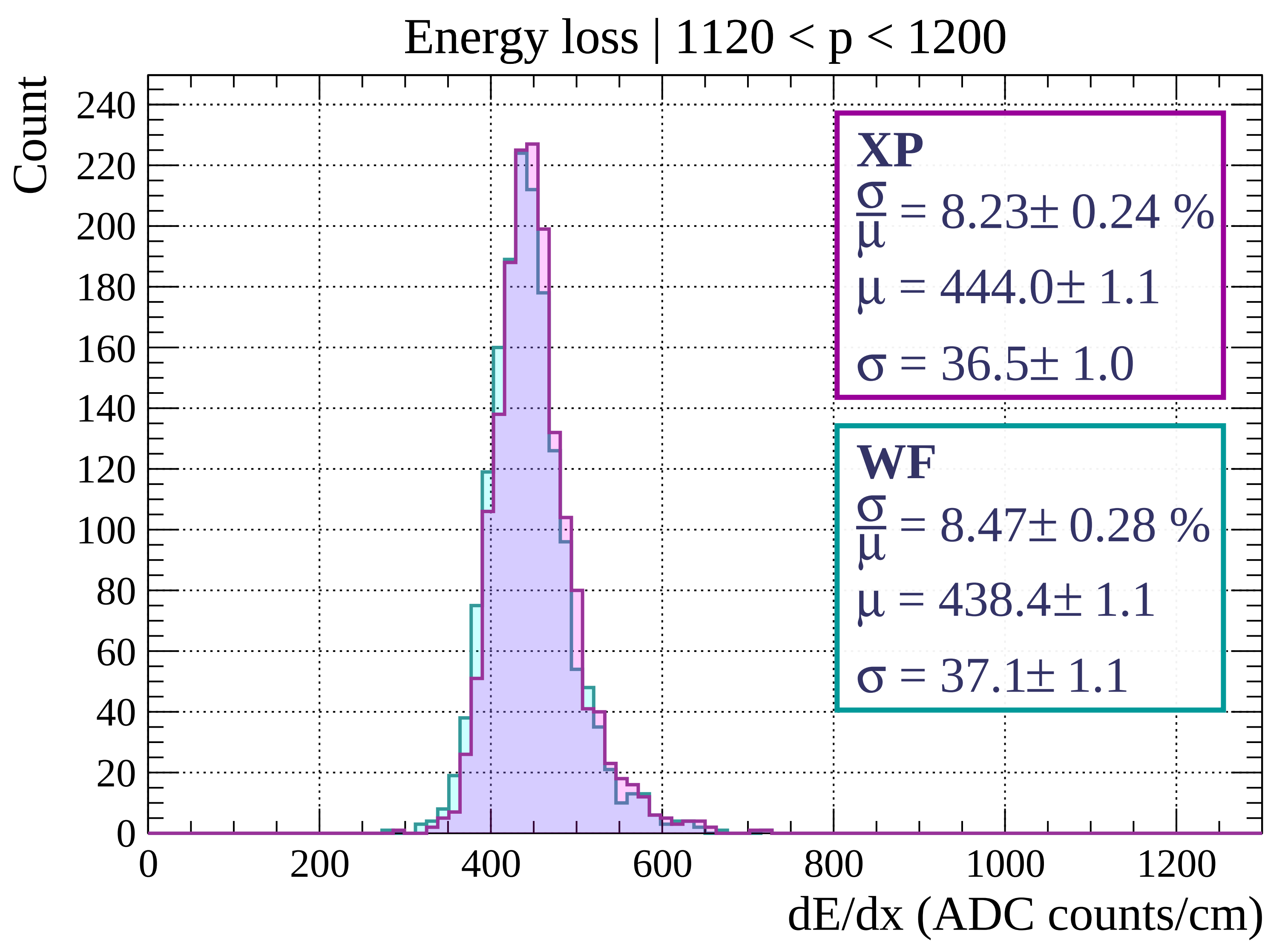
Energy loss | 720 < p < 800
Count
XP
σ/μ = 7.67± 0.26 %
μ = 421.4± 0.9
σ = 32.3± 1.0
WF
σ/μ = 8.02± 0.24 %
μ = 416.2± 0.9
σ = 33.4± 0.9
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)


Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 8.19± 0.23 %
μ = 426.5± 1.0
σ = 35.0± 0.9
WF
σ/μ = 8.12± 0.25 %
μ = 419.6± 1.0
σ = 34.1± 1.0

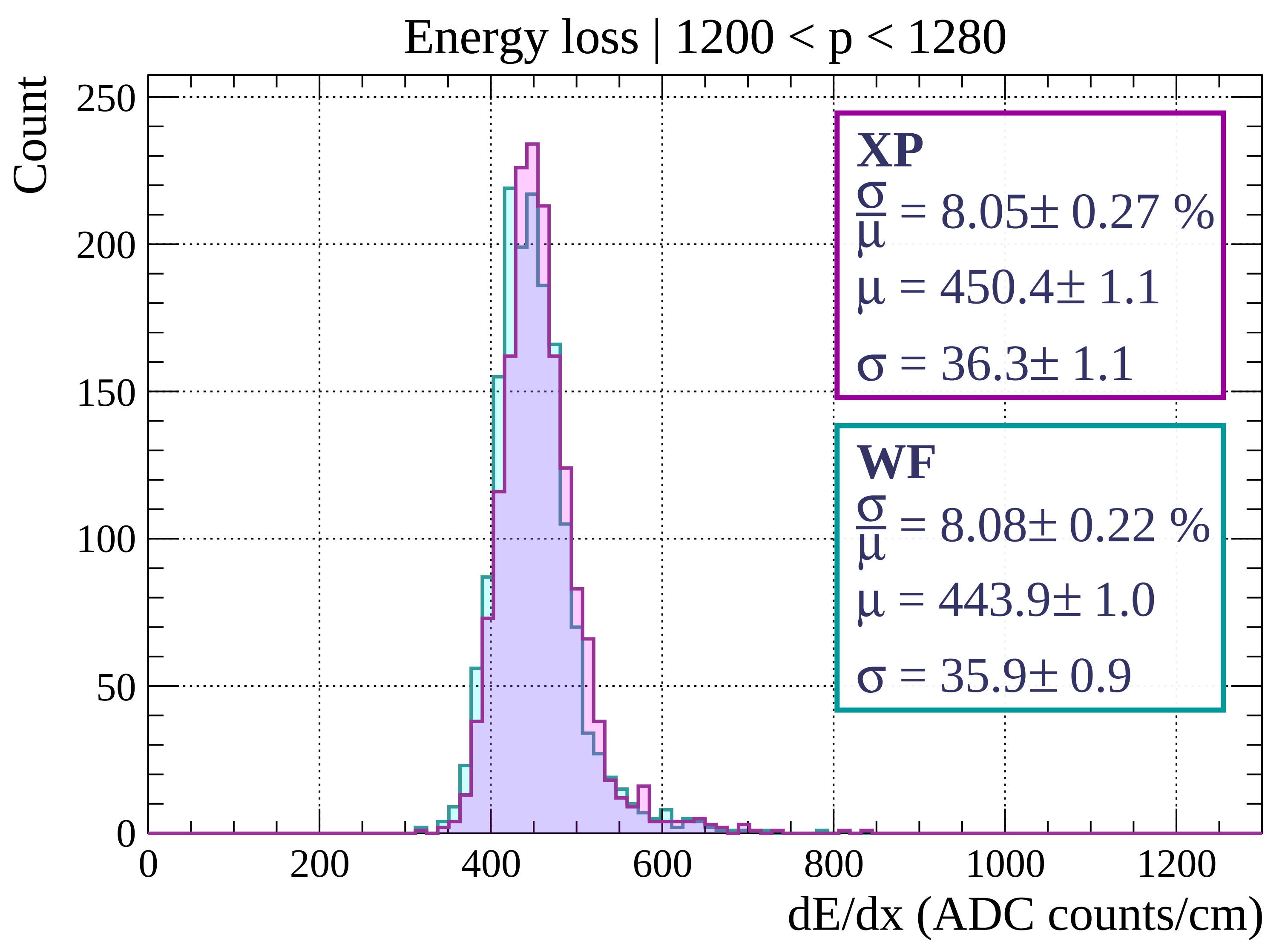

Energy loss | 880 < p < 960
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.84± 0.24 %
μ = 430.7± 1.0
σ = 33.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.33± 0.24 %
μ = 425.1± 1.0
σ = 35.4± 0.9


Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.80± 0.23 %
μ = 436.3± 1.0
σ = 34.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.93± 0.25 %
μ = 430.3± 1.0
σ = 34.1± 1.0

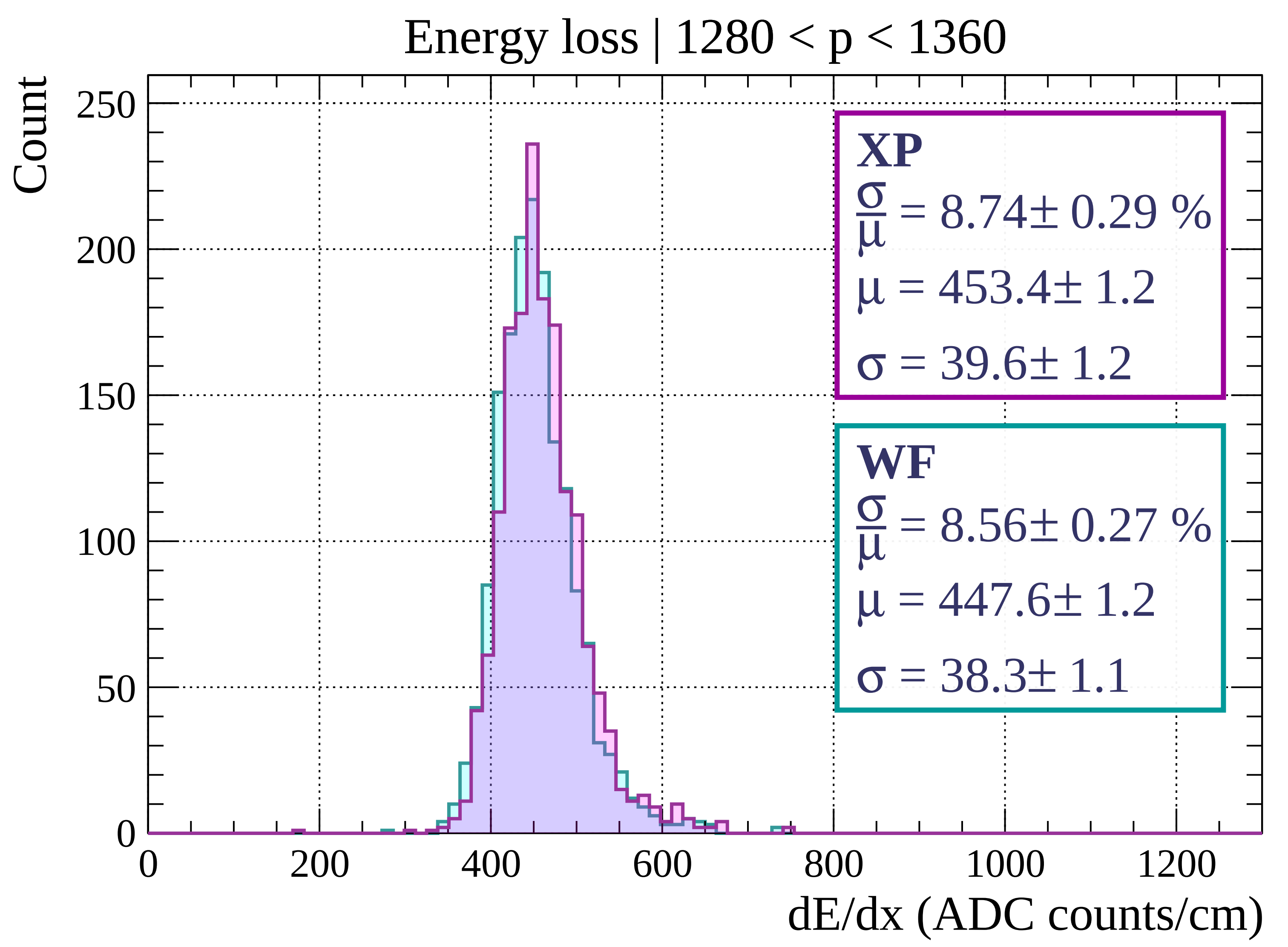

Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 7.97± 0.24 %
μ = 442.5± 1.0
σ = 35.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.14± 0.26 %
μ = 436.3± 1.0
σ = 35.5± 1.0



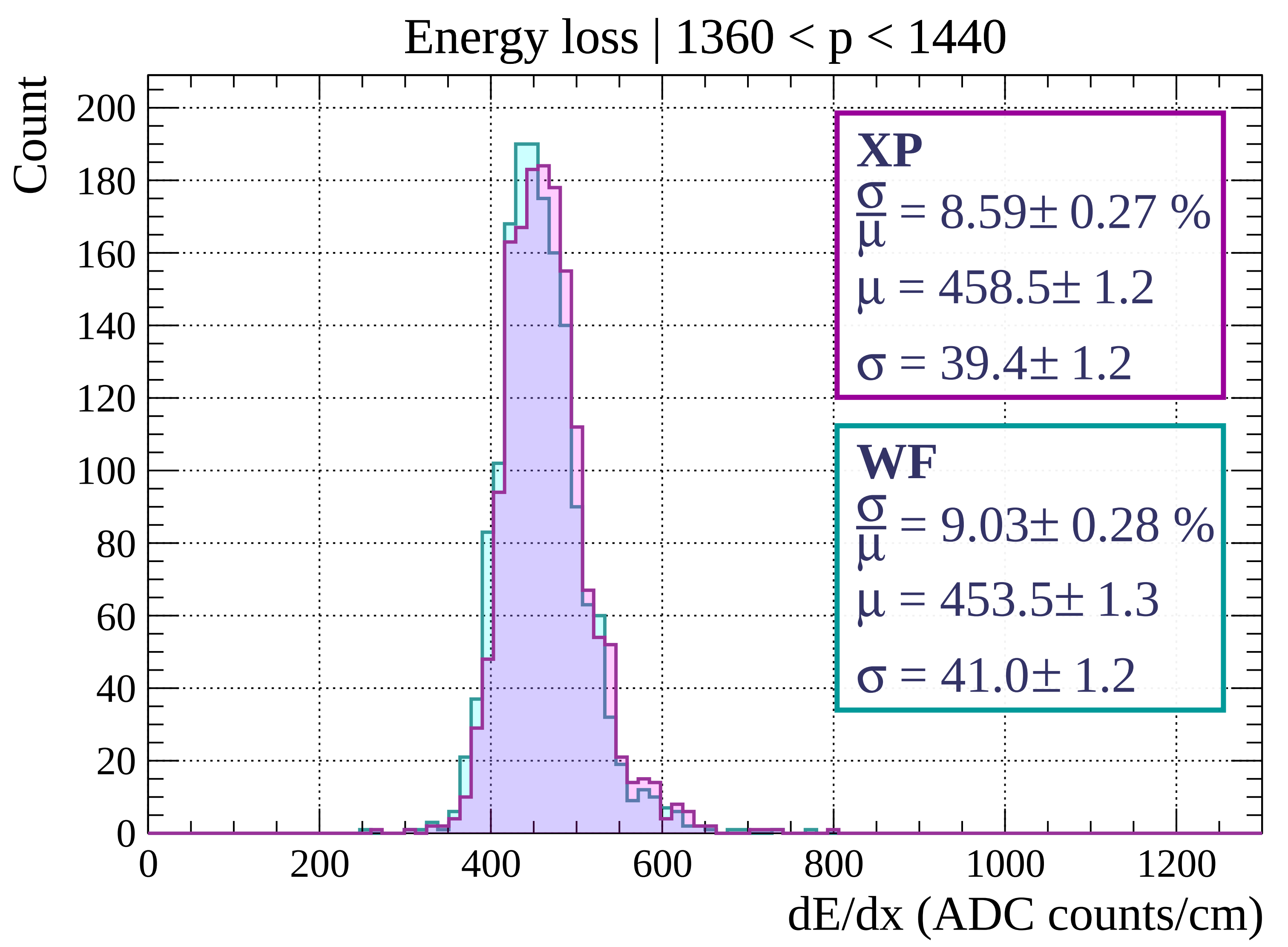
Energy loss | 1120 < p < 1200
Count
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.23± 0.24 %
μ = 444.0± 1.1
σ = 36.5± 1.0
WF
σ/μ = 8.47± 0.28 %
μ = 438.4± 1.1
σ = 37.1± 1.1



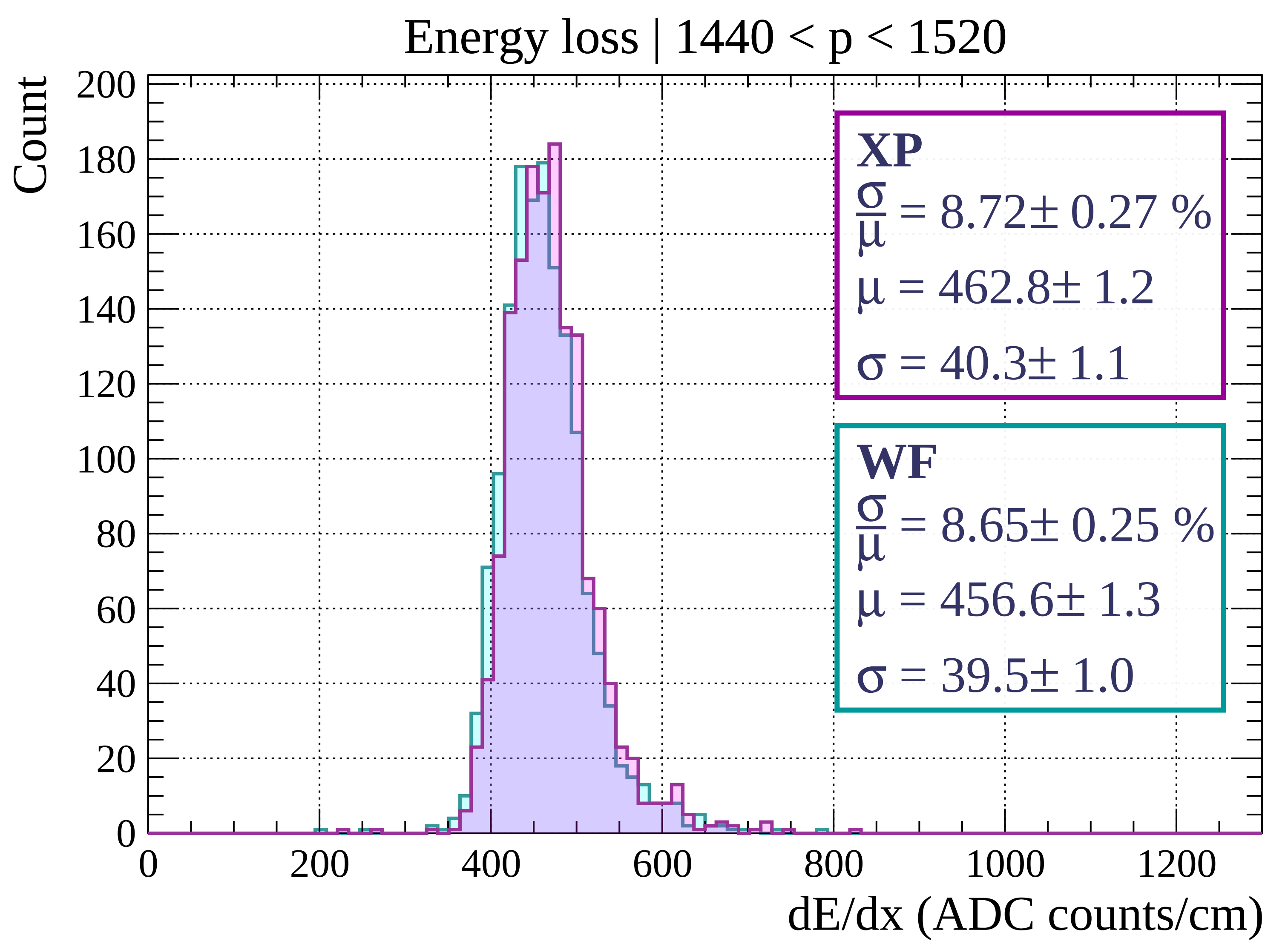
Energy loss | 1200 < p < 1280
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.05± 0.27 %
μ = 450.4± 1.1
σ = 36.3± 1.1
WF
$\frac{\sigma}{\mu}$ = 8.08± 0.22 %
μ = 443.9± 1.0
σ = 35.9± 0.9



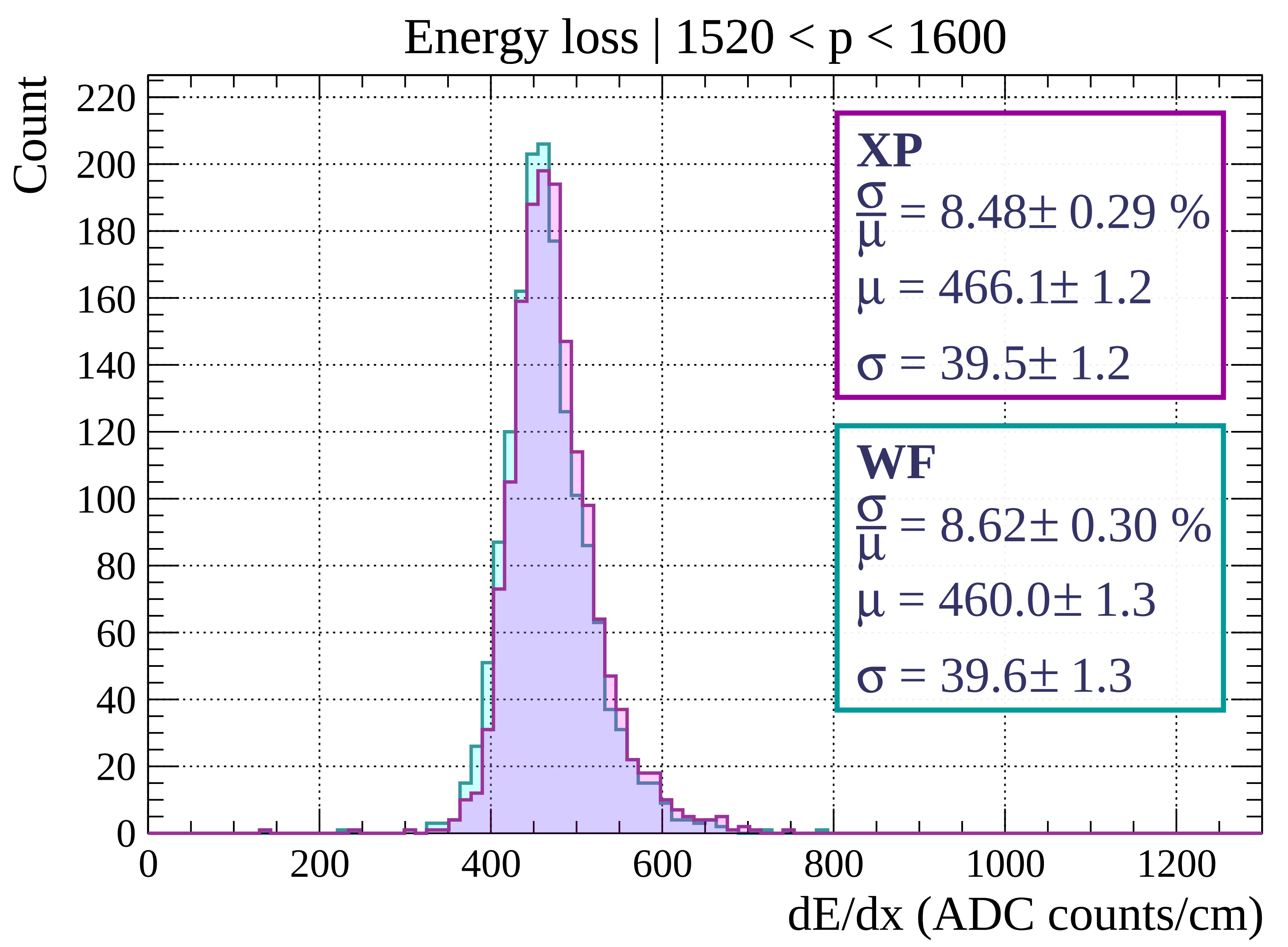
Energy loss | 1280 < p < 1360
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.74± 0.29 %
μ = 453.4± 1.2
σ = 39.6± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.56± 0.27 %
μ = 447.6± 1.2
σ = 38.3± 1.1

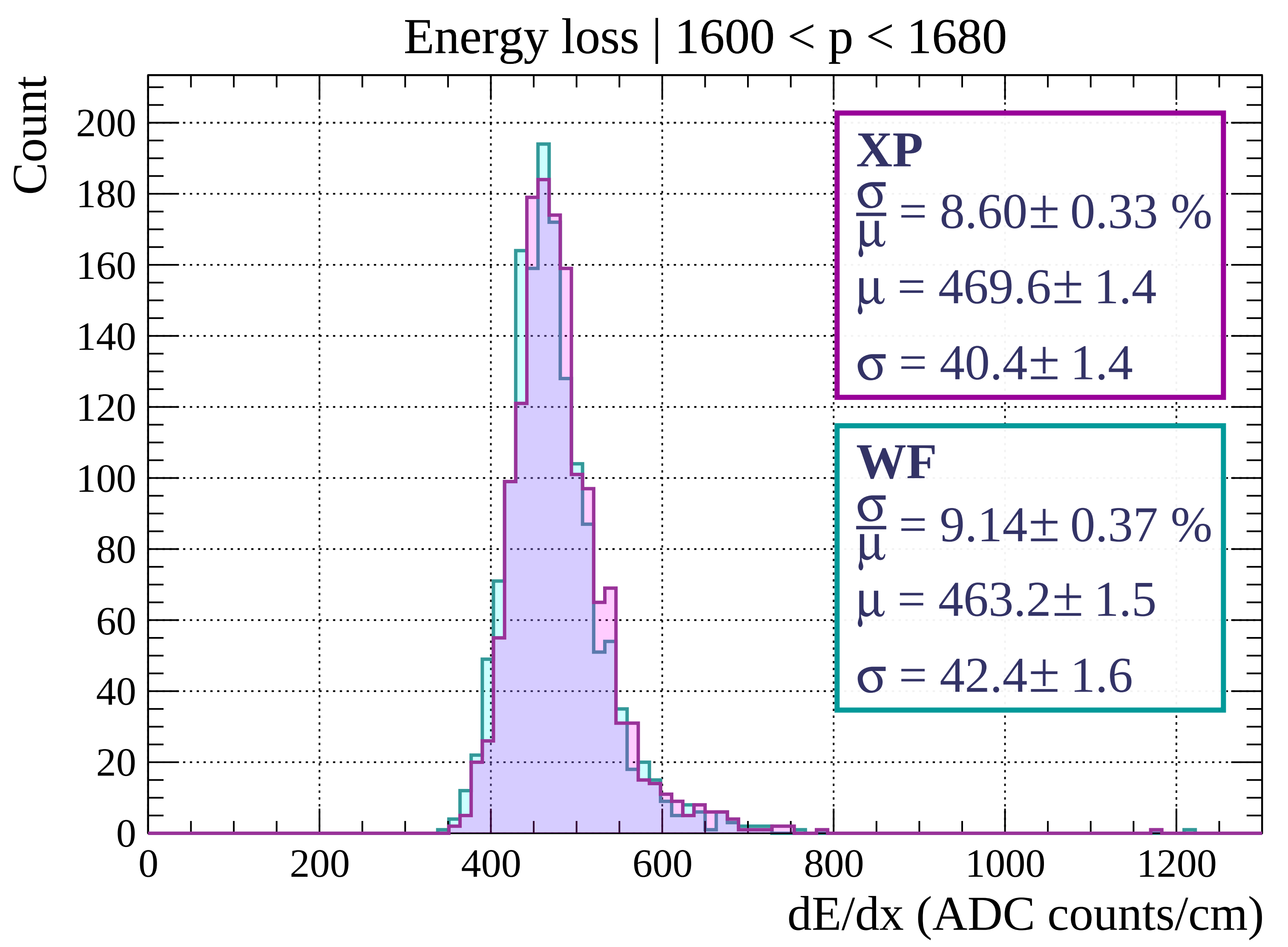


Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu}$ = 8.59± 0.27 %
μ = 458.5± 1.2
σ = 39.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.03± 0.28 %
μ = 453.5± 1.3
σ = 41.0± 1.2

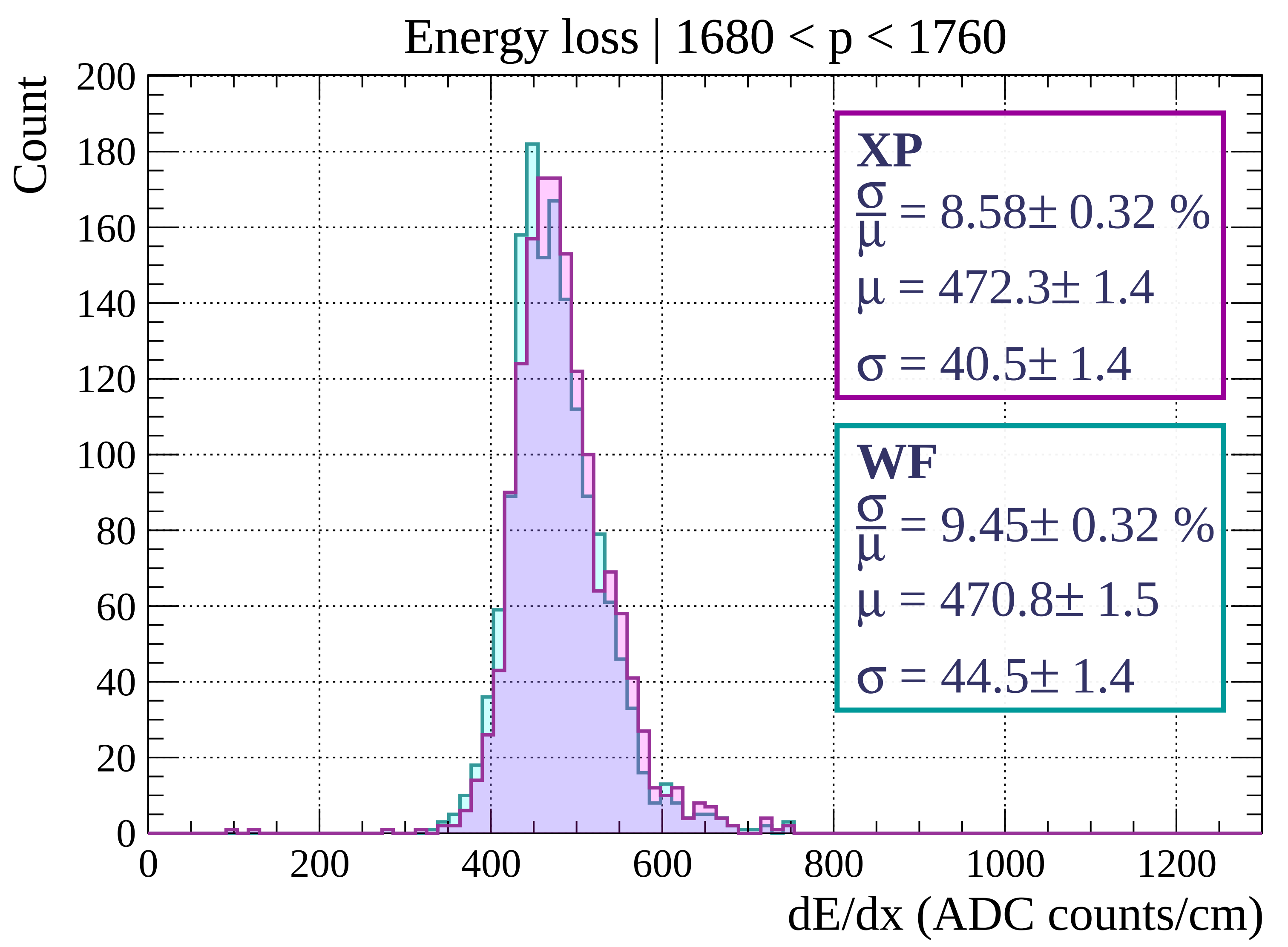


Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 8.72± 0.27 %
μ = 462.8± 1.2
σ = 40.3± 1.1
WF
σ/μ = 8.65± 0.25 %
μ = 456.6± 1.3
σ = 39.5± 1.0

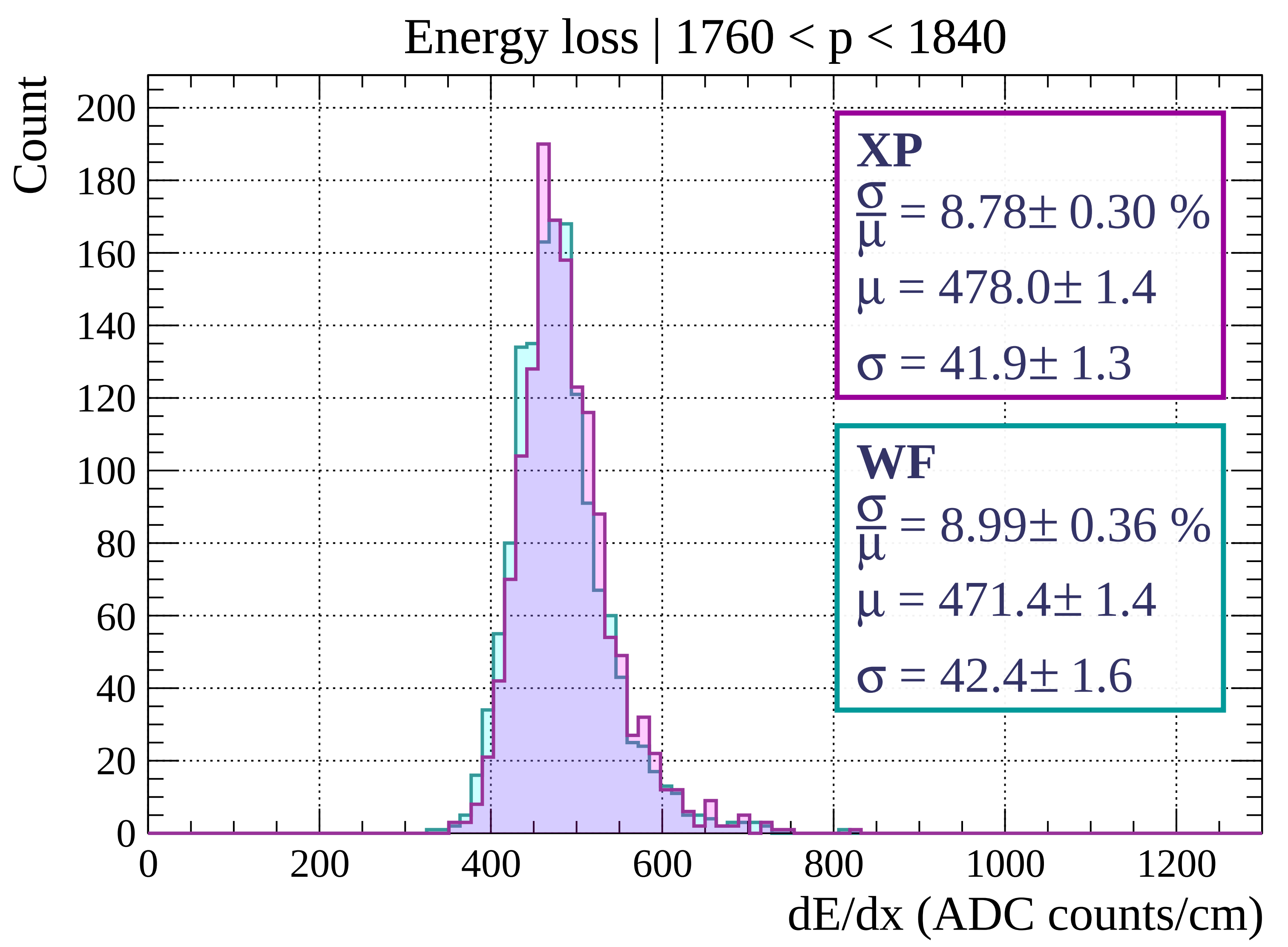

Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 8.48± 0.29 %
μ = 466.1± 1.2
σ = 39.5± 1.2
WF
σ/μ = 8.62± 0.30 %
μ = 460.0± 1.3
σ = 39.6± 1.3

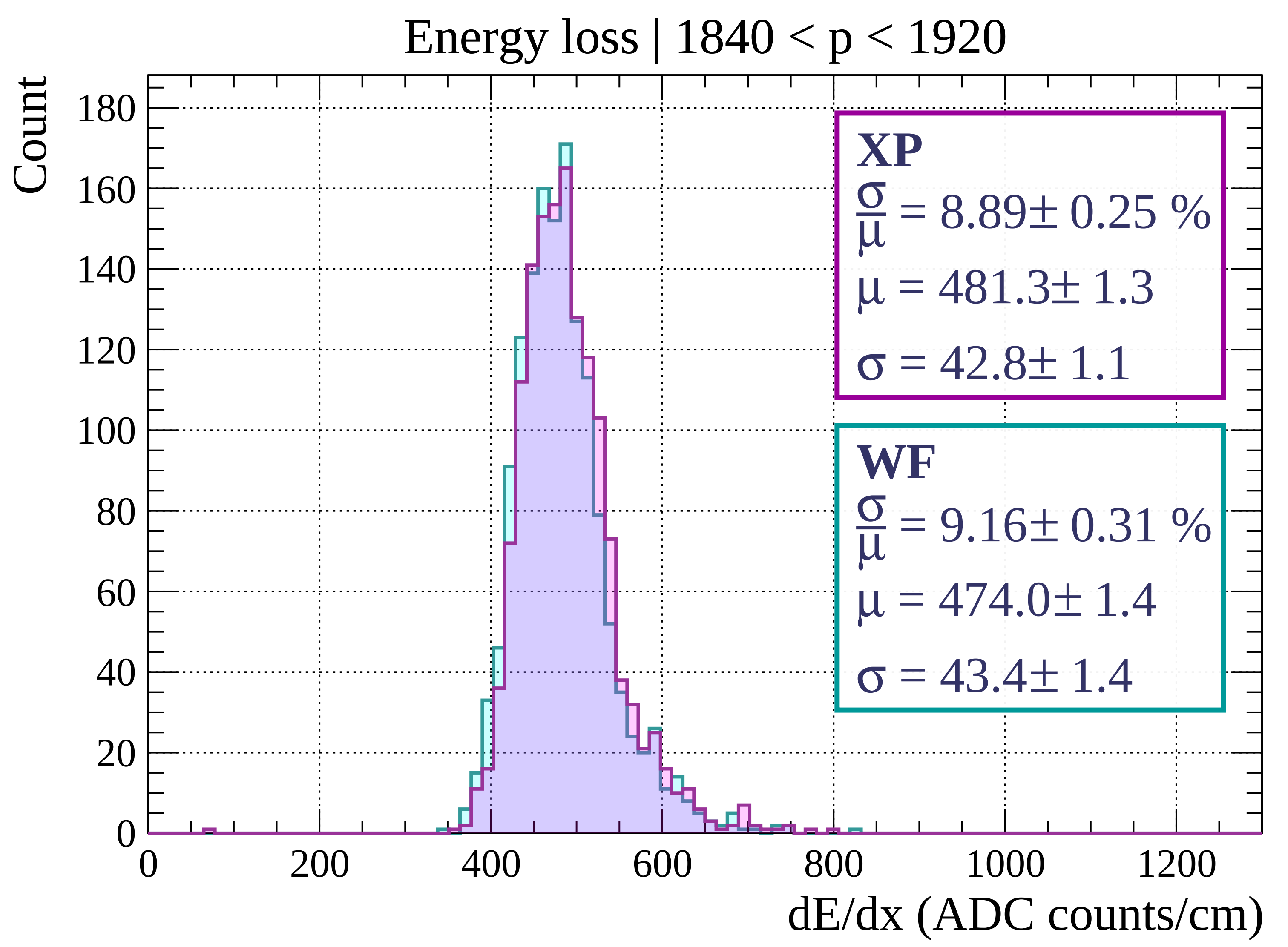


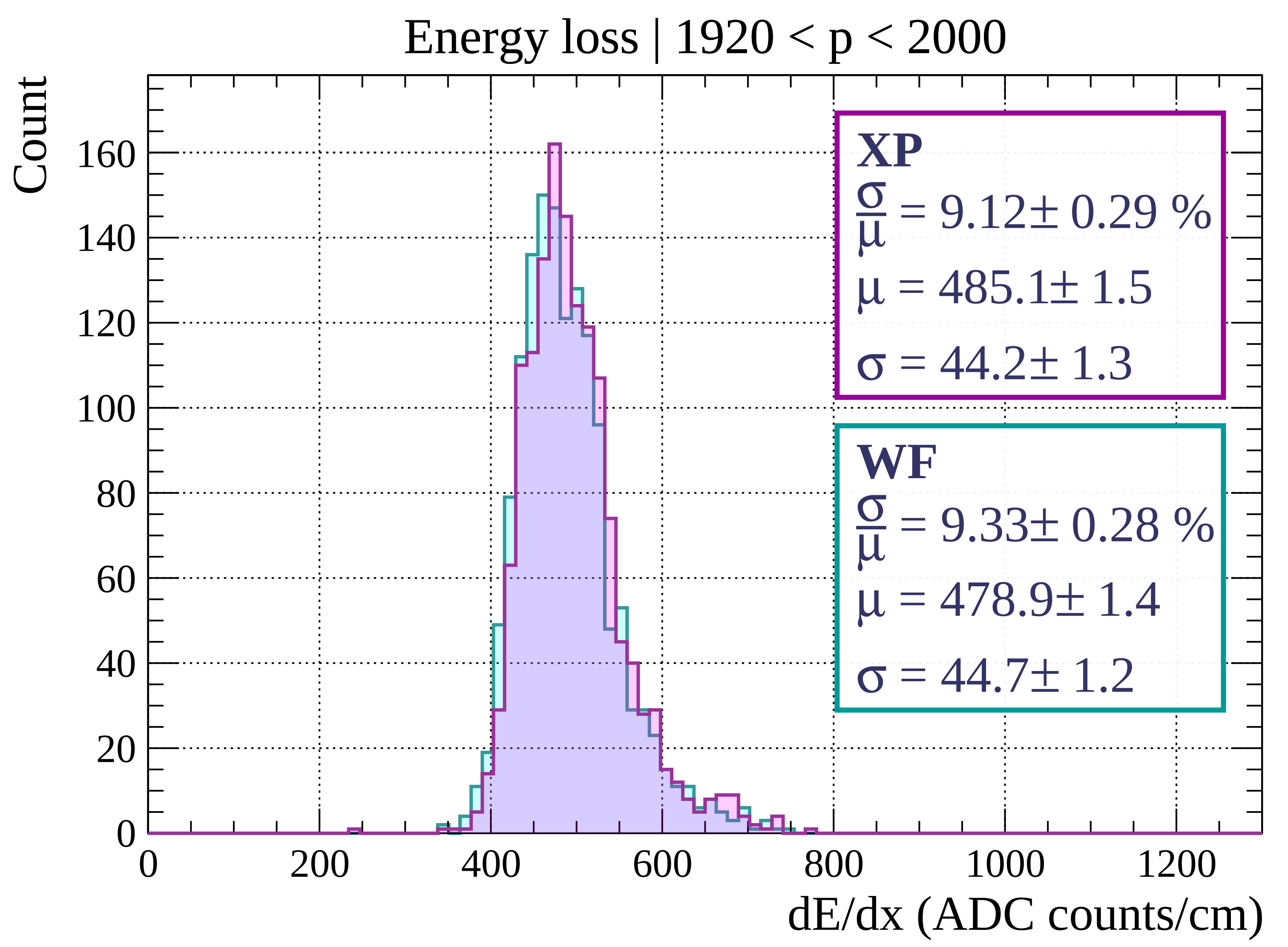
Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 8.60± 0.33 %
μ = 469.6± 1.4
σ = 40.4± 1.4
WF
σ/μ = 9.14± 0.37 %
μ = 463.2± 1.5
σ = 42.4± 1.6



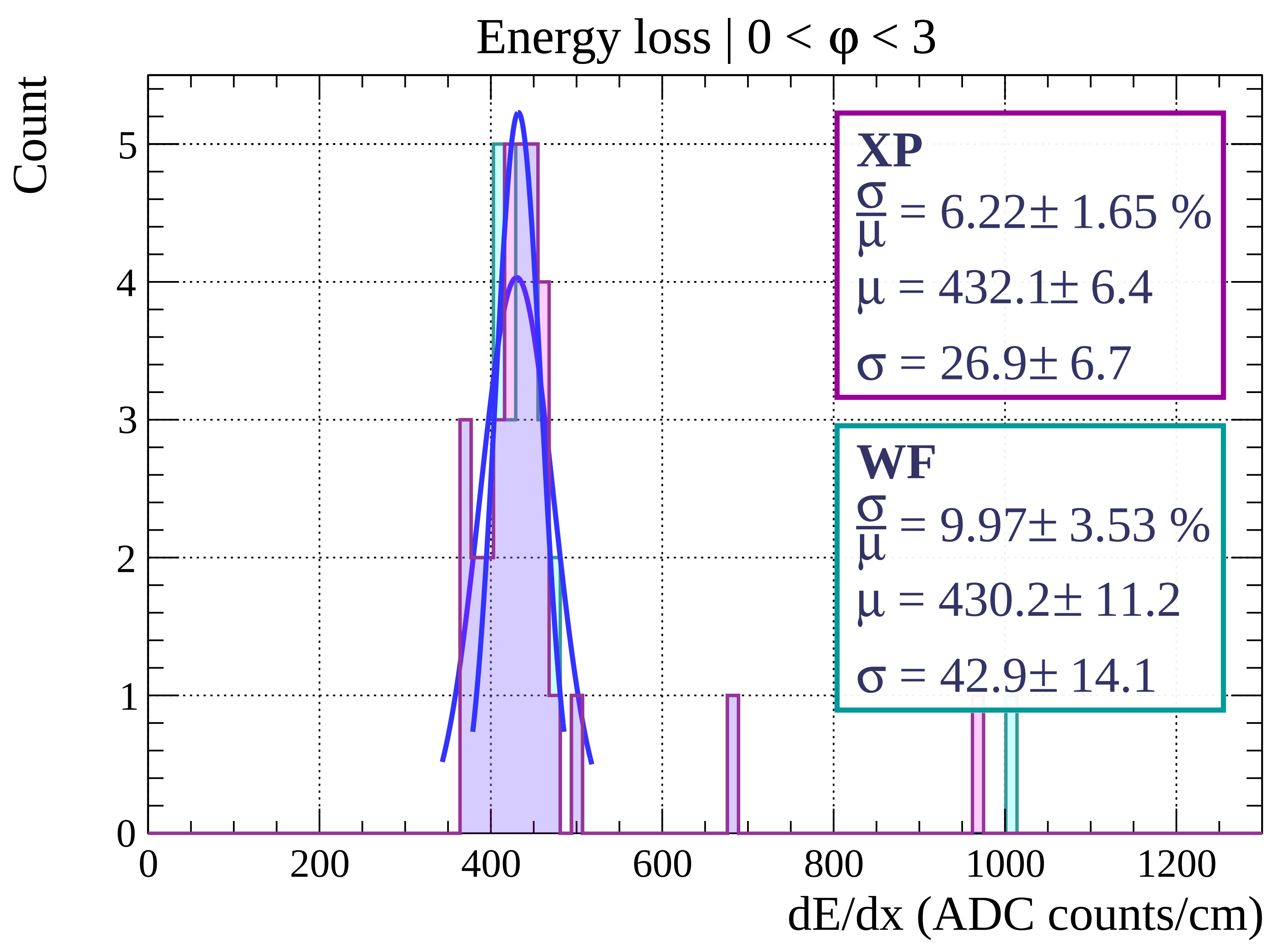
Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.58± 0.32 %
μ = 472.3± 1.4
σ = 40.5± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.45± 0.32 %
μ = 470.8± 1.5
σ = 44.5± 1.4

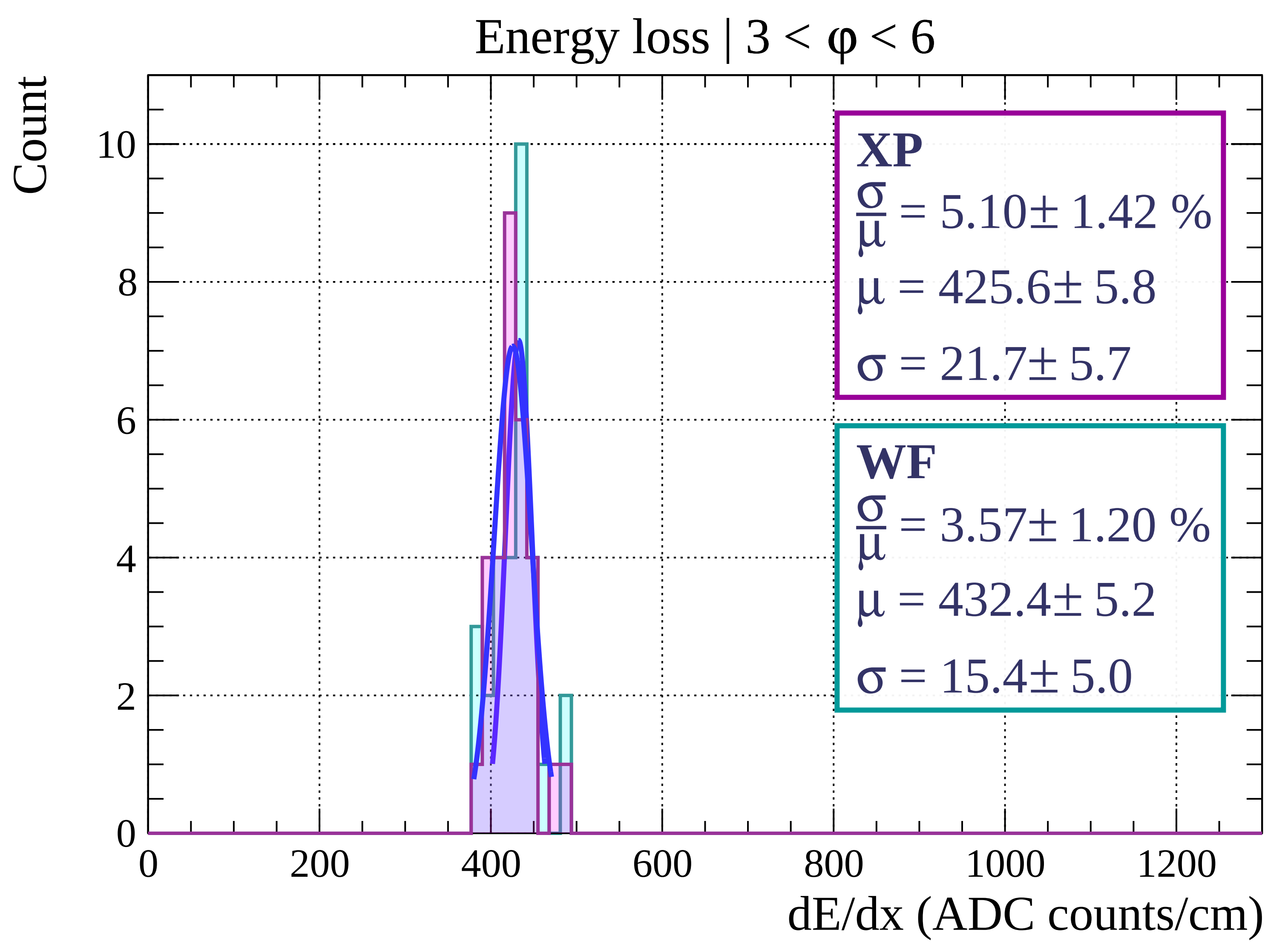

Energy loss | 1760 < p < 1840
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.78± 0.30 %
μ = 478.0± 1.4
σ = 41.9± 1.3
WF
σ/μ = 8.99± 0.36 %
μ = 471.4± 1.4
σ = 42.4± 1.6

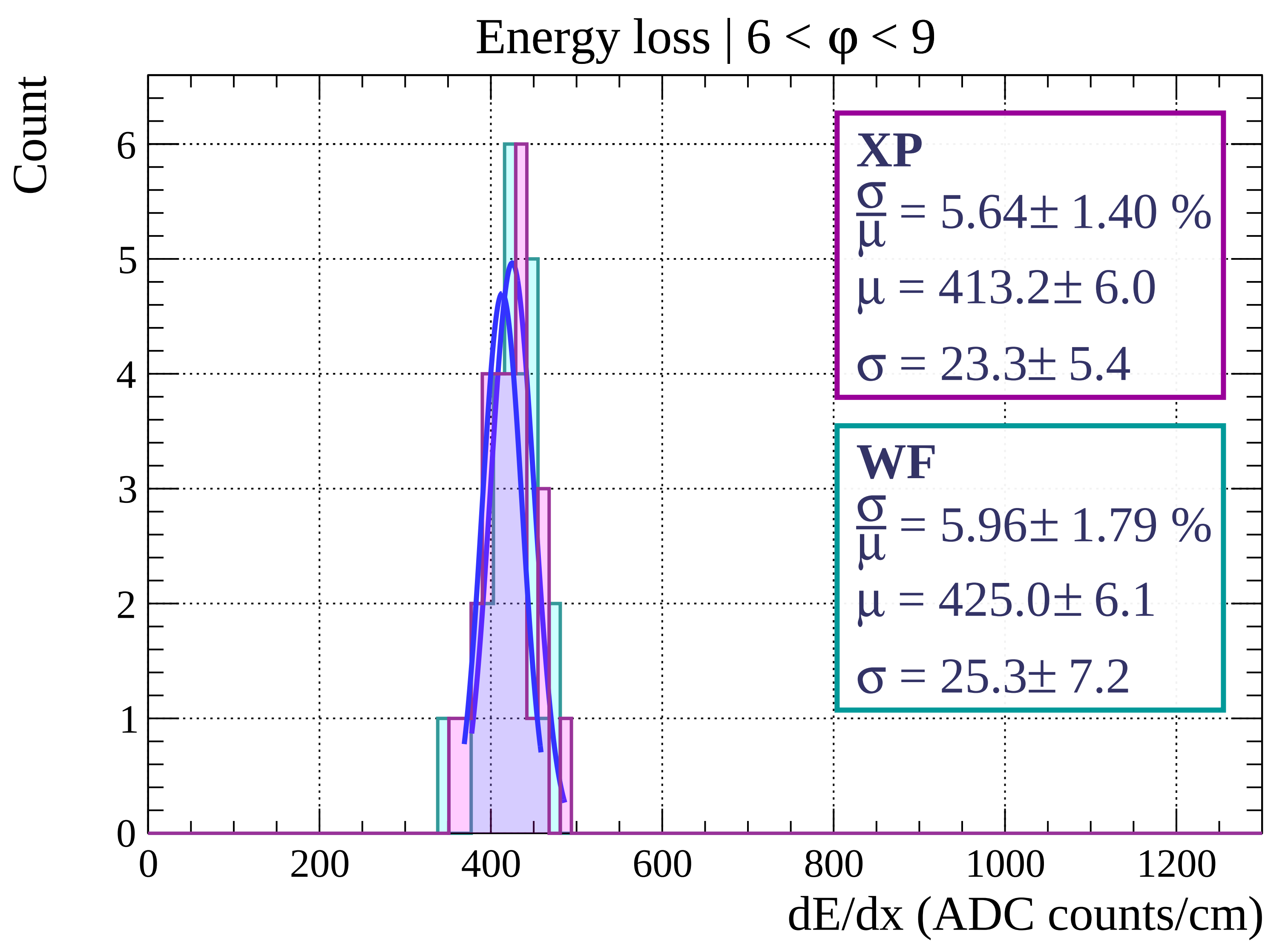

Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 8.89± 0.25 %
μ = 481.3± 1.3
σ = 42.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.16± 0.31 %
μ = 474.0± 1.4
σ = 43.4± 1.4



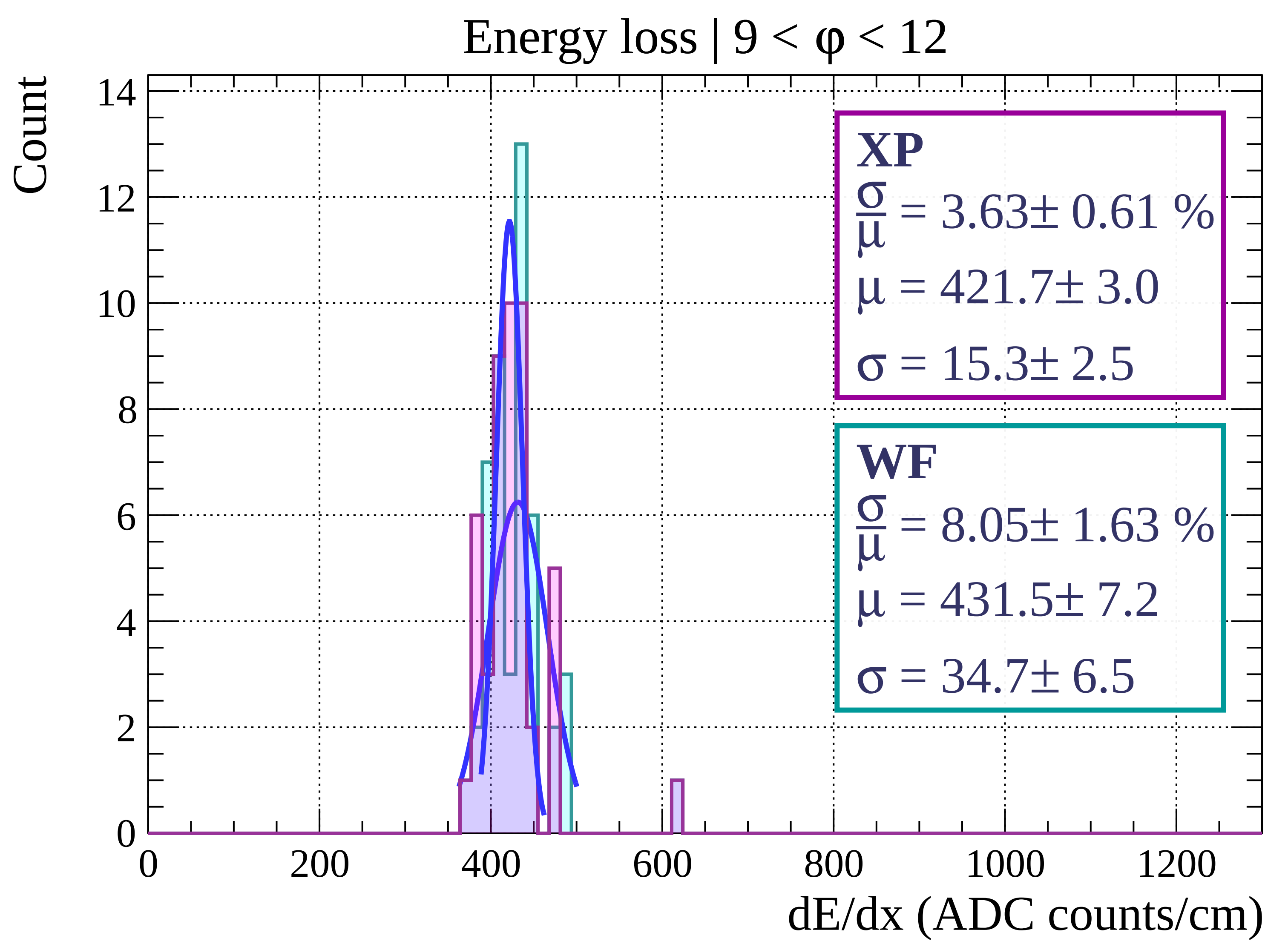
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 9.12± 0.29 %
μ = 485.1± 1.5
σ = 44.2± 1.3
WF
$\frac{\sigma}{\mu}$ = 9.33± 0.28 %
μ = 478.9± 1.4
σ = 44.7± 1.2


Energy loss | 0 < φ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
XP
σ/μ = 6.22± 1.65 %
μ = 432.1± 6.4
σ = 26.9± 6.7
WF
σ/μ = 9.97± 3.53 %
μ = 430.2± 11.2
σ = 42.9± 14.1

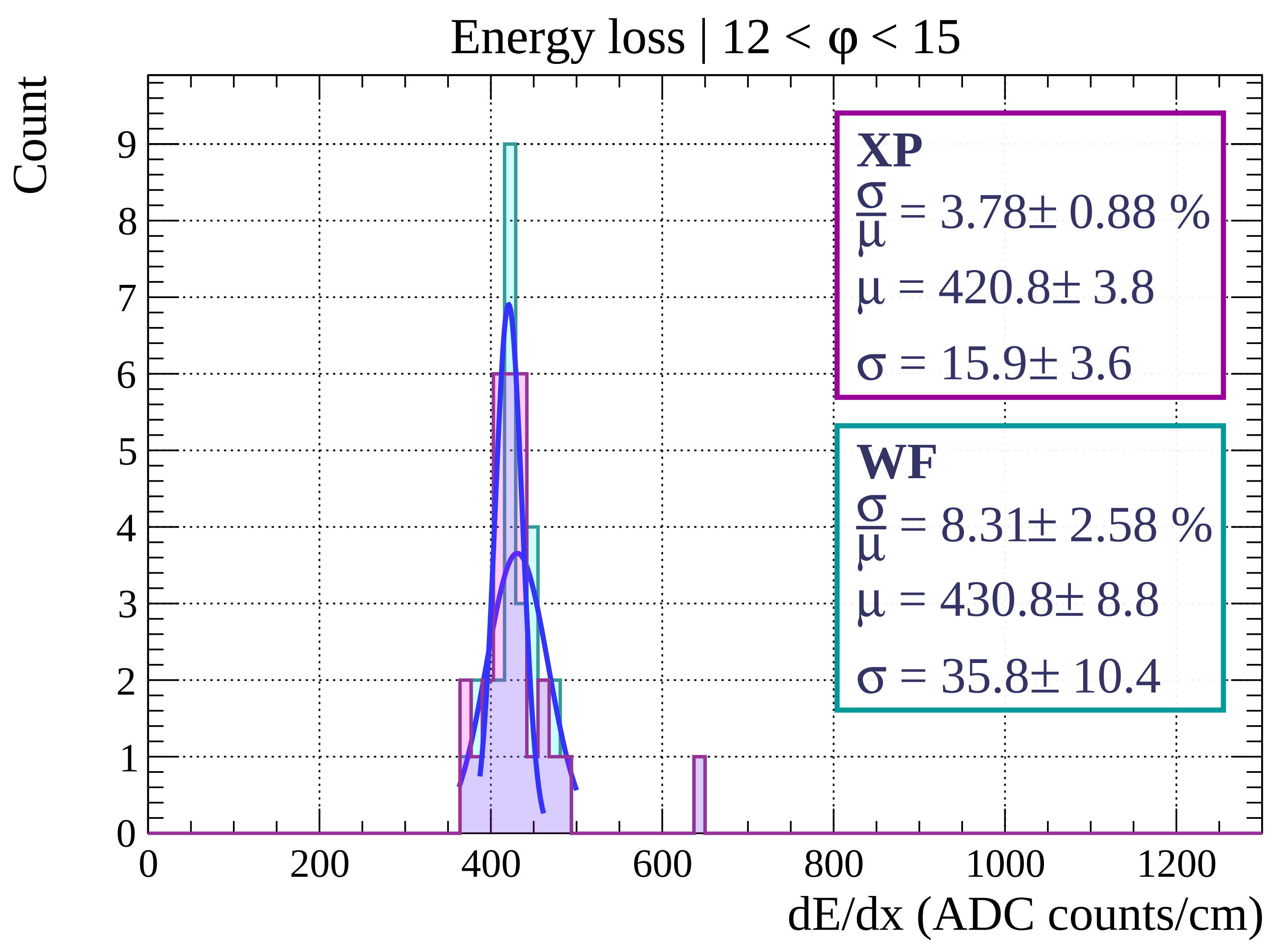

Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.10± 1.42 %
μ = 425.6± 5.8
σ = 21.7± 5.7
WF
σ/μ = 3.57± 1.20 %
μ = 432.4± 5.2
σ = 15.4± 5.0

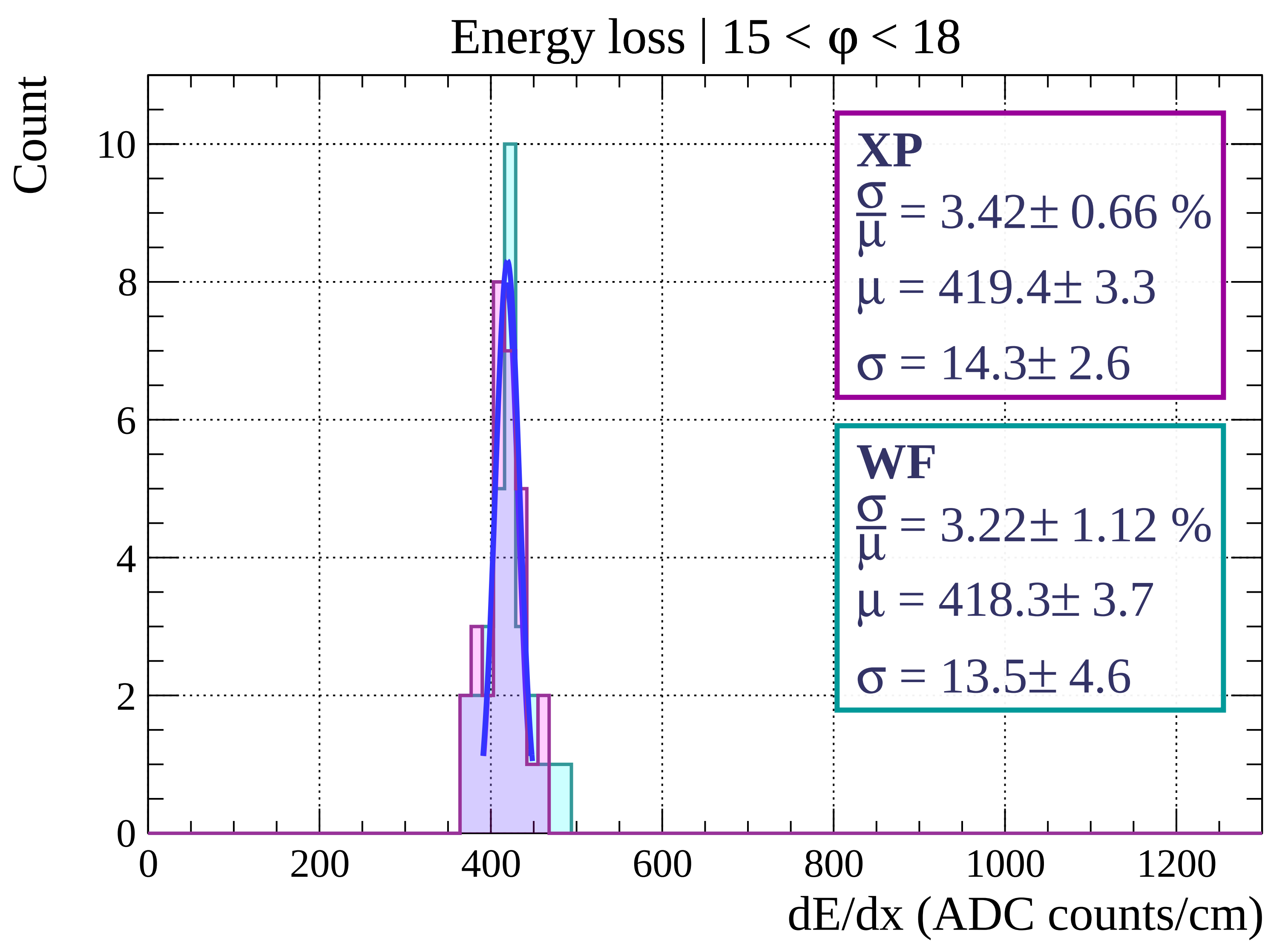


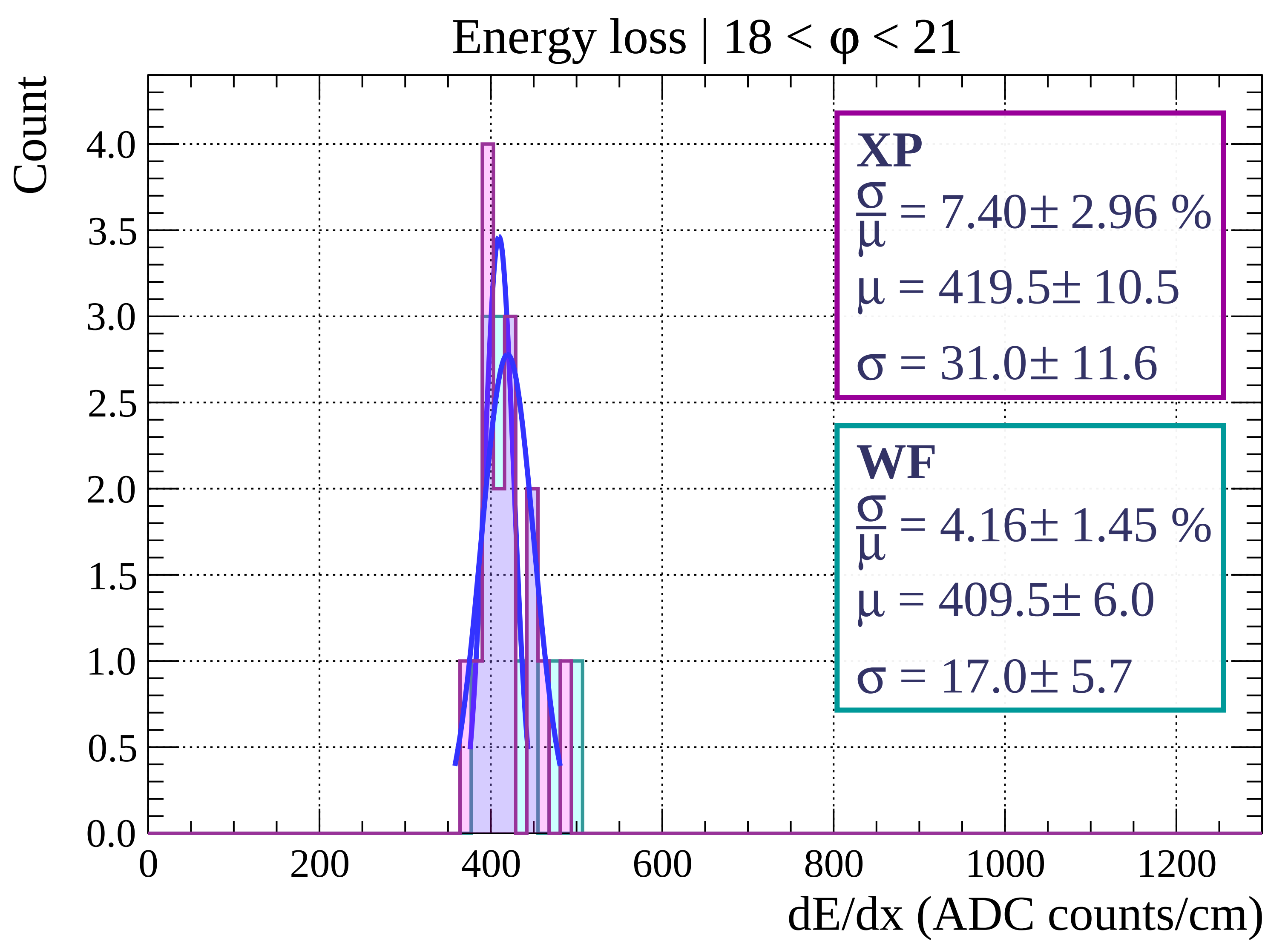
Energy loss | 6 < φ < 9
Count
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.64± 1.40 %
μ = 413.2± 6.0
σ = 23.3± 5.4
WF
σ/μ = 5.96± 1.79 %
μ = 425.0± 6.1
σ = 25.3± 7.2



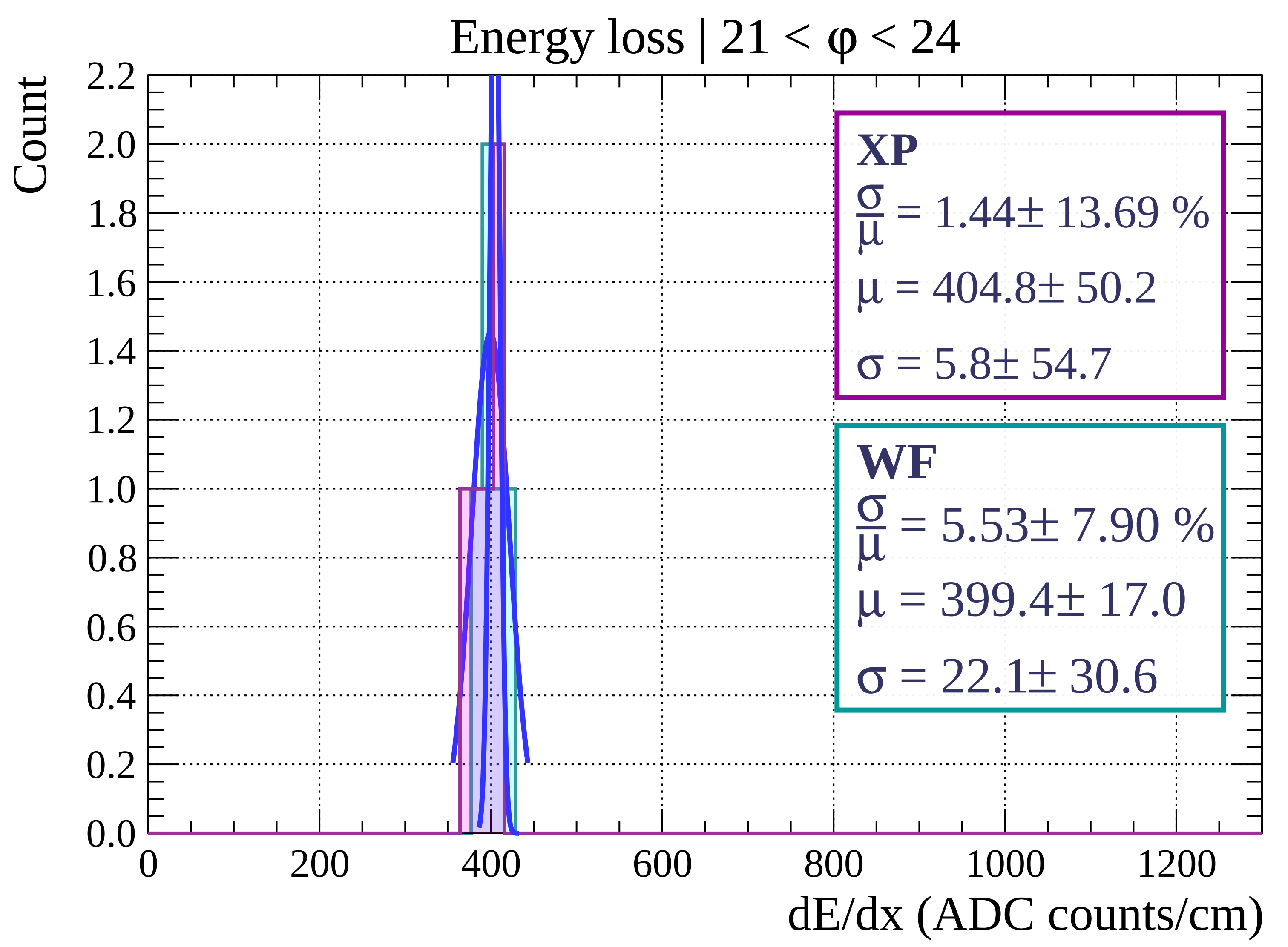
Energy loss | 9 < φ < 12
Count
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 3.63± 0.61 %
μ = 421.7± 3.0
σ = 15.3± 2.5
WF
σ/μ = 8.05± 1.63 %
μ = 431.5± 7.2
σ = 34.7± 6.5


Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 3.78± 0.88 %
μ = 420.8± 3.8
σ = 15.9± 3.6
WF
$\frac{\sigma}{\mu}$ = 8.31± 2.58 %
μ = 430.8± 8.8
σ = 35.8± 10.4

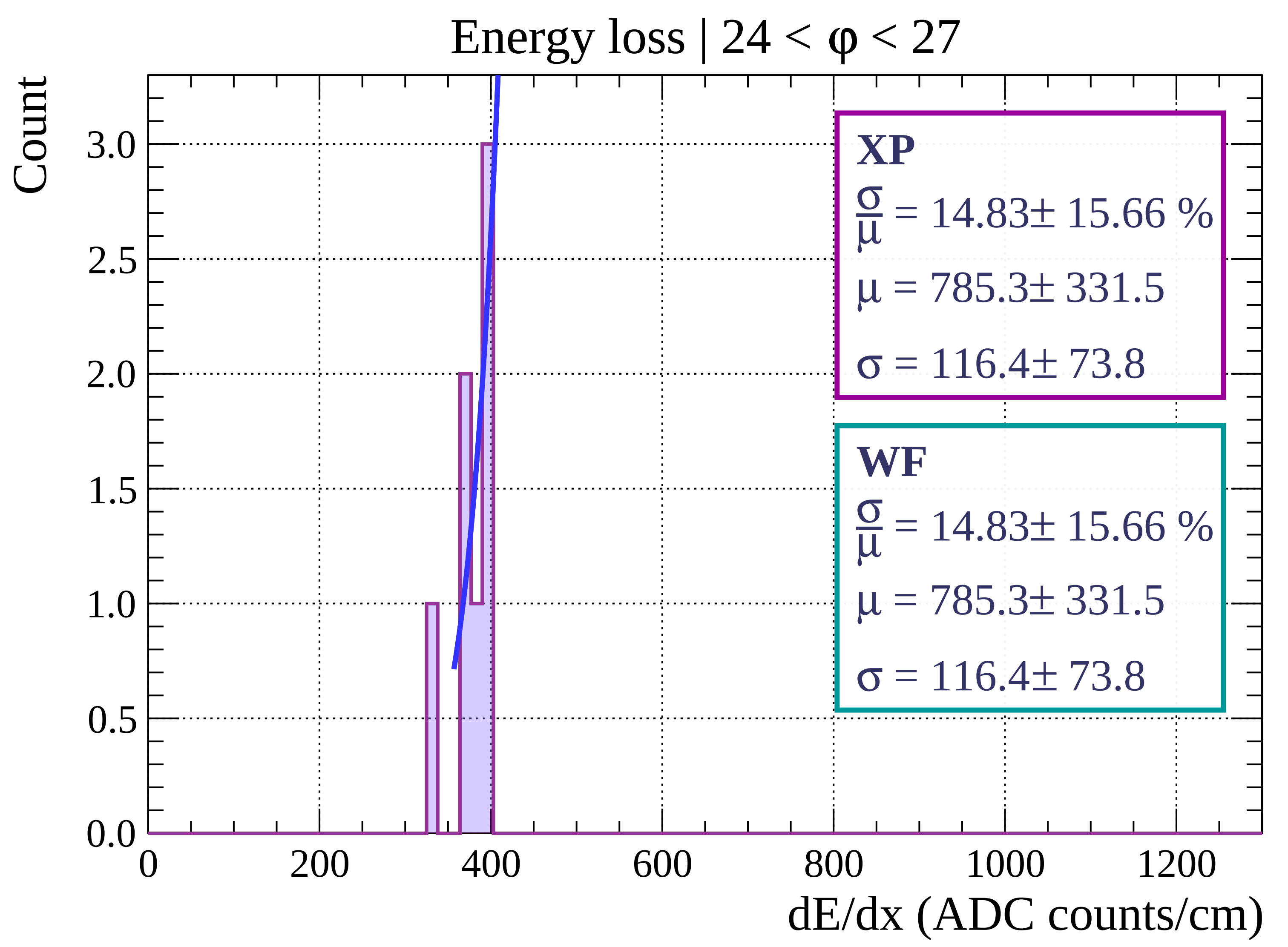

Energy loss | 15 < φ < 18
Count
0
2
4
6
8
10
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 3.42± 0.66 %
μ = 419.4± 3.3
σ = 14.3± 2.6
WF
σ/μ = 3.22± 1.12 %
μ = 418.3± 3.7
σ = 13.5± 4.6



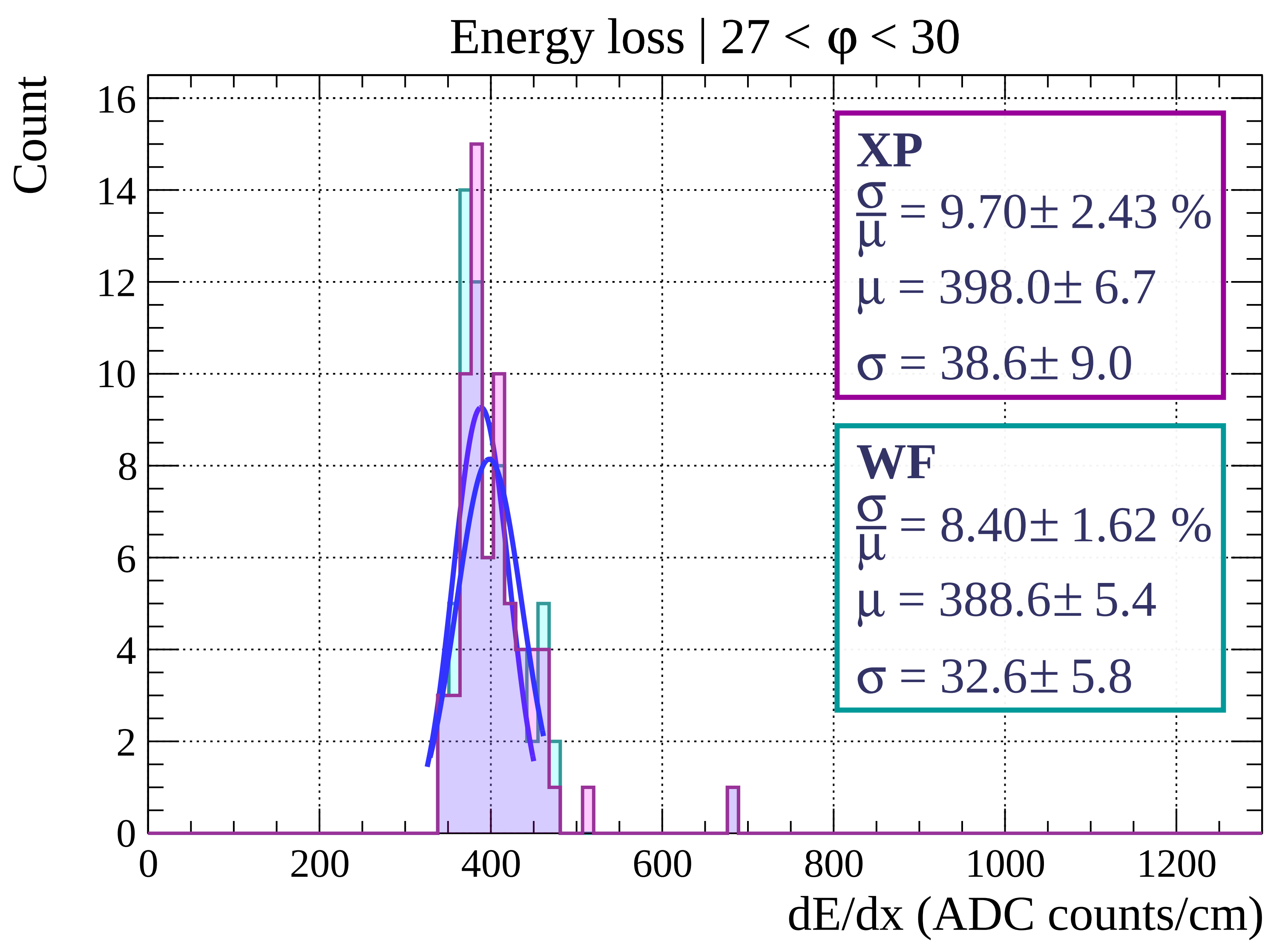
Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
XP
σ/μ = 7.40± 2.96 %
μ = 419.5± 10.5
σ = 31.0± 11.6
WF
σ/μ = 4.16± 1.45 %
μ = 409.5± 6.0
σ = 17.0± 5.7

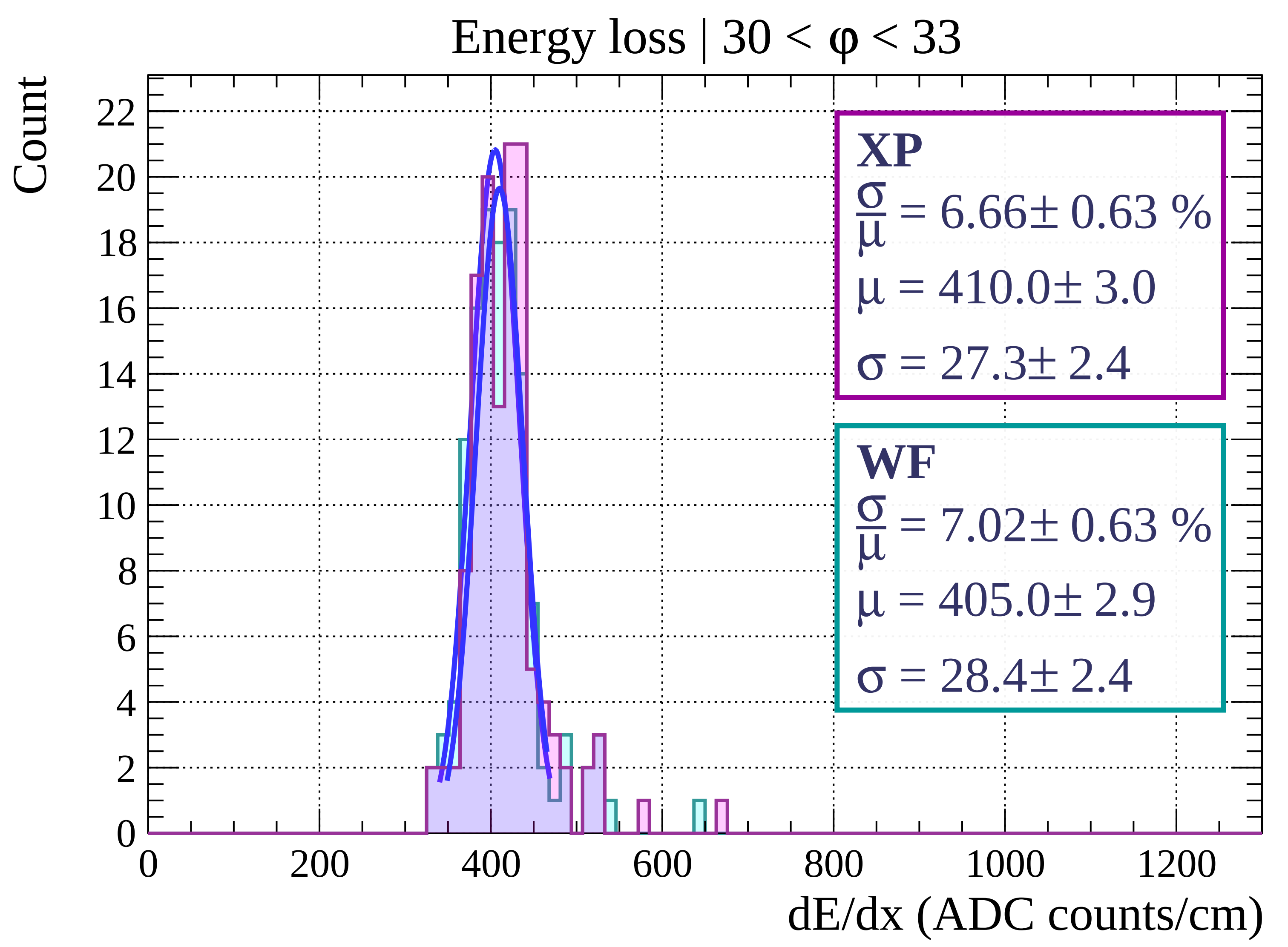


Energy loss | 21 < φ < 24
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.44± 13.69 %
μ = 404.8± 50.2
σ = 5.8± 54.7
WF
σ/μ = 5.53± 7.90 %
μ = 399.4± 17.0
σ = 22.1± 30.6

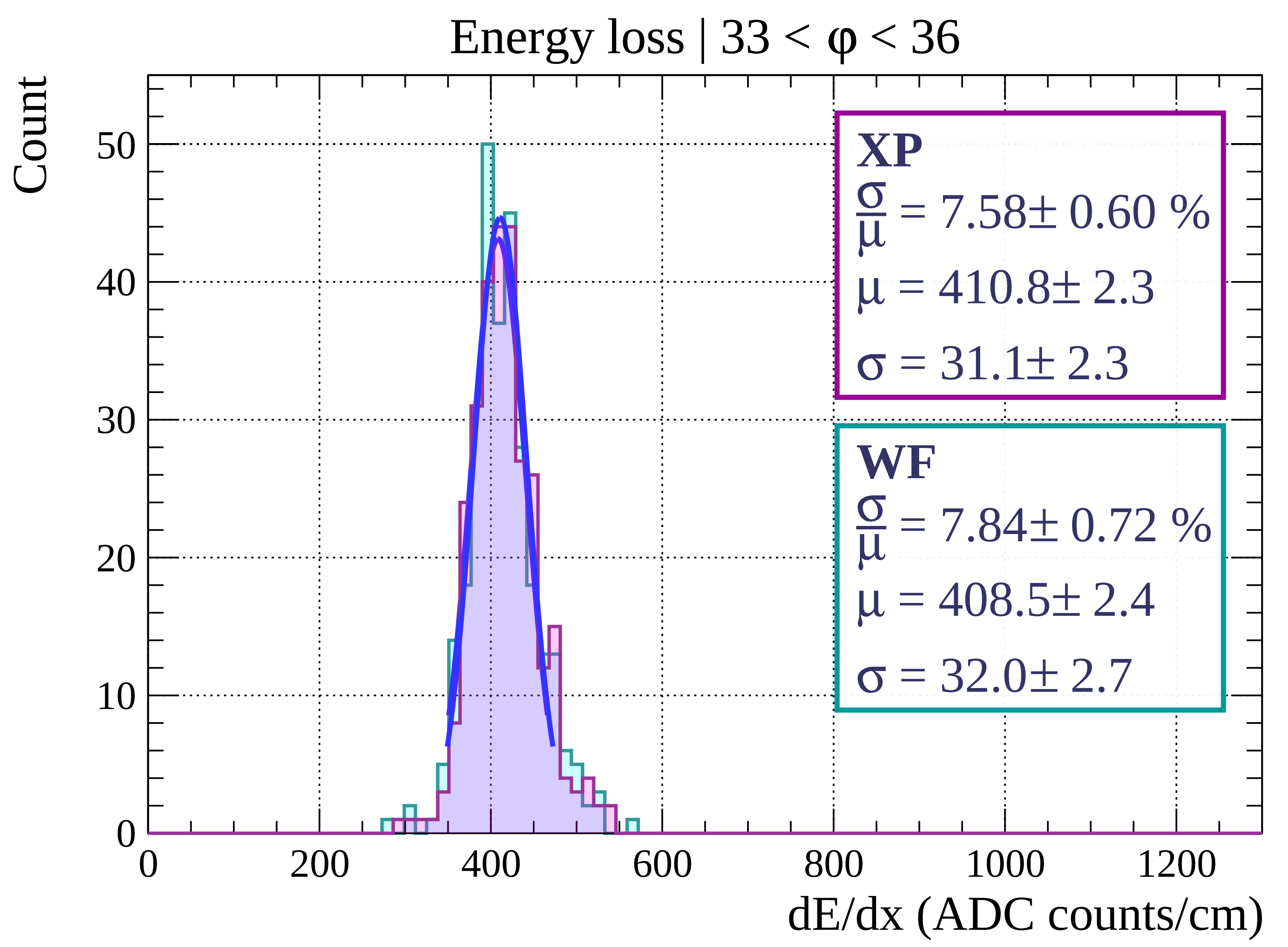


Energy loss | 24 < φ < 27
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.83± 15.66 %
μ = 785.3± 331.5
σ = 116.4± 73.8
WF
σ/μ = 14.83± 15.66 %
μ = 785.3± 331.5
σ = 116.4± 73.8

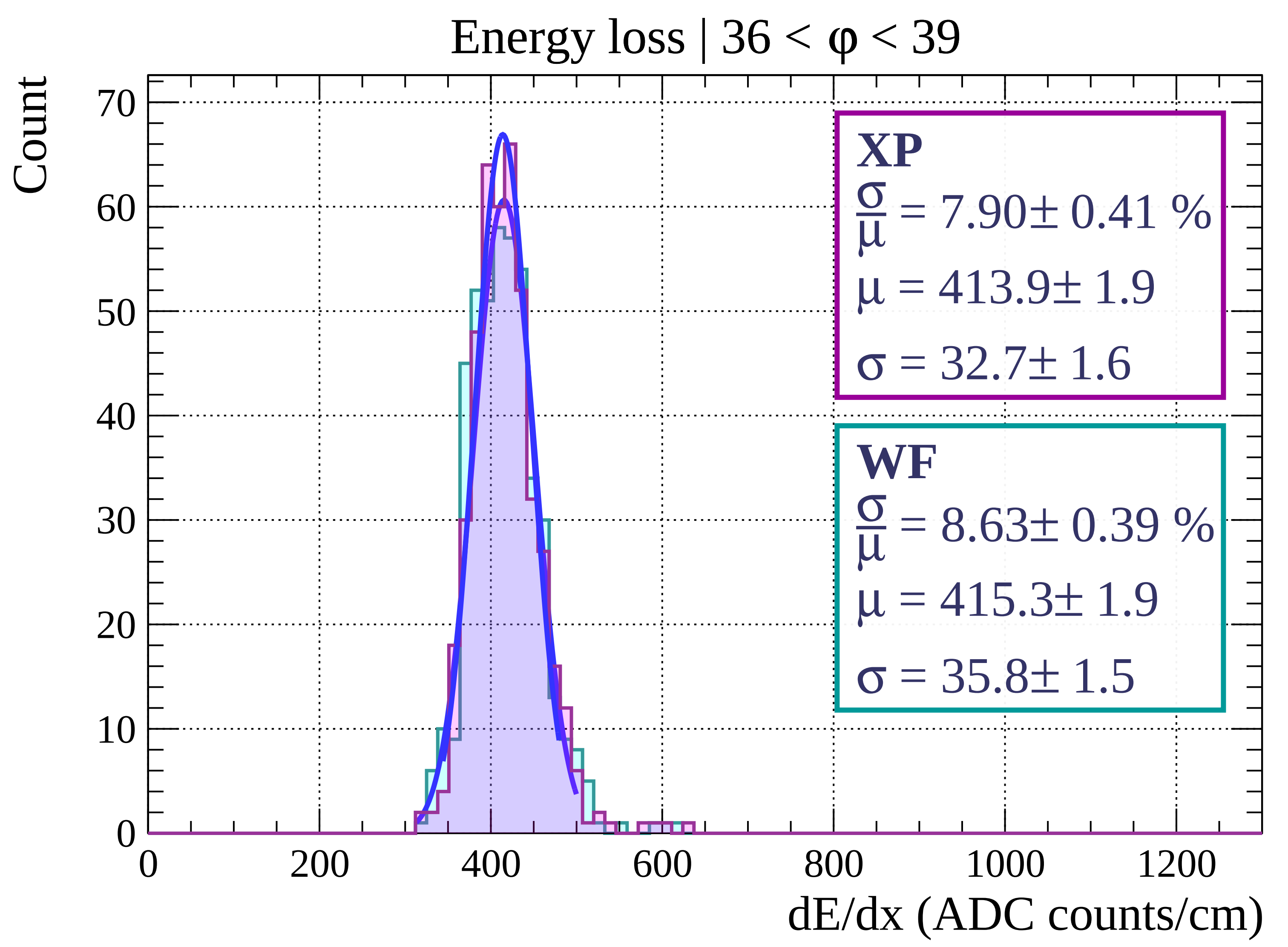

Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
XP
σ/μ = 9.70± 2.43 %
μ = 398.0± 6.7
σ = 38.6± 9.0
WF
σ/μ = 8.40± 1.62 %
μ = 388.6± 5.4
σ = 32.6± 5.8

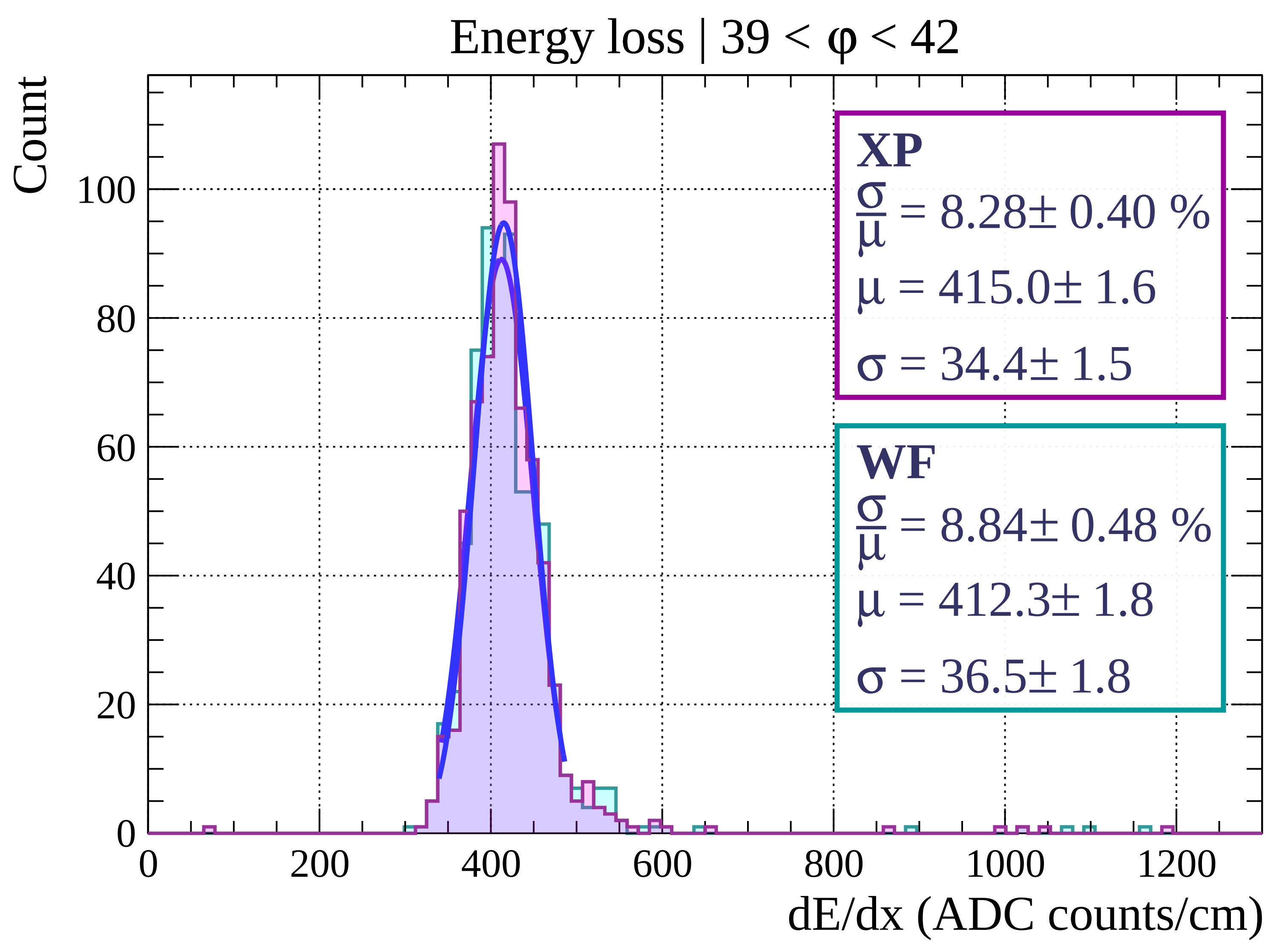


Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 6.66 \pm 0.63$ %
$\mu = 410.0 \pm 3.0$
$\sigma = 27.3 \pm 2.4$
WF
$\frac{\sigma}{\mu} = 7.02 \pm 0.63$ %
$\mu = 405.0 \pm 2.9$
$\sigma = 28.4 \pm 2.4$

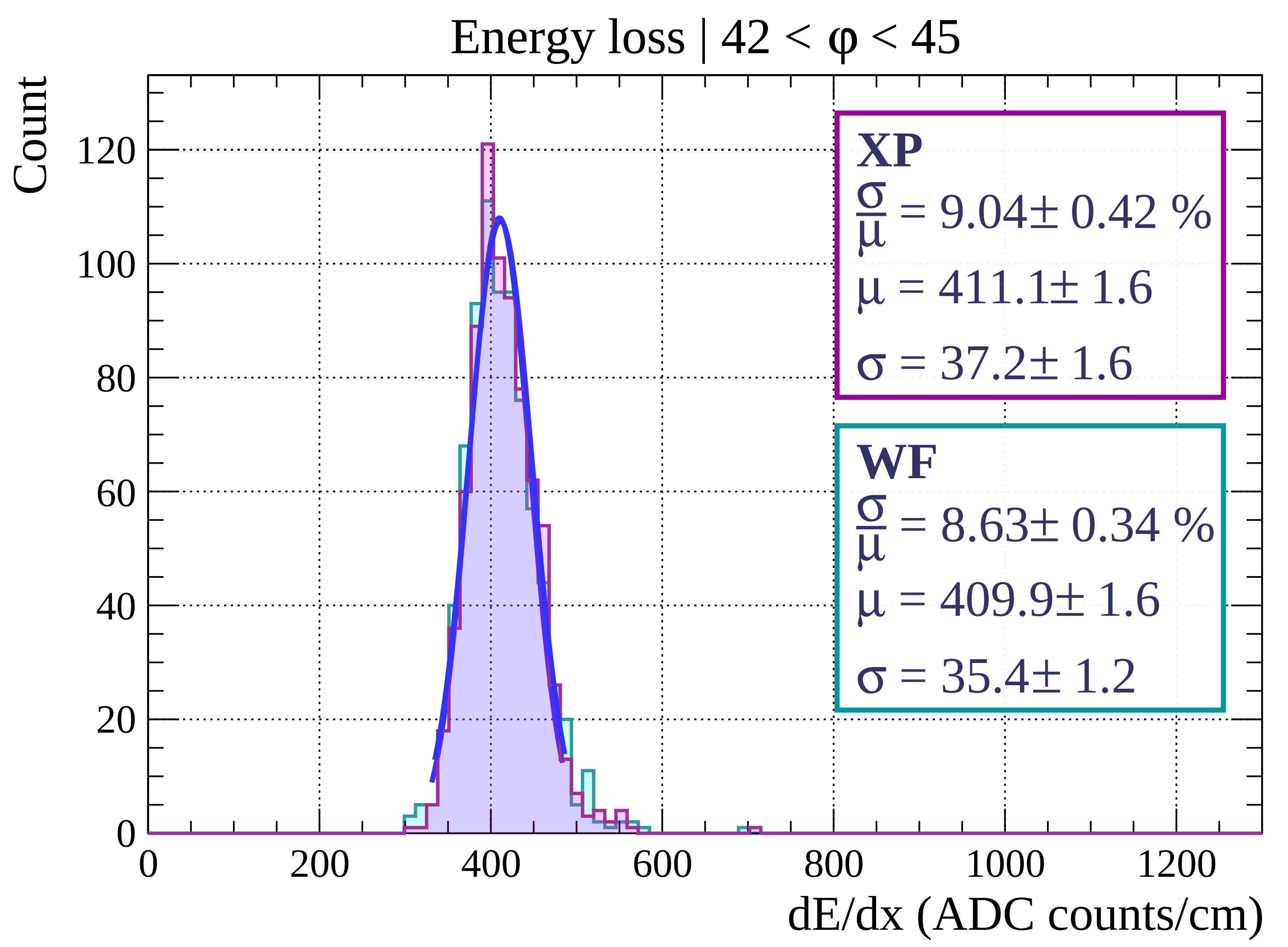


Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 7.58± 0.60 %
μ = 410.8± 2.3
σ = 31.1± 2.3
WF
σ/μ = 7.84± 0.72 %
μ = 408.5± 2.4
σ = 32.0± 2.7

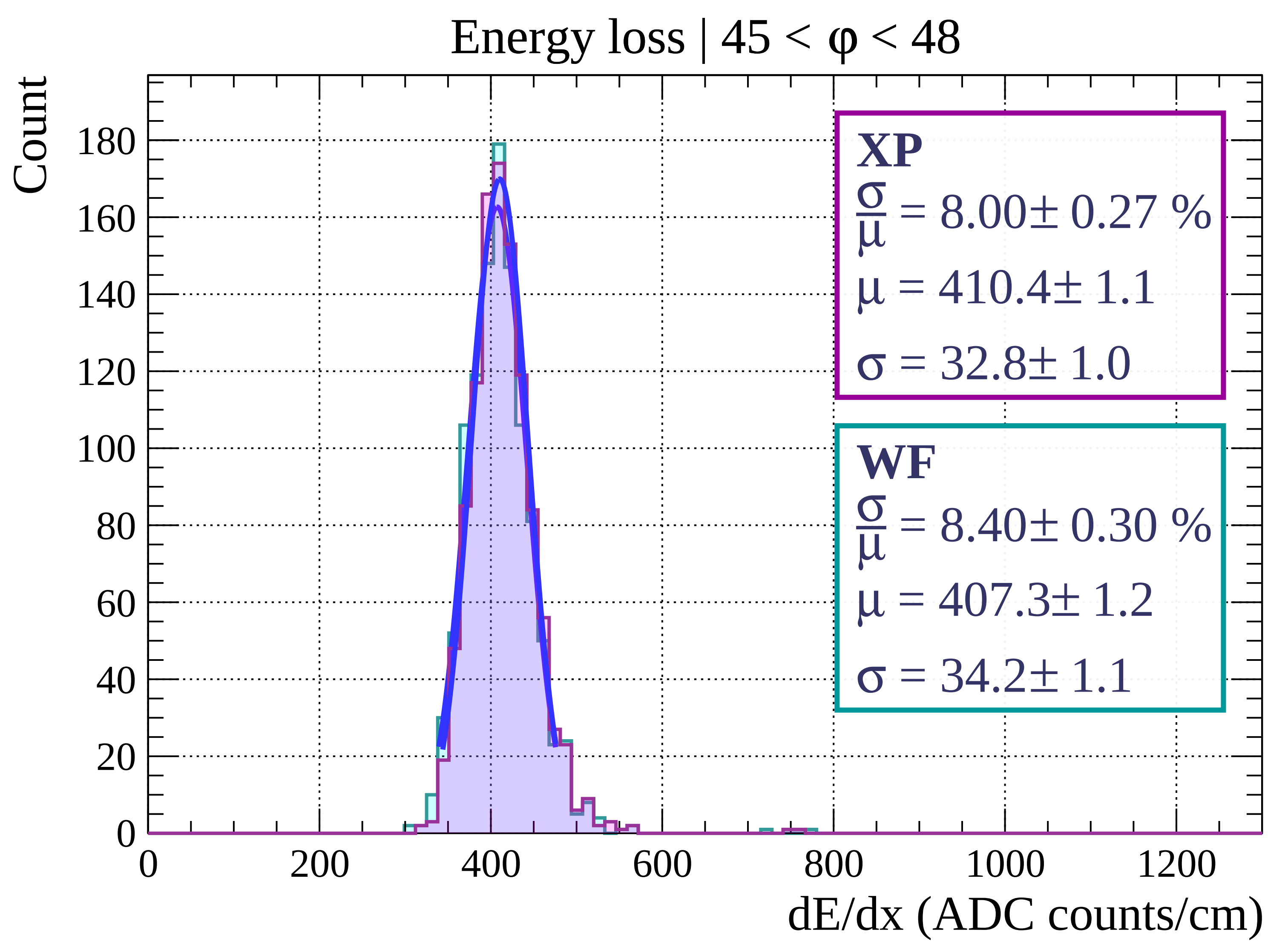

Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 7.90± 0.41 %
μ = 413.9± 1.9
σ = 32.7± 1.6
WF
σ/μ = 8.63± 0.39 %
μ = 415.3± 1.9
σ = 35.8± 1.5



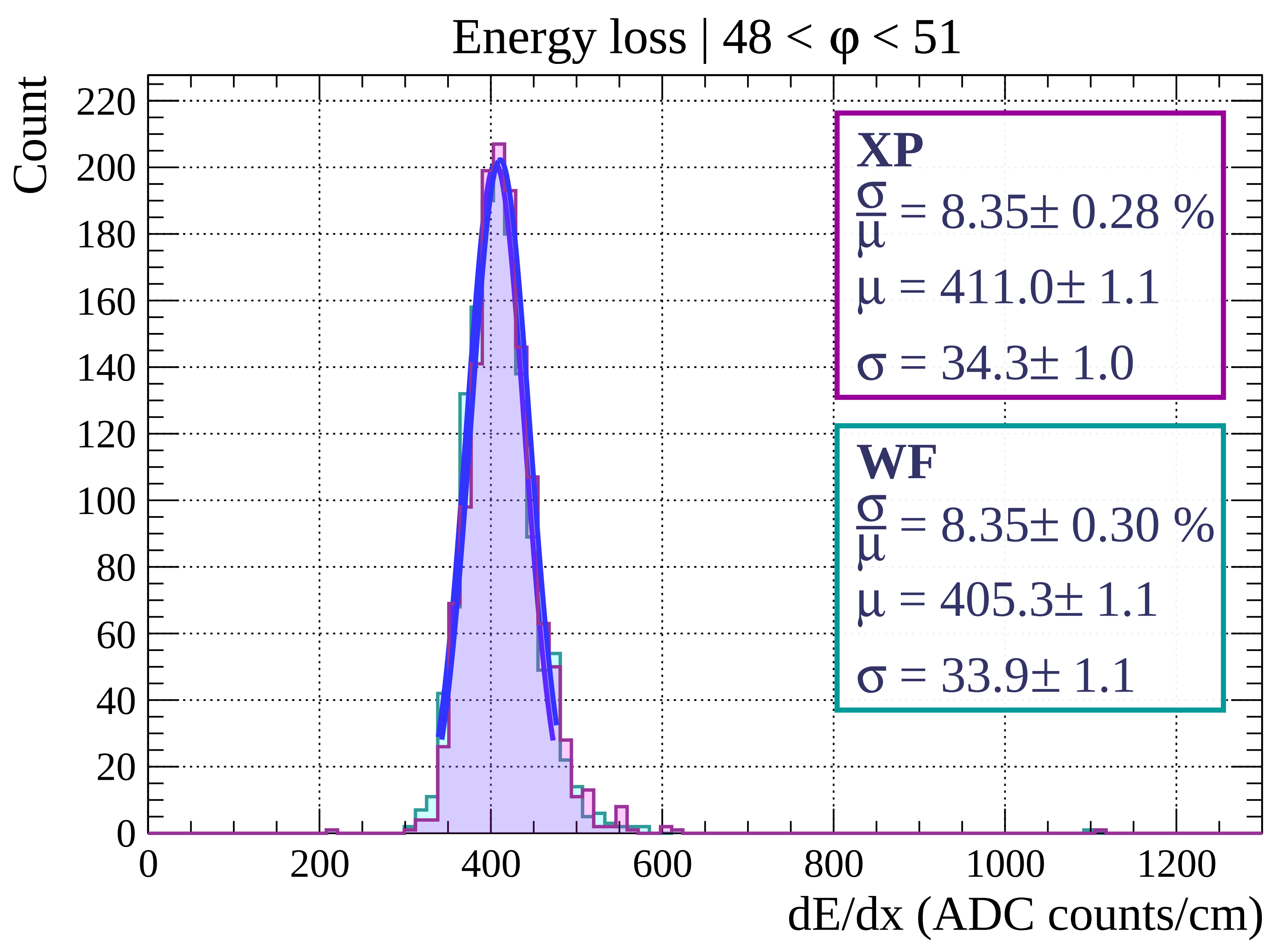
Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 8.28± 0.40 %
μ = 415.0± 1.6
σ = 34.4± 1.5
WF
σ/μ = 8.84± 0.48 %
μ = 412.3± 1.8
σ = 36.5± 1.8



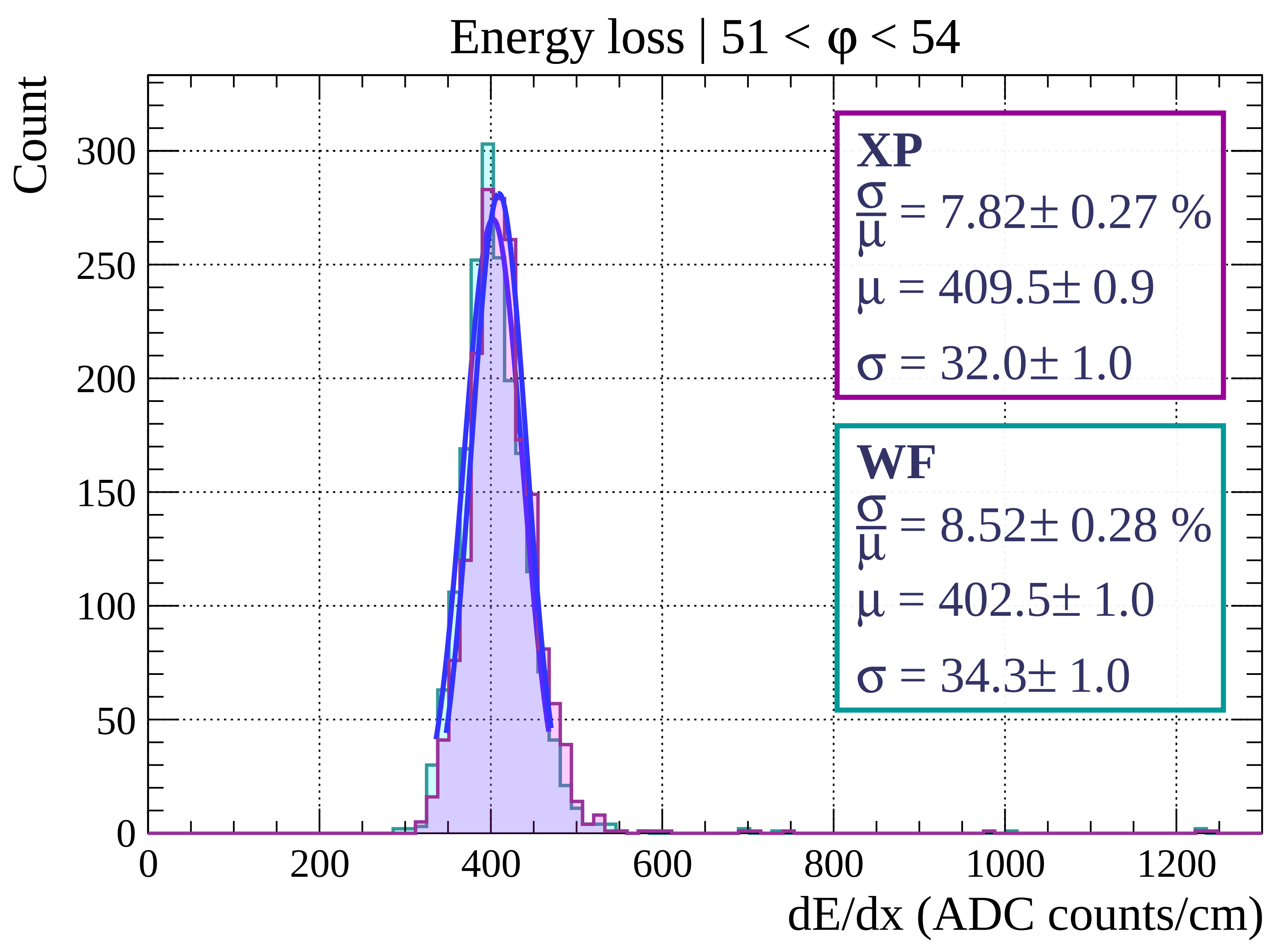
Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 9.04± 0.42 %
μ = 411.1± 1.6
σ = 37.2± 1.6
WF
σ/μ = 8.63± 0.34 %
μ = 409.9± 1.6
σ = 35.4± 1.2

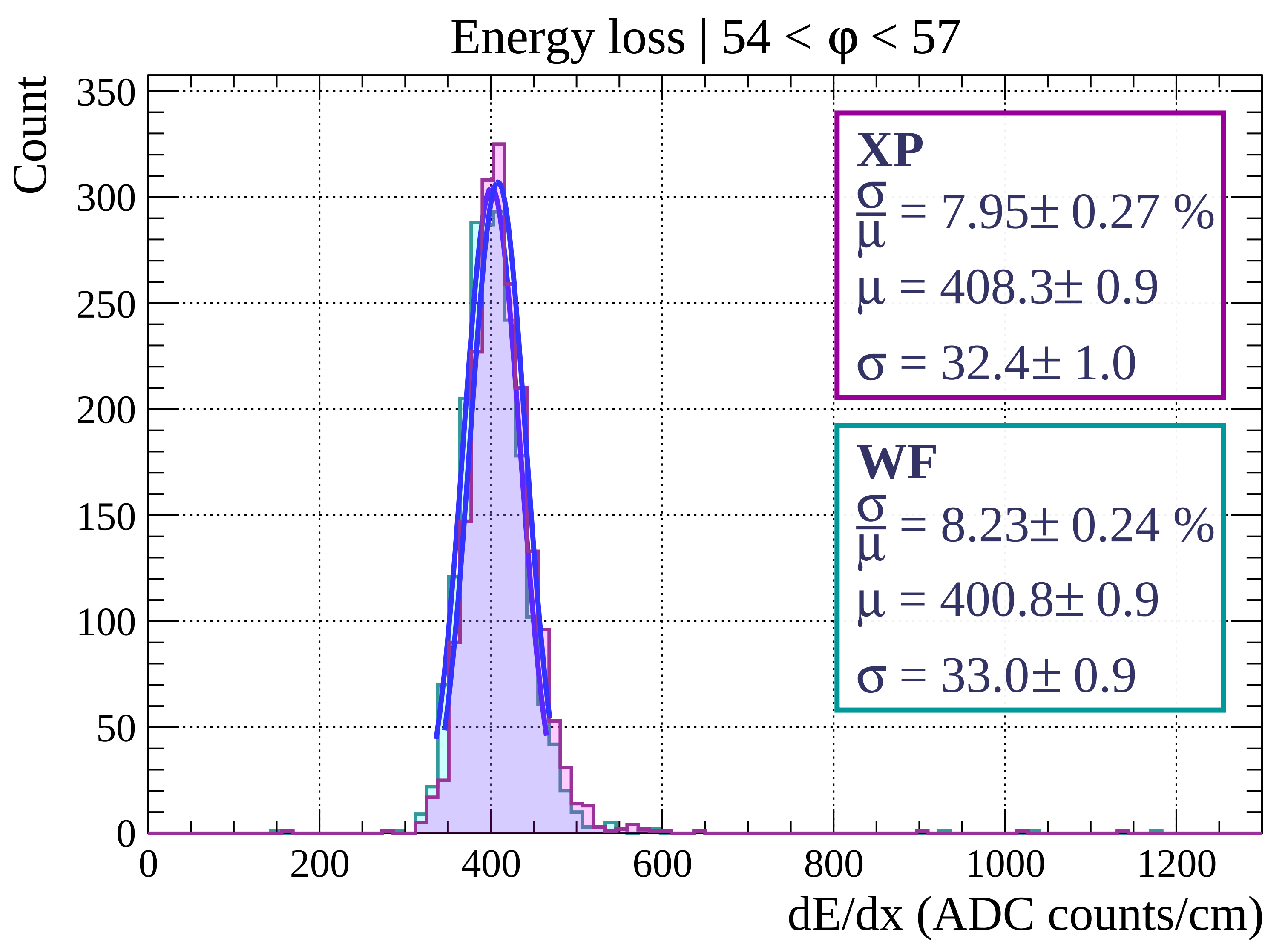


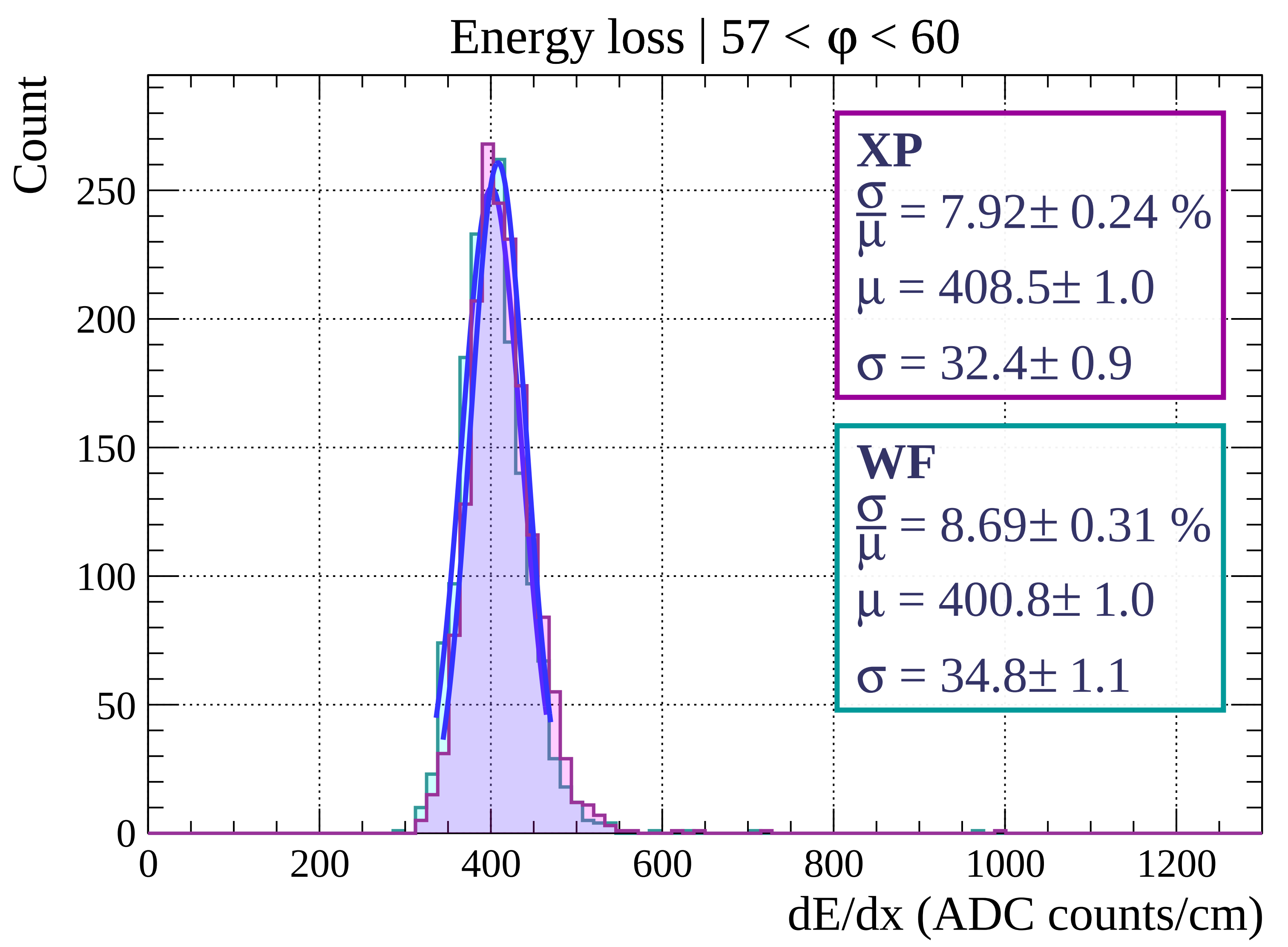
Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 8.00± 0.27 %
μ = 410.4± 1.1
σ = 32.8± 1.0
WF
σ/μ = 8.40± 0.30 %
μ = 407.3± 1.2
σ = 34.2± 1.1



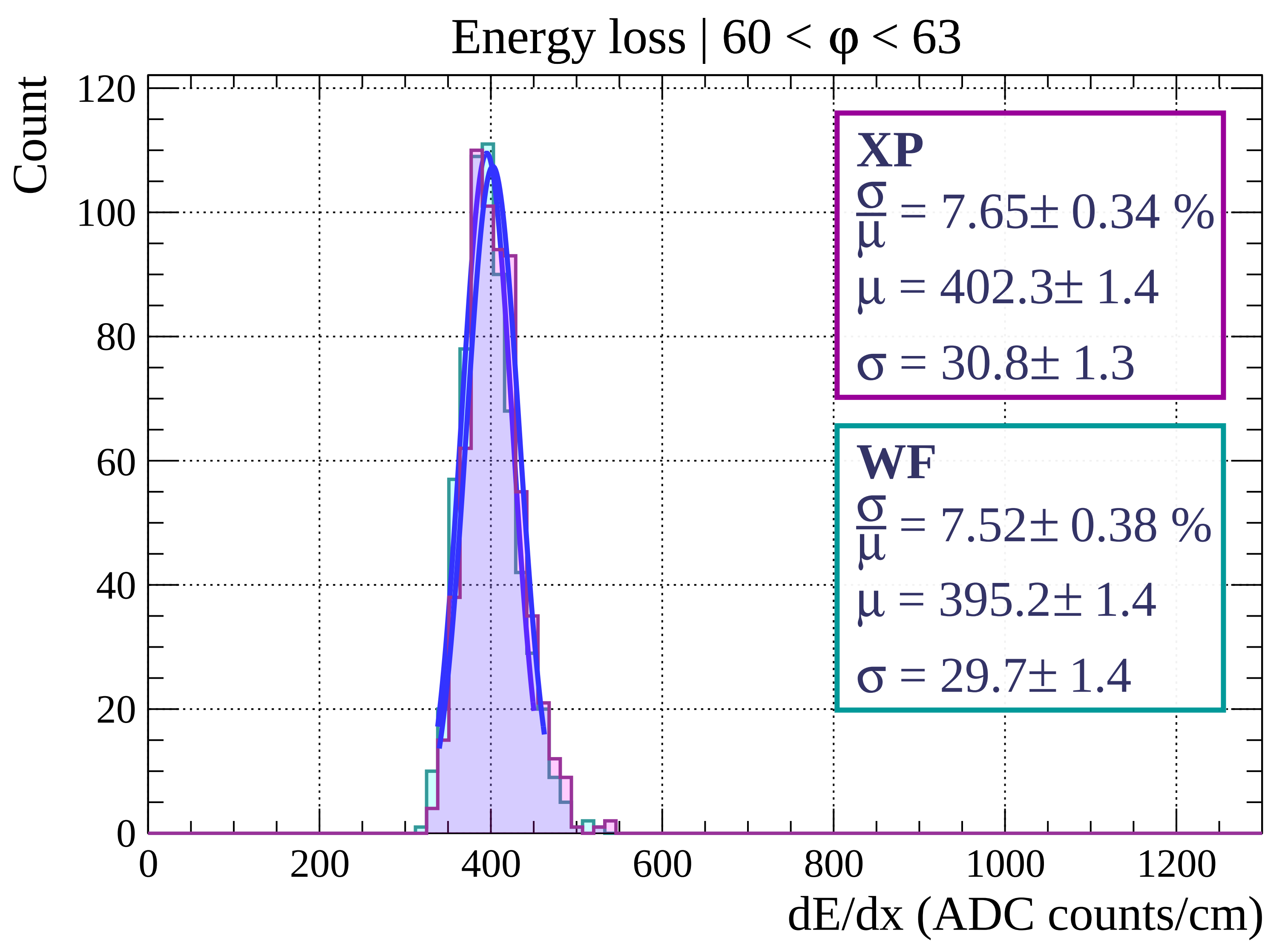
Energy loss | 48 < φ < 51
Count
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.35± 0.28 %
μ = 411.0± 1.1
σ = 34.3± 1.0
WF
σ/μ = 8.35± 0.30 %
μ = 405.3± 1.1
σ = 33.9± 1.1

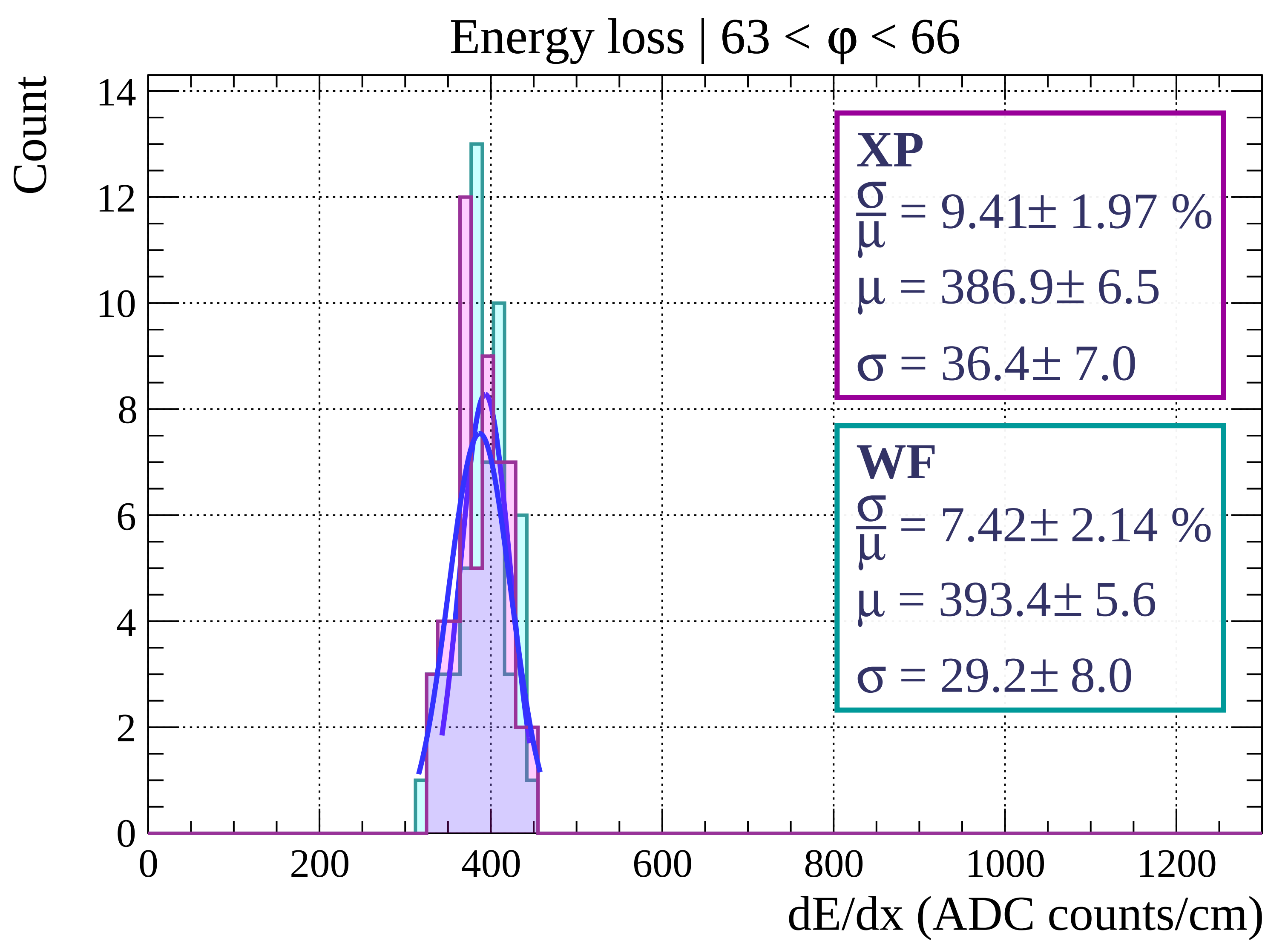

Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 7.82± 0.27 %
μ = 409.5± 0.9
σ = 32.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.52± 0.28 %
μ = 402.5± 1.0
σ = 34.3± 1.0


Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 7.95± 0.27 %
μ = 408.3± 0.9
σ = 32.4± 1.0
WF
σ/μ = 8.23± 0.24 %
μ = 400.8± 0.9
σ = 33.0± 0.9

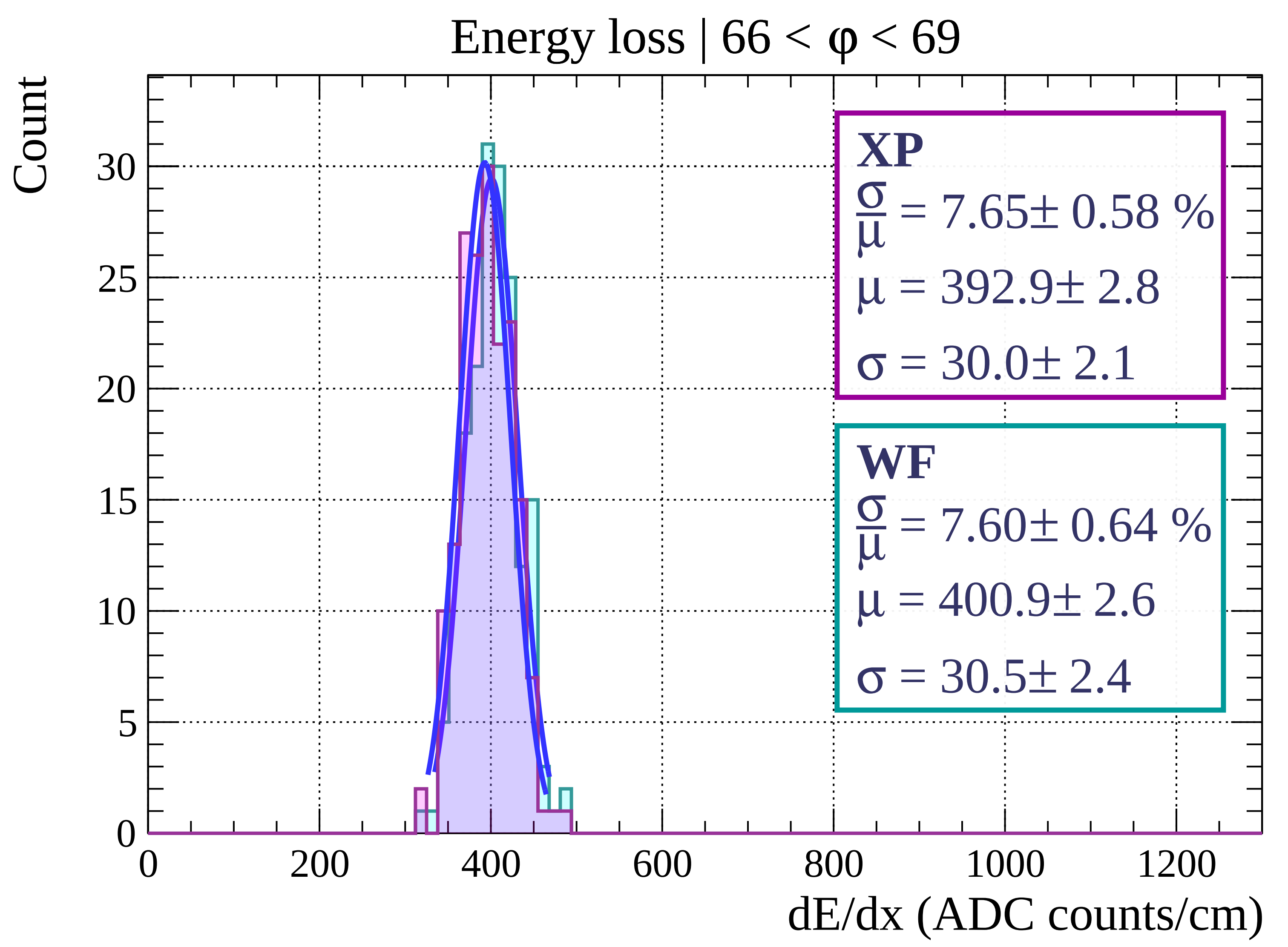


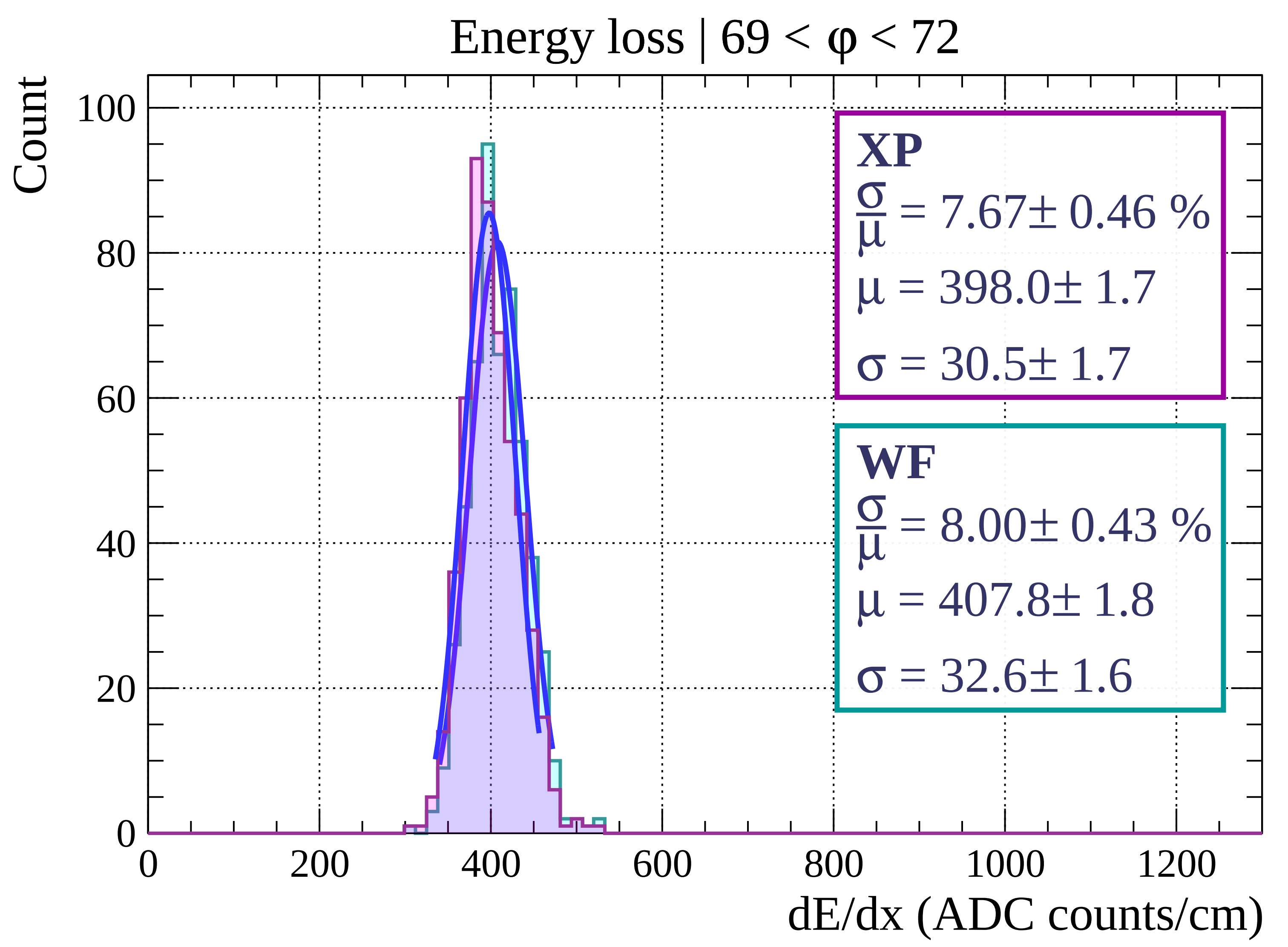
Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 7.92± 0.24 %
μ = 408.5± 1.0
σ = 32.4± 0.9
WF
σ/μ = 8.69± 0.31 %
μ = 400.8± 1.0
σ = 34.8± 1.1


Energy loss | 60 < φ < 63
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.65± 0.34 %
μ = 402.3± 1.4
σ = 30.8± 1.3
WF
σ/μ = 7.52± 0.38 %
μ = 395.2± 1.4
σ = 29.7± 1.4

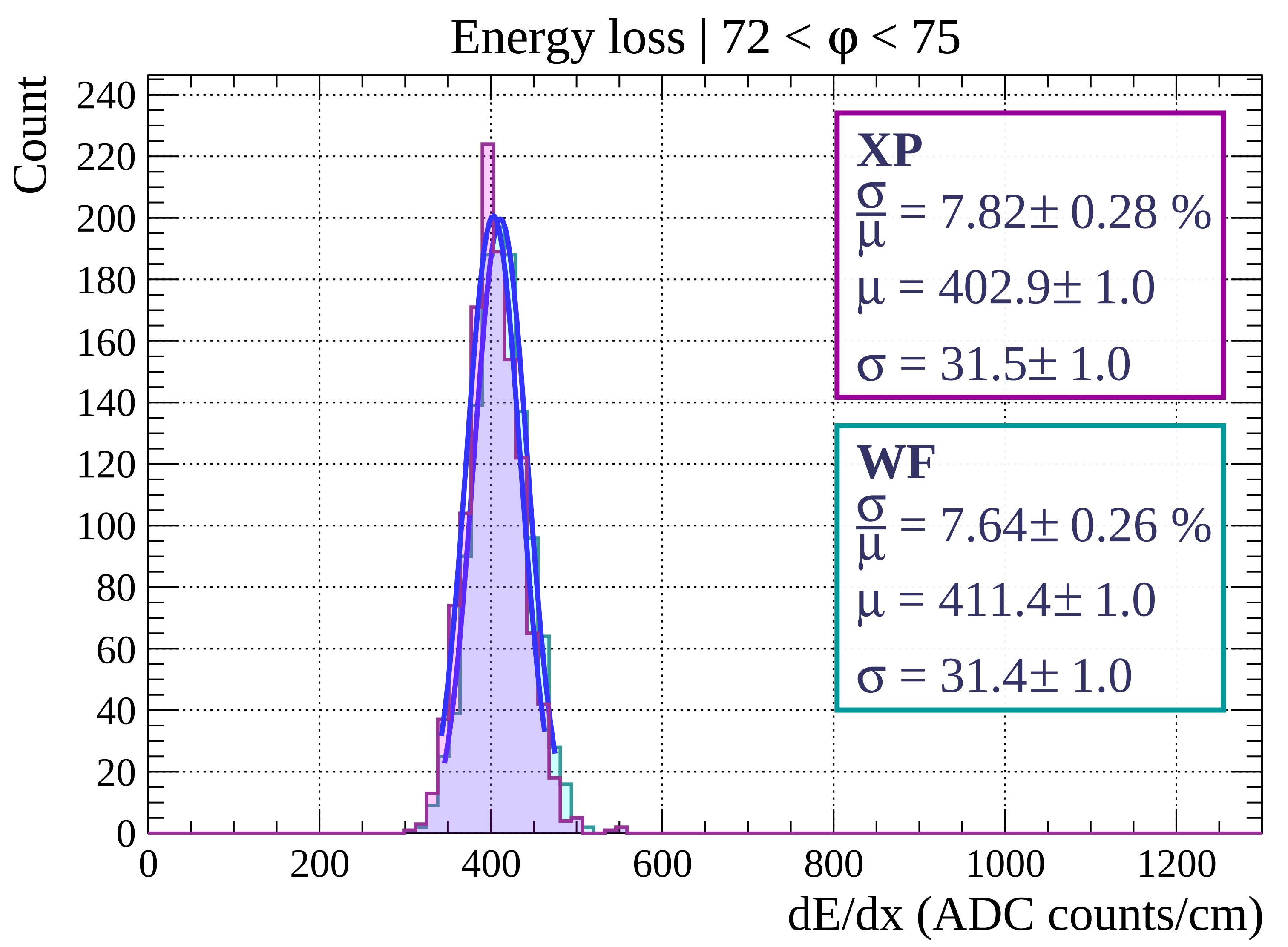


Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 9.41± 1.97 %
μ = 386.9± 6.5
σ = 36.4± 7.0
WF
σ/μ = 7.42± 2.14 %
μ = 393.4± 5.6
σ = 29.2± 8.0

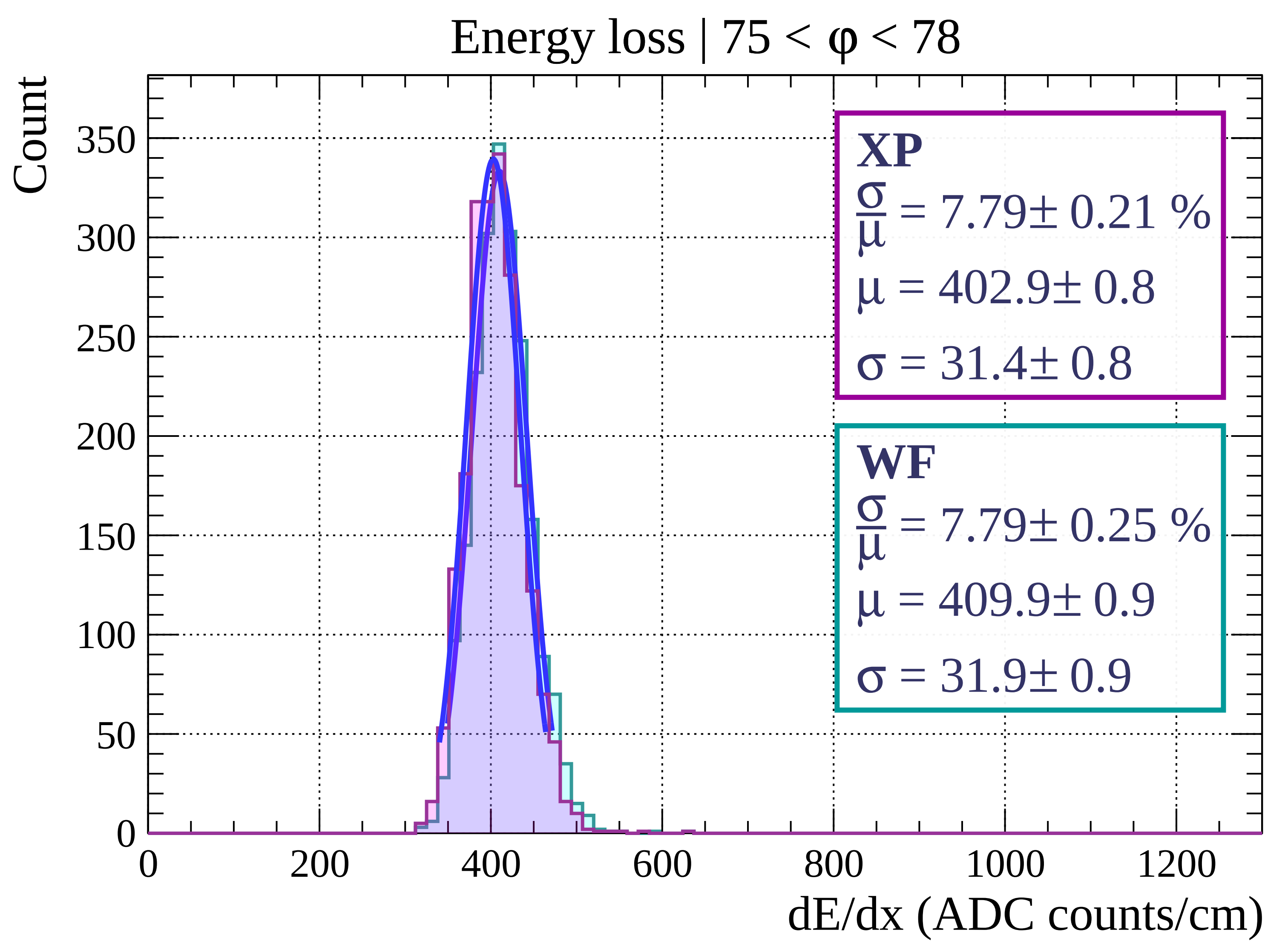

Energy loss | 66 < φ < 69
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.65± 0.58 %
μ = 392.9± 2.8
σ = 30.0± 2.1
WF
σ/μ = 7.60± 0.64 %
μ = 400.9± 2.6
σ = 30.5± 2.4

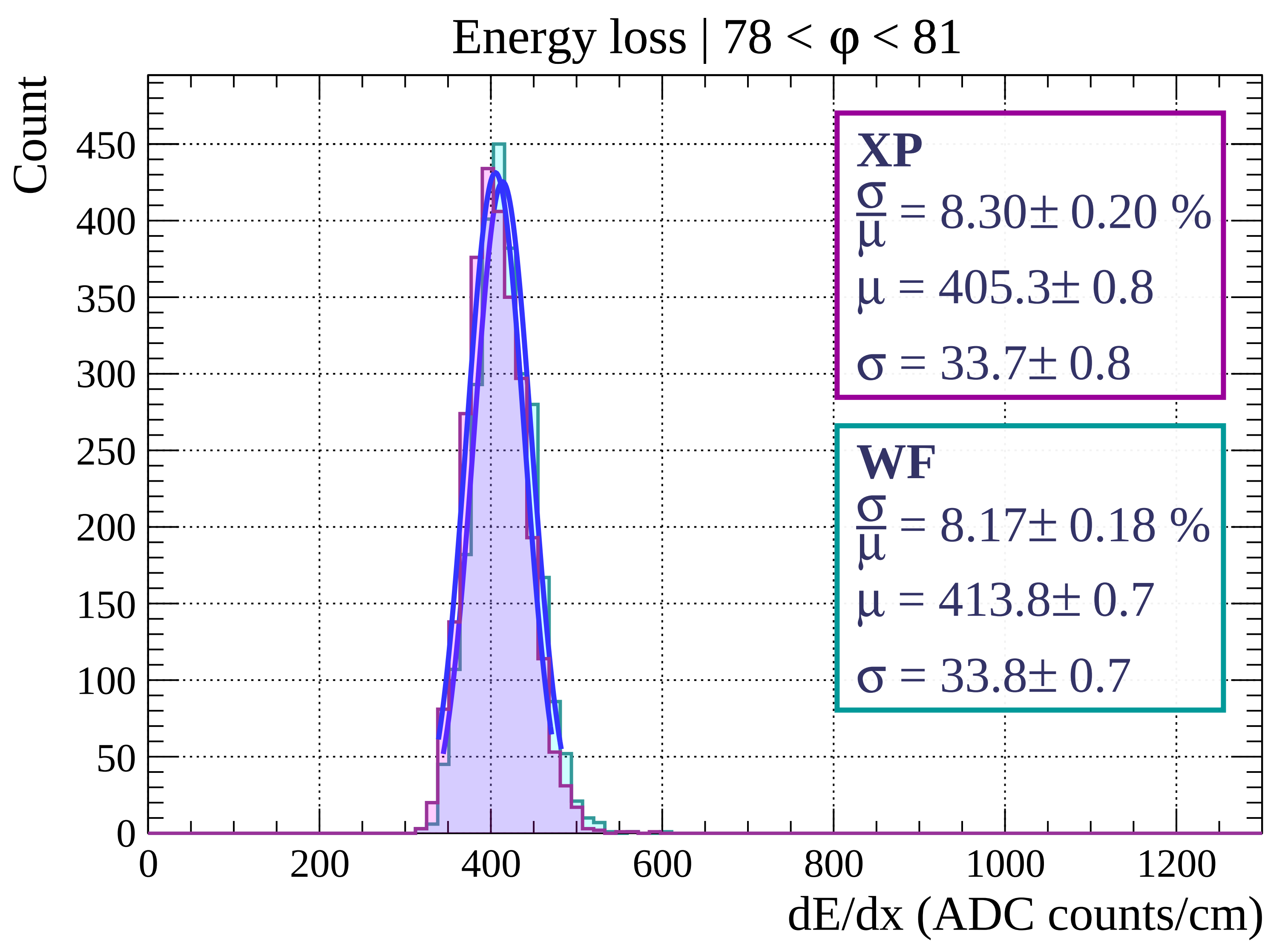


Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.67± 0.46 %
μ = 398.0± 1.7
σ = 30.5± 1.7
WF
σ/μ = 8.00± 0.43 %
μ = 407.8± 1.8
σ = 32.6± 1.6

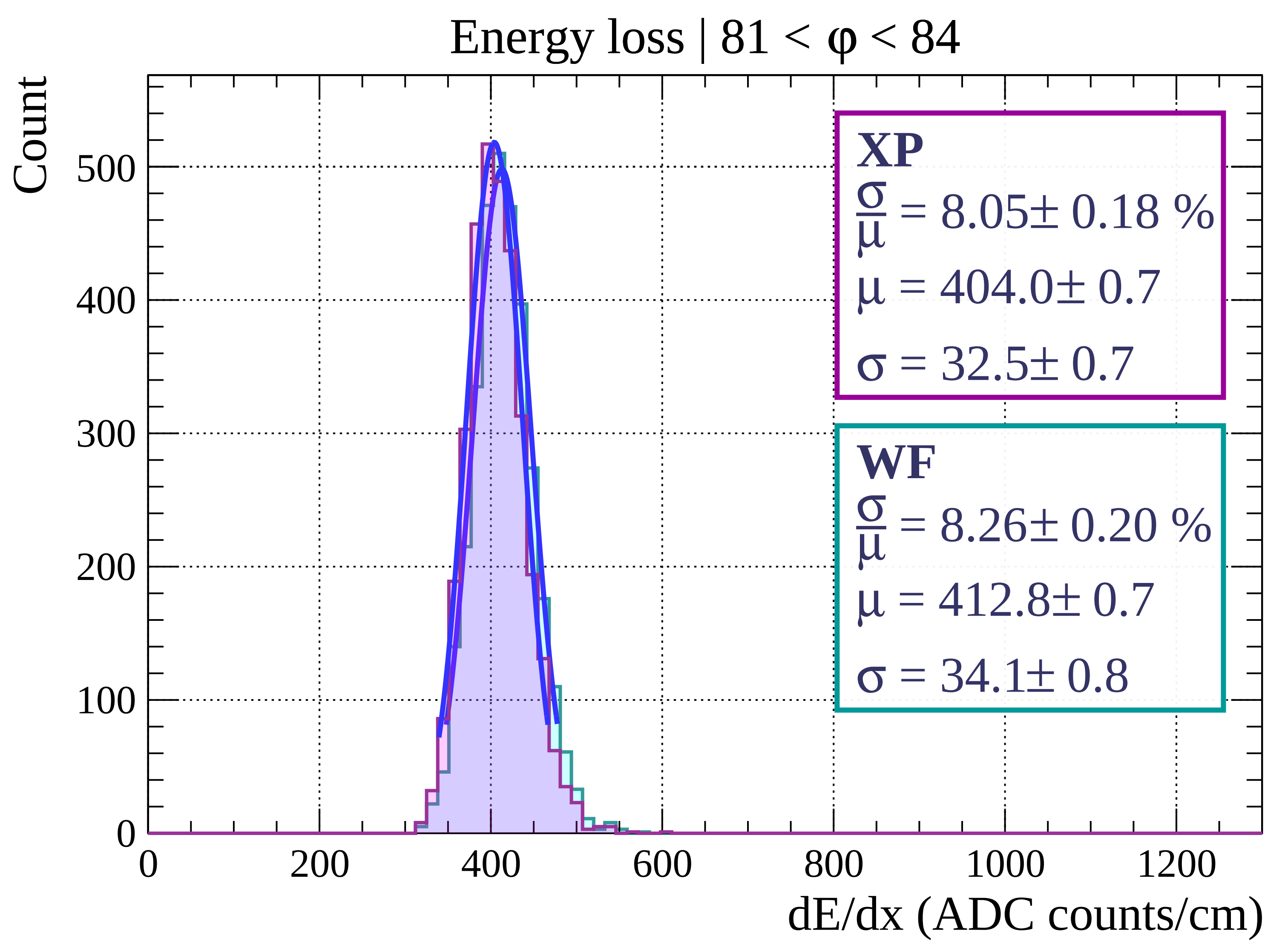


Energy loss | 72 < φ < 75
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
200
400
600
800
1000
1200
XP
σ/μ = 7.82± 0.28 %
μ = 402.9± 1.0
σ = 31.5± 1.0
WF
σ/μ = 7.64± 0.26 %
μ = 411.4± 1.0
σ = 31.4± 1.0

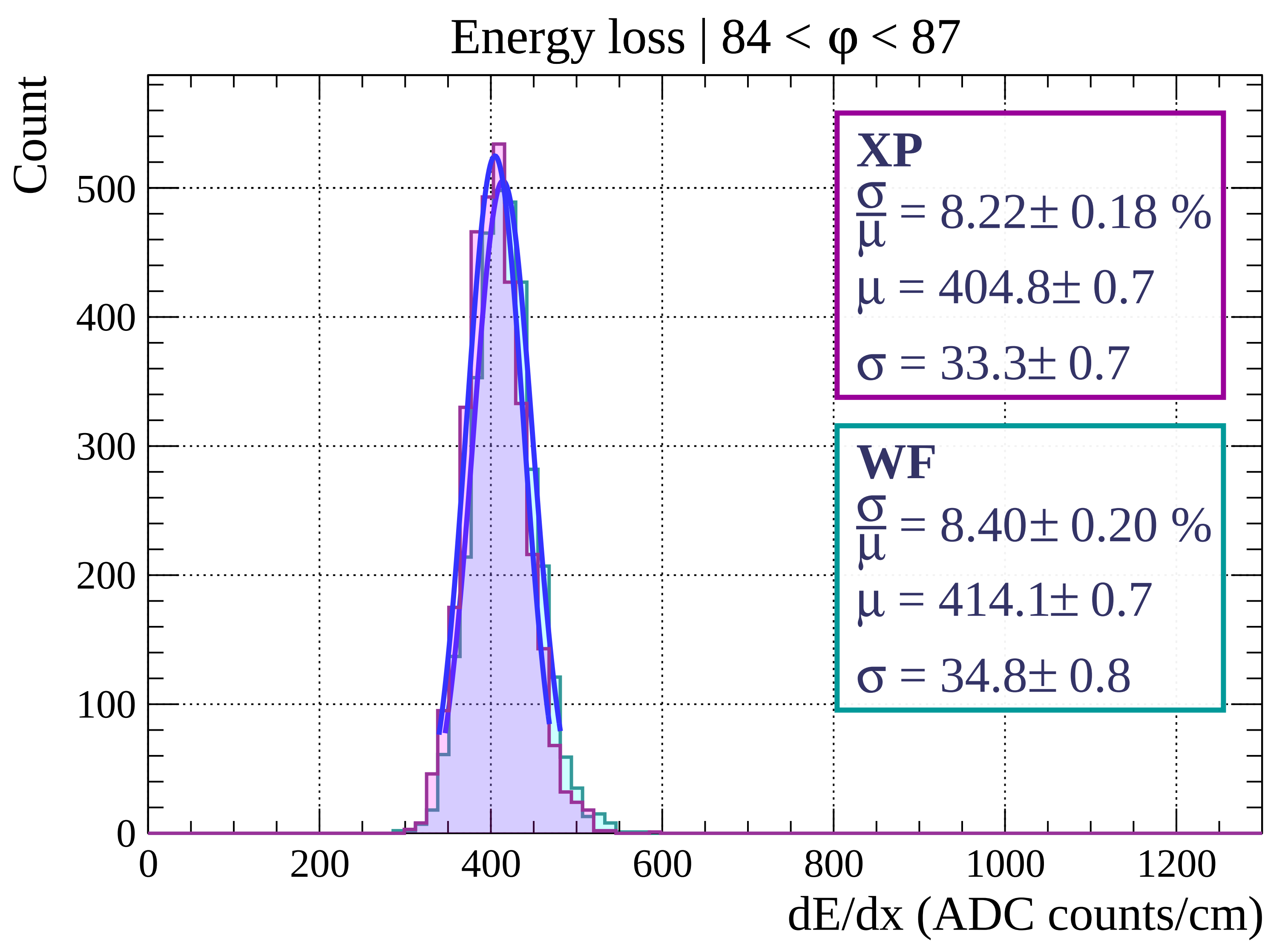

Energy loss | 75 < φ < 78
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.79± 0.21 %
μ = 402.9± 0.8
σ = 31.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.79± 0.25 %
μ = 409.9± 0.9
σ = 31.9± 0.9



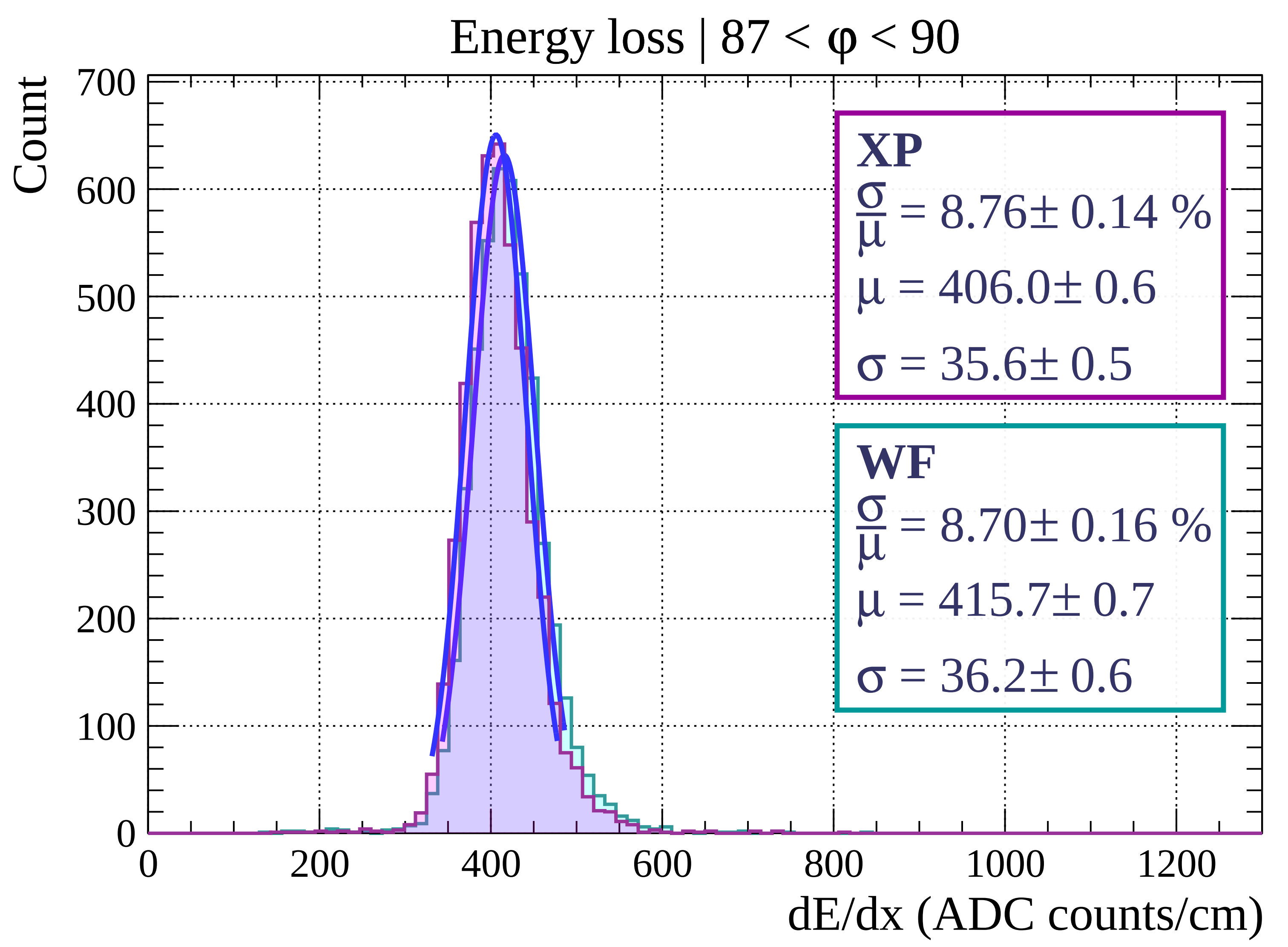
Energy loss | 78 < φ < 81
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.30± 0.20 %
μ = 405.3± 0.8
σ = 33.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.17± 0.18 %
μ = 413.8± 0.7
σ = 33.8± 0.7

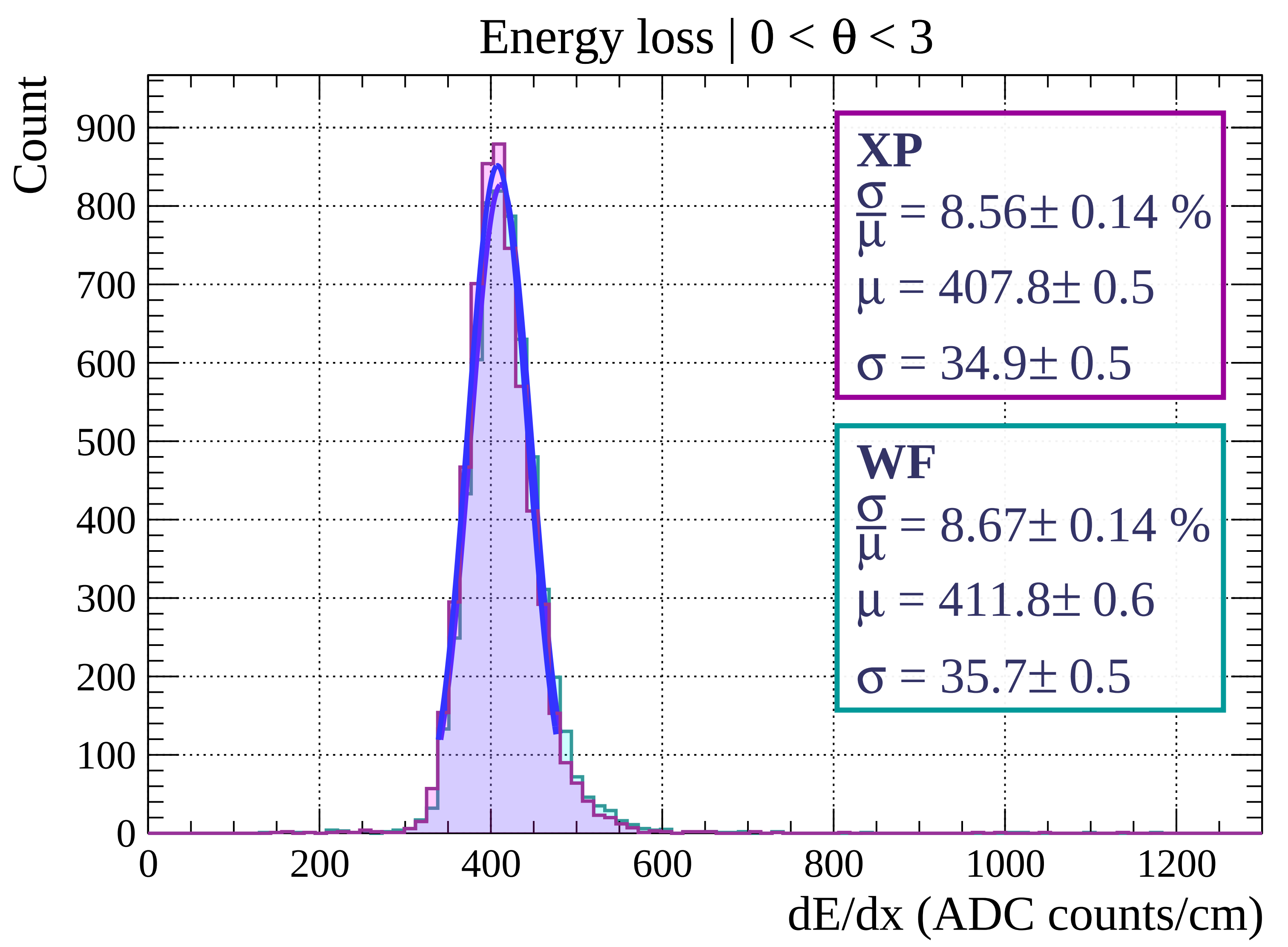


Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.05± 0.18 %
μ = 404.0± 0.7
σ = 32.5± 0.7
WF
σ/μ = 8.26± 0.20 %
μ = 412.8± 0.7
σ = 34.1± 0.8

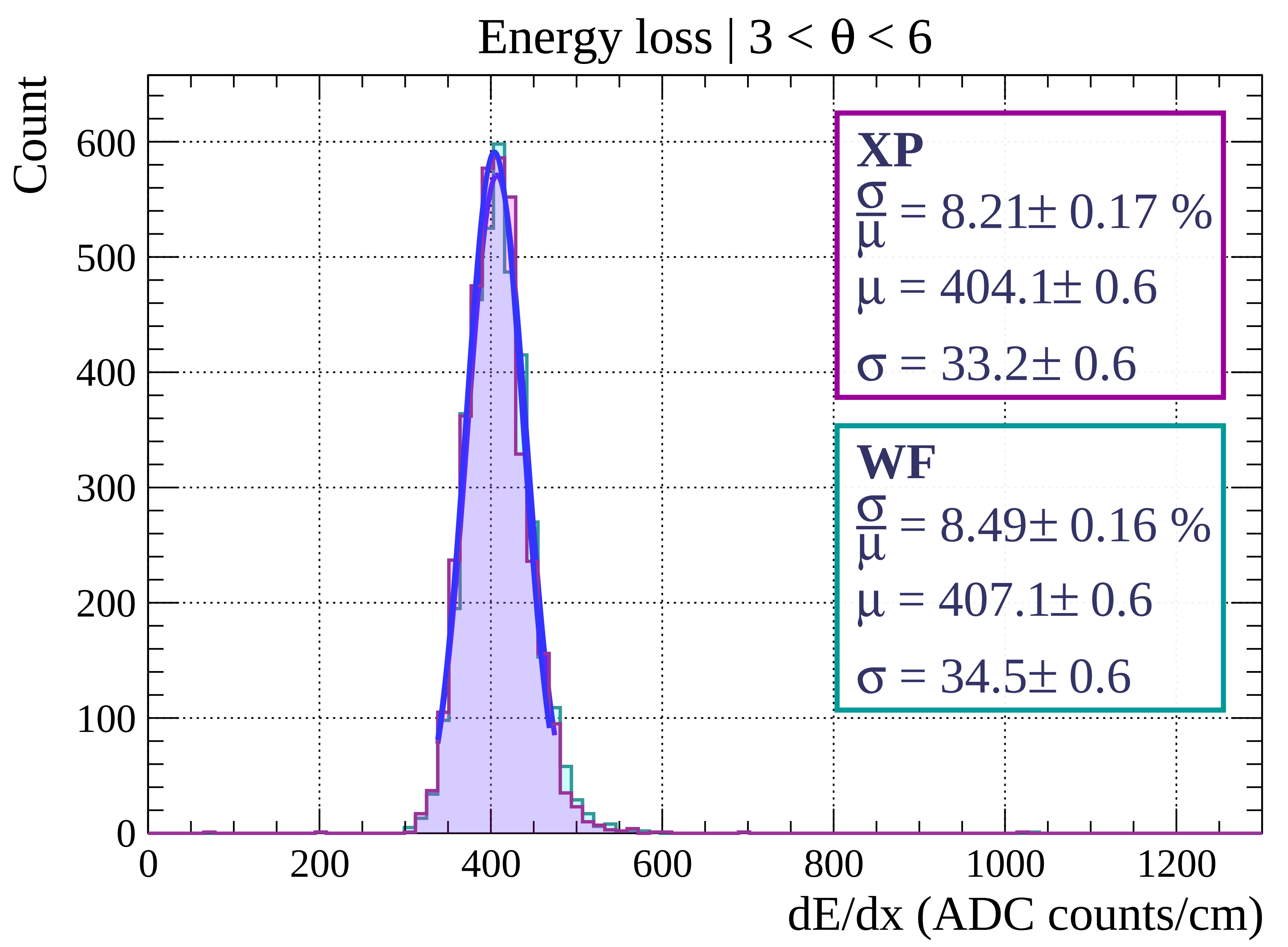


Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.22± 0.18 %
μ = 404.8± 0.7
σ = 33.3± 0.7
WF
σ/μ = 8.40± 0.20 %
μ = 414.1± 0.7
σ = 34.8± 0.8

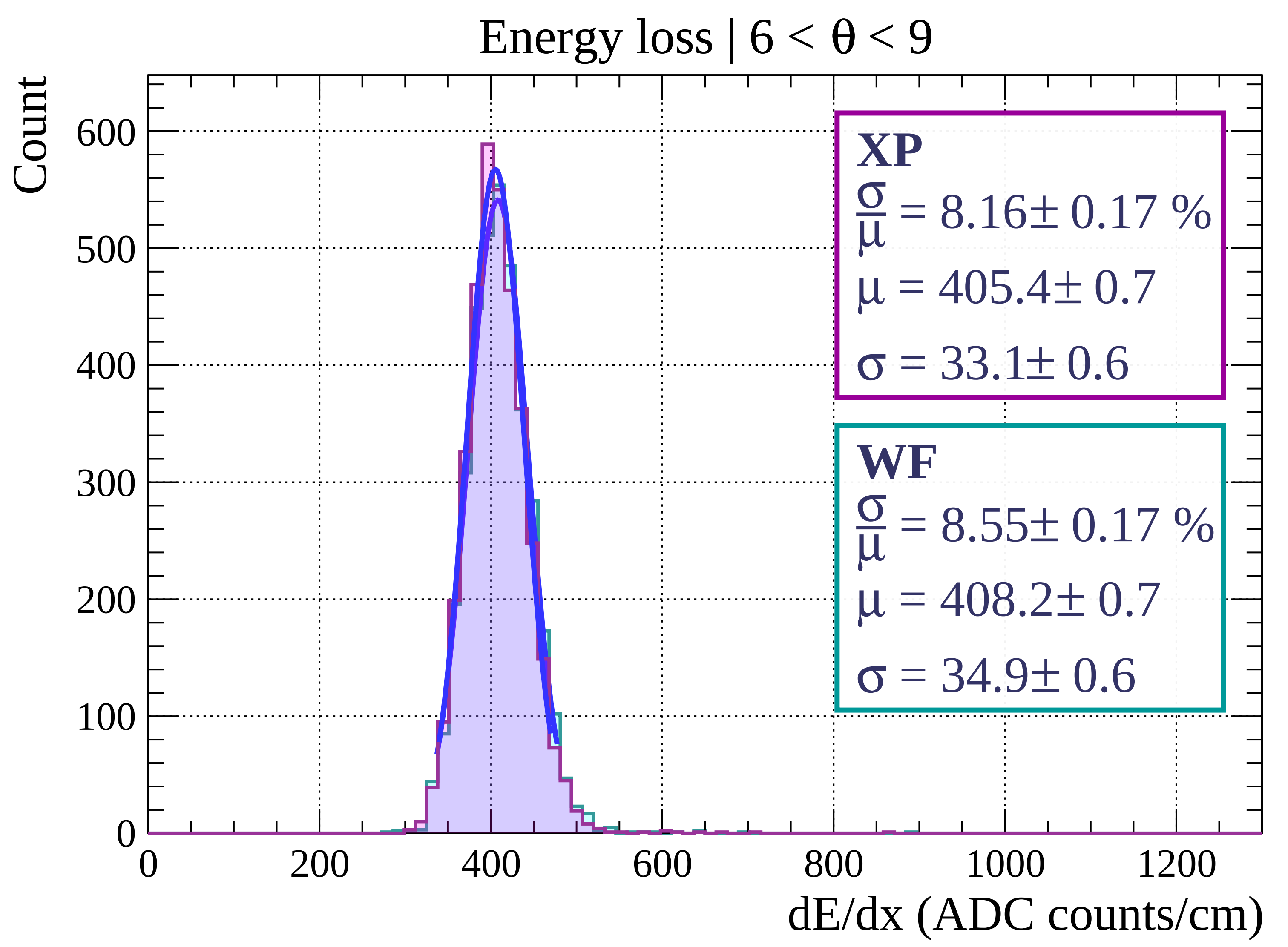

Energy loss | 87 < φ < 90
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.76± 0.14 %
μ = 406.0± 0.6
σ = 35.6± 0.5
WF
σ/μ = 8.70± 0.16 %
μ = 415.7± 0.7
σ = 36.2± 0.6



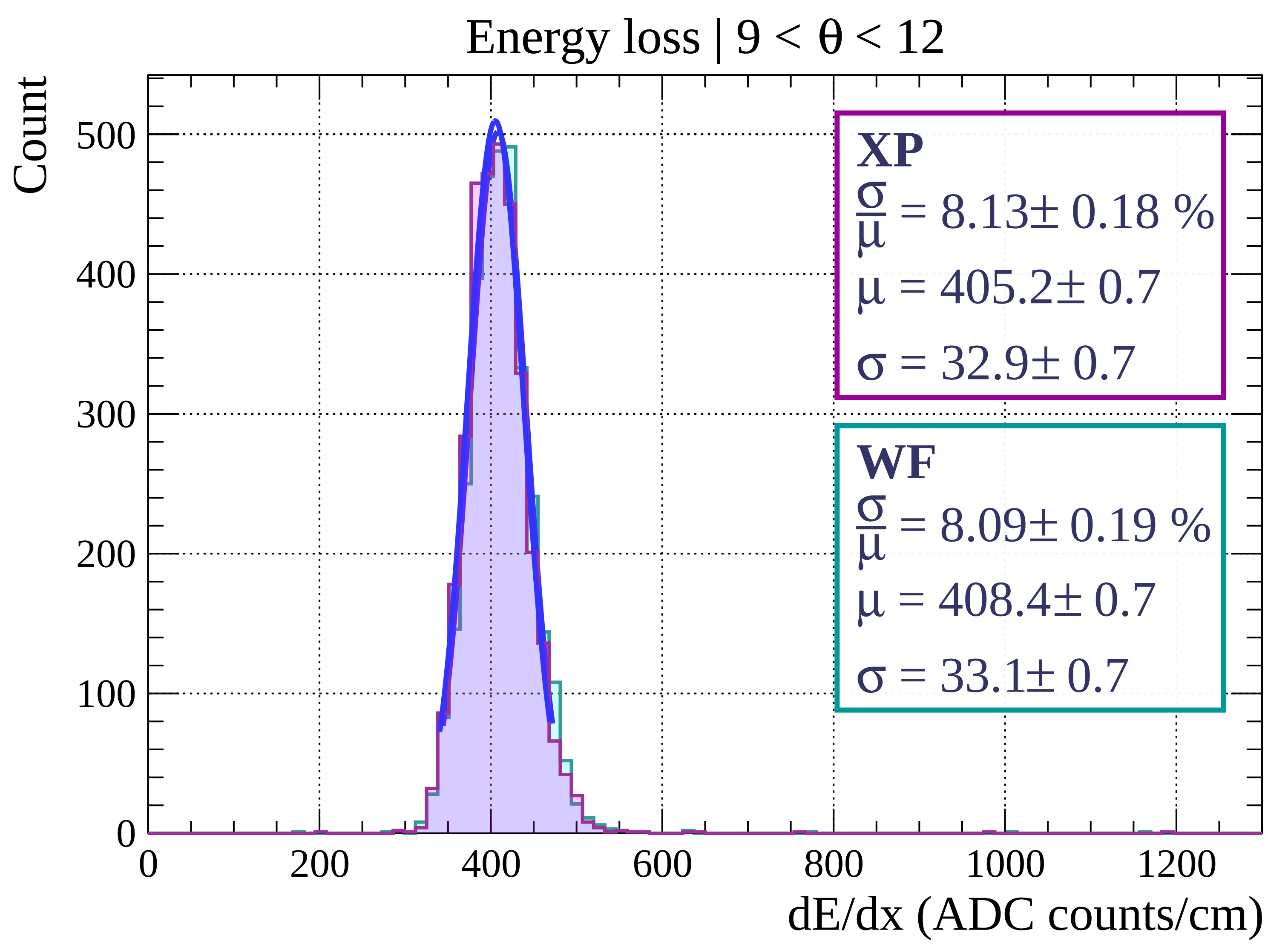
Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.56± 0.14 %
μ = 407.8± 0.5
σ = 34.9± 0.5
WF
σ/μ = 8.67± 0.14 %
μ = 411.8± 0.6
σ = 35.7± 0.5
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200



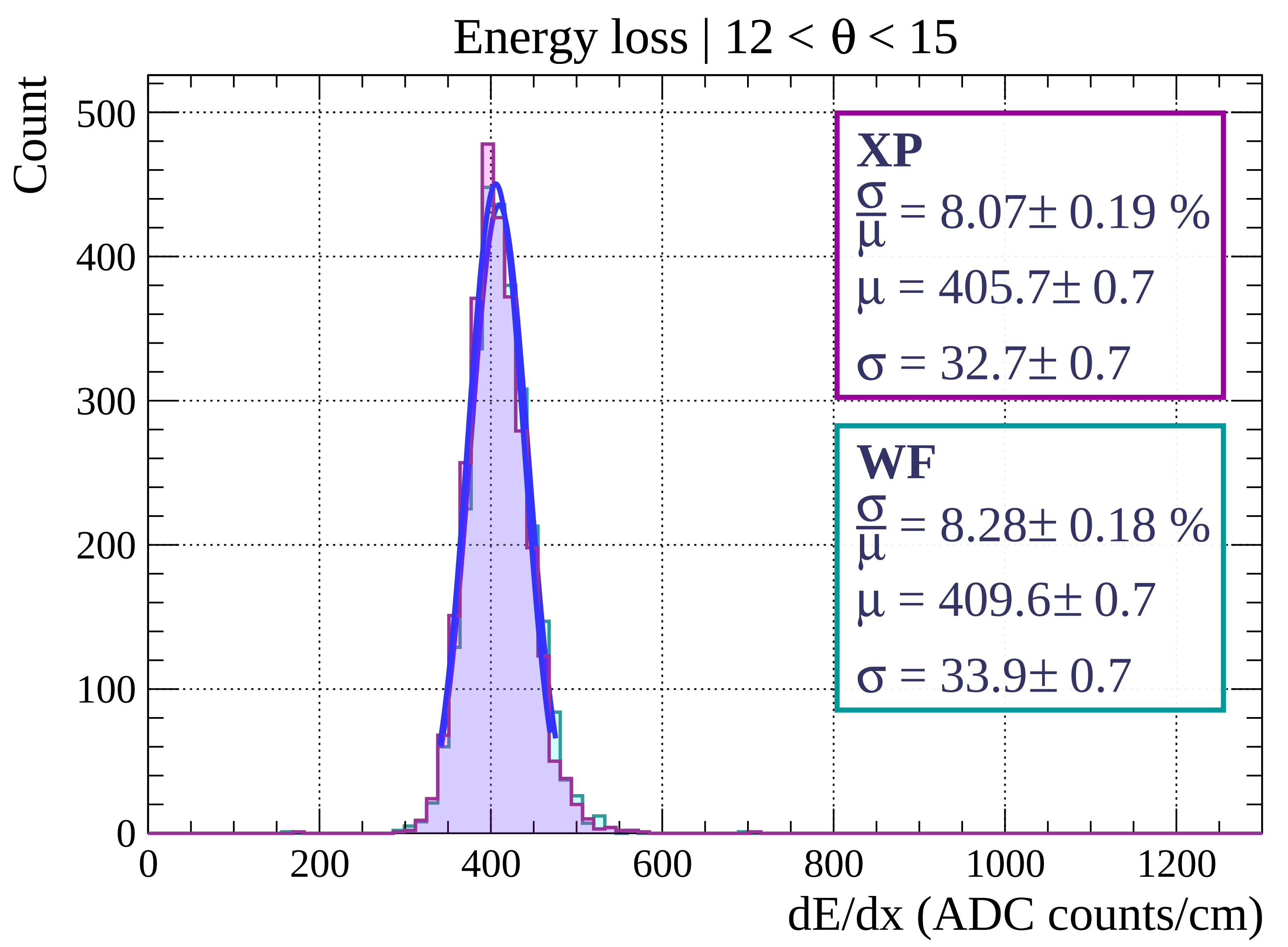
Energy loss | 3 < θ < 6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 8.21± 0.17 %
μ = 404.1± 0.6
σ = 33.2± 0.6
WF
σ/μ = 8.49± 0.16 %
μ = 407.1± 0.6
σ = 34.5± 0.6



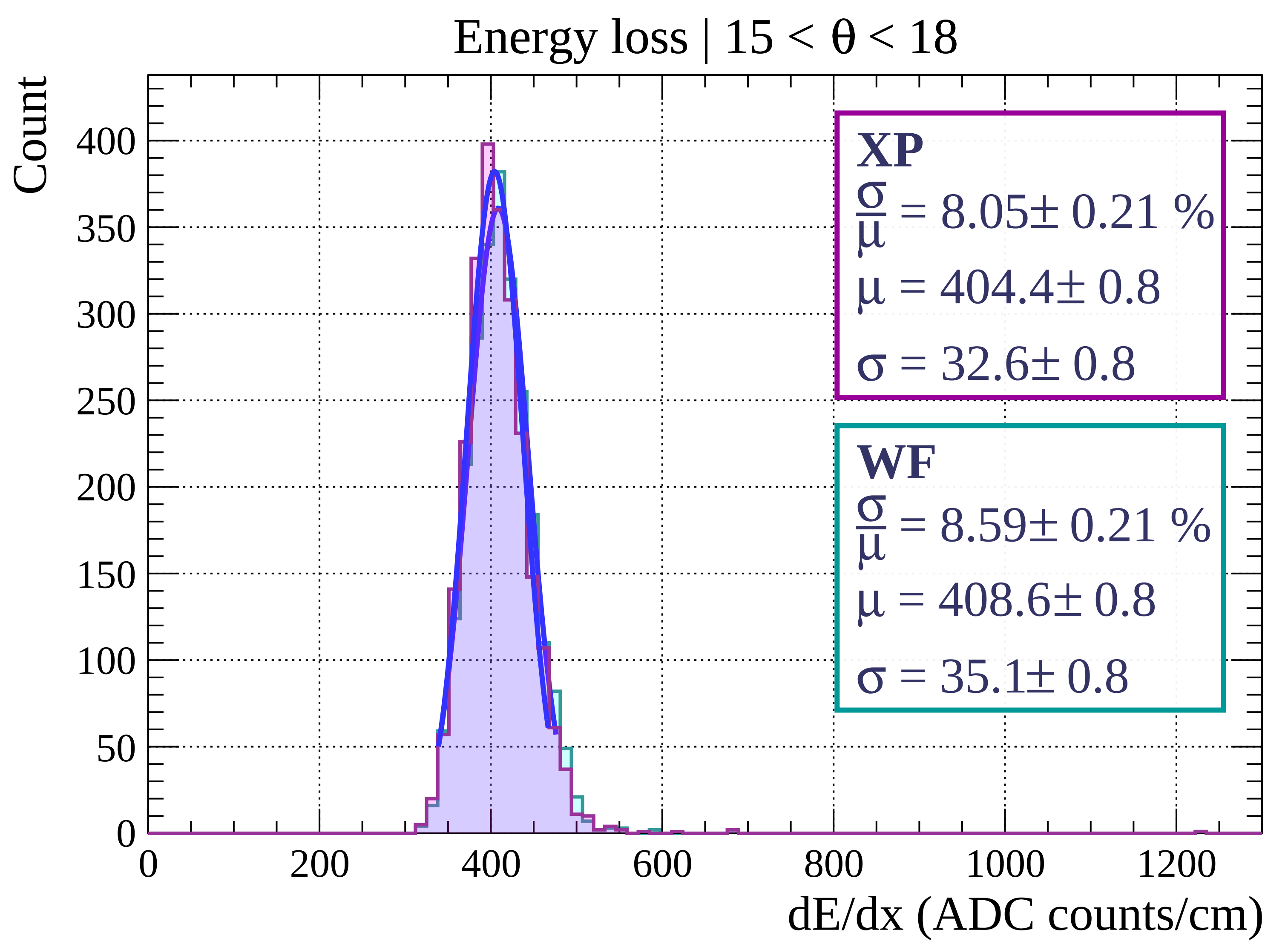
Energy loss | 6 < θ < 9
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 8.16± 0.17 %
μ = 405.4± 0.7
σ = 33.1± 0.6
WF
σ/μ = 8.55± 0.17 %
μ = 408.2± 0.7
σ = 34.9± 0.6



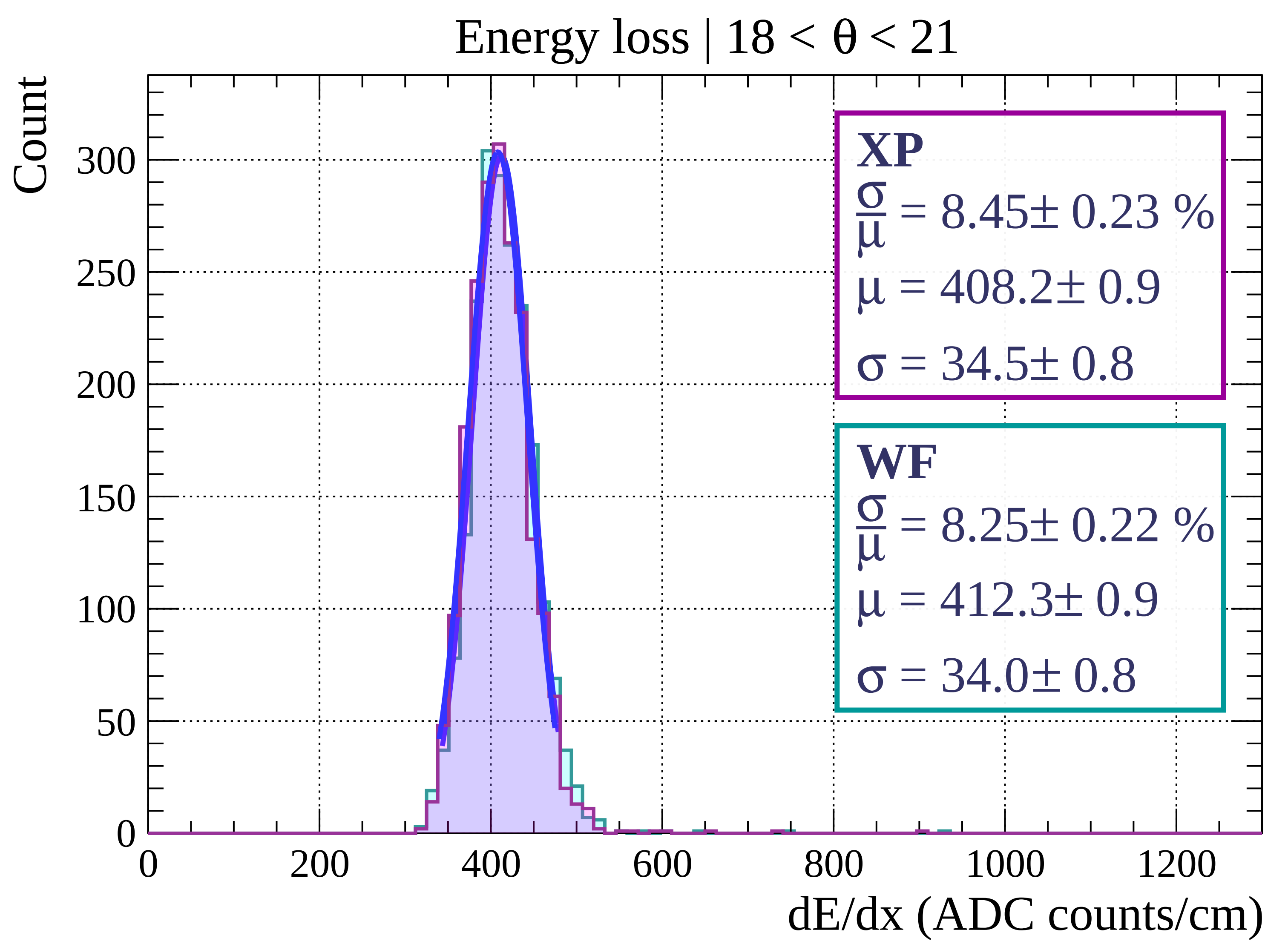
Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.13± 0.18 %
μ = 405.2± 0.7
σ = 32.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.09± 0.19 %
μ = 408.4± 0.7
σ = 33.1± 0.7



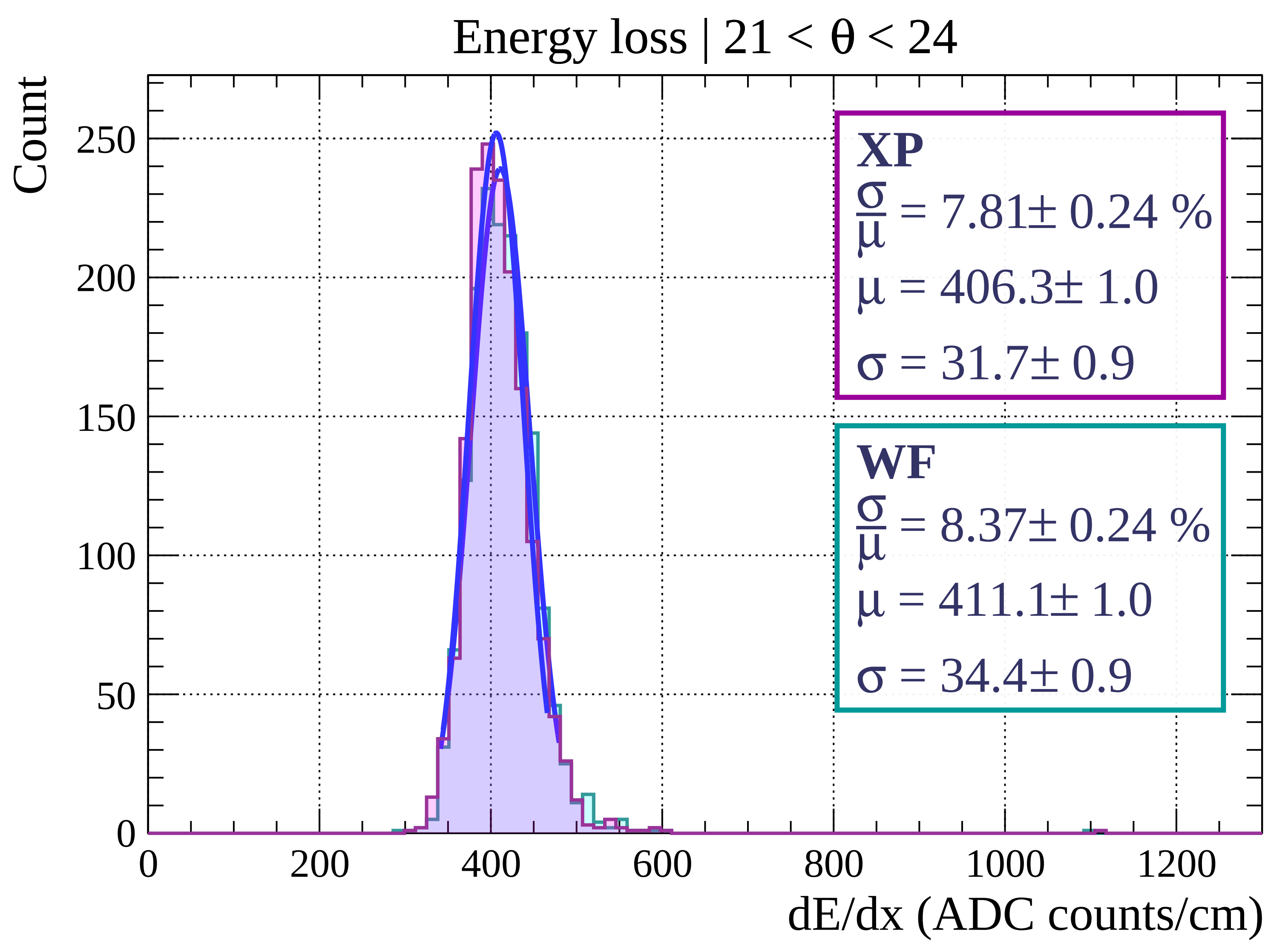
Energy loss | 12 < θ < 15
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.07± 0.19 %
μ = 405.7± 0.7
σ = 32.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.28± 0.18 %
μ = 409.6± 0.7
σ = 33.9± 0.7

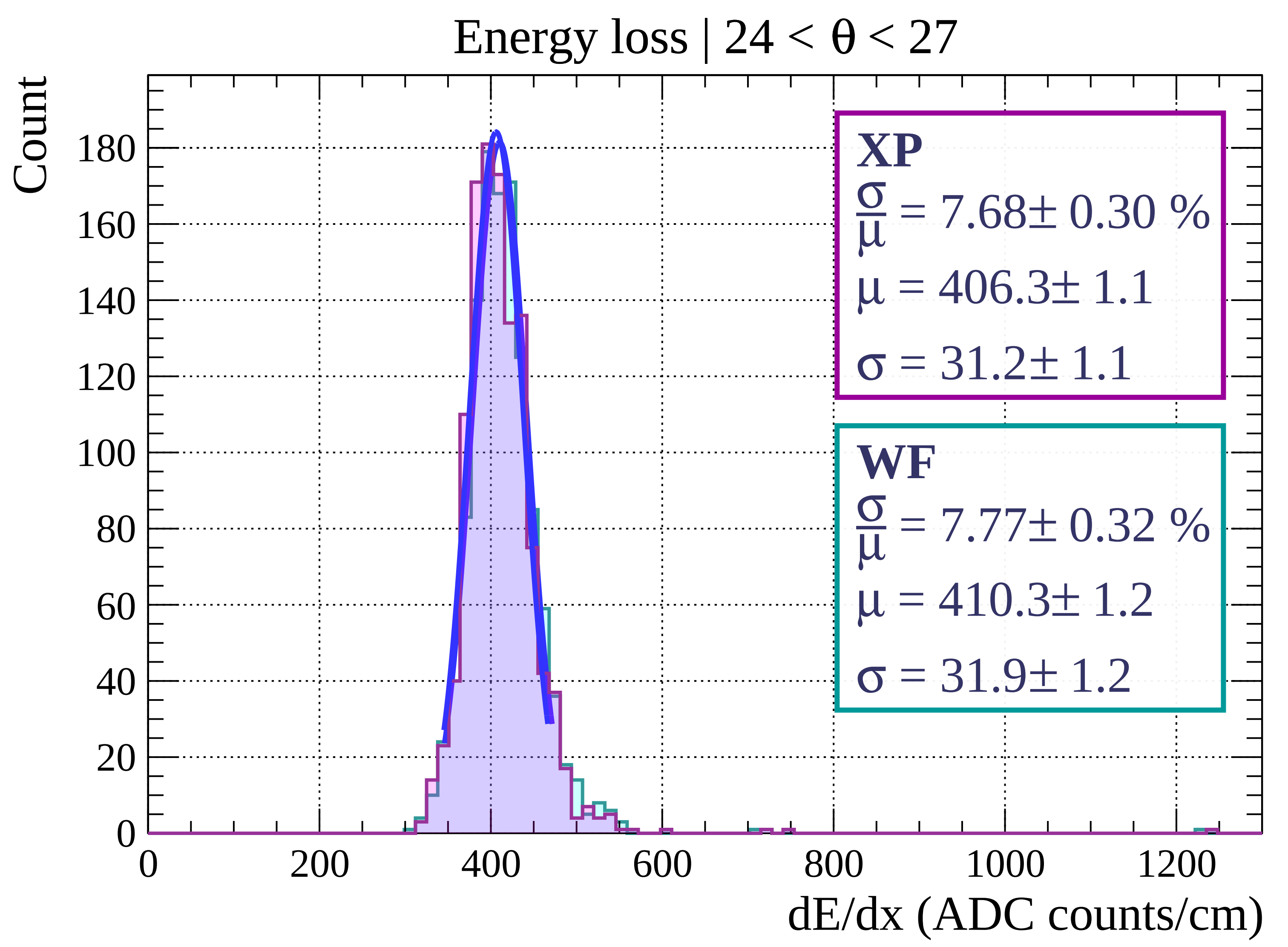


Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
σ/μ = 8.05± 0.21 %
μ = 404.4± 0.8
σ = 32.6± 0.8
WF
σ/μ = 8.59± 0.21 %
μ = 408.6± 0.8
σ = 35.1± 0.8

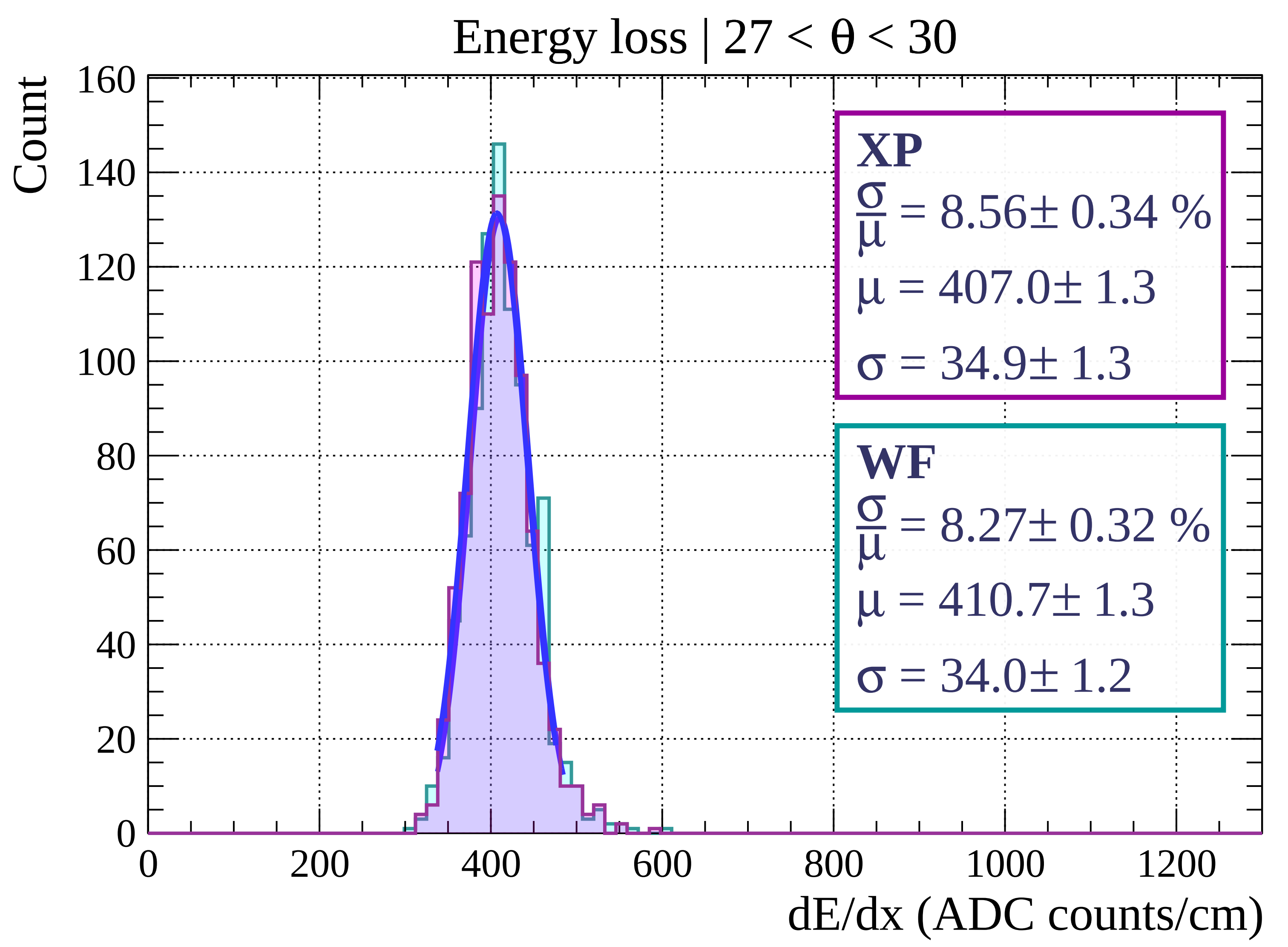


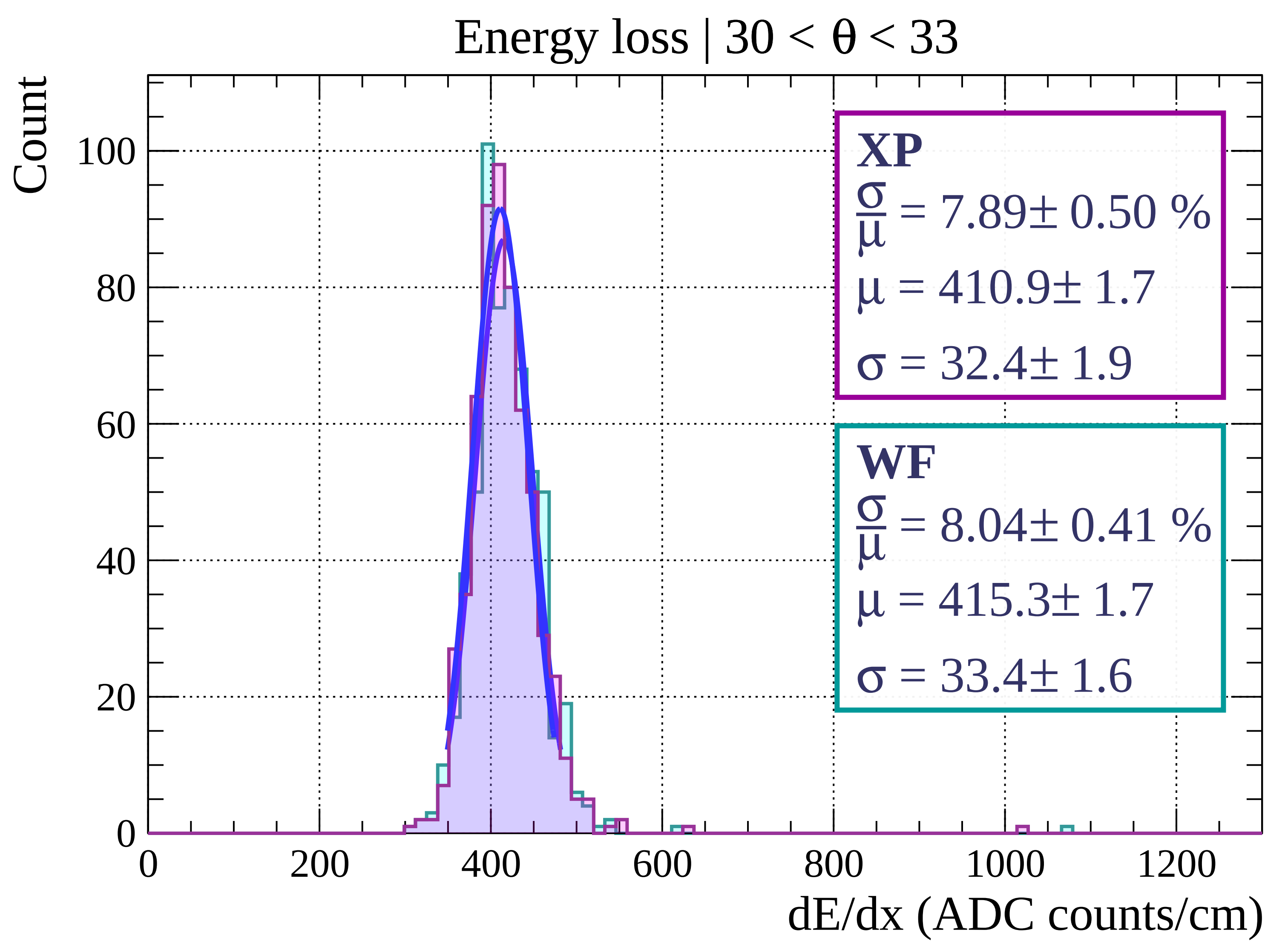
Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
200
400
600
800
1000
1200
XP
σ/μ = 8.45± 0.23 %
μ = 408.2± 0.9
σ = 34.5± 0.8
WF
σ/μ = 8.25± 0.22 %
μ = 412.3± 0.9
σ = 34.0± 0.8


Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 7.81± 0.24 %
μ = 406.3± 1.0
σ = 31.7± 0.9
WF
σ/μ = 8.37± 0.24 %
μ = 411.1± 1.0
σ = 34.4± 0.9

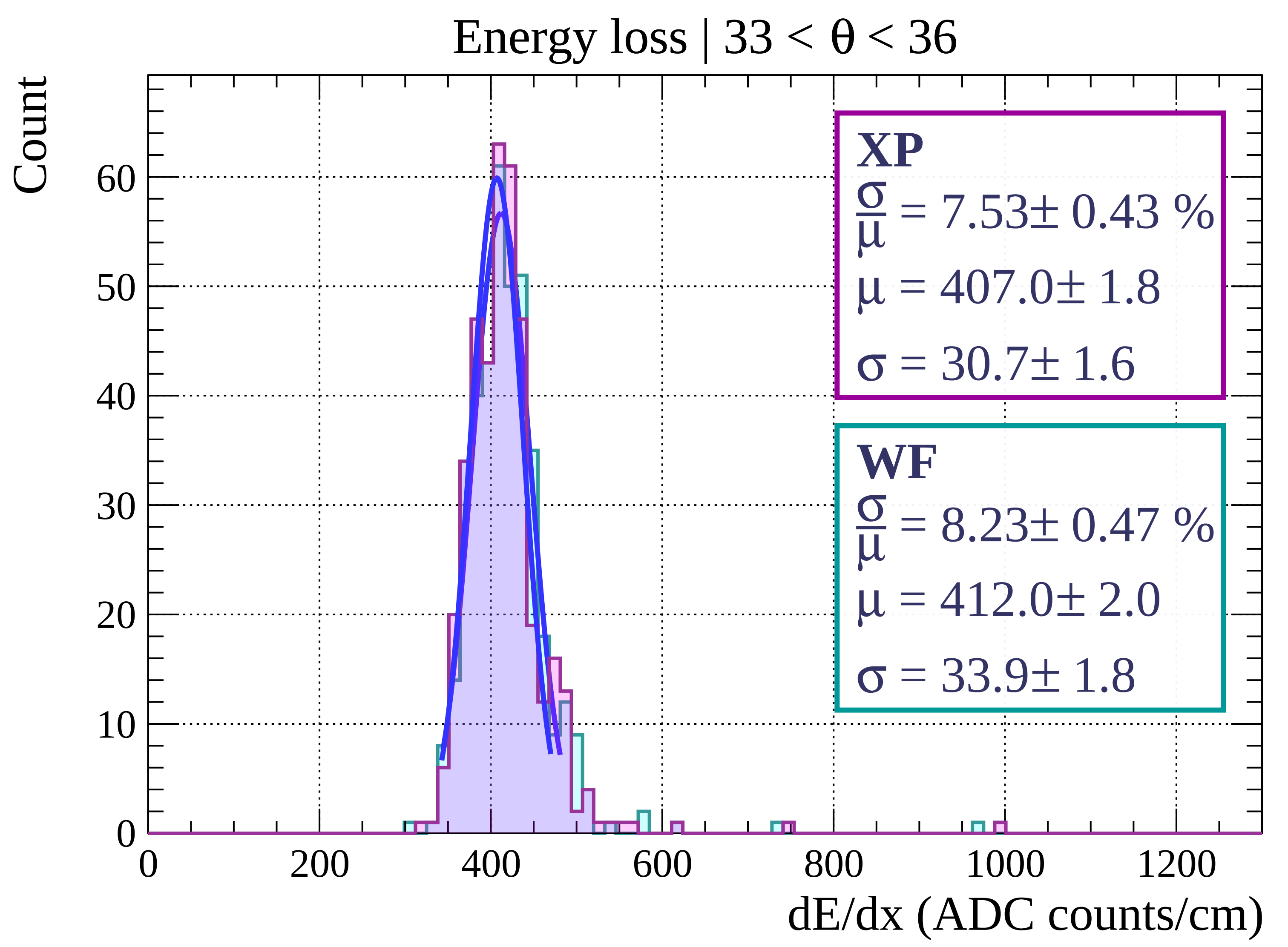


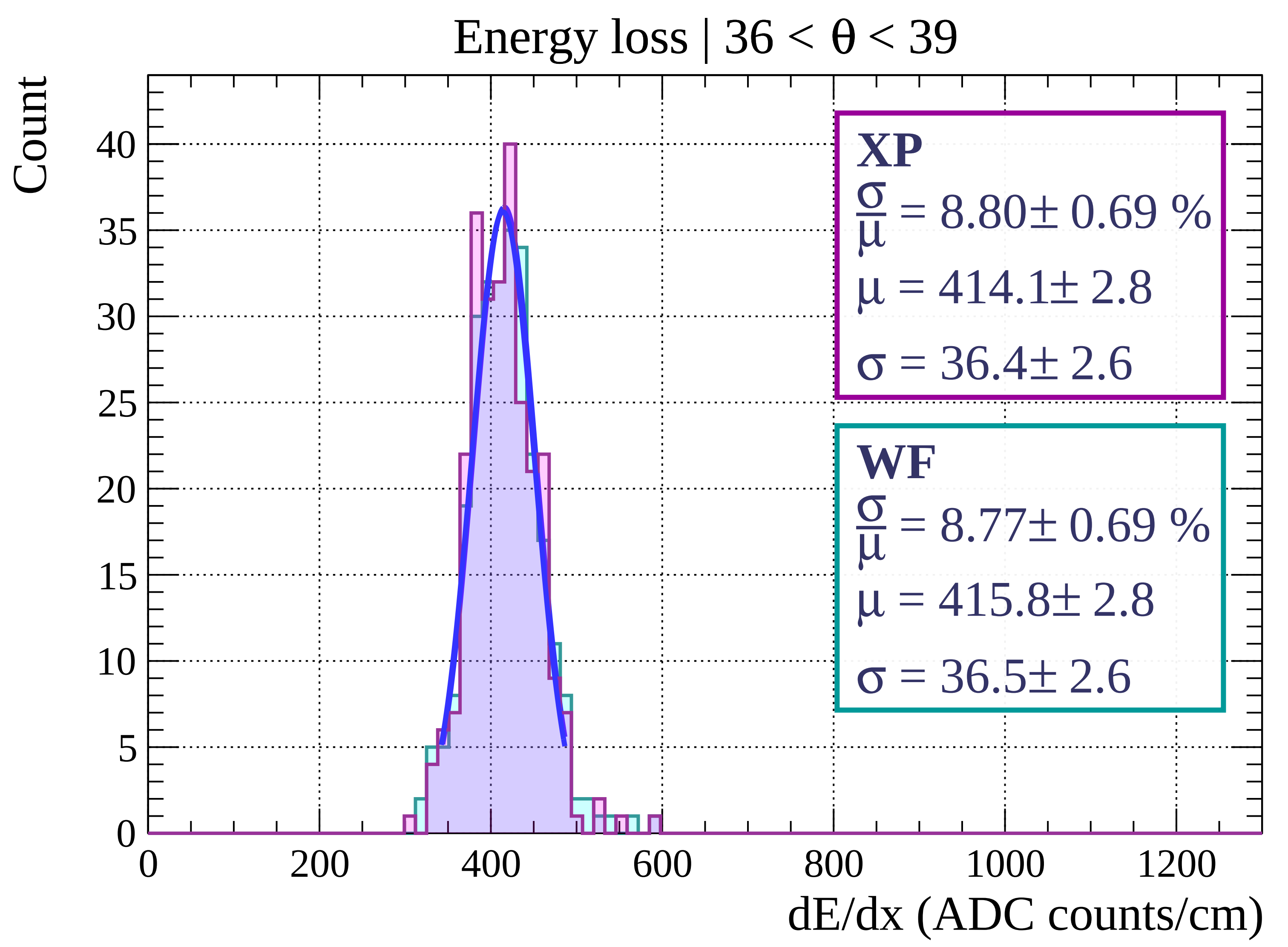
Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 7.68± 0.30 %
μ = 406.3± 1.1
σ = 31.2± 1.1
WF
σ/μ = 7.77± 0.32 %
μ = 410.3± 1.2
σ = 31.9± 1.2


Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 8.56± 0.34 %
μ = 407.0± 1.3
σ = 34.9± 1.3
WF
σ/μ = 8.27± 0.32 %
μ = 410.7± 1.3
σ = 34.0± 1.2

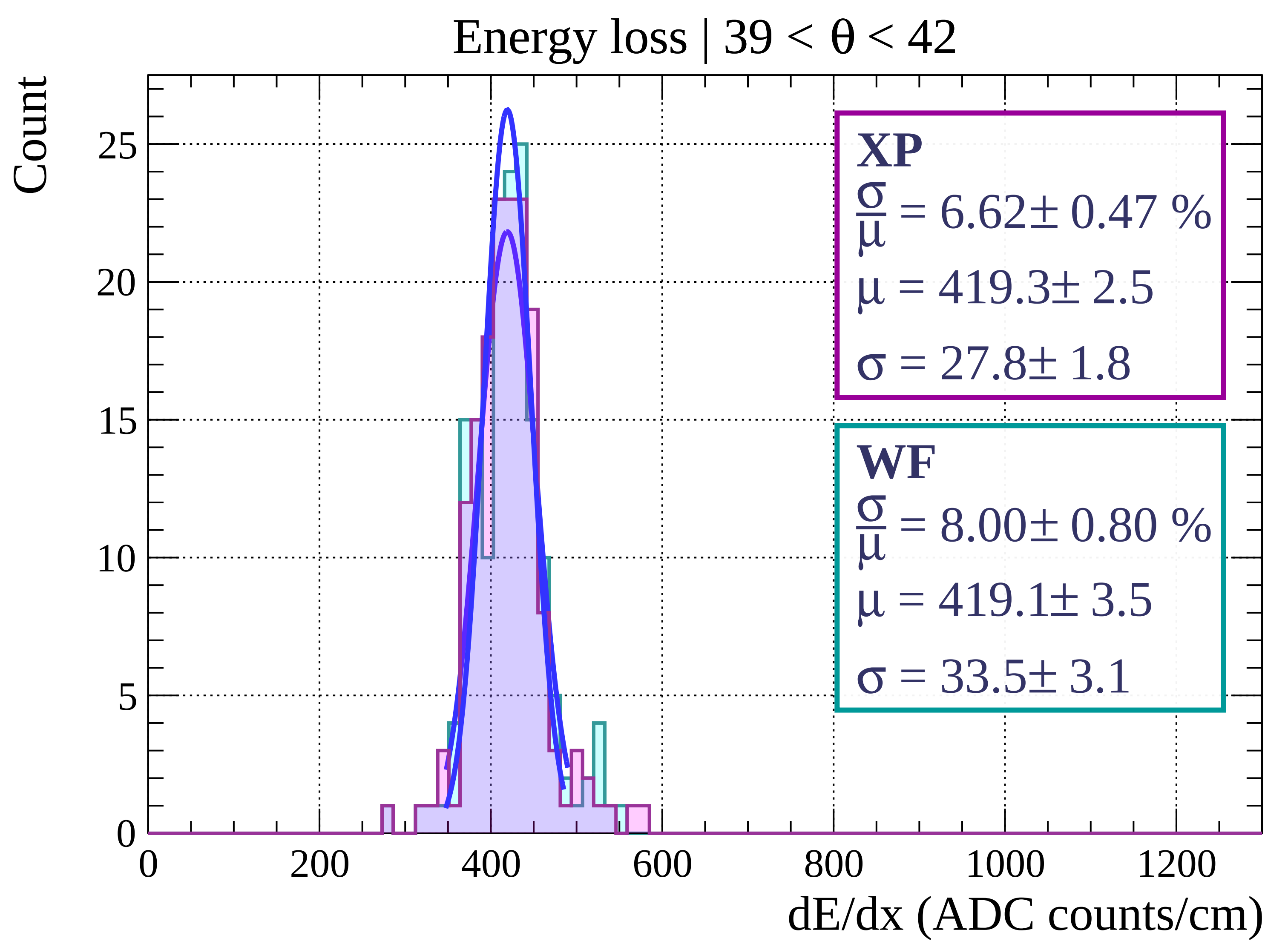


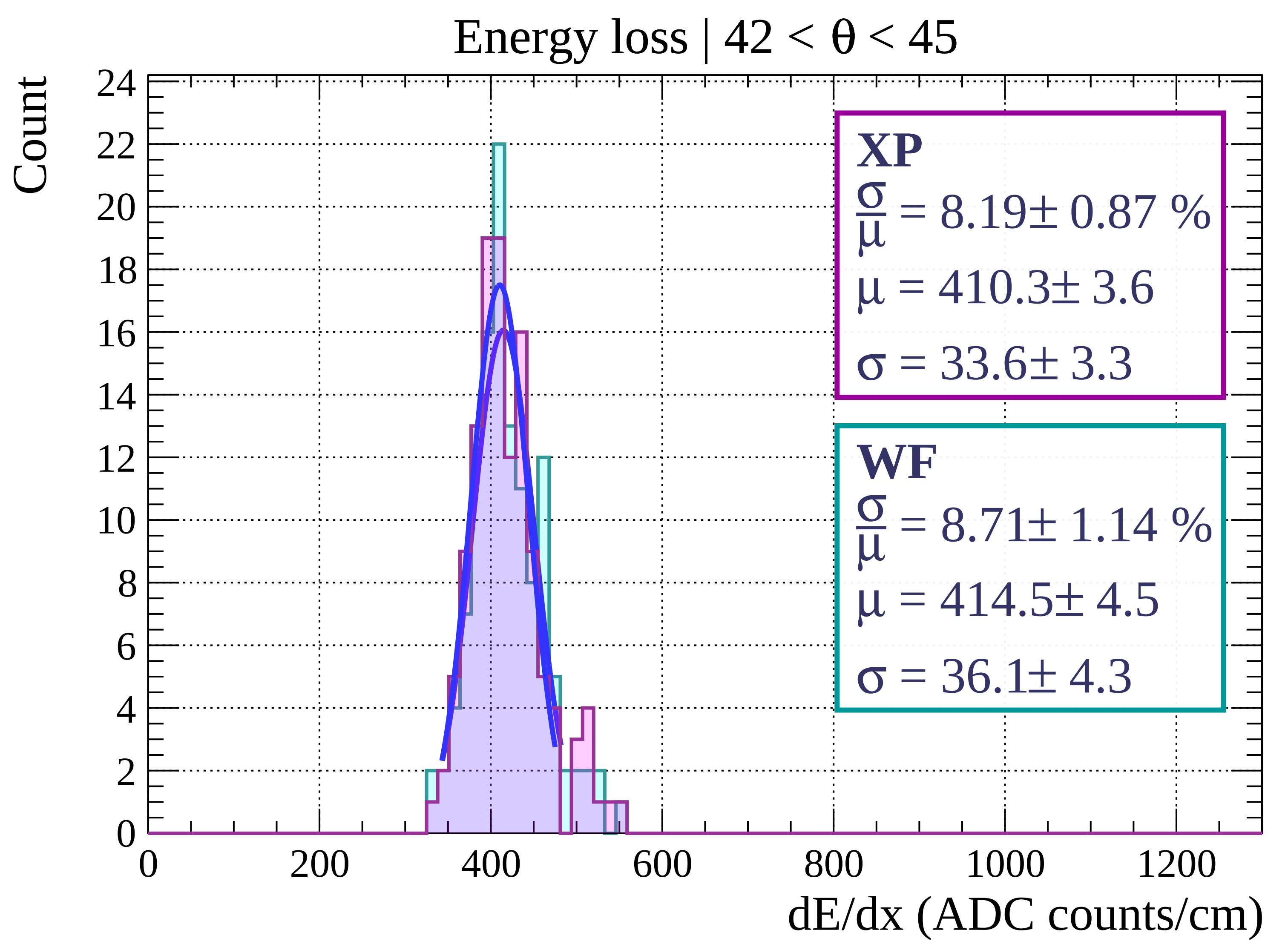
Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 7.89± 0.50 %
μ = 410.9± 1.7
σ = 32.4± 1.9
WF
σ/μ = 8.04± 0.41 %
μ = 415.3± 1.7
σ = 33.4± 1.6


Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 7.53± 0.43 %
μ = 407.0± 1.8
σ = 30.7± 1.6
WF
σ/μ = 8.23± 0.47 %
μ = 412.0± 2.0
σ = 33.9± 1.8

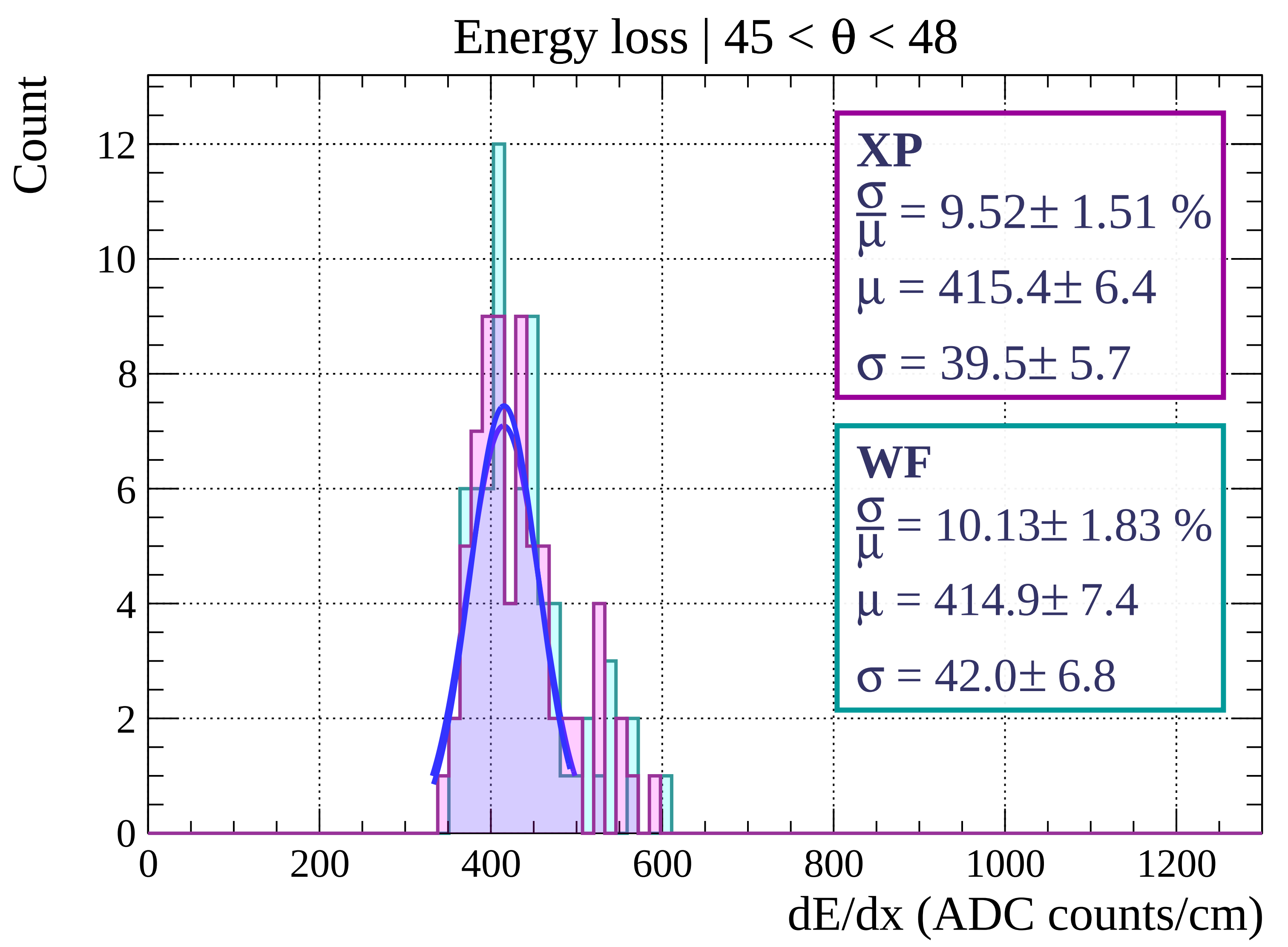


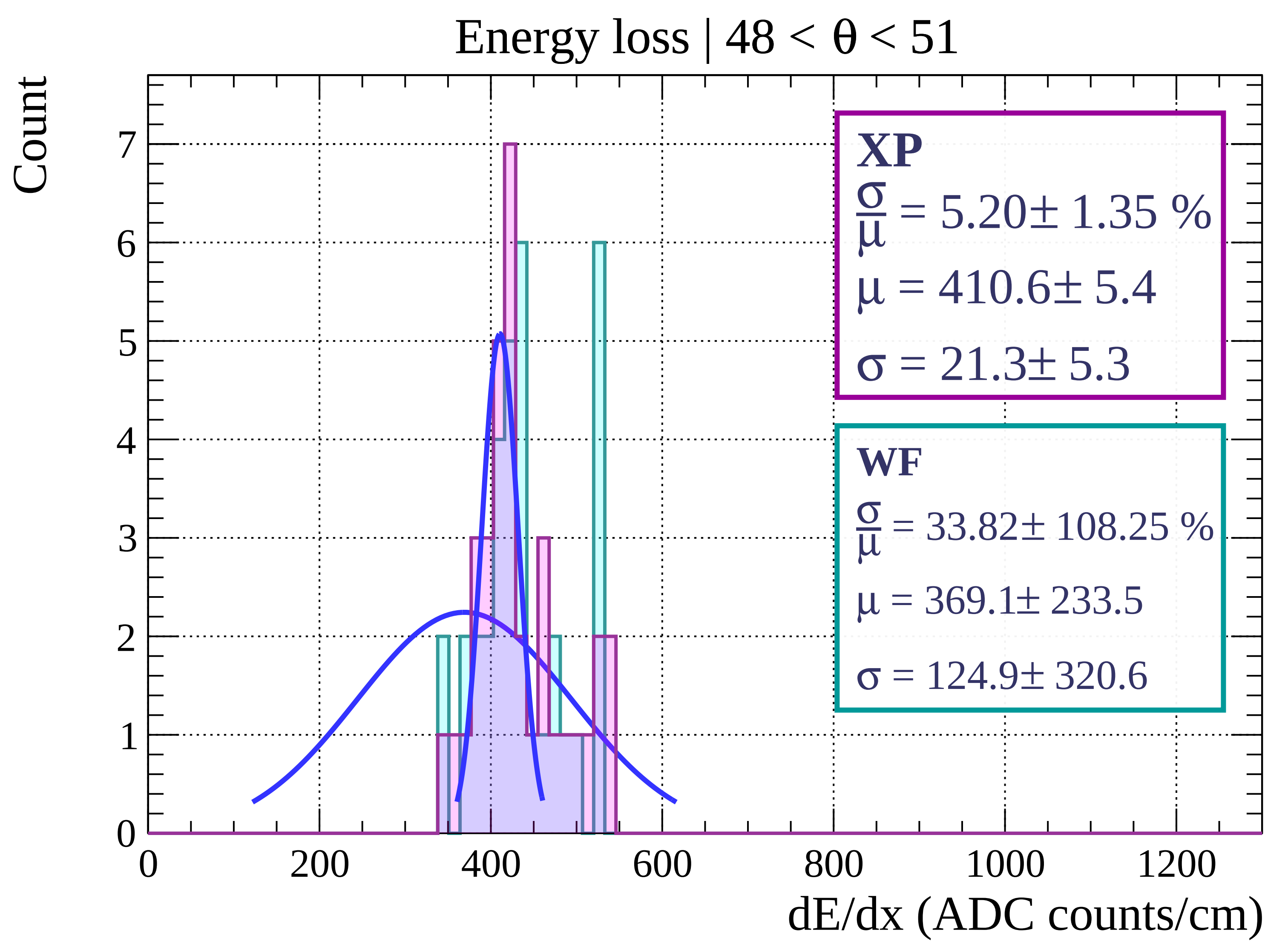
Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.80± 0.69 %
μ = 414.1± 2.8
σ = 36.4± 2.6
WF
σ/μ = 8.77± 0.69 %
μ = 415.8± 2.8
σ = 36.5± 2.6
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200


Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 6.62± 0.47 %
μ = 419.3± 2.5
σ = 27.8± 1.8
WF
σ/μ = 8.00± 0.80 %
μ = 419.1± 3.5
σ = 33.5± 3.1

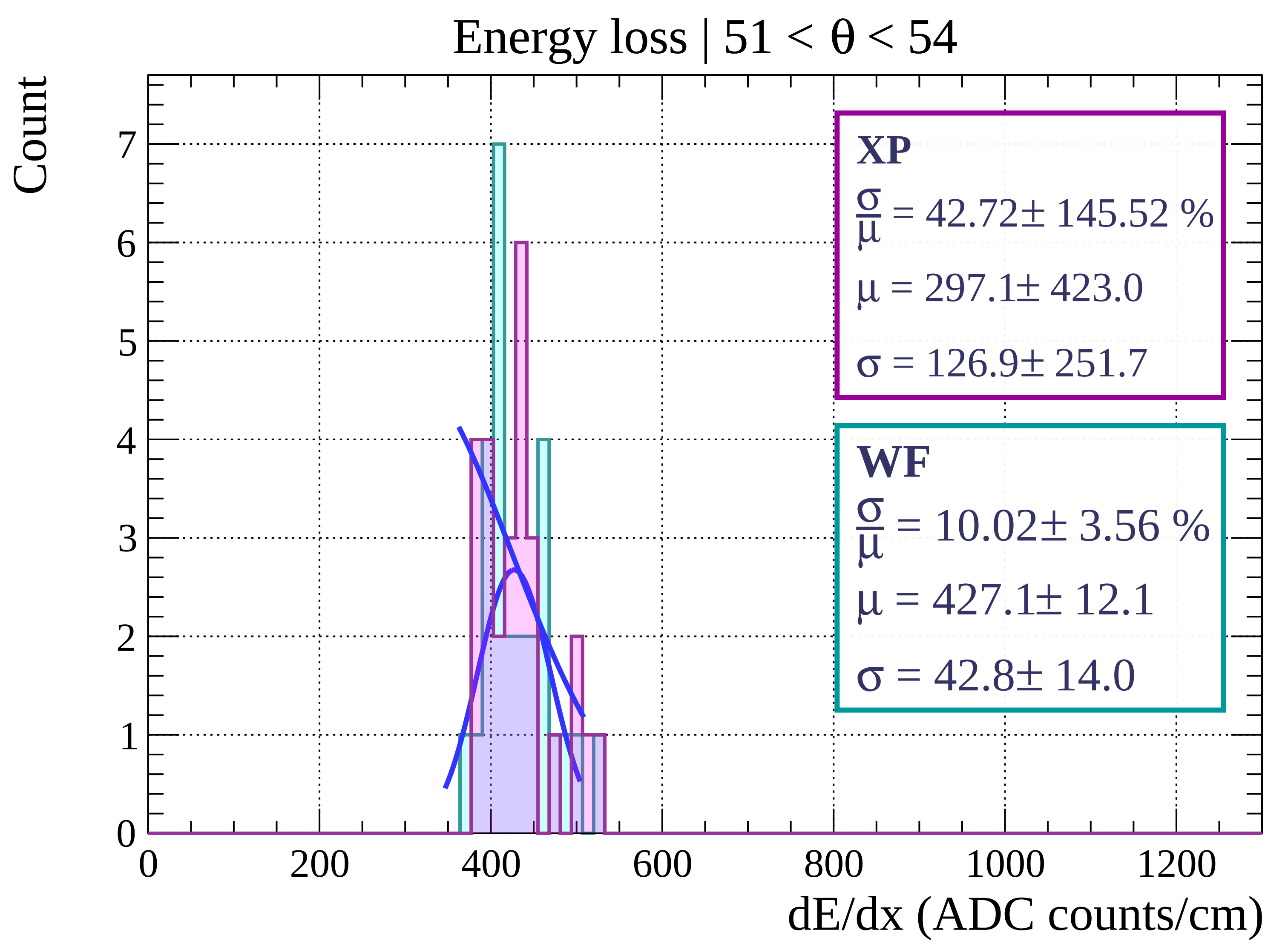


Energy loss | 42 < θ < 45
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
24
XP
σ/μ = 8.19± 0.87 %
μ = 410.3± 3.6
σ = 33.6± 3.3
WF
σ/μ = 8.71± 1.14 %
μ = 414.5± 4.5
σ = 36.1± 4.3

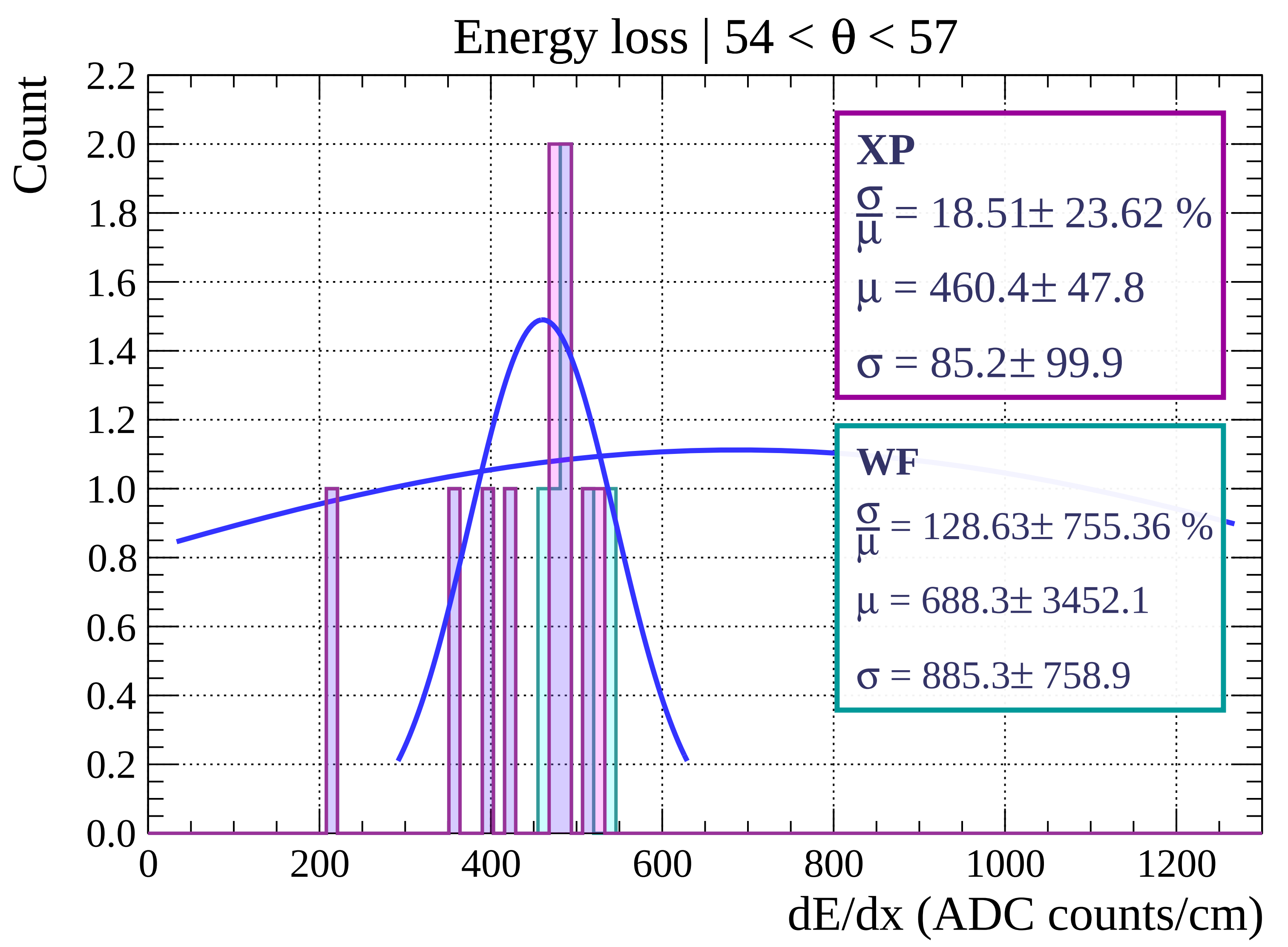

Energy loss | 45 < θ < 48
Count
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.52± 1.51 %
μ = 415.4± 6.4
σ = 39.5± 5.7
WF
σ/μ = 10.13± 1.83 %
μ = 414.9± 7.4
σ = 42.0± 6.8



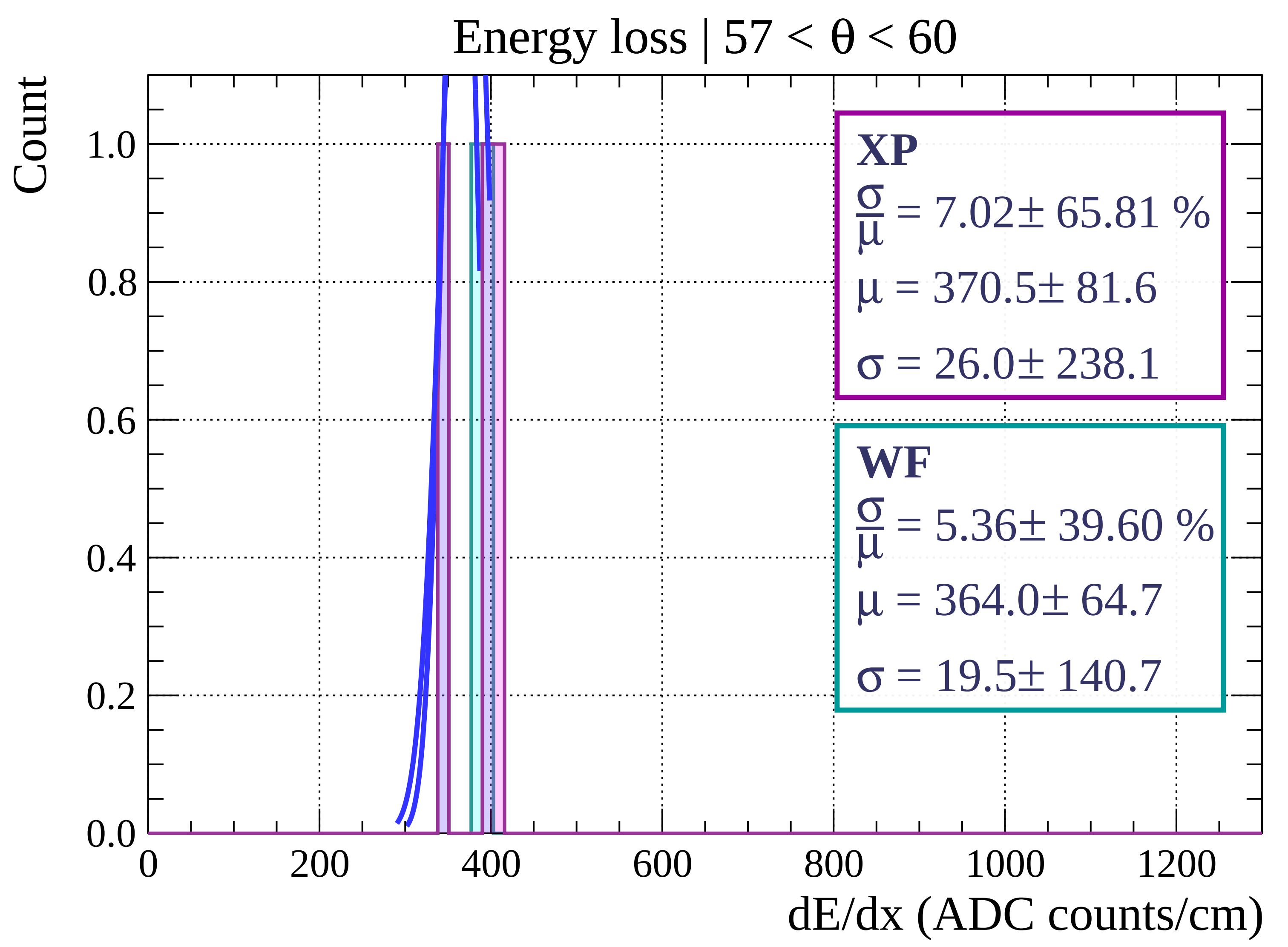
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
σ/μ = 5.20± 1.35 %
μ = 410.6± 5.4
σ = 21.3± 5.3
WF
σ/μ = 33.82± 108.25 %
μ = 369.1± 233.5
σ = 124.9± 320.6



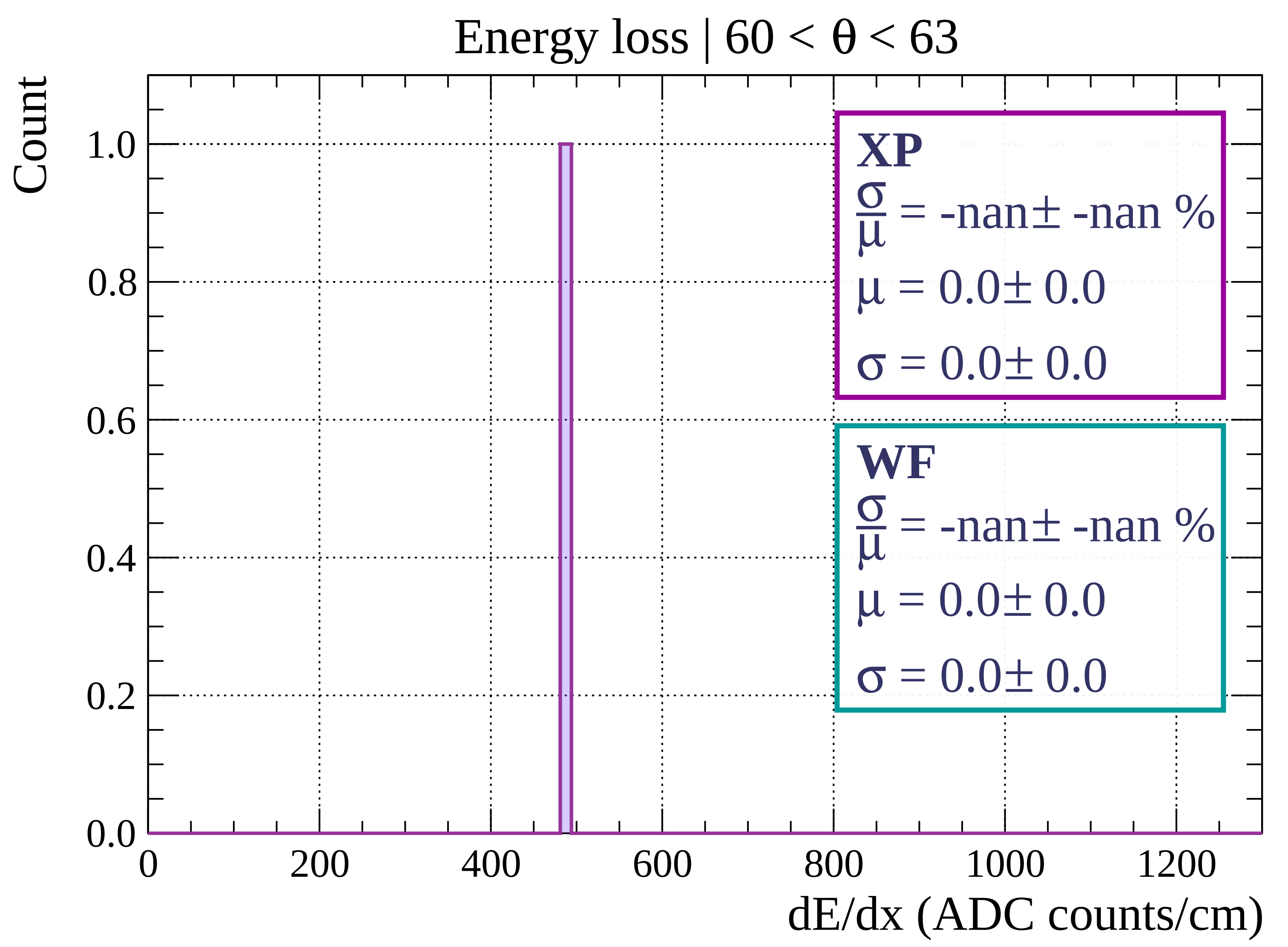
Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
σ/μ = 42.72± 145.52 %
μ = 297.1± 423.0
σ = 126.9± 251.7
WF
σ/μ = 10.02± 3.56 %
μ = 427.1± 12.1
σ = 42.8± 14.0

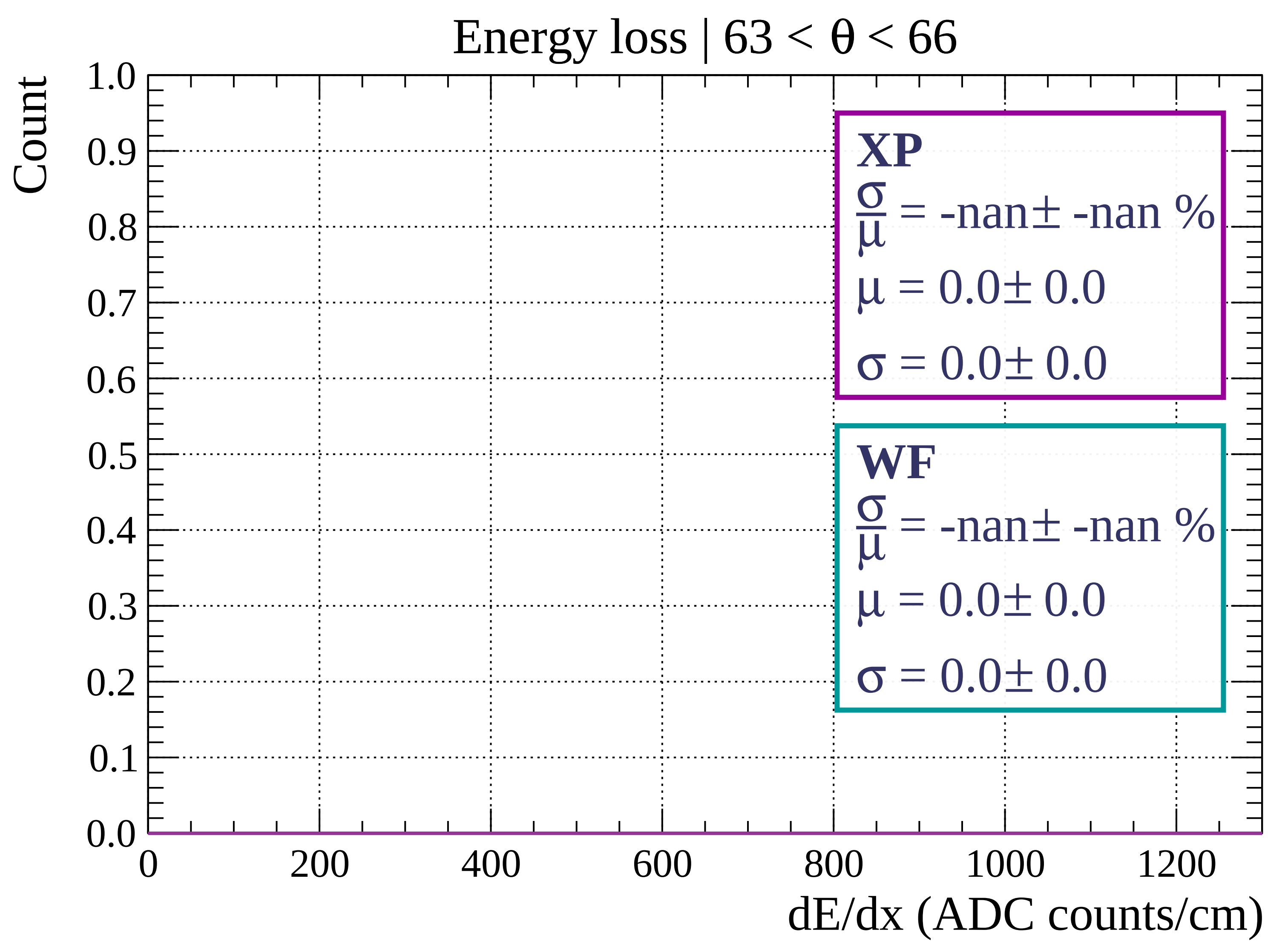


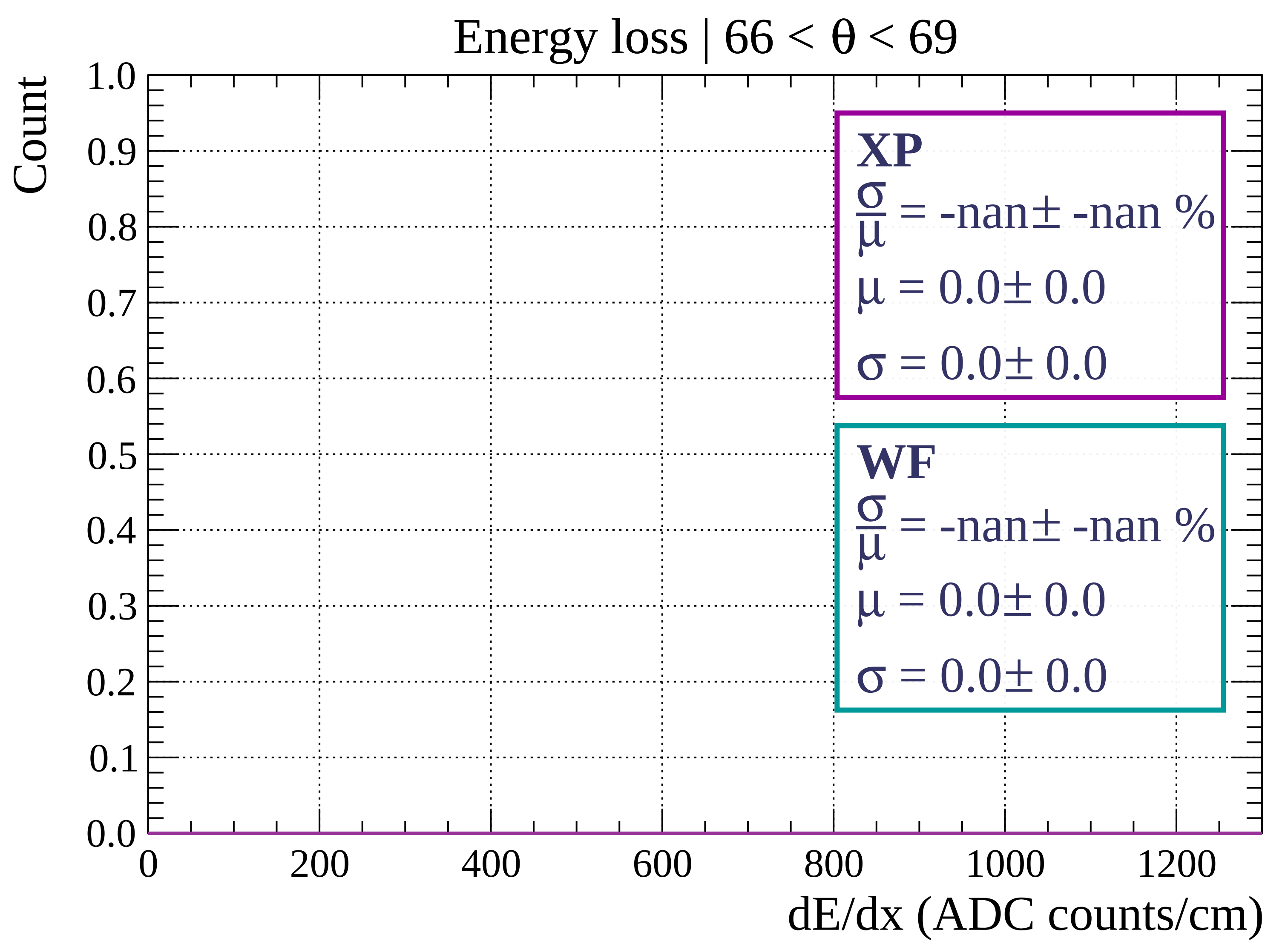
Energy loss | 54 < θ < 57
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 18.51± 23.62 %
μ = 460.4± 47.8
σ = 85.2± 99.9
WF
σ/μ = 128.63± 755.36 %
μ = 688.3± 3452.1
σ = 885.3± 758.9

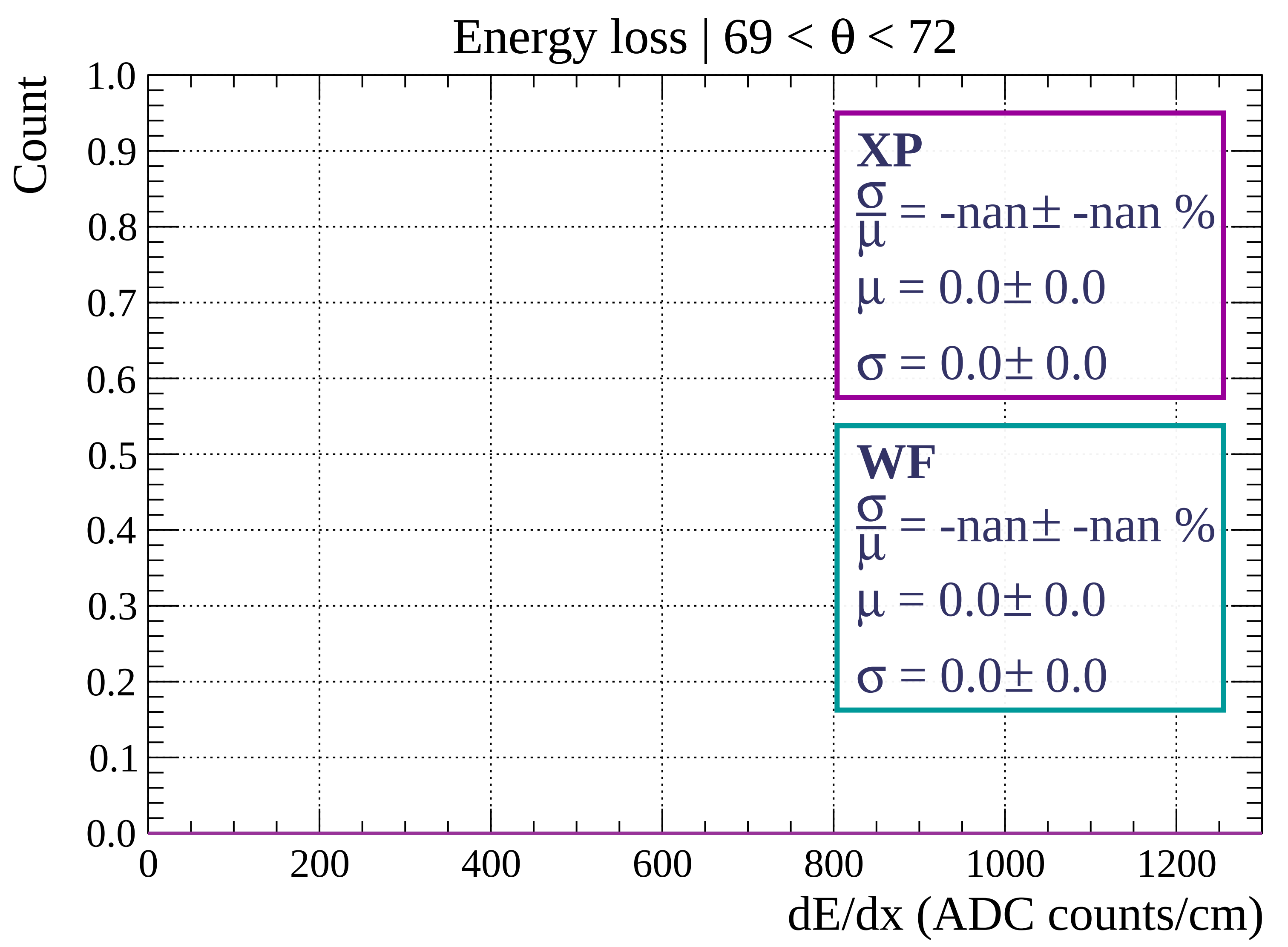


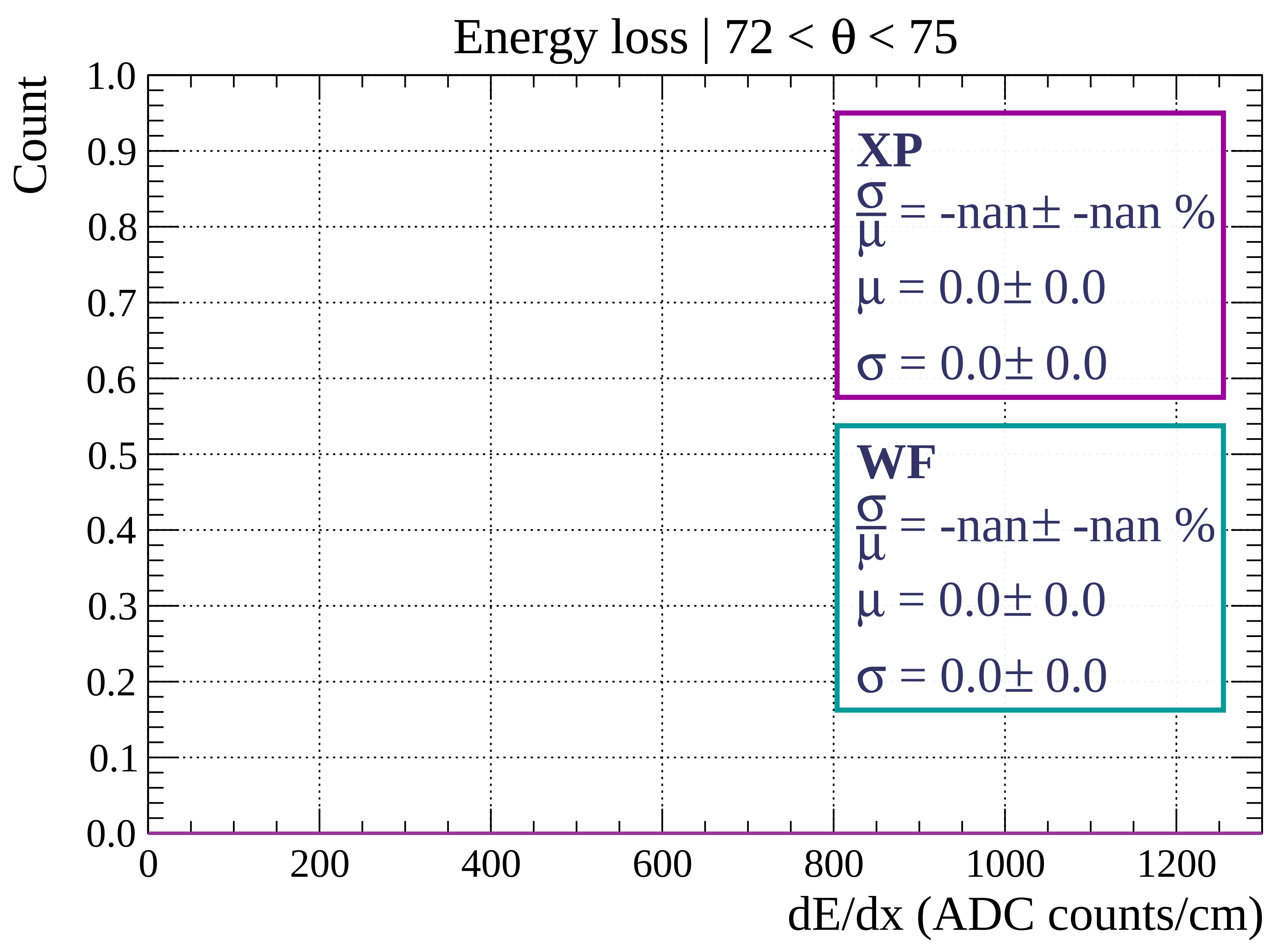
Energy loss | 57 < θ < 60
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.02± 65.81 %
μ = 370.5± 81.6
σ = 26.0± 238.1
WF
σ/μ = 5.36± 39.60 %
μ = 364.0± 64.7
σ = 19.5± 140.7

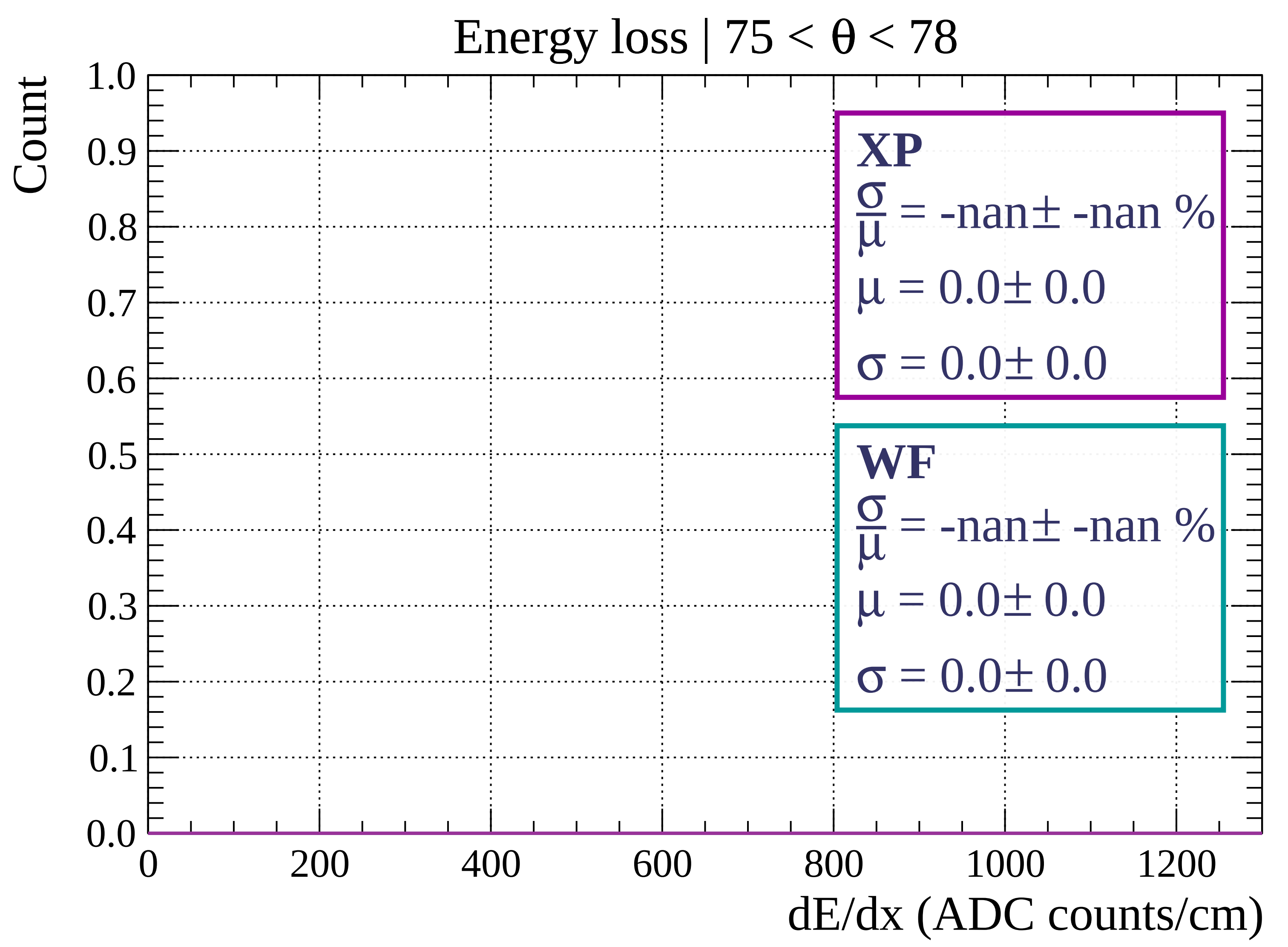

Energy loss | 60 < θ < 63
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

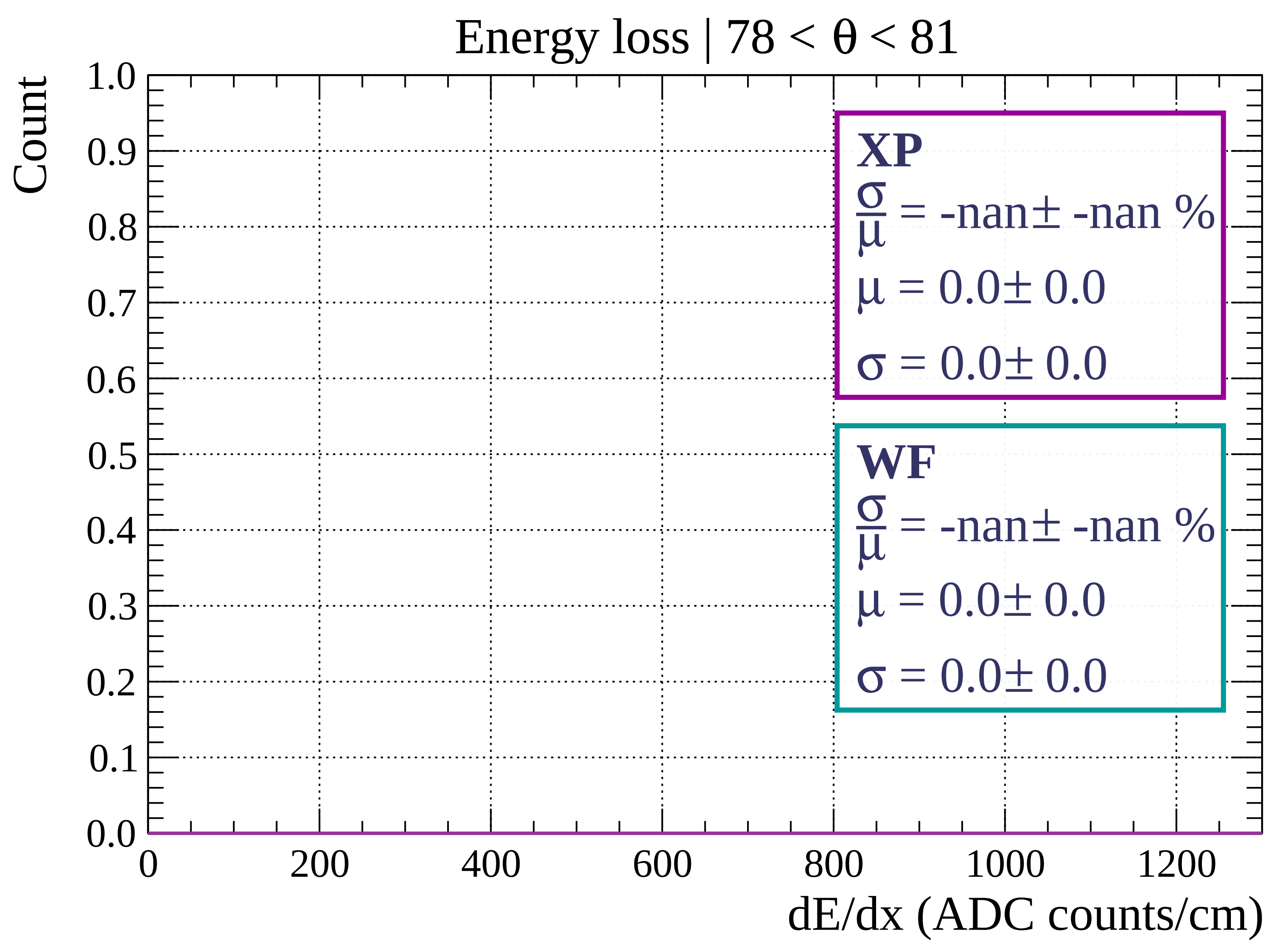

Energy loss | 63 < θ < 66
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200

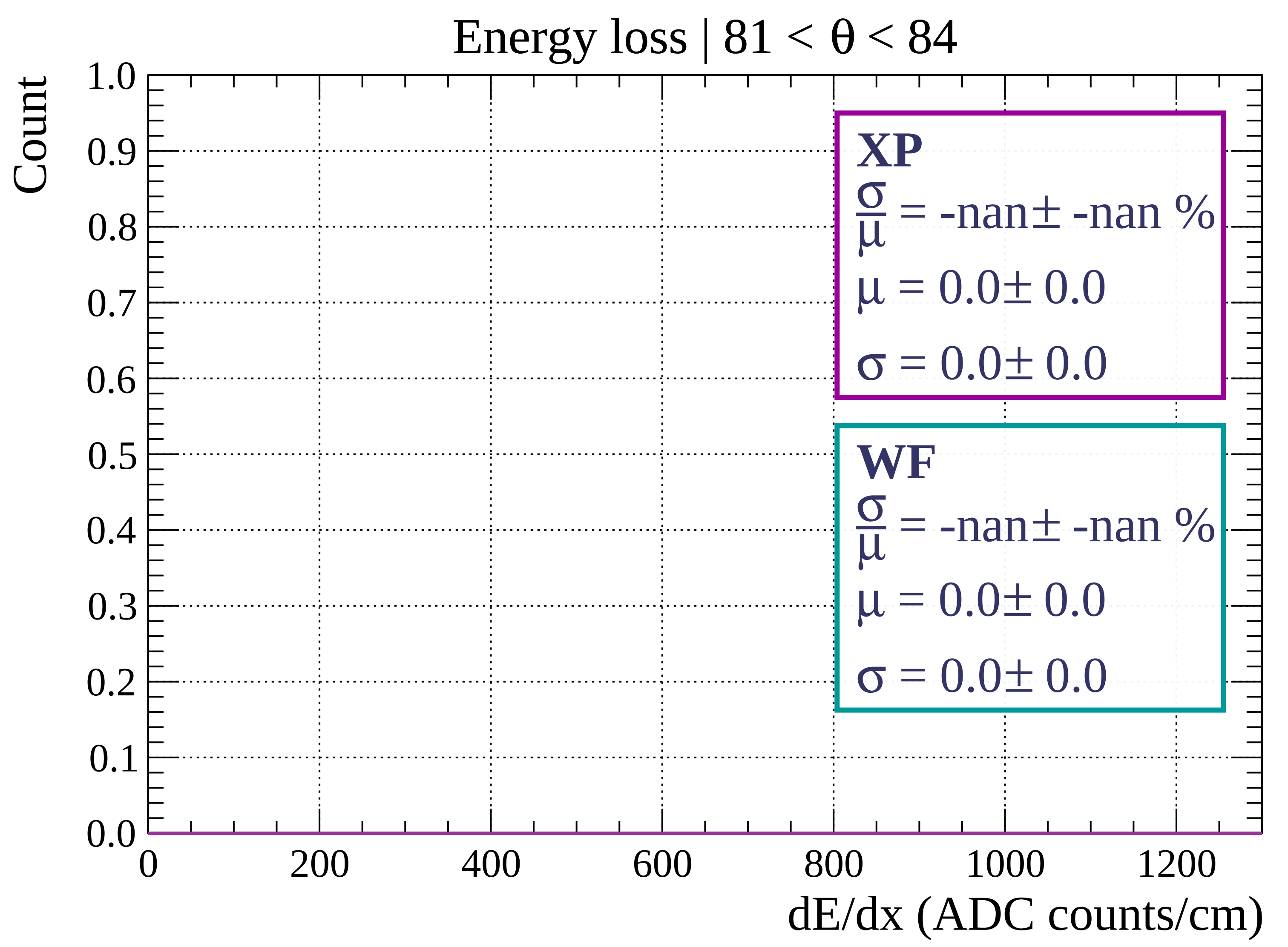

Energy loss | 66 < θ < 69
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 69 < θ < 72
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

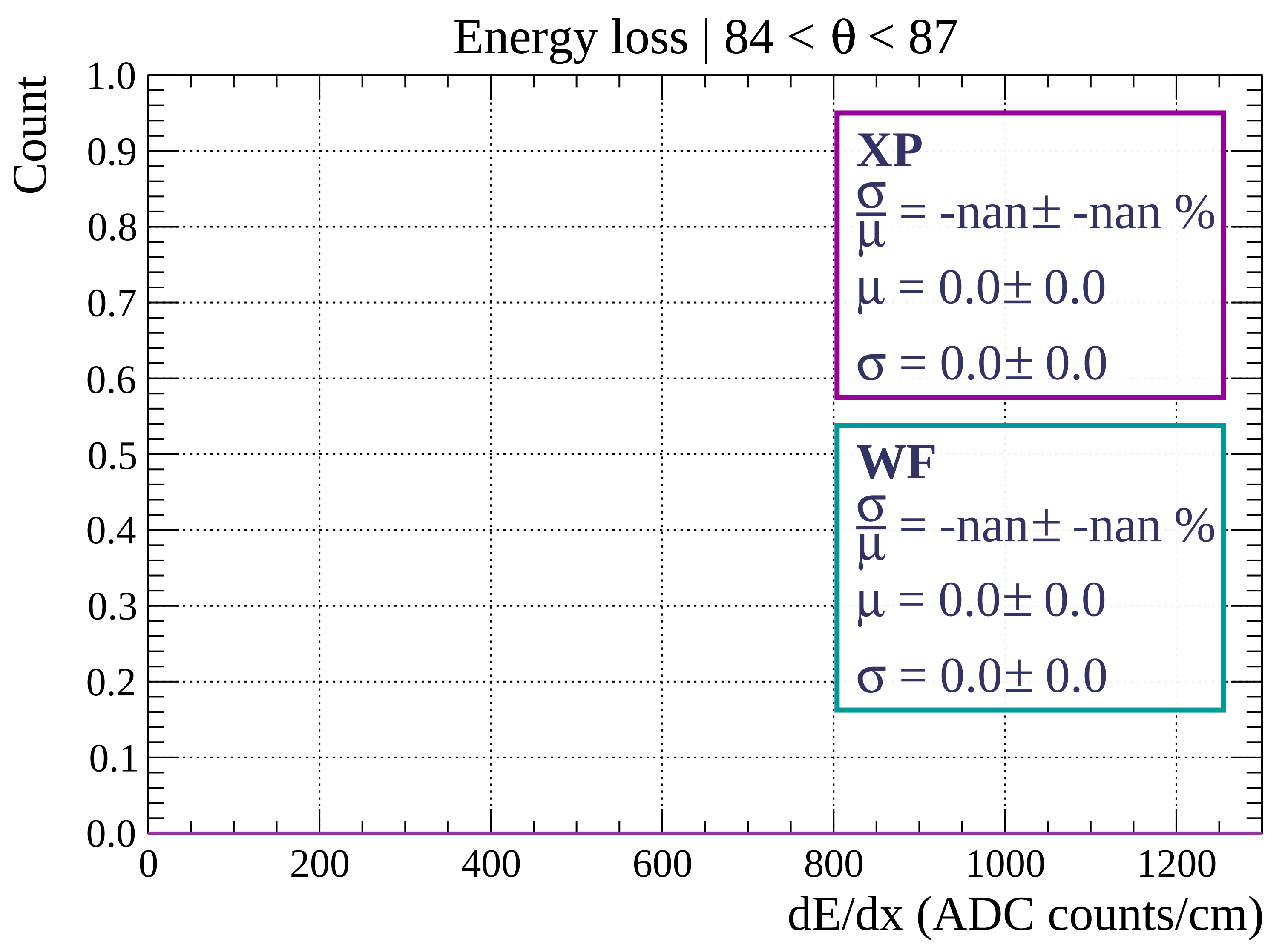

Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

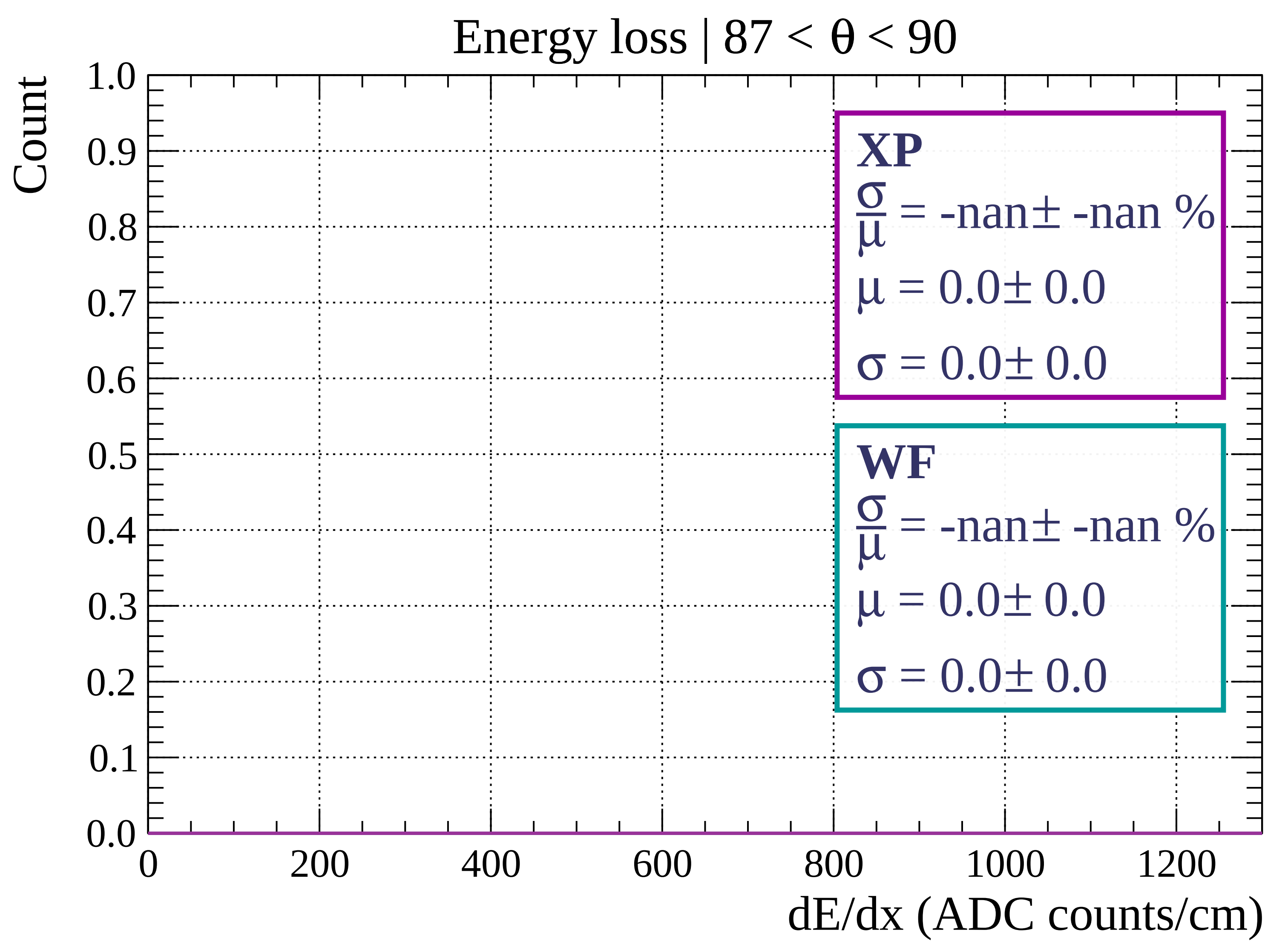

Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

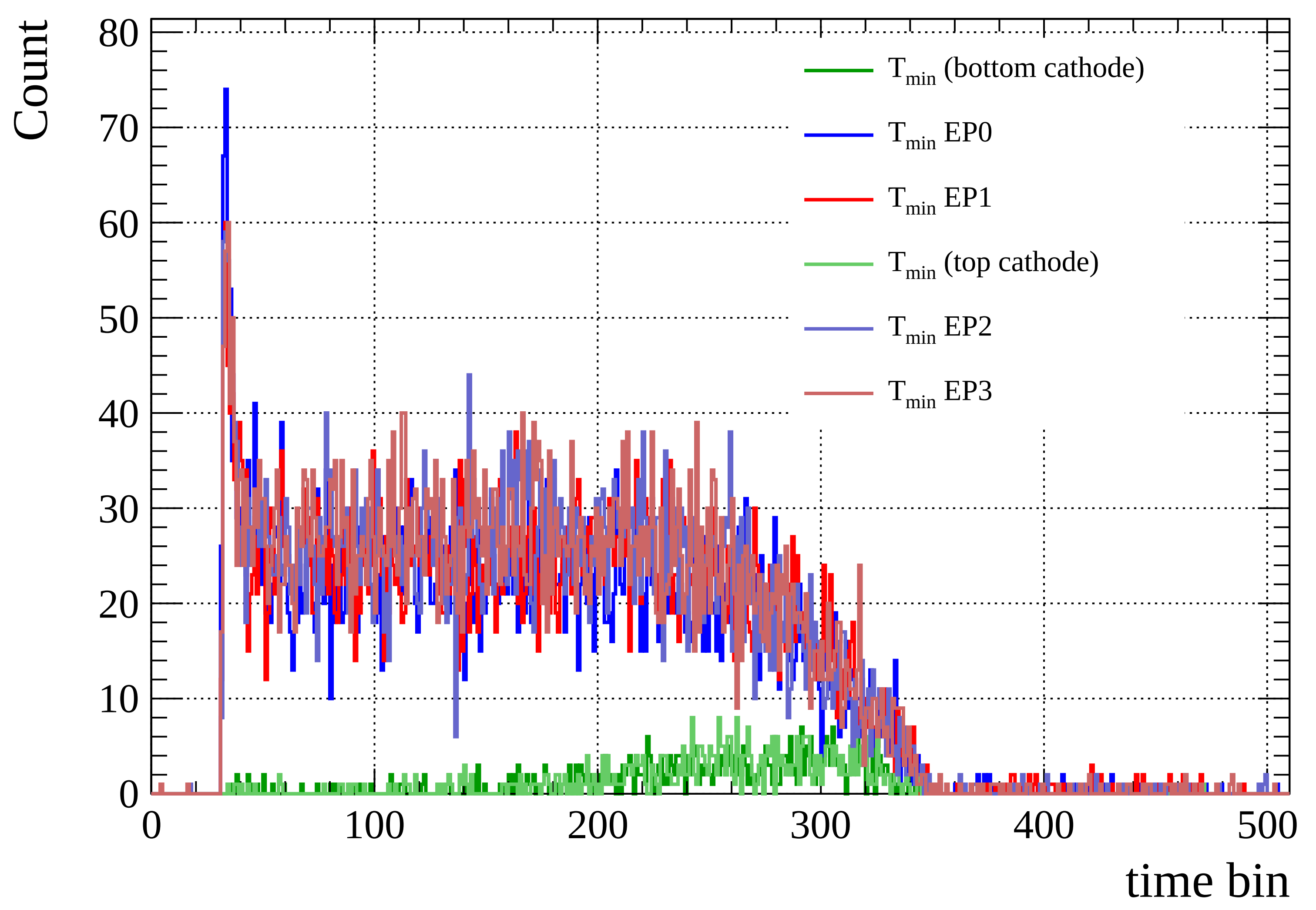

Energy loss | 78 < θ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



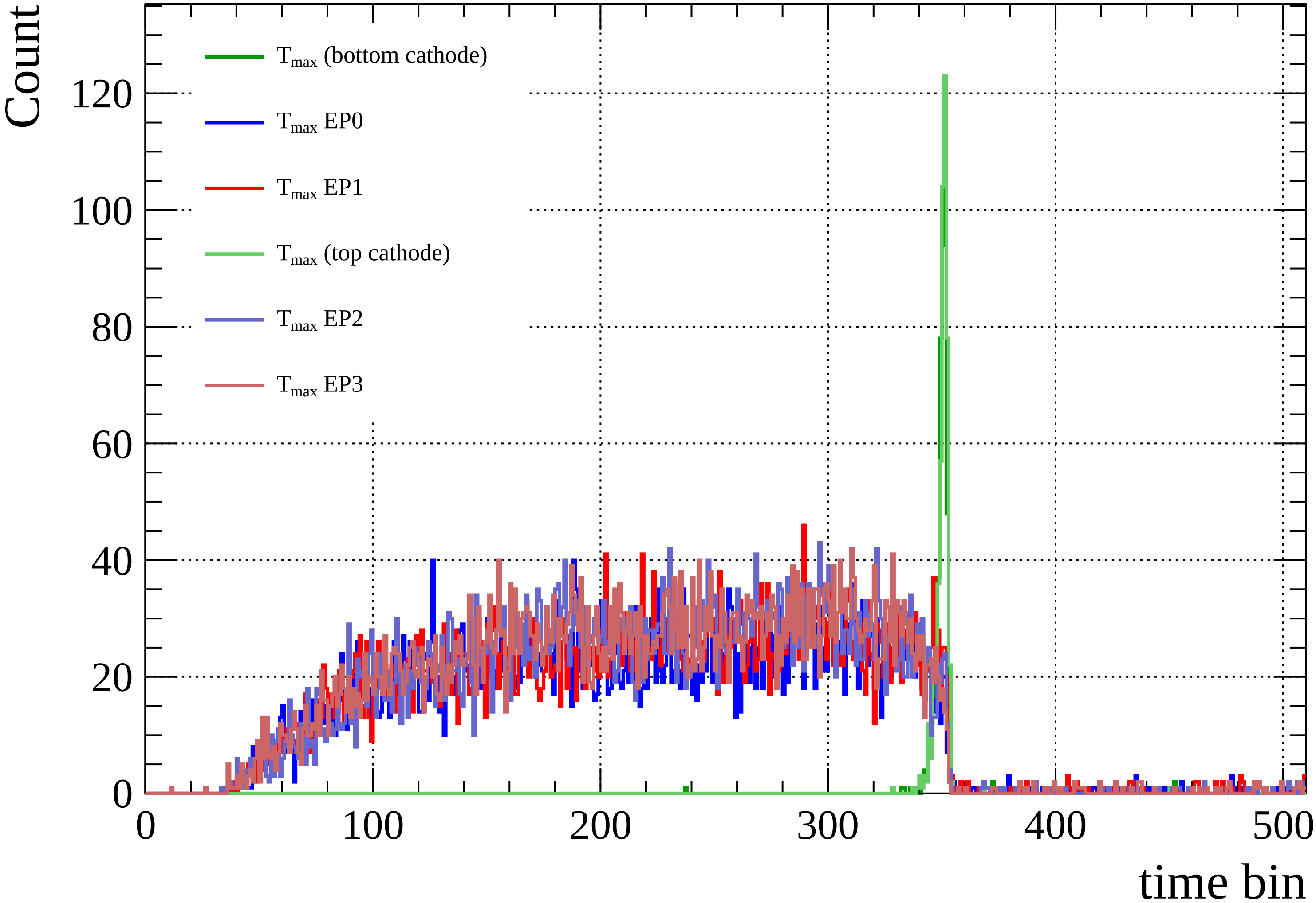
Energy loss | 81 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 84 < θ < 87
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 87 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
100
200
300
400
500
0
10
20
30
40
50
60
70
80
T min (bottom cathode)
T min EP0
T min EP1
T min (top cathode)
T min EP2
T min EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
20
40
60
80
100
120
0
100
200
300
400
500